FORM 1: INTRODUCTION AND EVALUATION FORM

Code : INS2322		**Course Name : STRENGTH of MATERIALS II**						
Year	Semester	Group(s)	Language	Theory	Appl.	Lab.	Credit	ECTS Credit
2015-2016	Autumn	1-2-3-4-5	Turkish, English	2	2	0	3	6
Course Type		Basic Sciences ☐	Basic Engineering ■	Technical Elective ☐		Non-Technical Elective ☐		
Prerequisite		Strength of Materials I						
Coordinator*		Prof. Dr. İrfan COŞKUN						
Instructors		Prof. Dr. Turgut KOCATÜRK, Prof. Dr. İrfan COŞKUN, Doç. Dr. Mesut ŞİMŞEK, Doç. Dr. Ayşe ERDÖLEN, Asist. Prof. Dr. Nihat İLHAN						
Objective		Design of structural systems. Introduction to energy theorems. Stability of bars.						
Content of Lesson		Normal Force with Bending/Core, Materials not Resistant to Tension under the Effect of Eccentric Normal Force/Bending with Shear/Principal Stress Trajectories/Study of Elastic Curve by Various Methods, Analysis of Statically Indeterminate Systems/Bending with Torsion/Energy Methods/Stability.						
Knowledge and Skills		Investigation of combined loading states. Determination of elastic curve. Energy methods, principle of virtual work. Stability of structural elements.						
References		1. Ferdinand P. Beer, E. Russell Johnston, Jr., John T. DeWolf, David F. Mazurek, Mechanics of Materials, Fifth edition, Mc Graw Hill. 2. Singer F. I, Strength of Materials, 3. Gere, J. M., Goodno B. J., Mechanics of Materials, Seventh edition, Cengage Learning. 4. İNAN, M., ‘‘Cisimlerin Mukavemeti ’’, İTÜ Vakfı, İstanbul, 2001. 5. BAKİOĞLU, M., ‘‘Cisimlerin Mukavemeti’’, Beta Yayınevi, İstanbul, 2001. 6. ÖZTÜRK, A. Z., ÇAĞDAŞ, S., ‘‘Mukavemet’’, Murat Ders Yayınları, İstanbul, 1982. 7. OMURTAG, M. H., ‘‘Mukavemet’’, Birsen Yayınevi, İstanbul, 2005.						
Homeworks and Projects		-						
Laboratory Experiment Topics		-						
Computer Codes		-						
Other Activities		-						

SUCCESS EVALUATION

Theoretical Courses			**Projects**		
	Number	Weight(%)		Number	Weight(%)
Midterms	2	50	Midterms	-	
Quizzes	-		Controls	-	
Homeworks	At least 2	10	Mid-submission	-	
Laboratory	-		Oral Examination	-	
Others	-		Others	-	
Final Exam	1	40	Final Exam	-	

Prepared by Prof. Dr. Turgut KOCATÜRK Date: 10.02.2016

FORM 2: COURSE SCHEDULE

Code : INS2322	**Course Name : STRENGTH of MATERIALS II**
Instructors	Prof. Dr. Turgut KOCATÜRK, Prof. Dr. İrfan COŞKUN, Doç. Dr. Mesut ŞİMŞEK, Doç. Dr. Ayşe ERDÖLEN, Asist. Prof. Dr. Nihat İLHAN
1. Week	Combined loadings, axial force and bending moment.
2. Week	Core of the sections, eccentric normal force in the case of materials which do not have tension strength.
3. Week	Determination of stresses and strains in the case of bending moments with shearing force.
4. Week	Design in the case of bending moments and shearing force. Directions of principal stresses.
5. Week	Elastic curve. Method of differential equation.
6. Week	Method of Mohr, Deflections by the method of superposition.
7. Week	Hiperstatic systems.
8. Week	**First midterm exam**
9. Week	Bending moment with torsion.
10. Week	Energy theorems.
11. Week	Energy theorems.
12. Week	Elastic stability. Buckling of bars.
13. Week	**Second midterm exam**
14. Week	Critical load for various end conditions. Euler hyperbola. Buckling in the plastic region.
15. Week	Critical load for various end conditions. Euler hyperbola. Buckling in the plastic region.

YILDIZ TECHNICAL UNIVERSITY
CIVIL ENGINEERING DEPARTMENT
MECHANICS DIVISION

STRENGTH OF MATERIALS II

COMBİNED LOADING

FIRST CHAPTER

NORMAL FORCE AND BENDING

3 Hours, First in Line

PROF. DR. TURGUT KOCATÜRK

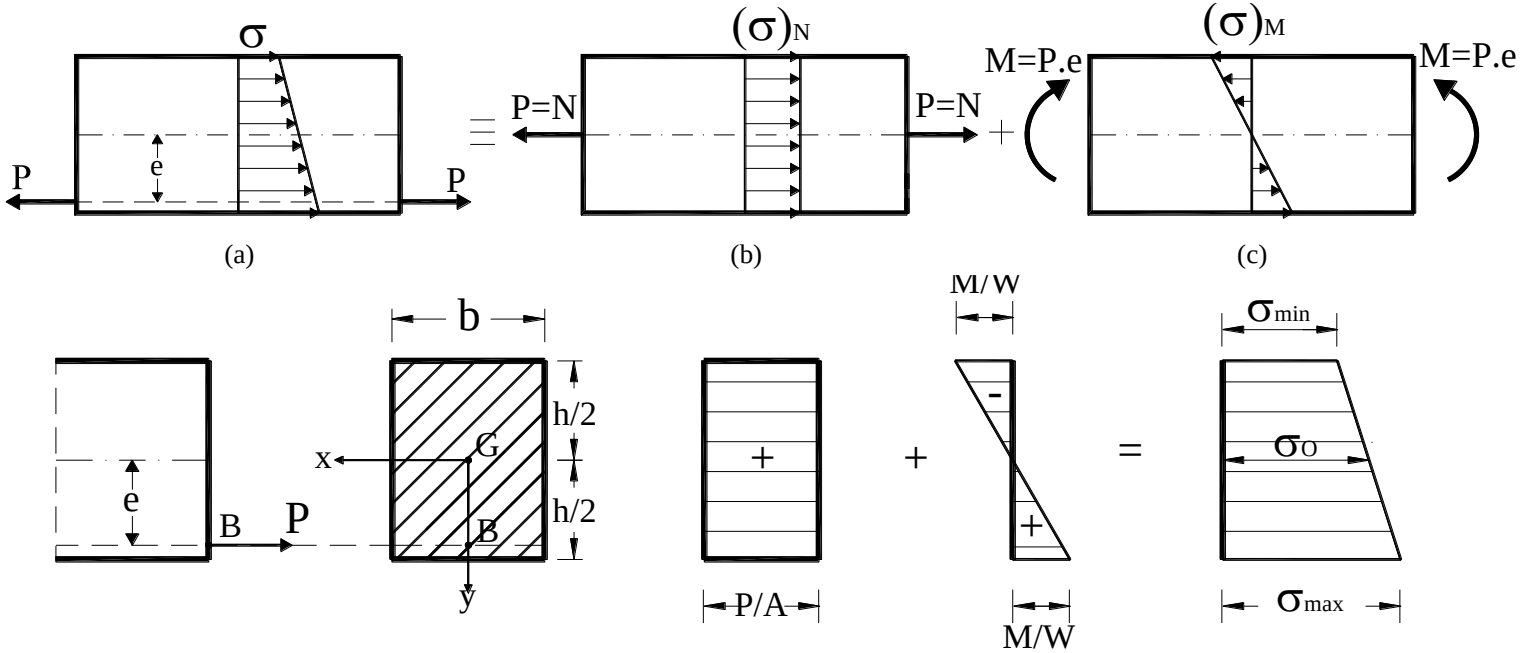
In the combined loading, there are more than one internal loading going on.

1- Eccentric Normal Force

Axial normal force (N) and bending moment (M) acting together in the cross-section of the beam occurs in the case of eccentric normal force.

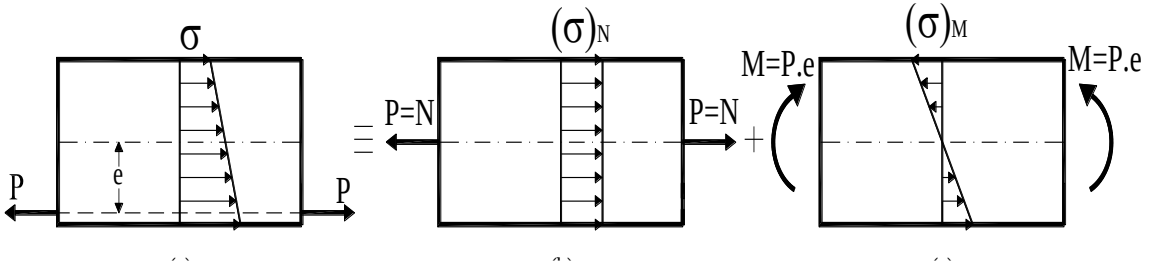
If the normal force acting on the beam is not on the axis of the beam, it is called the eccentric normal force case. Such a force when brought to the axis of the beam always produces an *axial normal force and a bending moment.*

As shown in the figure below, "e" is the distance between applied force point and the axis of the beam. When we move the force P to the axis of the beam, this process produces the bending moment P.e and the normal force P.



If the eccentric normal force is reduced to two simple states, then the stress can be calculated by using pure bending and normal force states. Although the stress in the case of axial normal force is uniformly distributed through the cross-section, in the case of pure bending moment, stresses have two signs as tensile stresses on one side of the cross-section and compressive stresses on the other side of the cross section.

Bending stresses + axial normal stresses = Single or double sign of stress distribution depending on the eccentricity.

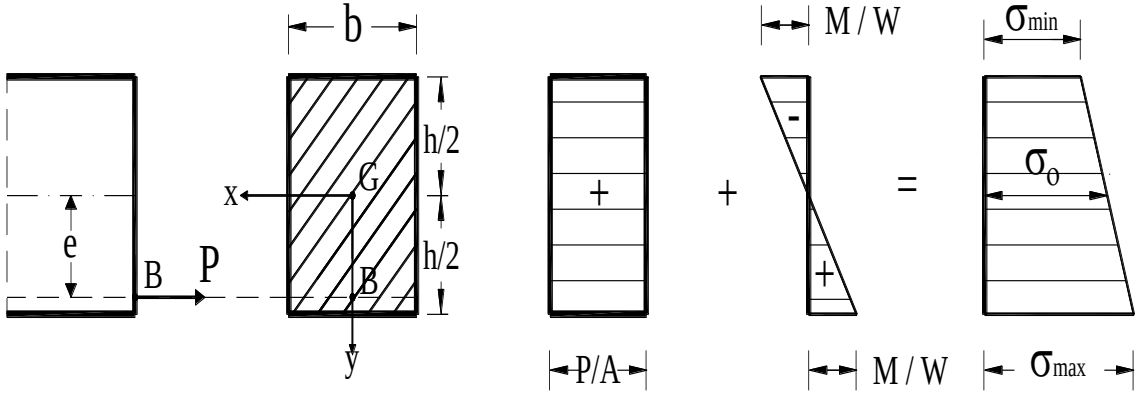


$$\sigma = \frac{N}{A} + \frac{M_x}{I_x}.y \qquad \sigma = \frac{P}{bh} + \frac{12Pe}{bh^3}.y$$

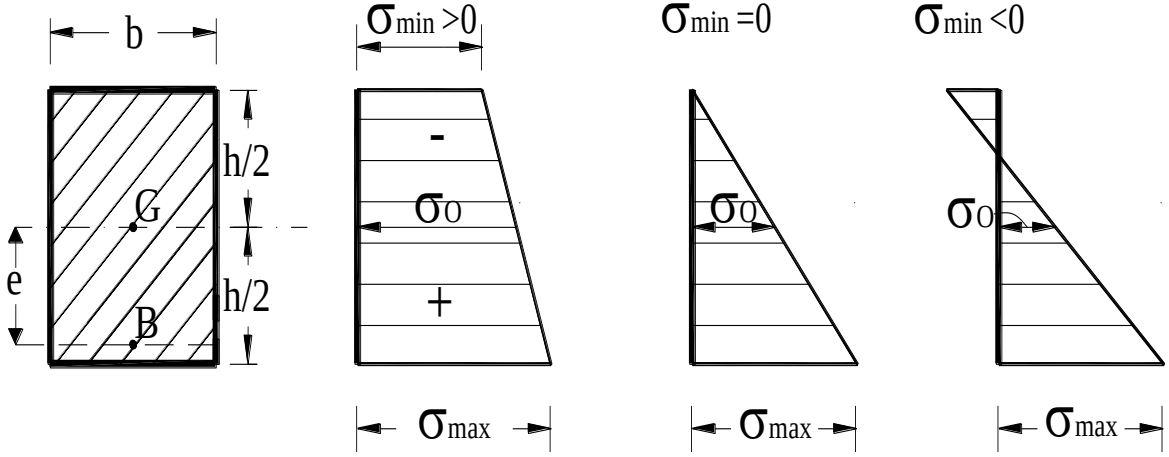
$y = \pm h/2$ **for**

$$\sigma_{max} = \frac{P}{bh}\left(1 + \frac{6e}{h}\right)$$

$$\sigma_{min} = \frac{P}{bh}\left(1 - \frac{6e}{h}\right)$$

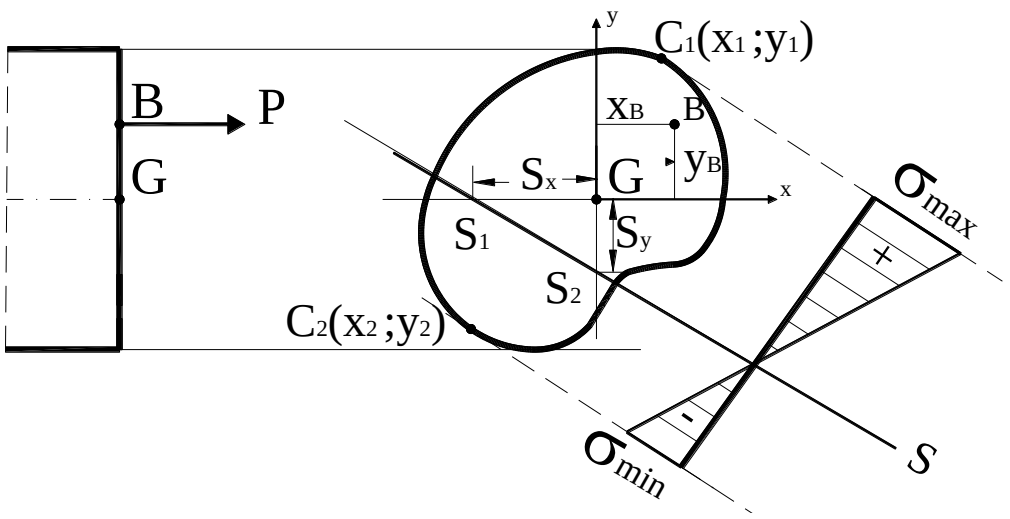


Stresses on neutral axis are always, $\sigma_0 = P/A$**, not depending on the eccentricity. Stress** σ_{min} **could be tension, zero, or compression**



2- General Calculation of State of Stress

In the previous section, an investigation was made for the special section of rectangular shape, and now the investigation will be made for an arbitrary cross-section (Figure 4).



$$i_x^2 = \frac{I_x}{A} \quad \text{and} \quad i_y^2 = \frac{I_y}{A}$$

x, y: principal axes

G: center of gravity

S-S: neutral axis

P eccentric force brought to the point B:

$$N = P;\ M_x = P.y_B;\ M_y = -P.x_B$$

any point x, y is

$$\sigma_z(x,y) = \frac{N}{A} + \frac{M_x}{I_x}y - \frac{M_y}{I_y}x = \frac{P}{A} + \frac{P.y_B.y}{I_x} + \frac{P.x_B.x}{I_y}$$

There is need to determine the position of the neutral axis.

$$\sigma_{max} = \frac{P}{A} + \frac{Py_B}{I_x}.y_1 + \frac{Px_B}{I_y}.x_1$$

$$\sigma_{min} = \frac{P}{A} + \frac{Py_B}{I_x}.y_2 + \frac{Px_B}{I_y}.x_2$$

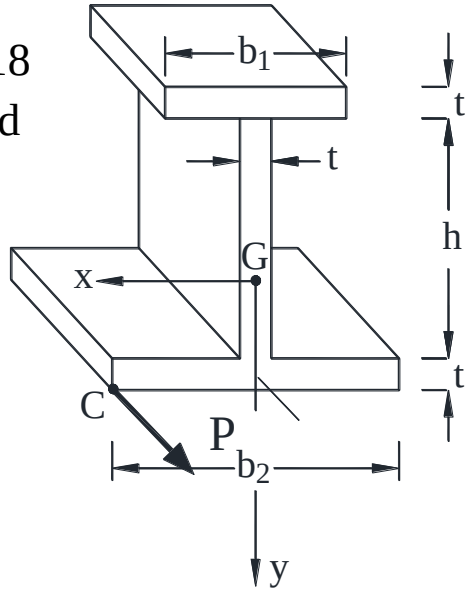
From $\sigma(x,y) = 0$ **, the equation of the neutral axis can be written as follows:**

$$\sigma_z(x,y) = \frac{P}{A} + \frac{P.y_B.y}{I_x} + \frac{P.x_B.x}{I_y} = \frac{P}{A}\left(1 + \frac{y_B y}{i_x^2} + \frac{x_B x}{i_y^2}\right) = 0$$

$$\left(1 + \frac{y_B y}{i_x^2} + \frac{x_B x}{i_y^2}\right) = 0 \rightarrow when\ x = 0\ then\ y\ is\ called\ S_y\ and\ \left(1 + \frac{y_B S_y}{i_x^2}\right) = 0 \rightarrow S_y = \frac{-i_x^2}{y_B}$$

$$\left(1 + \frac{y_B y}{i_x^2} + \frac{x_B x}{i_y^2}\right) = 0 \rightarrow when\ y = 0\ then\ x\ is\ called\ S_x\ and\ \left(1 + \frac{x_B S_x}{i_y^2}\right) = 0 \rightarrow S_x = \frac{-i_y^2}{x_B}$$

Problem: *P*=170 kN eccentric tensile force is acting at point C of the cross-section given below. Cross-sectional dimensions are b_1=12 cm, b_2=18 cm, h=18 cm, and t=2 cm, determine the max. and min. normal stresses at the section and draw the normal stress diagram.



Solution:

$$y_G = \frac{12.2.21 + 18.2.11 + 18.2.1}{12.2 + 2.18.2} = 9.75\ cm$$

$$I_x = \frac{18.2^3}{12} + 18.2.(1-9.75)^2 + \frac{2.18^3}{12} + 2.18.(11-9.75)^2 + \frac{12.2^3}{12} + 12.2.(21-9.75)^2 = 6842\ cm^4$$

$$I_y = \frac{2.12^3}{12} + \frac{2.18^3}{12} + \frac{18.2^3}{12} = 1272\ cm^4, \quad A = 96\ cm^4$$

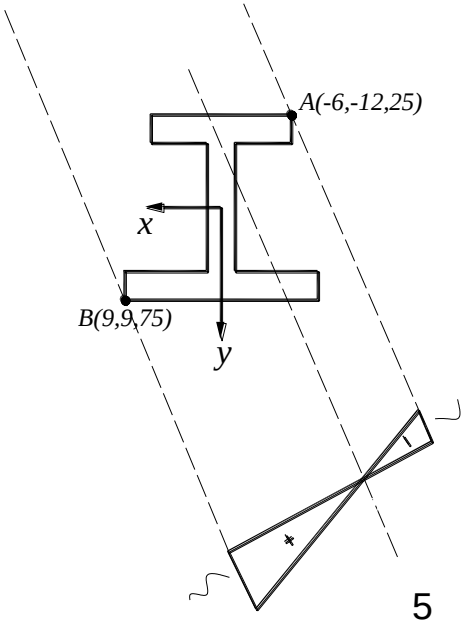
$$M_x = 170000.9,75 = 1657000\ Ncm$$

$$M_y = -170000.9 = -1530000\ Ncm$$

$$\sigma_z = \frac{170000}{96} + \frac{1657000}{6842}y + \frac{1530000}{1272}x \Rightarrow From\ \sigma_z = 0 \Rightarrow$$

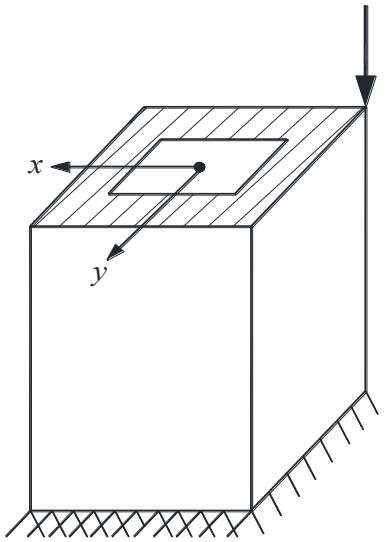
$$y = -4.96x - 7.314\ (Neutral\ axis)$$

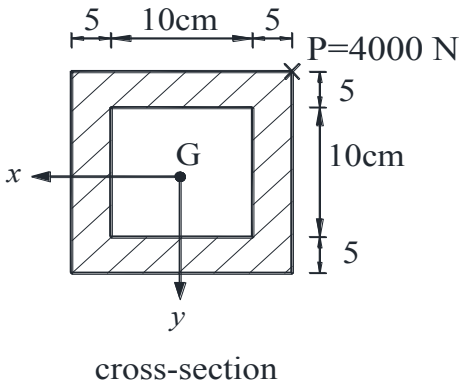
$$\sigma_z^A(-6;12,25) = \sigma_{\min} = -8411.69\ N/cm^2, \quad \sigma_z^B(9;9,75) = \sigma_{\max} = 14956\ N/cm^2$$



Problem: A column with known dimensions is under compression of eccentric load, *P=4000 N* as seen in the figure. Draw the normal stress distribution over cross section of the column.

$$\sigma_z(x,y) = \frac{N}{A} + \frac{M_x}{I_x}y - \frac{M_y}{I_y}x = \frac{P}{A} + \frac{P.y_B.y}{I_x} + \frac{P.x_B.x}{I_y}$$





cross-section

Solution:

$P = -4000N$

$M_x = +4000.10 = +40000Ncm$

$M_y = -4000.10 = -40000Ncm$

$I_x = I_y = \frac{20.20^3}{12} - \frac{10.10^3}{12} = 12500cm^4$

$A = 20^2 - 10^2 = 300cm^2$

$\sigma_z = \frac{-4000}{300} + \frac{40000}{12500}y - \frac{-40000}{12500}x$

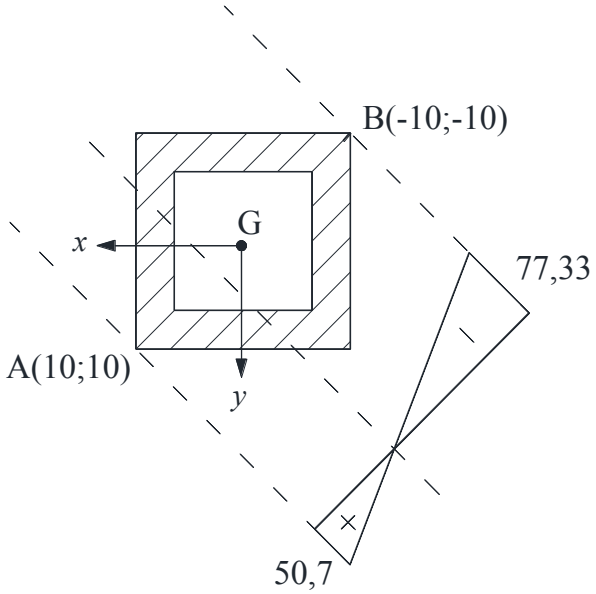
$\sigma_z = 0 \Rightarrow y = -x + 4{,}156 \Rightarrow$

neutral axis:

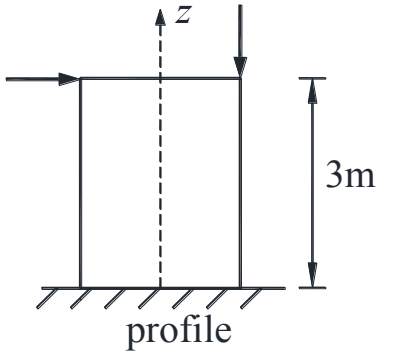
$(0;4{,}156),\ (4{,}156;0)$

$(\sigma_z)_A = 50{,}77N/cm^2$

$(\sigma_z)_B = -77{,}33N/cm^2$



Problem: P and Q forces are acting on the square cross section of the column as seen in the figure. Determine the neutral axis and draw the normal stress distribution of the cross section at the base of the column.



Solution

$$N = -200kN$$

$$M_x = 0$$

$$M_y = -3.50 - 0,5.200 = -250kNm$$

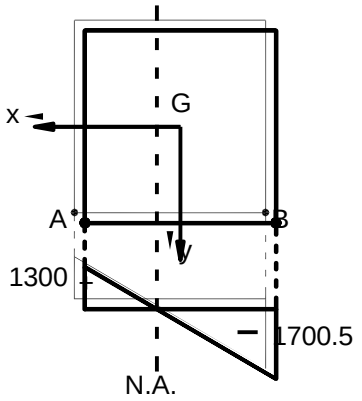
$$A = 1m^2 \qquad I = \frac{1.1^3}{12} = 0,0833m^4$$

$$\sigma_z = \frac{-200}{1} + \frac{250}{0,0833}x \quad \Rightarrow \quad \sigma_z = -200 + 3001x$$

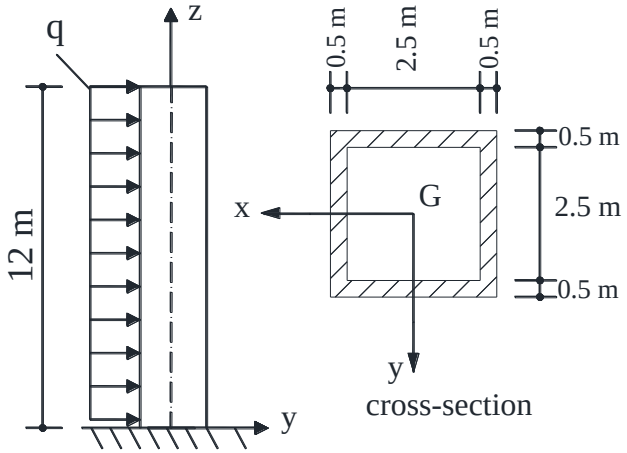
$$\sigma_z = 0 \quad \Rightarrow \quad x = +0,066$$

$$(\sigma_z)_A = -200 + 3001.0,5 = 1300\,kN/m^2$$

$$(\sigma_z)_B = -200 + 3001.(-0,5) = -1700,5\,kN/m^2$$



Problem: The weight density (specific weight) of the material used to construct the chimney shown in the figure is $\gamma = 18 kN/m^3$. Find the magnitude of the distributed load q (kN/m^2) such that no tension occurs at the base of chimney.



Çözüm:

$$W = (3,5^2 - 2,5^2).12.18 = 1296\,kN$$

$$M = q.12.3,5.6 = -252\,q\,kNm$$

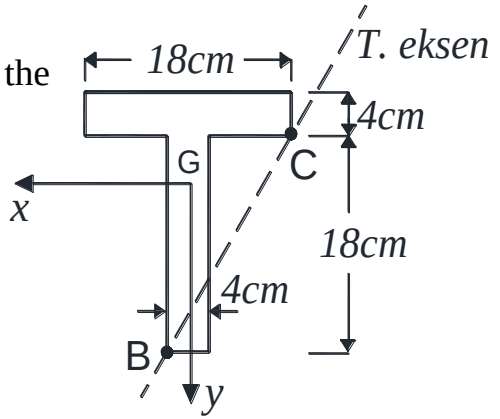
$$A = 3,5^2 - 2,5^2 = 6\,m^2,\ I_x = \frac{3,5^4 - 2,5^4}{12} = 9,25\,m^4$$

$$\sigma_z = \frac{-1296}{6} + \frac{252\,q}{9,25}(1,75) = 0 \rightarrow q = 4,51 kN/m^2$$

Problem: Knowing that the neutral axis passes from the corners B and C of the cross-section under the eccentric normal load;

a) Find the application point of the force.

b) If the force is 200 kN tension force, find the max. and min. normal stresses occur at the cross-section and show the stress distribution on the cross-section.



Solution:

$$a)\ y_G = \frac{18.4.9 + 18.4.20}{18.4 + 18.4} = 14,5\,cm$$

$$I_x = \frac{4.18^3}{12} + 4.18.(9-14,5)^2 + \frac{18.4^3}{12} + 18.4.(20-14,5)^2 = 6396\,cm^4$$

$$I_y = \frac{4.18^3}{12} + \frac{18.4^3}{12} = 2040\,cm^4$$

$$B(2,14,5),\quad C(-9;3,5)$$

$$\mathrm{A} = 2.18.4 = 144\,\mathrm{cm}^2$$

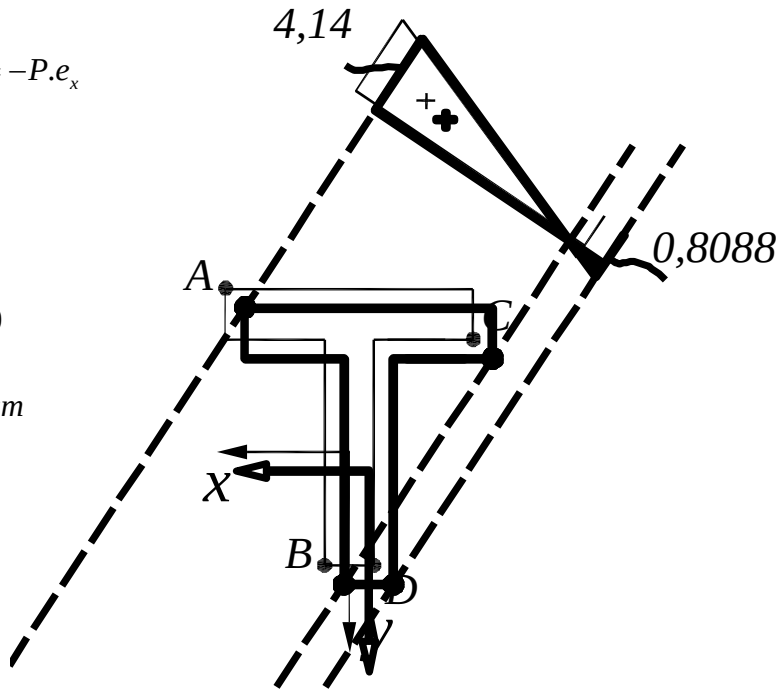
$$y - y_1 = \frac{y_2 - y_1}{x_2 - x_1}(x - x_1) \Rightarrow y = 1,636x + 11,228\ (\text{Neutral axis}),\quad M_x = P.e_y,\ M_y = -P.e_x$$

$$\sigma_z(x,y) = \frac{P}{A} + \frac{P.y_B.y}{I_x} + \frac{P.x_B.x}{I_y}$$

$$\sigma_z = \frac{P}{144} + \frac{P.e_y}{6396}y + \frac{P.e_x}{2040}x \Rightarrow for\ N.axis \Rightarrow \sigma_z = 0 \Rightarrow \frac{1}{144} + \frac{e_y}{6396}y + \frac{e_x}{2040}x = 0$$

$$y = -\frac{6396}{2040}.\frac{e_x}{e_y}x - \frac{6396}{144e_y} = 1,636x + 11,228 \Rightarrow e_y = -\frac{6396}{144.11,228} \Rightarrow e_y \cong -3,955\,cm$$

$$e_x = \frac{1,636.3,955.2040}{6396} \Rightarrow e_x \cong 2,063\,cm$$



$$b)\ M_x = 200.(-3,955) = -791\ \mathrm{kNcm},\quad A(9;-7,5),\ D(-2;14,5)$$

$$M_y = -200.(2,063) = -412,6\ \mathrm{kNcm}$$

$$\sigma_z = \frac{N}{A} + \frac{M_x}{I_x}y - \frac{M_y}{I_y}x = \frac{200}{144} - \frac{791}{6396}y + \frac{412,6}{2040}x \Rightarrow \sigma_A(9;-7,5) \cong 4,14\ kN/cm^2,\quad \sigma_D(-2;14,5) = -0,8088\ kN/cm^2$$

$$\sigma_z = 0 \Rightarrow y = 1,636x + 11,228\quad (\mathrm{T.E})$$

Problem: Draw the normal stresss diagram for the loading on a beam as shown in the right hand side figüre. P=5000N

Solution:

$$\sigma_z = \frac{N}{A} + \frac{M_x}{I_x}.y - \frac{M_y}{I_y}.x$$

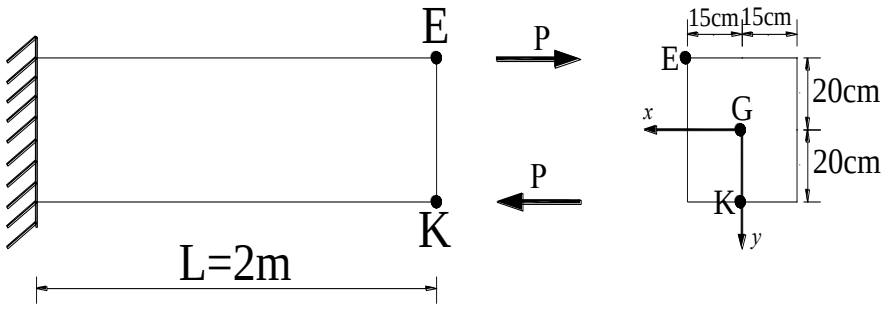
$$I_y = 90.10^3 cm^4$$

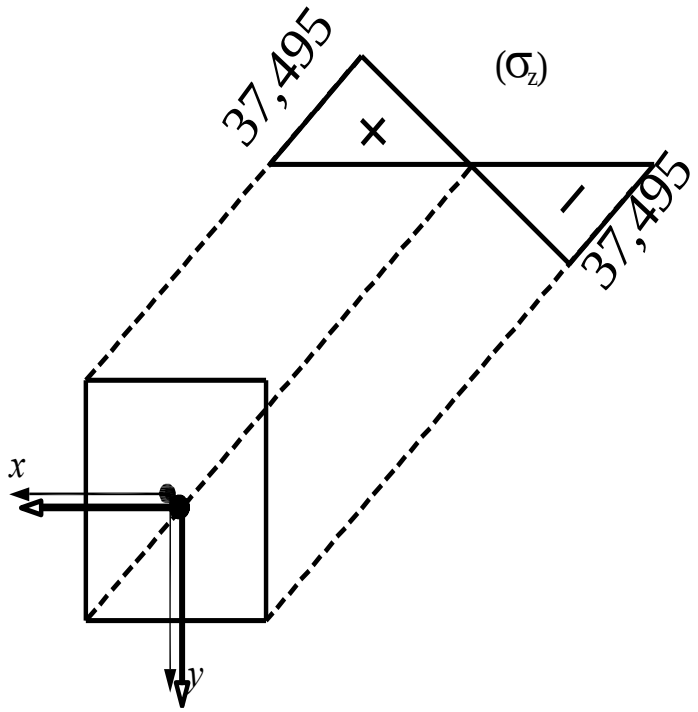
$$I_x = 160.10^3 cm^4$$

$$N = 0$$

$$M_x = -40.P = -200.10^3 N.cm$$

$$M_y = -15.P = -75.10^3 N.cm$$

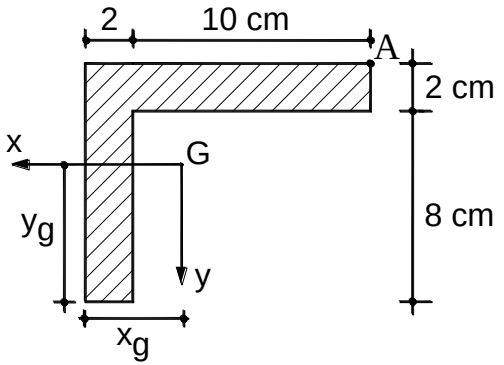




Problem: P=10000 N compression force is acting at point A of the cross-section in the **Figure 2**.

a) Find the equation of the neutral axis and show it on the section.

b) Find the max. and min. normal stresses and draw the normal stress diagram over the section. $x_g = 4\,cm$, $y_g = 7\,cm$



$$\sigma_z(x,y) = \frac{P}{A} + \frac{P.y_B.y}{I_x} + \frac{P.x_B.x}{I_y} = \frac{P}{A}\left(1 + \frac{y_B y}{i_x^2} + \frac{x_B x}{i_y^2}\right) \quad ,$$

$$\sigma_z = \frac{N}{A} + \frac{M_x I_y + M_y I_{xy}}{I_x I_y - I_{xy}^2} y - \frac{M_x I_{xy} + M_y I_x}{I_x I_y - I_{xy}^2} x$$

$x_g = 4\,cm$, $y_g = 7\,cm$,

$\mathrm{I}_x = 333,33\,cm^4$

$\mathrm{I}_y = 533,33\,cm^4$

$\mathrm{I}_{xy} = \bar{x}_1 \bar{y}_1 A_1 + \bar{x}_2 \bar{y}_2 A_2 = 3\times 2\times 2\times 10 + (-2)\times(-3)\times 10\times 2 = 240\,cm^4$,

$A = 40cm^2$

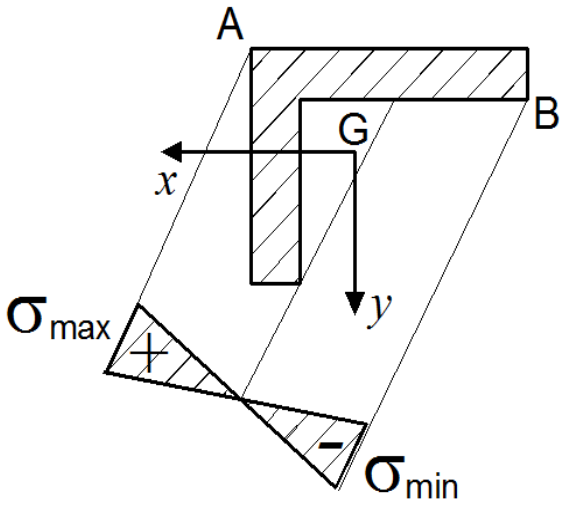
$N = -10000\,N$,

$\mathrm{M}_x = 30000\,Ncm$,

$\mathrm{M}_y = -80000\,Ncm$

$\sigma_z = -250 - 26,63y + 161,98x = 0 \quad \rightarrow T.\mathrm{E}$

$\sigma_A = 477,81\,N/cm^2 \rightarrow \sigma_{\max}$, $\sigma_B = -1519,21\,N/cm^2 \rightarrow \sigma_{\min}$



Problem: The cross section of a column with known dimensions is under compression of eccentric load, P as seen in the figure. The allowable tensile stress is $\sigma_{t,all} = 1200 N/cm^2$ and the allowable compressive stress is $\sigma_{c,all} = 6000 N/cm^2$. Find the maximum force P which can be carried by the cross section. Draw the normal stress distribution over the cross section of the column for this obtained force.

$$\sigma_z(x,y) = \frac{P}{A} + \frac{P.y_B.y}{I_x} + \frac{P.x_B.x}{I_y} = \frac{P}{A}\left(1 + \frac{y_B y}{i_x^2} + \frac{x_B x}{i_y^2}\right)$$

$$\sigma_z = \frac{N}{A} + \frac{M_x I_y + M_y I_{xy}}{I_x I_y - I_{xy}^2} y - \frac{M_x I_{xy} + M_y I_x}{I_x I_y - I_{xy}^2} x$$

$$e_x = \frac{(3\times18)\times1.5 + (9\times3)\times7.5}{3\times18 + 9\times3} = 3.5cm$$

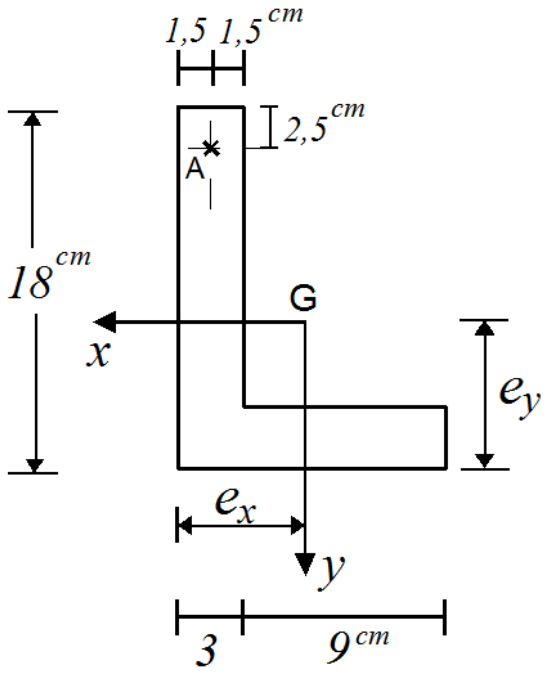
$$e_y = \frac{(3\times18)\times9.0 + (9\times3)\times1.5}{3\times18 + 9\times3} = 6.5cm$$

$$I_x = \frac{1}{12}\times3\times18^3 + (3\times18)\times(9.0-6.5)^2 + \frac{1}{12}\times9\times3^3 + (9\times3)\times(6.5-1.5)^2 = 2490,75cm^4$$

$$I_y = \frac{1}{12}\times18\times3^3 + (18\times3)\times(3.5-1.5)^2 + \frac{1}{12}\times3\times9^3 + (3\times9)\times(7.5-3.5)^2 = 870.75cm^4$$

$$I_{xy} = (18\times3)\times(3.5-1.5)\times(6.5-9.0) + (3\times9)\times(3.5-7.5)\times(6.5-1.5) = -810.00cm^4$$

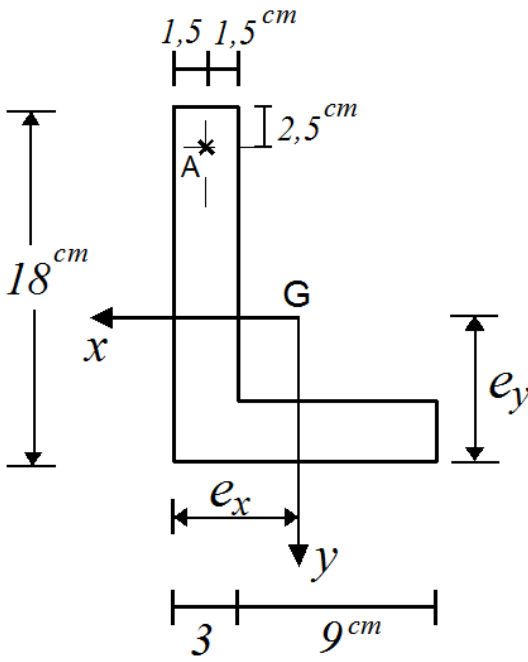
$$A = 3\times18 + 9\times3 = 81cm^2$$



$$\sigma_z = \frac{N}{A} + \frac{M_x I_y + M_y I_{xy}}{I_x I_y - I_{xy}^2} y - \frac{M_x I_{xy} + M_y I_x}{I_x I_y - I_{xy}^2} x \qquad N = P \quad M_x = 9P \quad M_y = 2P$$

$$I_x = 2490,75 cm^4 \qquad I_y = 870.75 cm^4 \qquad I_{xy} = -810.00 cm^4 \qquad A = 3\times 18 + 9\times 3 = 81 cm^2$$

$$\sigma_z = P\left(-\frac{1}{81} + \frac{9\times 870.75 + 2\times(-810)}{2490.75\times 870.75 - (-810)^2} y - \frac{9\times(-810) + 2\times 2490.75}{2490.75\times 870.75 - (-810)^2} x \right)$$



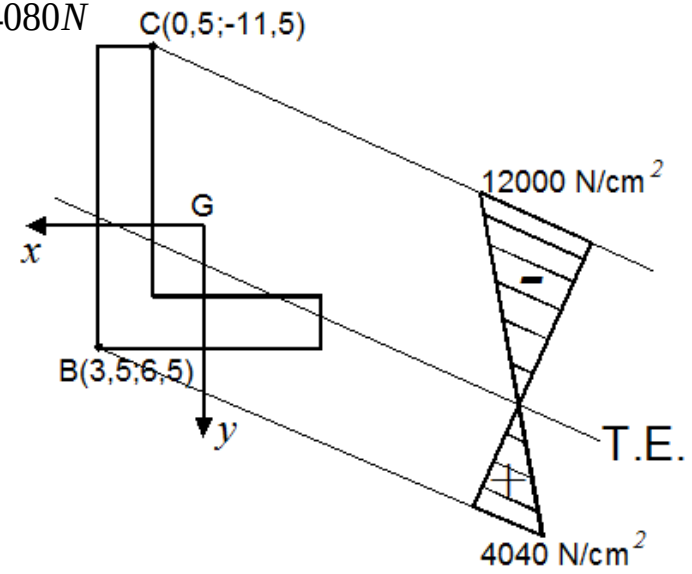
$$\sigma_z = P(-0.0123 + 0.00411y + 0.00153x)$$

$$Neutral\ axis \rightarrow \sigma_z = 0 \rightarrow y = 3 - 0.37x$$

$$\sigma_z = P(-0.0123 + 0.00411y + 0.00153x)$$

$$\sigma_D = P(-0.0123 + 0.00411(6.5) + 0.00153(3.5)) \le 6000 \rightarrow P \le 303490N$$

$$\sigma_C = P(-0.0123 + 0.00411(-11.5) + 0.00153(0.5)) \le -12000 \rightarrow P \le 204080N$$



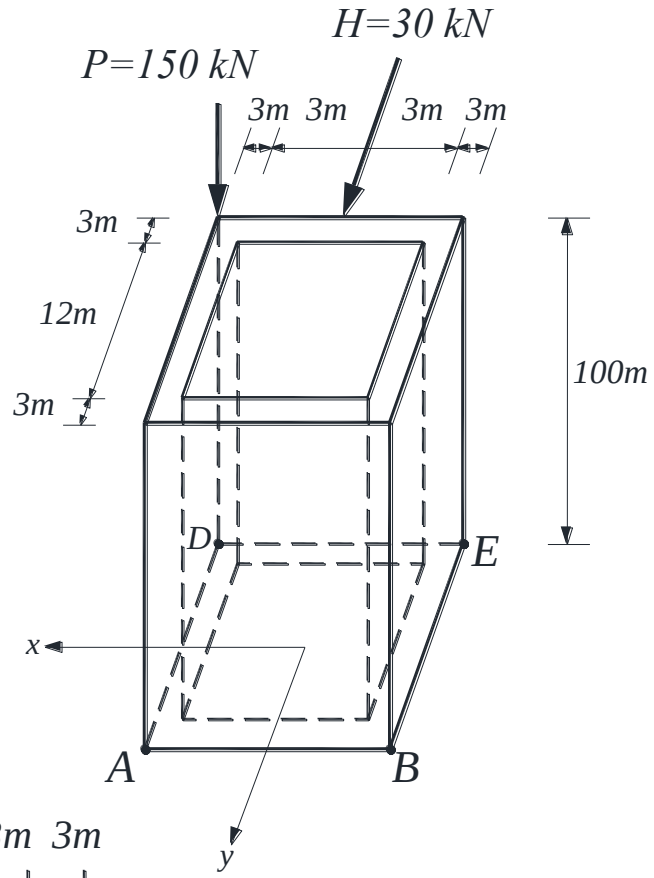
Smaller load must be chosen $P \le 204080N$

$$\sigma_D = 0.01977 \times 204080 \rightarrow \sigma_D = 4040 N / cm^2$$

$$\sigma_C = -0.0588 \times 204080 \rightarrow \sigma_C = -12000 N / cm^2$$

Problem: A box column with known dimensions is under the effect of eccentric compression load *P=150 kN* and a load which is perpendicular to the axis of the column *H=30 kN* as seen in the figure.

a) Find the normal stresses at the points A, B, D, E,
b) write the equation of the neutral axis and draw the diagram of the normal stress distribution.



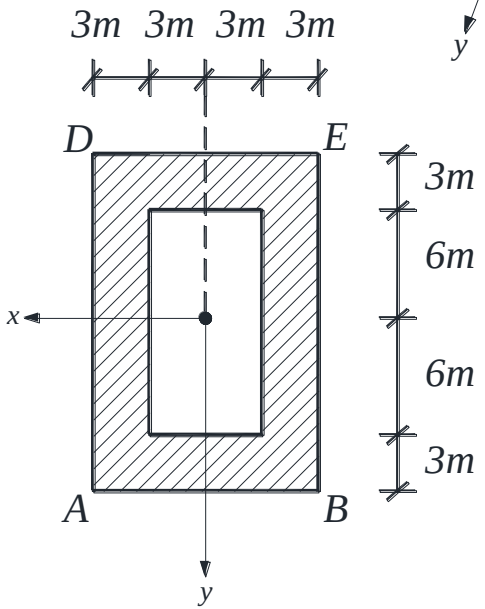
$$M_x = 150.9 - 30.100 = -1650\,kNm$$

$$M_y = 150.6 = 900\,kNm$$

$$A = 12.18 - 6.12 = 144\,cm^2$$

$$I_x = \frac{1}{12}.12.18^3 - \frac{1}{12}.6.12^3 = 4968\,m^4$$

$$I_y = \frac{1}{12}.18.12^3 - \frac{1}{12}.12.6^3 = 2376\,m^4$$



$$\sigma_z(x,y) = \frac{P}{A} + \frac{P.y_B.y}{I_x} + \frac{P.x_B.x}{I_y} = \frac{P}{A}\left(1 + \frac{y_B y}{i_x^2} + \frac{x_B x}{i_y^2}\right)$$

$$\sigma_z = \frac{N}{A} + \frac{M_x}{I_x}y - \frac{M_y}{I_y}x \rightarrow \sigma_z = \frac{-150}{144} + \frac{-1650}{4968}y - \frac{900}{2376}x$$

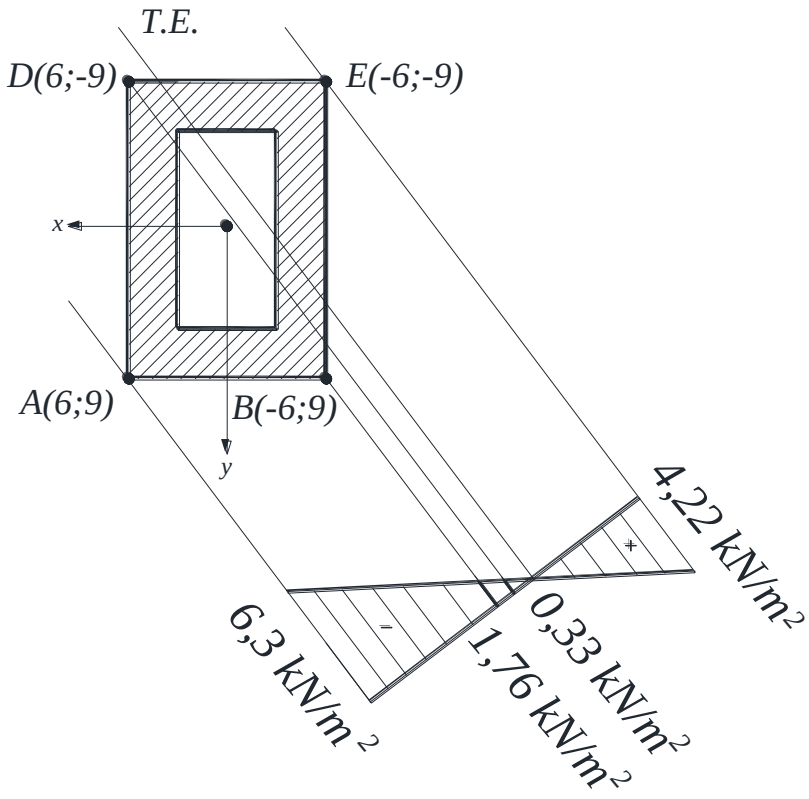
$$\sigma_z = -1,0417 - 0,3321y - 0,3788x$$

$$\sigma_A = -1,0417 - 0,3321.9 - 0,3788.6 = -6,3\,kN/m^2$$

$$\sigma_B = -1,0417 - 0,3321.9 - 0,3788.(-6) = -1,76\,kN/m^2$$

$$\sigma_D = -1,0417 - 0,3321.(-9) - 0,3788.6 = -0,33\,kN/m^2$$

$$\sigma_E = -1,0417 - 0,3321.(-9) - 0,3788.(-6) = 4,22\,kN/m^2$$



YILDIZ TECHNICAL UNIVERSITY
CIVIL ENGINEERING DEPARTMENT
MECHANICS DIVISION

STRENGTH OF MATERIALS II

COMBİNED LOADING

FIRST CHAPTER

CORE AND TENSIONLESS MATERIAL

3 Hours First in Line

PROF. DR. TURGUT KOCATÜRK

3- Application point and Neutral Axis

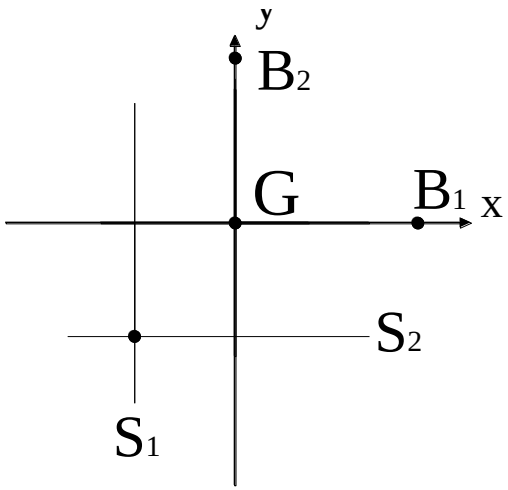
a) When application point approaches to the center of gravity, neutral axis goes away from it.

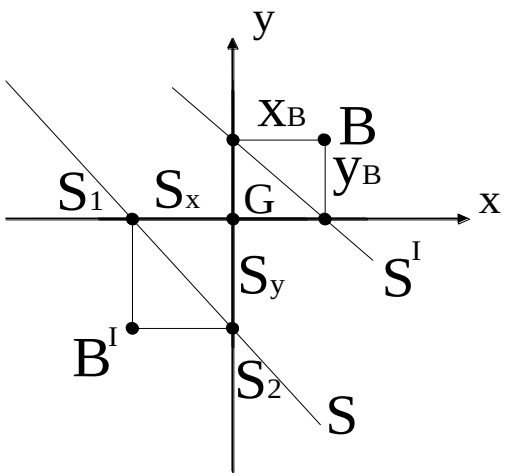
b) When application point is on one of the principal axis, which is x_B or y_B equals zero, neutral axis is parallel to the other principal axis.

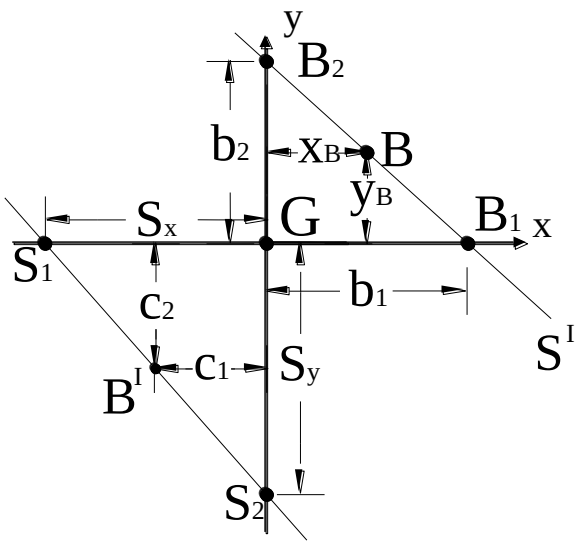
c) Due to the symmetry between x_B, S_x and y_B, S_y, if the neutral axis S corresponds to the application point B, the neutral axis S' corresponds to the application point B' as seen in Fig. 6

$$S_x = \frac{-i_y^2}{x_B} \qquad S_y = \frac{-i_x^2}{y_B}$$

d) If the application points B is on a fixed line, then neutral axes belonging to these application points passes from a fixed point.

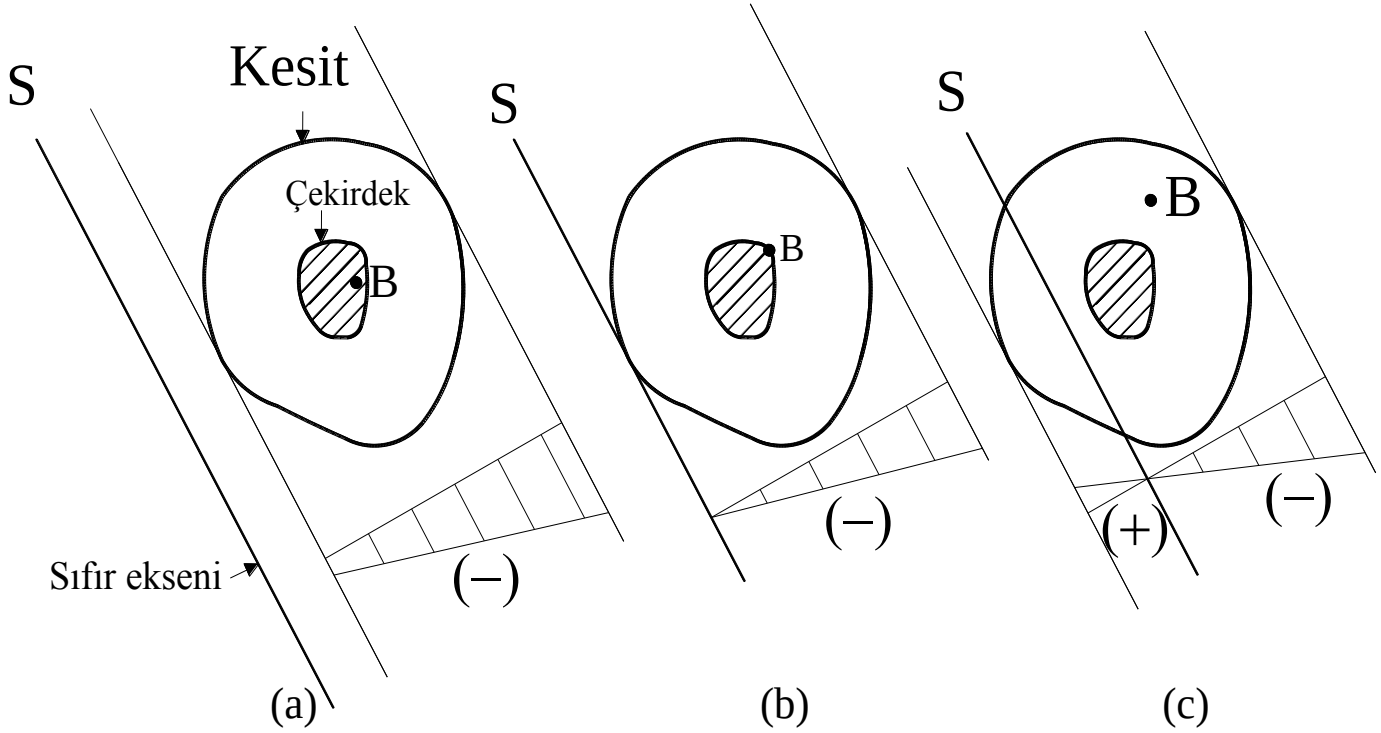






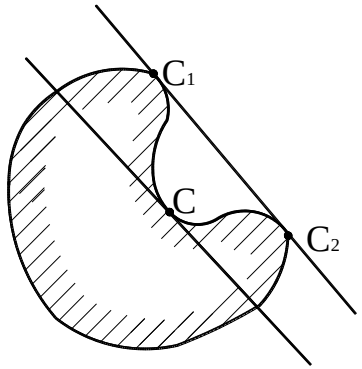
4- Core

In which position of the eccentric normal force do all the normal stresses have the same sign, in other words, don't all the neutral axes cut the section and do they leave the whole section on one side and in the limit case do they become tangent to the section? The application points which satisfy these conditions are located in a specific region, and this region is called the core of the cross section. For example, let the force acting at the point B be a compressive force. In this case, the position of neutral axis for the forces in the core region, on the border of the core region and outside the core region, are shown below.

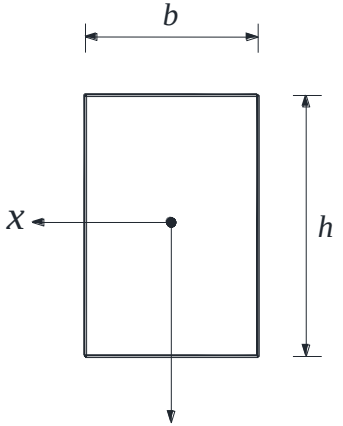


The coordinates of the application points B for which the neutral axes touches the border of the cross section and leaves the cross section at one side can be calculated by formulas $S_x = -i_y^2 / x_B$ and $S_y = -i_x^2 / y_B$ (5). Moving S as staying tangent to the border of the cross section, the corresponding application points B indicates the perimeter of the core of the cross section.

If the cross-section is not convex, some tangents cut the cross-section area and both sides of the neutral axis have some part of the section. For example, tangent at point C is such a point shown in the figure given below. These types of tangents mustn't be considered in the calculation of the core of the section; because the requirement of having a single sign is not satisfied. In Figure 10, C_1CC2 part of the perimeter will not be considered in the calculation of the core of the section, instead common tangent C1 C2 will be considered. According to this, even if the cross-section is not always convex, the core of the section will always be convex.



Problem: Find the core of rectangular cross section given on the right hand side.



Solution:

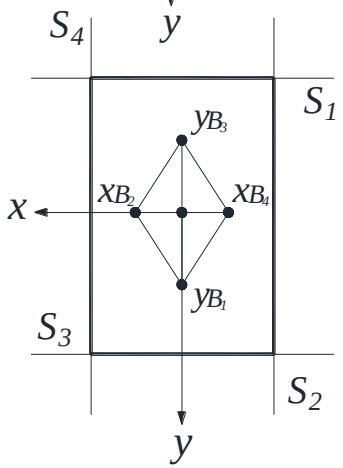
$$S_x = \frac{-i_y^{\,2}}{x_B} \quad ; \quad S_y = \frac{-i_x^{\,2}}{y_B} \qquad i_x^{\,2} = \frac{I_x}{A} = \frac{bh^3}{12 \cdot bh} = \frac{h^2}{12} \qquad i_y^{\,2} = \frac{I_y}{A} = \frac{hb^3}{12 \cdot bh} = \frac{b^2}{12}$$

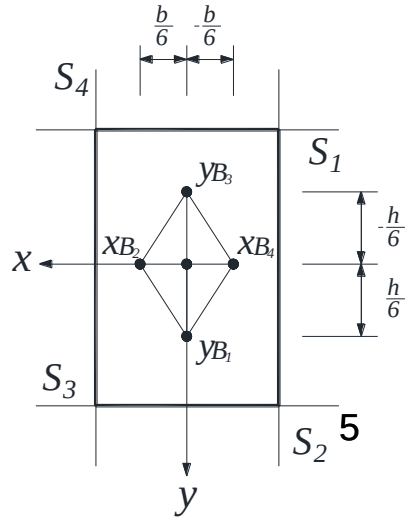
$$S_{1_x} = \infty \quad ; \quad S_{1_y} = \frac{-h}{2} \qquad S_{1y} = -\frac{h}{2} = -\frac{h^2}{12\, y_{B1}} \rightarrow y_{B1} = +\frac{h}{6}$$

$$S_{2x} = -\frac{b}{2} \quad ; \quad S_{2y} = \infty \qquad S_{2x} = -\frac{b}{2} = -\frac{b^2}{12\, x_{B2}} \rightarrow x_{B2} = +\frac{b}{6}$$

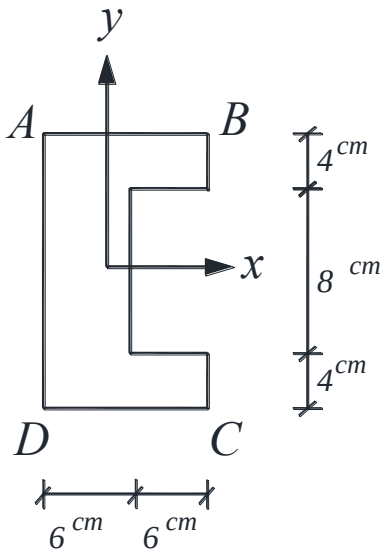
$$S_{3x} = \infty \quad ; \quad S_{3y} = -\frac{h}{2} \qquad S_{3y} = \frac{h}{2} = -\frac{h^2}{12\, y_{B3}} \rightarrow y_{B3} = -\frac{h}{6}$$

$$S_{4x} = \frac{b}{2} \quad ; \quad S_{4y} = \infty \qquad S_{4x} = \frac{b}{2} = -\frac{b^2}{12 x_{B4}} \rightarrow x_{B4} = -\frac{b}{6}$$





EXAMPLE Find the core of the cross-section given on the left.



SOLUTION

$$x_G = \frac{16\cdot 6\cdot 3 + 2(4\cdot 6\cdot 9)}{16\cdot 6 + 2(4\cdot 6)} = \frac{720}{144} = 5cm$$

$$I_x = \frac{1}{12}\cdot 12\cdot 16^3 - \frac{1}{12}\cdot 6\cdot 8^3 = 4096 - 256 = 3840cm^4$$

$$I_y = \frac{1}{12}\cdot 16\cdot 6^3 + 16\cdot 6\cdot (5-3)^2 + 2(\frac{1}{12}\cdot 4\cdot 6^3 + 4\cdot 6\cdot (9-5)^2)$$

$$I_y = 288 + 384 + 912 = 1584cm^4 \quad ; \quad A = 144cm^2$$

In order to find the core of the section, the points corresponding to the tangents of the section being as a neutral axis are connected. This tangents are AB,BC,CD and AD edges of the section. Because the section is symmetric with respect to the x axis, it will be enough to find three points of the core.

$$i_x^{\,2} = \frac{I_x}{A} = \frac{3840}{144} = 26,67cm^2 \; ; \; i_y^{\,2} = \frac{I_y}{A} = \frac{1584}{144} = 11cm^2$$

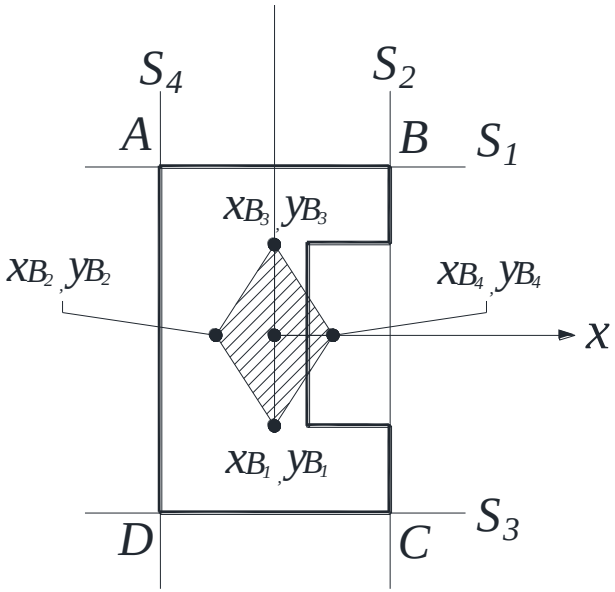
$$AB \rightarrow S_{1x} = \infty \quad ; \quad S_{1y} = 8 \qquad S_x = \frac{-i_y^{\,2}}{x_B} \quad ; \quad S_y = \frac{-i_x^{\,2}}{y_B}$$

$$x_{B1} = -\frac{11}{\infty} = 0 \quad ; \quad y_{B1} = -\frac{26,67}{8} = -3,33$$

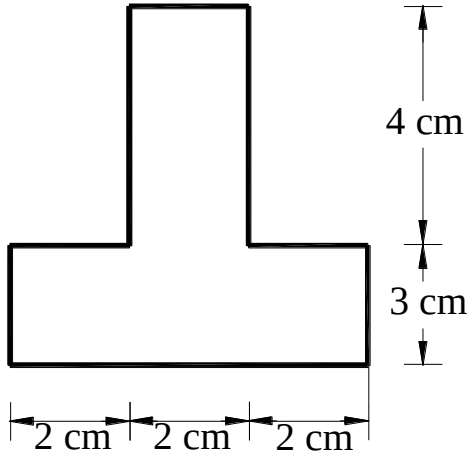
$$BC \rightarrow S_{2x} = 7 \quad ; \quad S_{2y} = \infty \qquad x_{B2} = -\frac{11}{7} = -1,57 \quad ; \quad y_{B2} = -\frac{26,67}{8} = 0$$

$$CD \rightarrow from\,symmetry \rightarrow x_{B3} = 0 \quad ; \quad y_{B3} = 3,33$$

$$AD \rightarrow S_{4x} = -5 \quad ; \quad S_{4y} = \infty \rightarrow x_{B4} = \frac{-11}{-5} = 2,2 \quad ; \quad y_{B4} = \frac{-26,67}{\infty} = 0$$



Example: **Find the core of the given section.**



$$i_x = \sqrt{\frac{I_x}{A}} \; ; \; S_x = -i_y^2 / x_B \; ; \; S_y = -i_x^2 / y_B$$

$$y_g = \frac{6.3.1{,}5 + 2.4.5}{6.3 + 2.4} = \frac{67}{26} = 2{,}58^{cm}$$

$$I_x = \frac{6.3^3}{12} + 6.3\left(1{,}5 - 2{,}58\right)^2 + \frac{2.4^3}{12} + 2.4\left(5 - 2{,}58\right)$$

$$I_x = 20{,}25 + 31{,}5 + 16 + 29{,}04 = 92.02cm^4$$

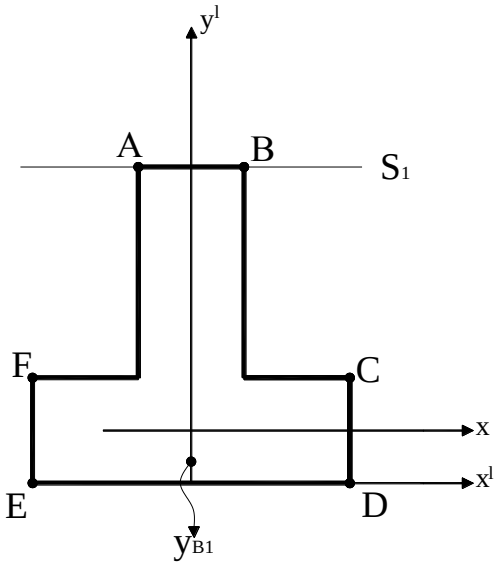
$$I_y = \frac{3.6^3}{12} + \frac{4.2^3}{12} = 56{,}67cm^4$$

$$A = 6.3 + 2.4 = 26cm^2$$

$$i_x^2 = \frac{92{,}02}{26} = 3{,}54cm^2 \quad i_y^2 = \frac{56{,}67}{26} = 2{,}18cm^2$$

$$AB \rightarrow S_{1x} = \infty \; ; \; S_{1y} = 4{,}42$$

$$x_{B1} = \frac{-2{,}18}{\infty} = 0 \; ; \; y_{B1} = -\frac{3{,}54}{4{,}42} = -0{,}8cm$$



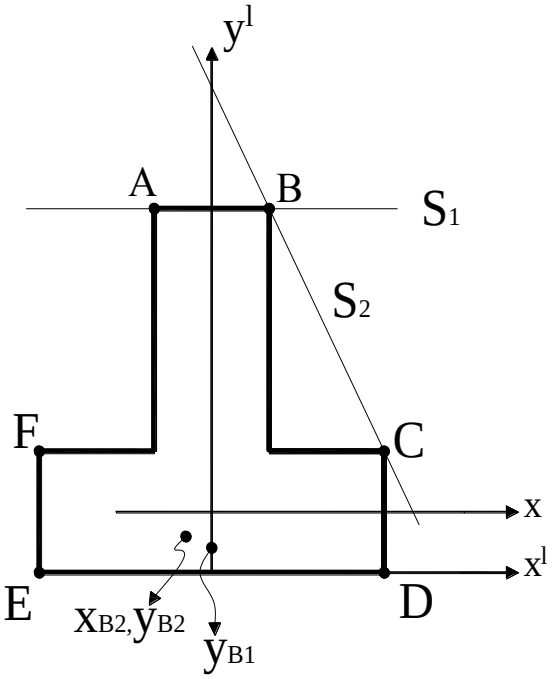
$BC \rightarrow S_2$ **equation** $B(+1;+4,42)\ \ C(+3;+0,42)$

B and C are the points on S_2

Equation of S_2 **neutral axis ;** $y_{S_2} = mx_{S_2} + n$

$$B \rightarrow +4,42 = +m.1 + n \quad \textbf{(1)}$$

$$C \rightarrow +0,42 = +m.3 + n \quad \textbf{(2)}$$



From (1) and (2) it is found that m=-2, n=6,42.

$$y_{S_2} = -2x_{S_2} + 6,42 \rightarrow S_{2y} = 6,42;\ y_{s2} = 0 \rightarrow S_{2x} = +3,21$$

$$x_{B2} = -\frac{i_y^2}{S_{x2}} = -\frac{2,18}{3,21} = -0,679;\ y_{B2} = -\frac{i_x^2}{S_{y2}} = -\frac{3,54}{6,42} = -0,551$$

The most general equation for the neutral axis $\sigma_z = \frac{N}{A} + \frac{M_x I_y + M_y I_{xy}}{I_x I_y - I_{xy}^2} y - \frac{M_x I_{xy} + M_y I_x}{I_x I_y - I_{xy}^2} x = 0$

When the system of axes is principal, then we can write for the neutral axis

$$\sigma_z = \frac{N}{A} + \frac{M_x I_y + M_y I_{xy}}{I_x I_y - I_{xy}^2} y - \frac{M_x I_{xy} + M_y I_x}{I_x I_y - I_{xy}^2} x = \frac{N}{A} + \frac{M_x I_y + M_y \times 0}{I_x I_y - 0} y - \frac{M_x \times 0 + M_y I_x}{I_x I_y - 0} x = \frac{N}{A} + \frac{M_x}{I_x} y - \frac{M_y}{I_y} x = 6,42 - y_{S_2} - 2x_{S_2}$$

$$\sigma_z(x,y) = 0 \rightarrow \left(1 + \frac{y_B y_{S_2}}{i_x^2} + \frac{x_B x_{S_2}}{i_y^2}\right) = 6,42 - y_{S_2} - 2x_{S_2}$$

$$\left(1 + \frac{y_B y_{S_2}}{3,54} + \frac{x_B x_{S_2}}{2,18}\right) = 6,42 - y_{S_2} - 2x_{S_2} = 1 - 0,156 y_{S2} - 0,312 x_{S2} = ($$

$$\left(1 + 0,282 y_B y_{S_2} + 0,459 x_B x_{S_2}\right) = 1 - 0,156 y_{S2} - 0,312 x_S$$

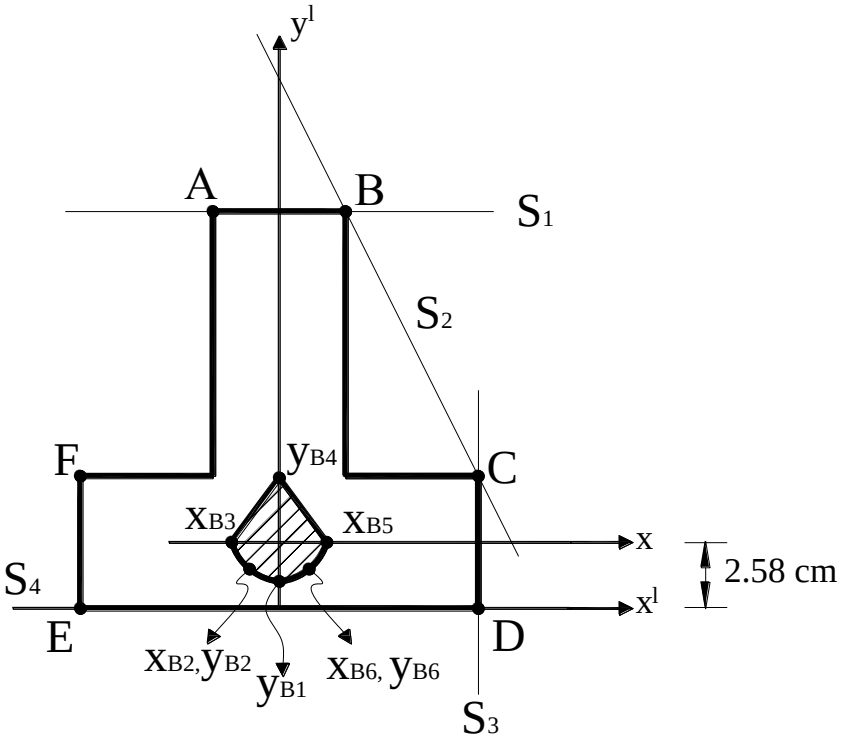
$$0,282 y_B = -0,156 \rightarrow y_B = -0,55cm \qquad 0,459 x_B = -0,312 \rightarrow x_B = 0,679cm$$

$$CD \rightarrow S_{3x} = 3, S_{3y} = \infty$$

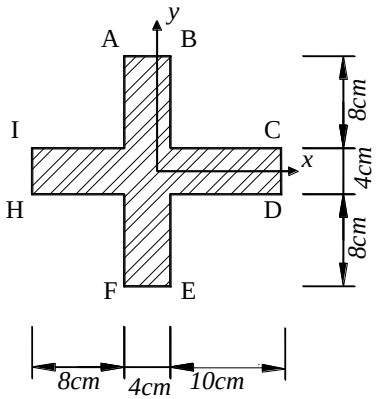
$$x_{B3} = -\frac{2{,}18}{3} = -0{,}73; y_{B3} = -\frac{3{,}54}{\infty} = 0$$

$$DE \rightarrow S_{4x} = \infty, S_{4y} = -2{,}58$$

$$x_{B4} = -\frac{2{,}18}{\infty} = 0; y_{B4} = \frac{-3{,}54}{-2{,}58} = 1{,}37$$



Problem: Draw the core of the cross-section given below.



$$x_G = \frac{2.4.8.10 + 22.4.11}{2.4.8 + 22.4} = 10,57\ cm, \quad I_x = \frac{4.20^3}{12} + \frac{8.4^3}{12} + \frac{10.4^3}{12} = 2762,6\ cm^4$$

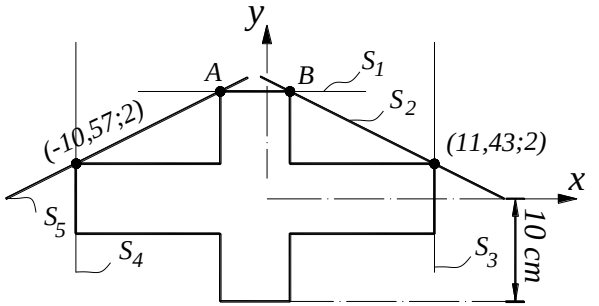
$$I_y = \frac{4^3.20}{12} + 20.4.(10-10,57)^2 + \frac{4.10^3}{12} + 4.10.(17-10,57)^2 + \frac{4.8^3}{12} + 4.8.(4-10,57)^2 = 3671\ cm^4,$$

$$A = 152\ cm^2,\ i_x^2 = 2762/152 = 18,17\ cm, \quad i_y^2 = 3671/152 = 24,15\ cm^4$$

$$i_x^2 = \frac{I_x}{A} \text{ and } i_y^2 = \frac{I_y}{A} \qquad S_x = -\frac{i_y^2}{x_B} \text{ and } S_y = -\frac{i_x^2}{y_B} \quad \textbf{(1)}$$

$$\frac{y-y_1}{(x-x_1)} = \frac{y_2-y_1}{x_2-x_1} \rightarrow (y-y_1) - \frac{y_2-y_1}{x_2-x_1}(x-x_1) = 0 \quad \textbf{(2)}$$

$$\sigma_z(x,y) = \frac{P}{A}\left(1 + \frac{y_B y}{i_x^2} + \frac{x_B x}{i_y^2}\right) = 0 \rightarrow \left(1 + \frac{y_B y}{i_x^2} + \frac{x_B x}{i_y^2}\right) = 0 \quad \textbf{(3)}$$



A(-2,57;10)
B(1,43;10)

If S_1 **is neutral axis,**

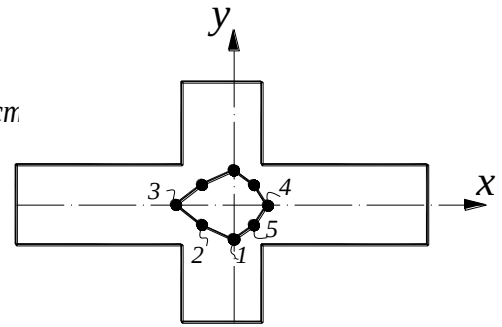
$$S_{1x} = \infty, \quad S_{1y} = 10$$

$$x_{B1} = 0, \quad y_{B1} = -18,17/10 = -1,817$$

If S_2 **is neutral axis,** $(y-y_1) - \frac{y_2-y_1}{x_2-x_1}(x-x_1) = 0 \rightarrow (y-2) - \frac{(10-2)}{(1,43-11,43)}(x-11,43) = 0 \rightarrow y = -0,8x + 11,144$

$$y = -0,8x + 11,144 \quad x = 0 \rightarrow S_{2y} = 11,144cm \quad y = -0,8x + 11,144 \quad y = 0 \rightarrow S_{2x} = 13,93cm$$

$$S_{2x} = 13,93, \quad S_{2y} = 11,144 \quad S_{x2} = -\frac{i_y^2}{x_{B2}} \rightarrow x_{B2} = -\frac{i_y^2}{S_{x2}} = -\frac{24,15}{13,93} = -1,73cm \quad S_y = -\frac{i_x^2}{y_B} \rightarrow y_{B2} = -\frac{i_x^2}{S_{y2}} = -\frac{18,17}{11,144} = -1,63cm$$



If S_3 **is neutral axis,**

$$S_{3x} = 11,43; \quad S_{3y} = \infty$$

$$x_{B3} = -2,11 \quad y_{B3} = 0$$

If S_4 **is neutral axis,**

$$S_{4x} = 10,57; \quad S_{4y} = \infty$$

$$x_{B4} = 2,284 \quad y_{B4} = 0$$

If S_5 **is neutral axis,**

$$y = x + 12,57$$

$$S_{5x} = -12,57, \quad S_{5y} = 12,57 \Rightarrow$$

$$x_{B5} = 1,92\ cm, \quad y_{B5} = -1,445\ cm$$

Second Way:

If S_2 is neutral axis, $(y-y_1)-\frac{y_2-y_1}{x_2-x_1}(x-x_1)=0\rightarrow(y-2)-\frac{(10-2)}{(1,43-11,43)}(x-11,43)=0\rightarrow y=-0,8x+11,144$

$$y=-0,8x+11,144\rightarrow -y-0,8x+11,144=0\rightarrow 1-0.0897y-0.0718x=0$$

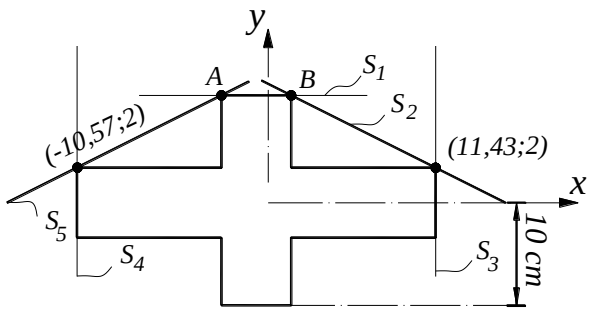
$$\left(1+\frac{y_B y}{i_x^2}+\frac{x_B x}{i_y^2}\right)=0\rightarrow 1+\frac{y_{B2}y}{18,17}+\frac{x_{B2}x}{24,15}=1+0.055y_{B2}y+0,0414x_{B2}x=0$$

$$1+0.055y_{B2}y+0,0414x_{B2}x=0$$

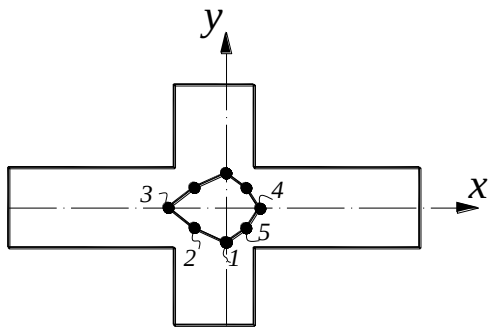
$$1-0.0897y-0.0718x=1+0.055y_{B2}y+0,0414x_{B2}x=0$$

$$-0.0897y=0.055y_{B2}y\rightarrow y_{B2}=-1,63cm$$

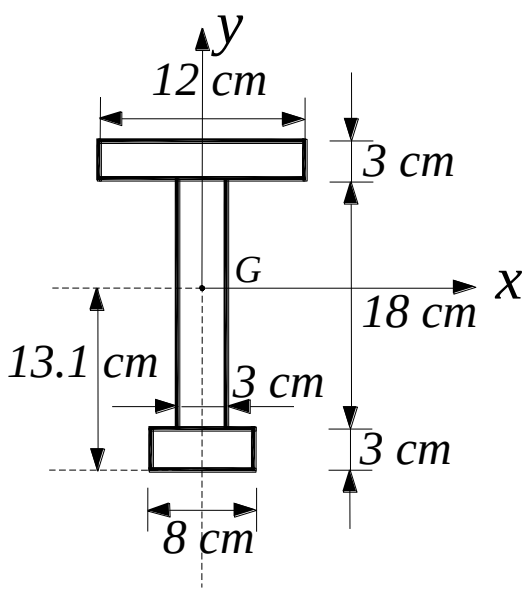
$$-0.0718x=0,0414x_{B2}x\rightarrow x_{B2}=1,73cm$$



A(-2,57;10)
B(1,43;10)



Problem 1: Obtain the core of the cross section given on the right side.
Gravity center of the cros section is obtained as follows:



$$y_G = \frac{8.3.1,5 + 3.18.12 + 12.3.22,5}{8.3 + 3.18 + 12.3} = \frac{1494}{114} = 13,1 \text{ cm}$$

Moment of inertias with respect to x and y axes which pass from the gravity center of the cross section are obtained below:

$$I_Y = \frac{1}{12}.3.8^3 + \frac{1}{12}.18.3^3 + \frac{1}{12}.3.12^3 = 128 + 40,5 + 432 = 600,5 \text{ cm}^4$$

$$I_X = \frac{1}{12}.8.3^3 + 8.3.(13,1-1,5)^2 + \frac{1}{12}.3.18^3 + 3.18.(13,1-12)^2 + \frac{1}{12}.12.3^3 + 12.3.(22,5-13,1)^2 = 7978,74 \text{ cm}^4$$

$$i_x^2 = \frac{I_X}{A} = \frac{7978,74}{114} = 69,99 \text{ cm}^2 \qquad i_Y^2 = \frac{I_Y}{A} = \frac{600,5}{114} = 5,27 \text{ cm}^2$$

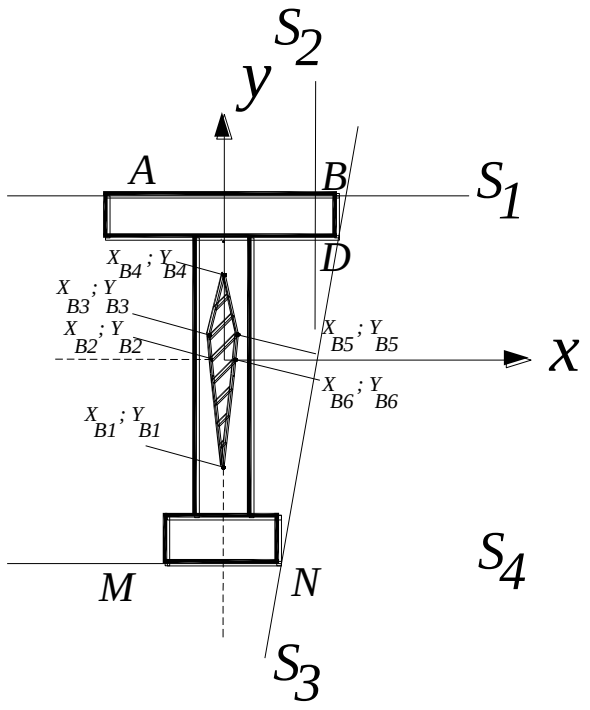
We define the core of the cross section by considering the tangents that don't cut the cross section and obtain the corresponding application points and later by drawing straight lines for connecting these points. These tangents are AB, BD, DN, NM, MC ve AC. It is sufficient to find only four points of the core due to the symmetry of the section with respect to the y axis.

AB ; $S_{1X} = \infty$; $S_{1Y} = 10,9$ cm

$s_X = -\frac{i_y^2}{x_B}$; $s_Y = -\frac{i_X^2}{y_B}$ $x_{B1} = -\frac{5,27}{\infty} = 0$ $y_{B1} = -\frac{69,99}{10,9} = -6,42$ cm

BD ; $S_{2X} = 6$; $S_{2Y} = \infty$

$x_{B2} = -\frac{5,27}{6} = -0,88$ cm $y_{B2} = -\frac{69,99}{\infty} = 0$ cm

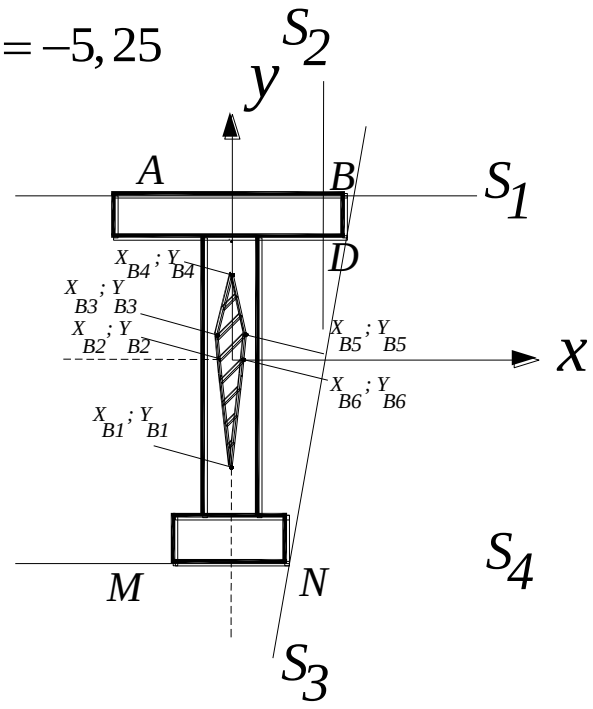


DN; D(6;7,9); N(4;-13,1), D and N are the two points on the neutral axis S_3. The equation of the neutral axis S_3 is given below:

$$y_{s3} = mx_{s3} + n$$

D; 7,9=6m+n **N;** -13,1=4m+n 21=2m m=10,5 n=-55,1

$$y_{s3} = 10,5x_{33} - 55,1 \quad x_{s3} = 0 \rightarrow s_{3y} = -55,1 \quad y_{s3} = 0 \rightarrow s_{3x} = -5,25$$



$$x_{B3} = -\frac{5,27}{5,25} = -1 \quad y_{B3} = -\frac{69,99}{55,1} = 1,27$$

MN; $S_{4X} = \infty$; $S_{4Y} = -13,1$

$$x_{B4} = -\frac{5,27}{\infty} = 0 \; ; \quad y_{B4} = -\frac{69,99}{-13,1} = 5,34 \text{ cm}$$

From symmetry, we obtain the following values:

$$x_{B5} = 1; \; y_{B5} = 1,27; \; x_{B6} = 0,88; \; y_{B6} = 0$$

MATERIAL WHICH IS WEAK IN TENSION

If the applied force is outside of the core, the stress distribution of the cross-section is double-signed. If the material is weak in tension, the tensile stresses shouldn't be considered while calculating the stresses:

- **The compression region and the region without stress are divided by a straight line which is called neutral axis or zero axis.**
- **Compressive stresses of the cross-section increase linearly with respect to the increasing distance from the neutral axis.**

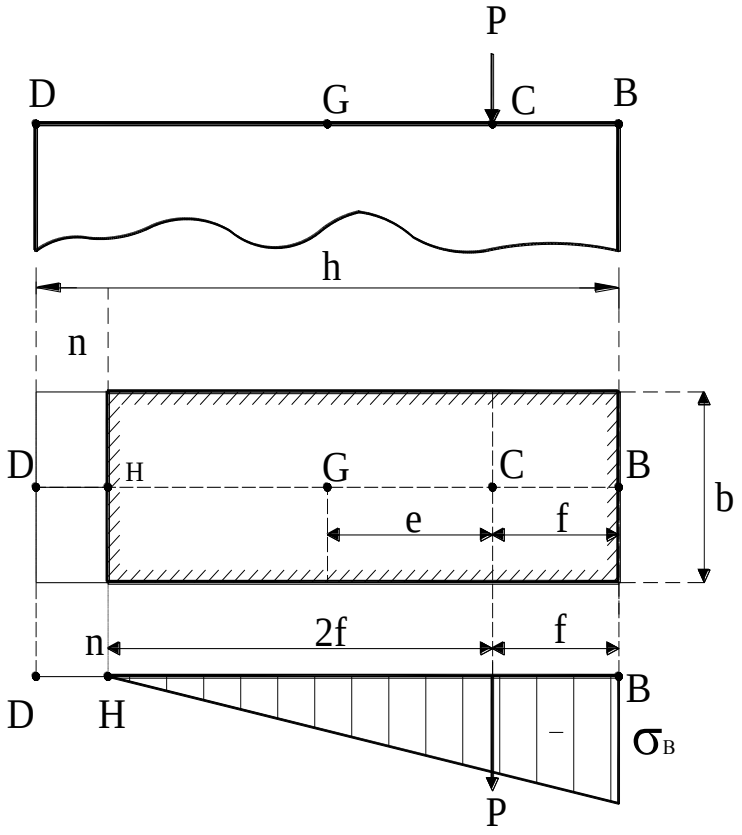
If the application point is outside of the core, by using the assumptions given above and equilibrium equations $\sum F_y = 0$, $\sum M = 0$, **the location of the neutral axis and compression stresses can be defined.**

For the material which is weak in tensile and has a rectangular cross section, (figure 11); when the force is on the x axis which is the symmetry axis for the cross-section, then the eccentricity is also on the x axis and it is explicit that the neutral axis is parralel to the y axis.

If the cross-section has't got a symmetry with respect to the axis which passes from the application point of the force, the calculation proses must be trial and error and because of this the solution will be too long. The resultant of stress prism must be at the point C because of equilibrium necessities for the rectangular section. From this condition we get

$$\frac{3 f\, b\, \sigma_B}{2} = P \rightarrow \sigma_B = -2\frac{P}{3bf} \qquad \textbf{(15)}$$

Concrete can not withstand tensile forces. Also the foundations of the structures are put on the ground freely and therefore they only can transmit compressive forces and can't transmit the tensile forces to the ground.



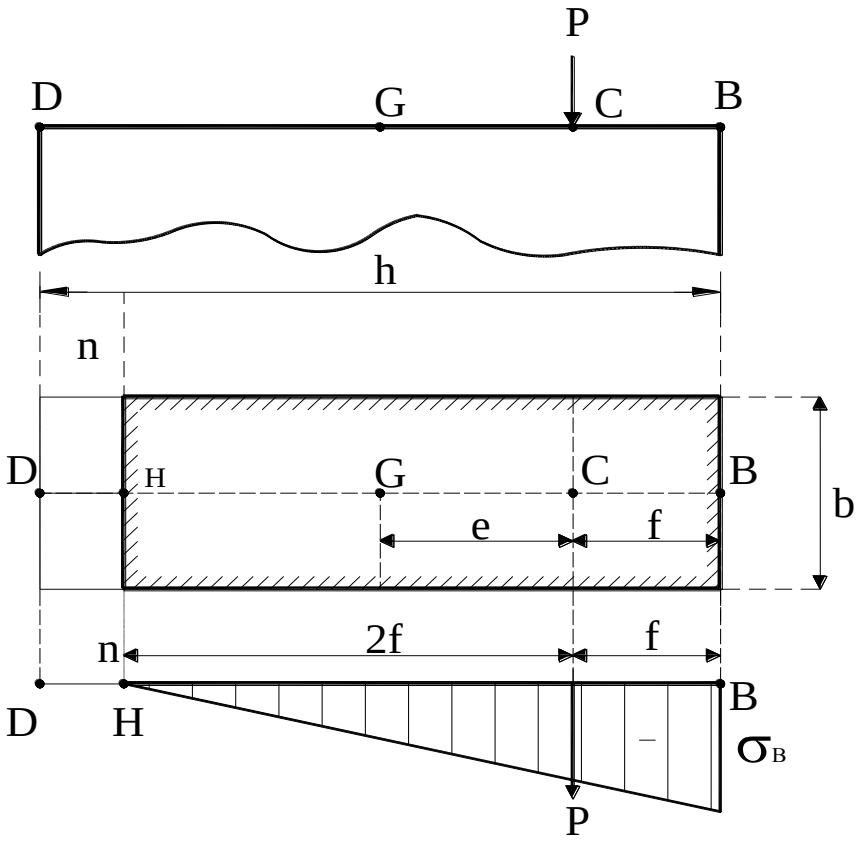
It can be seen from figure that $f = \frac{h}{2} - e$

If $h/2 - e$ is substituted for f, then the stress equation can be be written as

$$\sigma_B = -2\frac{P}{3bf} \rightarrow \sigma_B = -\frac{4P}{3bh(1-2e/h)} \qquad (16)$$

It can be seen that this equation is valid for $(h/6 \le e < h/2)$. If If e<h/6, then the stress distribution becomes trapezoidal, therefore Equation (16) is not valid any more, and when e>h/2 then the stress becomes infinite. If we omit the tensile stresses which can occur in the section, then the results of this calculation stay on the safe side.

For the material capable of carrying tensile stresses, for the investigated example, the stres distribution is double-signed and the maximum compression stress σ'_B is



$$\sigma' = \frac{P}{A} + \frac{M}{I} y_{\max} = \frac{P}{bh} + \frac{Pe}{bh^3/12}\frac{h}{2} \rightarrow \sigma'_B = \frac{P(e+h/6)}{bh^2/6} \qquad (17)$$

The ratio of the resulting stresses from the two theories is written as

$$\frac{\sigma_B}{\sigma'_B} = \frac{4}{(3-6e/h)(1+6e/h)} \ge 1 \quad (h/6 \le e < h/2) \qquad (18).$$

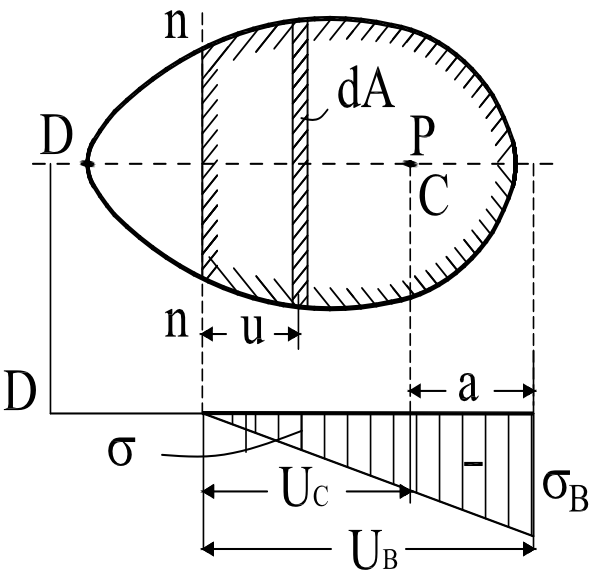
for $e = h/2$ $\sigma_B \rightarrow \infty$.

It is possible to generalize this theory for any cross-section. Consider a cross-section that has symmetry with respect to the plane of forces. Let's consider that the compression force P is applied at the point C which is outside of the core, and the corresponding neutral axis is n-n. In the figure compression effective region is hatched: the other region is not effective because of the tension weakness of the material that means there is no stress in this region.

For the linear law, the stress at the distance (u) from the neutral axis is $\sigma = k.u$ **(19)**. $k = proportion\ coefficient$

$$-P = \iint_{A'} \sigma.dA \quad -P.U_C = \iint_{A'} u.\sigma.dA \qquad (20)$$

$A' = Effective\ area\ of\ cross\ section$



$$-P = k\int_0^{U_B} u.dA = k.S_n \rightarrow k = \frac{-P}{S_n} \qquad (21a)$$

$S_n = static\ moment\ with\ respect\ to\ the\ neutral\ axis\ n-n$

$$-P.U_C = k\int_0^{U_B} u^2.dA. = k.I_n \qquad (21b)$$

$I_n = moment\ of\ inertia\ with\ respect\ to\ the\ neutral\ axis\ n-n$

$$-P.U_C = k\ I_n = -\frac{P}{S_n} I_n \rightarrow U_C = \frac{I_n}{S_n} \qquad (22)$$

$$\sigma = k.u = \frac{-P.u}{S_n} \quad ve \quad \sigma_B = \frac{-P.U_B}{S_n} \qquad \textbf{(23)}$$

We calculate I_n and S_n in terms of U_B.

$$U_C = \frac{I_n}{S_n}$$

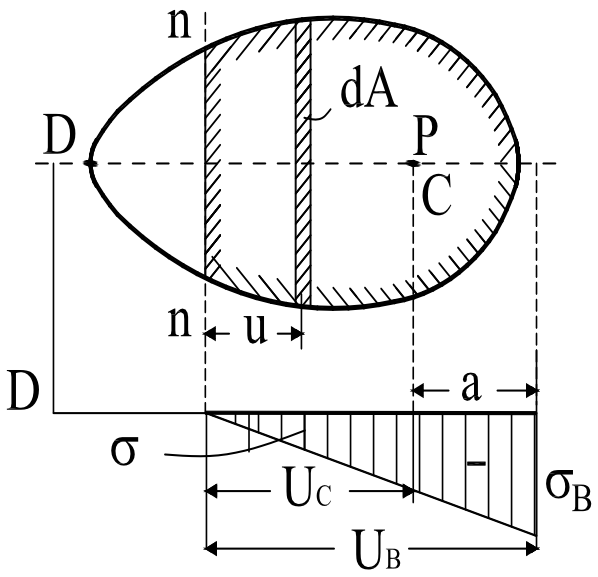
Also, from the geometry of the section

$$U_C = U_B - a$$

These two U_c must be equal to each other

$$\frac{I_n}{S_n} = U_B - a$$

We solve this equation for U_B.



Problem: The force P is applied perpendicularly on an isosceles triangle at point C. It is known that the materail is tensionless. Namely it can not withstand tensile stresses. Find the location of the neutral axis.

Second solution way: There is no need to trial and error. Because the section is a special one. It has a symmetry axis and it is an isosceles triangle. Hence this problem can be solved directly as follows.

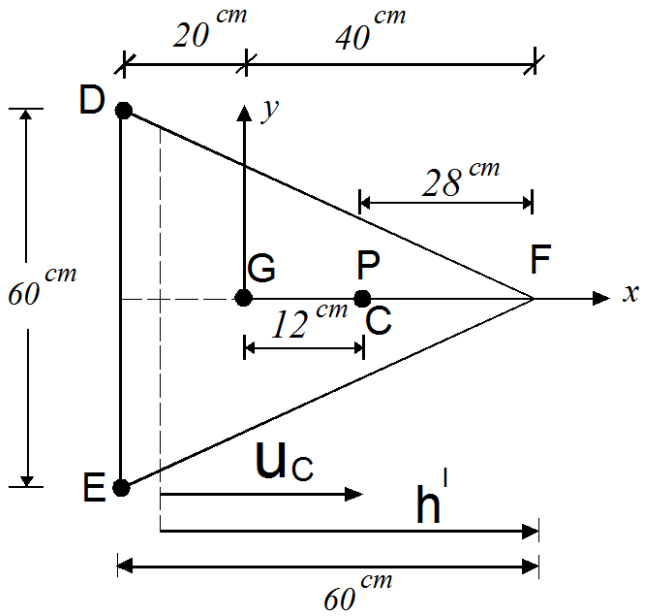
$$U_c = \frac{I_n}{S_n} = \frac{\frac{h'.h'^3}{12}}{\frac{h'.h'}{2}.\frac{h'}{3}} = h' - 28$$

$$h' - 28 = \frac{h'^4.6}{12.h'^3} = \frac{h'}{2}$$

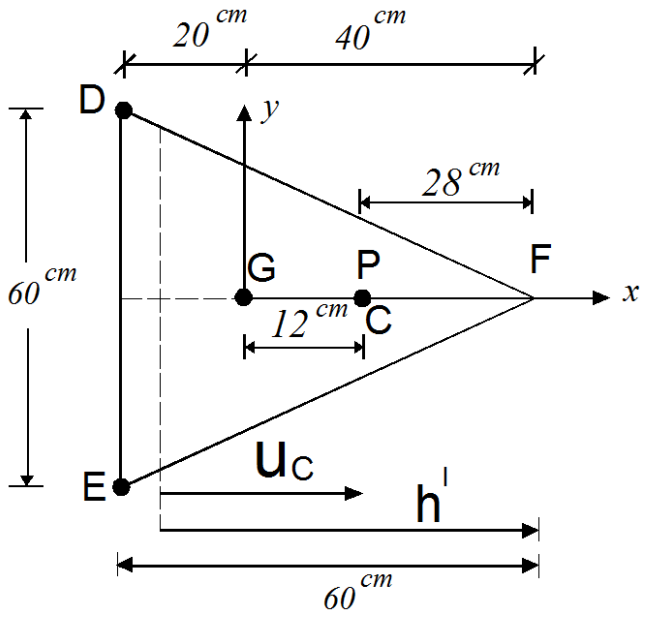
$$h' - 28 = \frac{h'}{2} \rightarrow (h' - 28).2 = h'$$

$$2h' - 2.28 = h'$$

$$h' = 56\ cm$$



Problem: The force P is applied perpendicularly on an isosceles triangle at point C. It is known that the materail is tensionless. Namely it can not withstand tensile stresses. Find the location of the neutral axis.



Solution: First of all we will check whether or not the application point C of the force is in the core of the section.

$$s_x = -\frac{i_y^2}{x_c};\ i_y^2 = \frac{I_y}{A};\ A = \frac{bh}{2} = \frac{60.60}{2} = 1800\,cm^2$$

$$I_y = \frac{bh^3}{36} = \frac{60.60^3}{36} = 360000\,cm^4,\ i_y^2 = \frac{360.000}{1800} = 200\,cm^2$$

By assuming that S_C is tangent to DE, we obtain

$S_{Cx} = -20cm$. In this case

$x_C = -i_y^2 / S_{Cx} = -200/(-20) = 10cm$. $10cm < |GC| = 12cm$.

Therefore the stress distribution has double sign. It means that the application point is out of the core. We can find the location of the neutral axis by trial and error. Lets assume that the neutral axis is at the left side of the point G and the distance between the neutral axis and the point F is h' and the point C is U_C.

1. *Step* $h' = 57 \rightarrow I_n = \frac{1}{12}.57.57^3 = 879666,75\ cm^4;\ S_n = \frac{57.57}{2}.\frac{57}{3} = 30865,5\ cm^3$

$$U_c = \frac{I_n}{S_n} = \frac{879666,75}{30865,5} = 28,5\ cm \neq U_c = 57 - 28 = 29\ cm$$

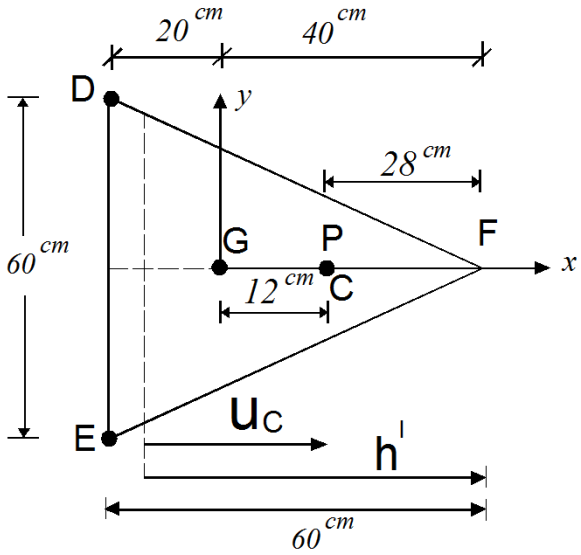
2. *Step* $h' = 55 \rightarrow \frac{I_n}{S_n} = \frac{55.55^3}{12}.\frac{6}{(55.55.55)} = 27,5 \neq U_c = 55 - 28 = 27\ cm$

$Step\,1 \quad h' = 57 \rightarrow I_n = \frac{1}{12}.57.57^3 = 879666,75\ cm^4;\ S_n = \frac{57.57}{2}.\frac{57}{3} = 30865,5\ cm^3$

$U_c = \frac{I_n}{S_n} = \frac{879666,75}{30865,5} = 28,5\ cm \neq U_c = 57 - 28 = 29\ cm$

$Step\,2 \quad h' = 55 \rightarrow \frac{I_n}{S_n} = \frac{55.55^3}{12}.\frac{6}{(55.55.55)} = 27,5 \neq U_c = 55 - 28 = 27\ cm$

$Step\,3 \quad h' = 56 \rightarrow \frac{I_n}{S_n} = \frac{56.56^3}{12}.\frac{6}{(56.56.56)} = 28 = U_c = 56 - 28 = 28\ cm$



Neutral axis is 56-40=16cm to the left side of the point G for the force P.

Problem: If we invert the problem, namely if the location of the neutral axis is known, then what is the application point of the force $P = 5000\,N$ and the maximum value of the compresive stress?

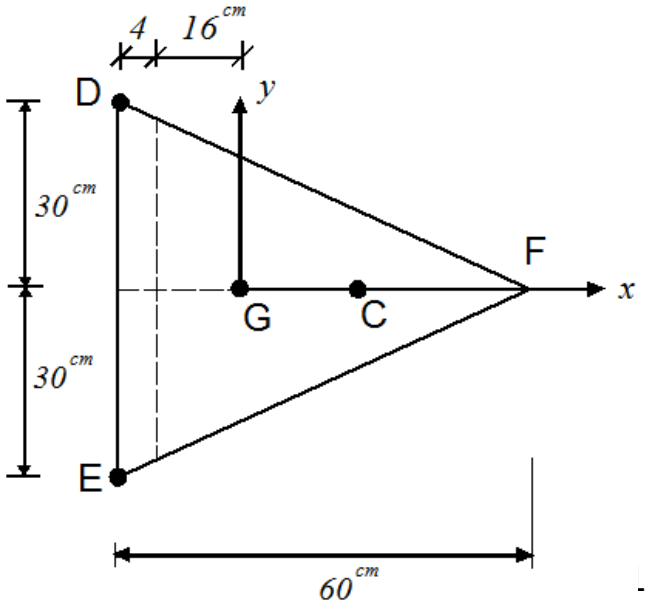
Solution:

$$-P.U_C = k\ I_n = -\frac{P}{S_n} I_n \rightarrow U_C = \frac{I_n}{S_n} \quad (22)$$

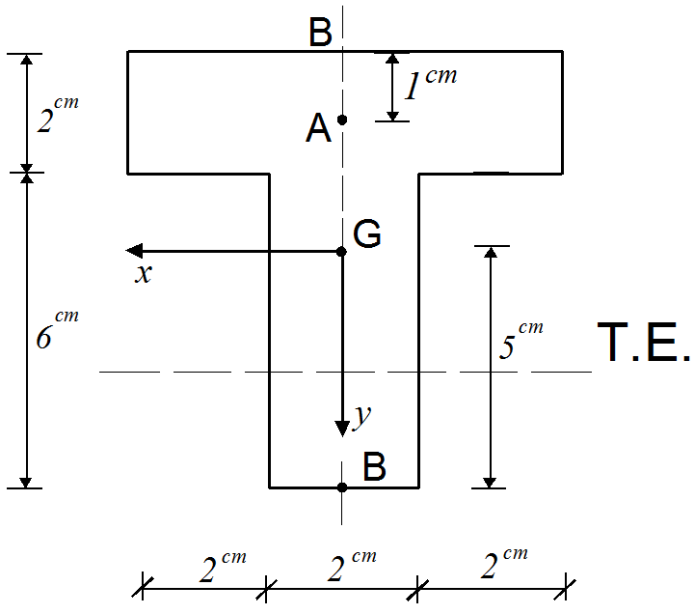
$$\sigma = k.u = \frac{-P.u}{S_n} \quad ve \quad \sigma_B = \frac{-P.U_B}{S_n} \quad (23)$$

$$U_c = \frac{I_n}{S_n} = \frac{56.56^3}{12}.\frac{6}{56.56.56} = 28$$

$$\sigma_B = -\frac{5000.6.56}{56.56.56} = -9,57\ kN/m^2$$



Problem: The force P is applied perpendicularly on the given cross section at the point A. It is known that the materail is tensionless. Namely it can not withstand tensile stresses. Also we know that the allowable compressive stress is $\sigma_{c,\,all} = 5000\,N/cm^2$. a) Find the location of the neutral axis. b) Find the maximum force P which can be carried by the cross section.



$$y_G = \frac{2.6.3 + 6.2.7}{2.6 + 6.2} = 5cm \qquad A = 2.6 + 6.2 = 24\ cm^2$$

$$I_x = \frac{2.6^3}{12} + 2.6(5-3)^2 + \frac{6.2^3}{12} + 6.2.(7-5)^2 = 136\ cm^2$$

$$Neutral\ axis: S_y = -\frac{i_x^2}{y_A}; \quad i_x^2 = \frac{I_x}{A}; \quad S_y = -\frac{136}{24}.\left(\frac{1}{-2}\right) = 2,83\ cm$$

According to this result, the neutral axis is between the points G and B. Hence the stress distribution has double sign. It means that the application point is out of the core. Also, we know that the material is tensionless. Namely there can't be double sign in the cross section. In this case, we have to determine the new stress distribution which has only one sign.

$$1.\ Adım \quad h' = 6; \quad I_n = \frac{2.4^3}{3} + \frac{6.2^3}{12} + 6.2.5^2 = 346,67\ cm^4$$

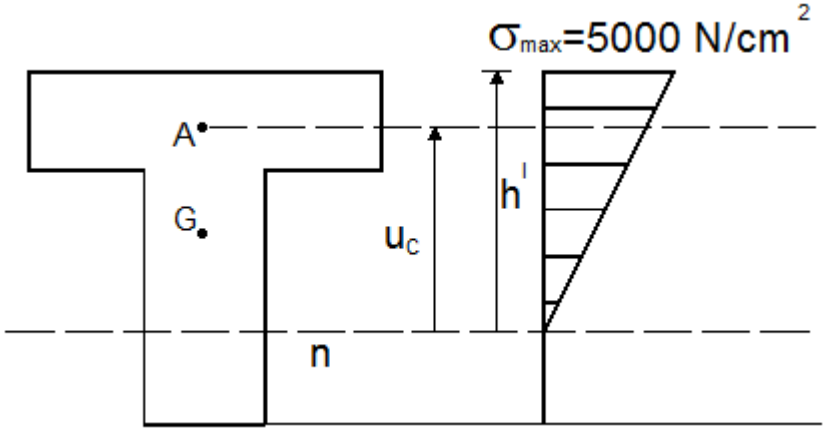
$$S_n = 6.2.5 + 2.4.2 = 76\ cm^3$$

Formülden;

$$U_c = \frac{I_n}{S_n} = \frac{346,67}{76} = 4,56\ cm;$$

Geometriden;

$$U_c = 6 - 1 = 5\ cm$$



2. *Adım* $\quad h' = 4; \quad I_n = \frac{2.2^3}{3} + \frac{6.2^3}{12} + 6.2.3^2 = 117,33\ cm^4$

$S_n = 6.2.3 + 2.2.1 = 40\ cm^3$

3. *Adım* $\quad h' = 3,5; \quad I_n = \frac{2.(1,5)^3}{3} + \frac{6.2^3}{12} + 6.2.(2,5)^2 = 81,25\ cm^4$

$S_n = 6.2.(2,5) + 2.(1,5).(0,75) = 32,25\ cm^3$

$\frac{I_n}{S_n} = \frac{81,25}{32,25} = 2,52 \qquad U_c = 3,5 - 1 = 2,5\ cm$

4. *Adım* $\quad h' = 3,61; \quad I_n = \frac{2.(1,61)^3}{3} + \frac{6.2^3}{12} + 6.2.(2,61)^2 = 88,53\ cm^4$

$S_n = 6.2.(2,61) + 2.(1,61).(\frac{1,61}{2}) = 33,91\ cm^3$

$\frac{I_n}{S_n} = \frac{88,53}{33,91} = 2,61 \qquad U_c = 3,61 - 1 = 2,61\ cm$

$\sigma_B = -\frac{P.U_B}{S_n} \rightarrow \sigma_B = -5000 = -\frac{P.(3,61)}{33,91} \rightarrow P = 46966,78\ N$

Second solution way: This problem could be solved directly by means of the stresss distribution and the equilibrium equations.

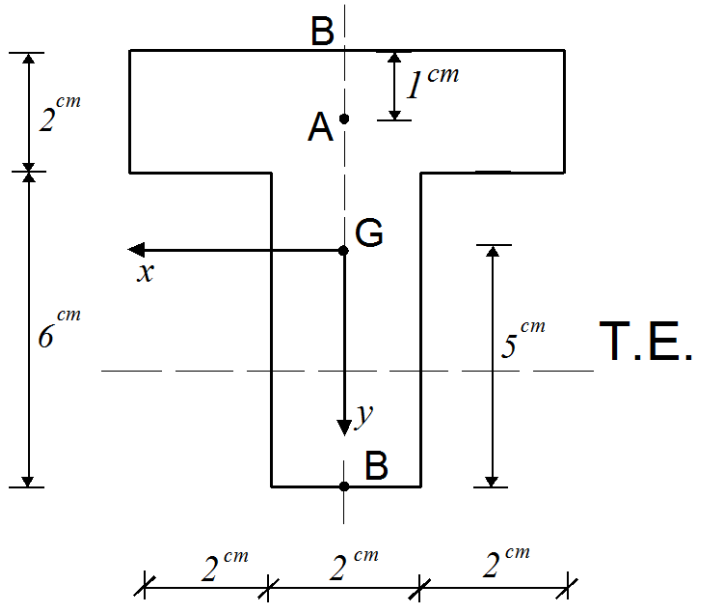
$$U_c = \frac{I_n}{S_n}$$

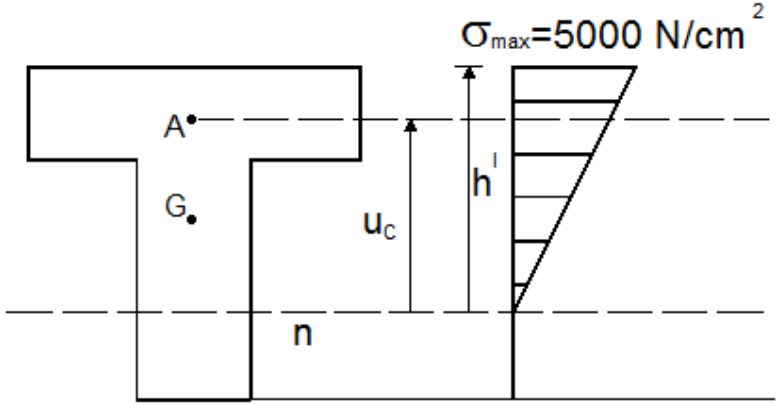
$$I_n = \frac{6.2^3}{12} + 6.2.(h^{'} - 1)^2 + \frac{2.(h^{'} - 2)^3}{3}$$

$$S_n = 6.2.(h^{'} - 1) + \frac{2.(h^{'} - 2).(h^{'} - 2)}{2} \rightarrow S_n = h^{'2} + 8h^{'} - 8$$

$$U_c = h^{'} - 1 = \frac{4 + 12(h^{'} - 1)^2 + \frac{2}{3}(h^{'} - 2)^3}{12(h^{'} - 1) + (h^{'} - 2)^2}$$

$$4 + \frac{2}{3}(h^{'} - 2)^3 - (h^{'} - 1)(h^{'} - 2)^2 = 0 \rightarrow h^{'} = 3,61\, bulunur.$$





YILDIZ TECHNICAL UNIVERSITY
DEPARTMENT OF CIVIL ENGINEERING
MECHANICS DIVISION

STRENGTH OF MATERIALS II

SECOND CHAPTER
COMBİNED LOADING-BENDİNG WITH SHEAR
1.5 weeks, Second in Line

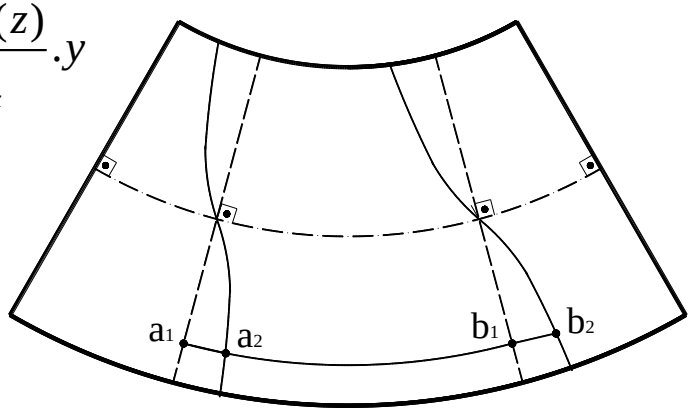
PROF. DR. TURGUT KOCATÜRK

1- BENDING with SHEAR

$$\sigma_z = E\varepsilon_z = \frac{E}{\rho_x} y \qquad \frac{1}{\rho_x} = \frac{M_x}{E I_x}$$

In the case of shear with bending, shear force and bending moment are acting together. It can be considered that $a_1 b_1 \cong a_2 b_2$

Also as with the simple bending it can be assumed that $\sigma_z(y,z) = \frac{M_x(z)}{I_x}.y$



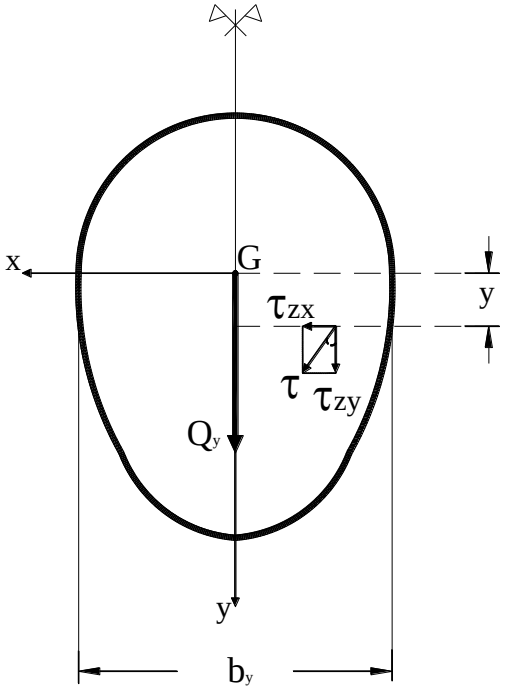
In the case bending with shear the cause of shear stress is the shear force, Q_y

Due to the boundary conditions, shear stress distribution is not uniform through the thickness of the beam. The components of shear stress along the coordinates are

τ_{zx} and τ_{zy} as shown in the figure. From the equilibrium conditions the resultants of the shear stresses are as follows:

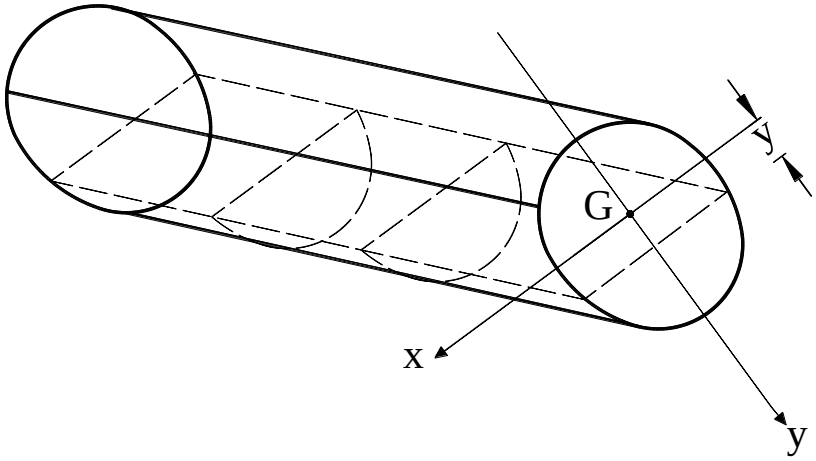
$$\iint \tau_{zx} \mathrm{d}A = 0$$

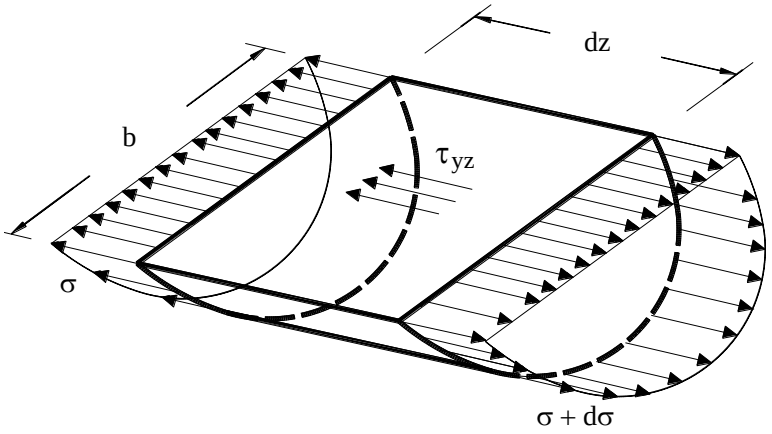
$$\iint \tau_{zy} \mathrm{d}A = Q_y$$



In general $\tau_{zy}(x,y)$ depends on x and y coordinates.

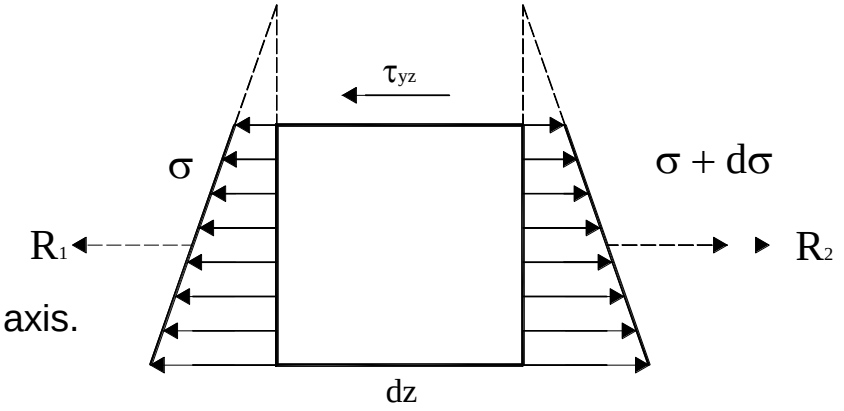
However, it is assumed that this function only depends on y axis. In other words the shear streses remain constant through the width b of the cross-section which is parallel to x axis. After this assumption, the shear stresses $\tau_{zy}(y)$ can be obtained by means of normal stress formula $\sigma_z = \frac{M_x}{I_x}.y$





$$R_1 = \int_{A_{alt}} \sigma dA = \int_{A_{alt}} \frac{M_x}{I_x} y dA = \frac{M_x}{I_x} \int_{A_{alt}} y dA = \frac{M_x}{I_x} S_x$$

$$R_2 = \int_{A_{alt}} (\sigma + d\sigma).dA = \int_{A_{alt}} \frac{M_x + dM_x}{I_x} y dA = \frac{M_x + dM_x}{I_x} S_x$$



S_x Static moment of the lower part of the section according to x axis.

After calculation of the stress resultants, from the equilibrium condition of the lower part we get

$$-\tau_{yz}.b.dz - R_1 + R_2 = 0 \rightarrow -\tau_{yz}.b.dz - \frac{M_x}{I_x}.S_x + \frac{M_x + dM_x}{I_x}.S_x = 0 \quad \tau = \frac{dM}{dz}.\frac{S_x}{b.I_x} \quad ; \quad \frac{\mathrm{dM}}{\mathrm{dz}} = Q \rightarrow \tau_{yz} = \frac{QS_x}{bI_x} \rightarrow (\tau_{zy})_{\max} = \frac{Q}{I_x}\left(\frac{S_x}{b(y)}\right)_{\max}$$

Q : Shear force in the investigated section

S_x : Static moment of the area which is below the y axis at which we want to calculate the shear stress

b : Width of cross-section at the distance y,

$$S_x = \int_{A_{alt}} y dA = \bar{y}.A_{alt}$$

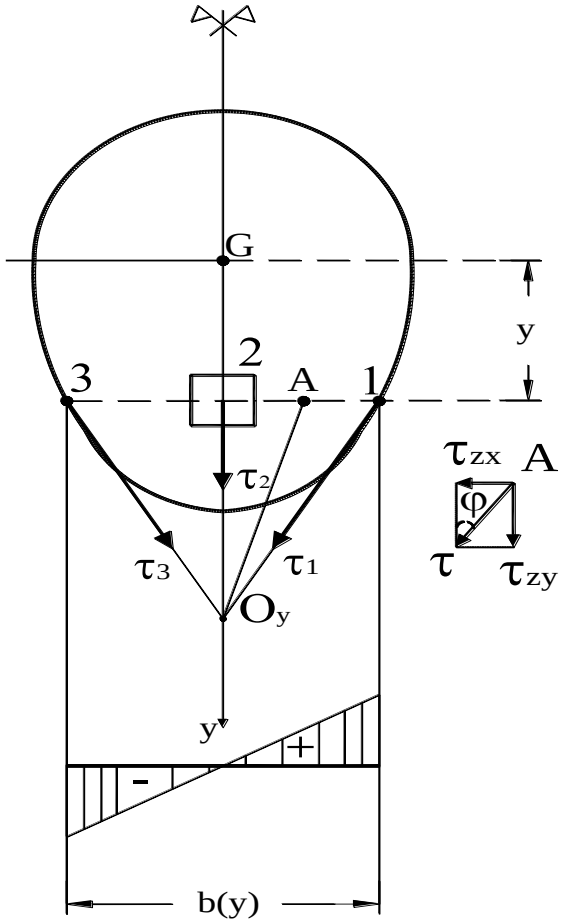
$\bar{y}$: distance between the gravity center of the lower part and gravity center of the whole section

I_x : Second moment of area (moment of inertia) of the whole cross-section with respect to x axis,

When it comes to the other shear stress component τ_{zx} this problem will be solved by reducing it to a vector whose direction and a coordinate is known and the other coordinate is sought.

The assumption for directions of the shear stress is as follows: Shear stress directions for all the points on a line at y distance away from the neutral axis pass through the same point. This point is O_y as shown in the figure.

Although this assumption is obvious for the points 1,2, 3 of the cross-section, for any point on the width of the cross section such as the point A, it is approximately correct.



On this basis, the sought component of shear is $\tau_{zx} = \tau_{zy}.tg\varphi$ which is obtained depending on the other component. The τ_{zx} components on the width b(y) is distributed linearly as shown in the figure.

In general,

τ_{zx} components are small. In some cases when

τ_{zy} is maximum, they even could be zero. The resultant shear stress is

$$\tau = \sqrt{\tau_{zx}^{\ 2} + \tau_{zy}^{\ 2}}$$

After obtaining the maximum shear stress in the cross-section, this value are compared with the average shear stress in the cross-section.

$$\tau_{max} = k_y.\tau_m \quad ; \quad \tau_m = Q/A \qquad \tau_m : \text{average shear stress}$$

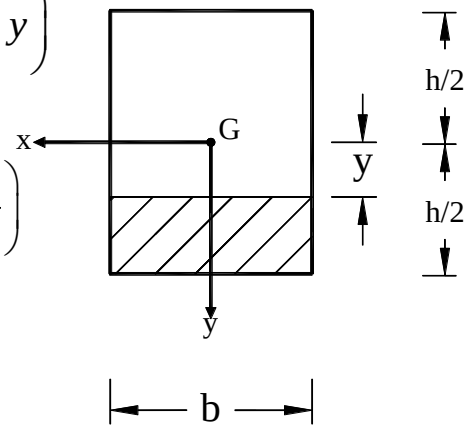
For some symetric cross-sections, calculation of multiplier ky

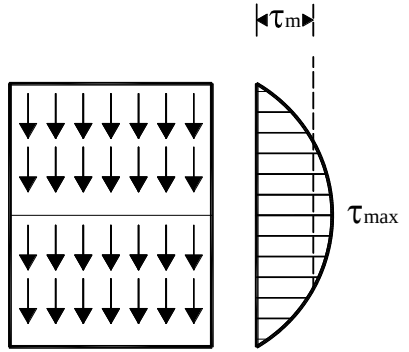
Rectangular cross-section:

$$S_x = \bar{y}.A_{alt} \; ; \; \bar{y} = y + \frac{1}{2}\left(\frac{h}{2} - y\right) \qquad \bar{y} = \frac{h}{4} + \frac{y}{2} \quad ; \quad A_{alt} = b\left(\frac{h}{2} - y\right)$$

$$S_x = \bar{y}.A_{alt} = \frac{b}{2}\left(\frac{h^2}{4} - y^2\right) \qquad \tau = \frac{Q.\frac{b}{2}\left(\frac{h^2}{4} - y^2\right)}{b.\frac{bh^3}{12}} = \frac{Q}{bh}.1{,}5\left(1 - \frac{4y^2}{h^2}\right)$$

$$y = 0 \rightarrow \tau = \tau_{max} = 1.5\tau_m$$



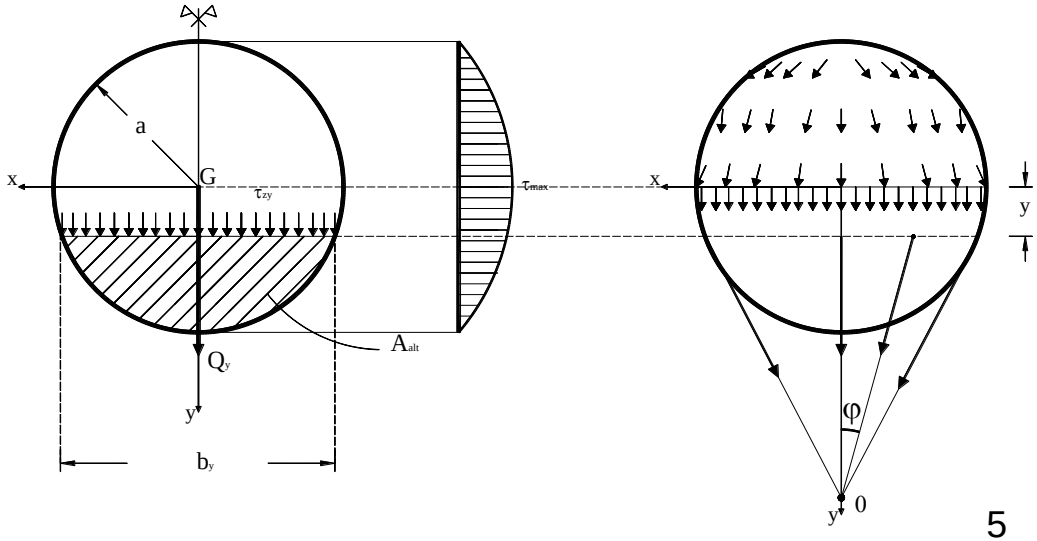


Circular cross-section:

$$\tau_{zy} = \frac{4}{3}\tau_m\left[1 - \left(\frac{y}{a}\right)^2\right] \qquad \left(\tau_{zy}\right)_{max} = \frac{4}{3}\tau_m$$

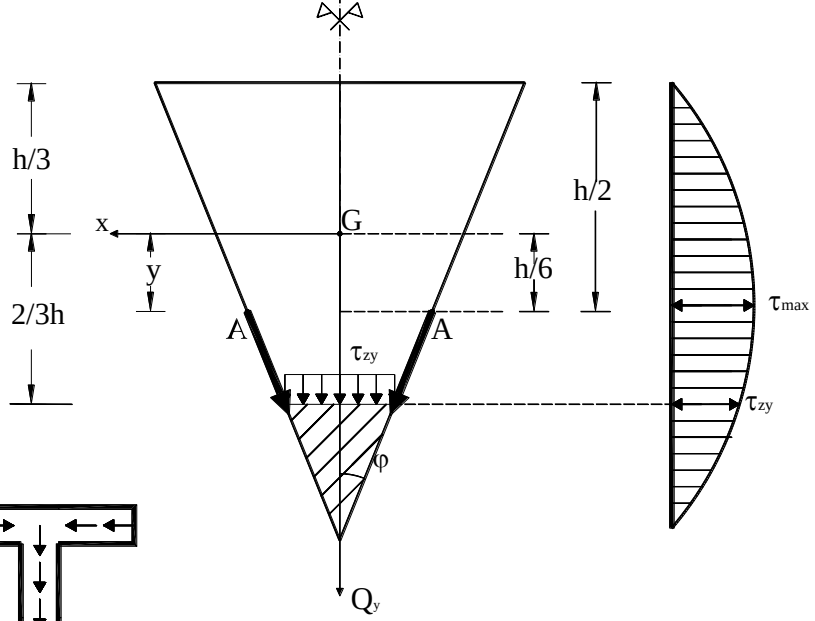
$$k_y = 4/3 \qquad \tau_{zx} = \tau_{zy}.\mathrm{t}\,g\varphi$$

$$\tau_{zx} = \frac{-xy}{\left(a^2 - y^2\right)}\tau_{zy} = -\frac{4}{3}\tau_m\frac{xy}{a^2}$$

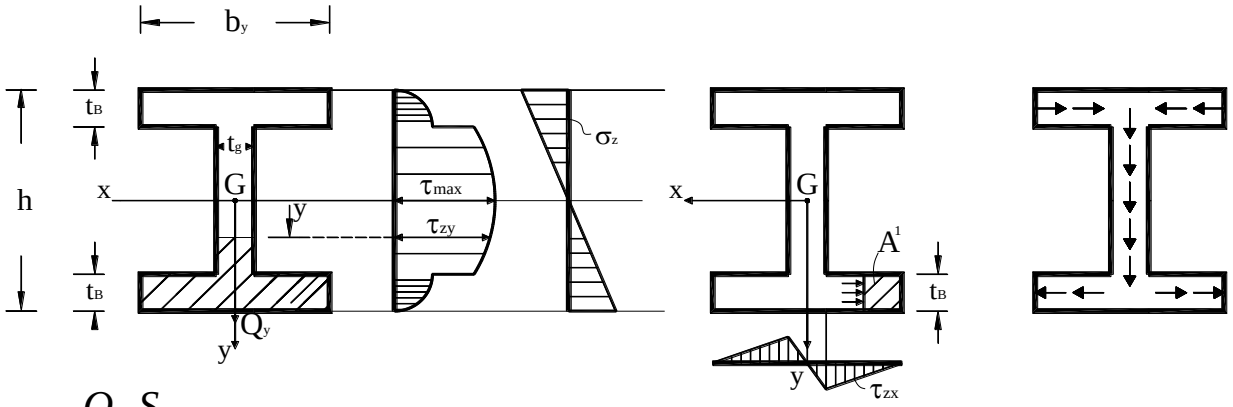


Isosceles Triangle:

$$(\tau_{\max})_A = \frac{3}{2}\tau_m \sqrt{1+\left(\frac{b}{2h}\right)^2} \qquad ky = \frac{3}{2}\sqrt{1+\left(\frac{b}{2h}\right)^2}$$



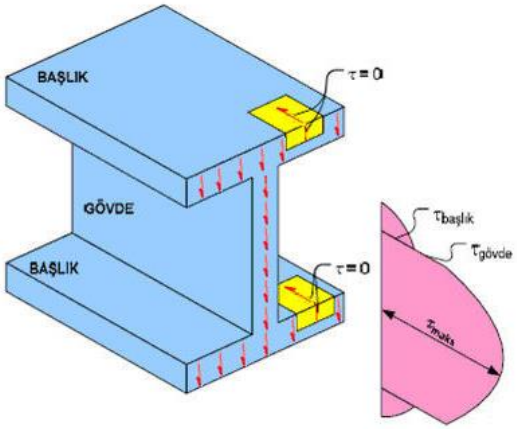
I Profile:

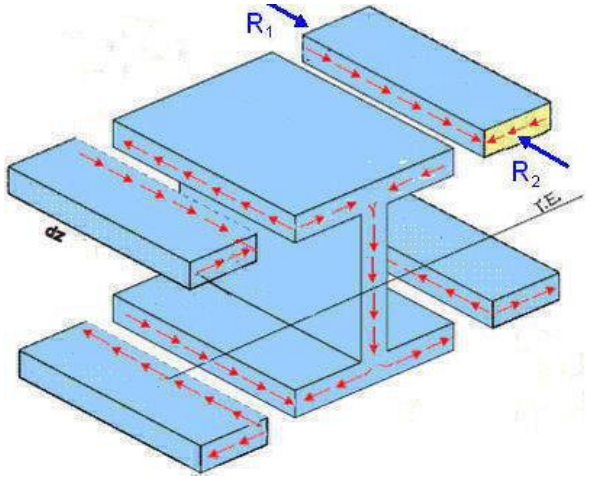


$$\tau_{zy} = \frac{Q_y . S_x}{b_{(y)} . I_x}$$

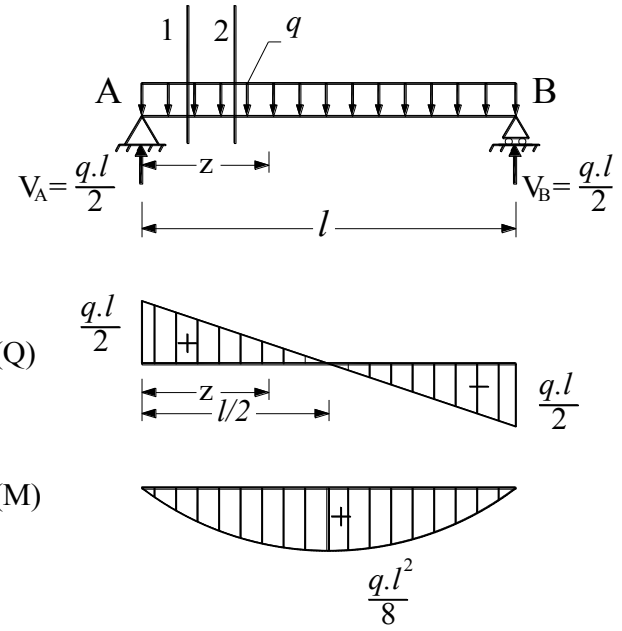
In calculating the shear stress component τ_{zy} , t_b must be considered instead of b(y) and A' must be considered instead of A.

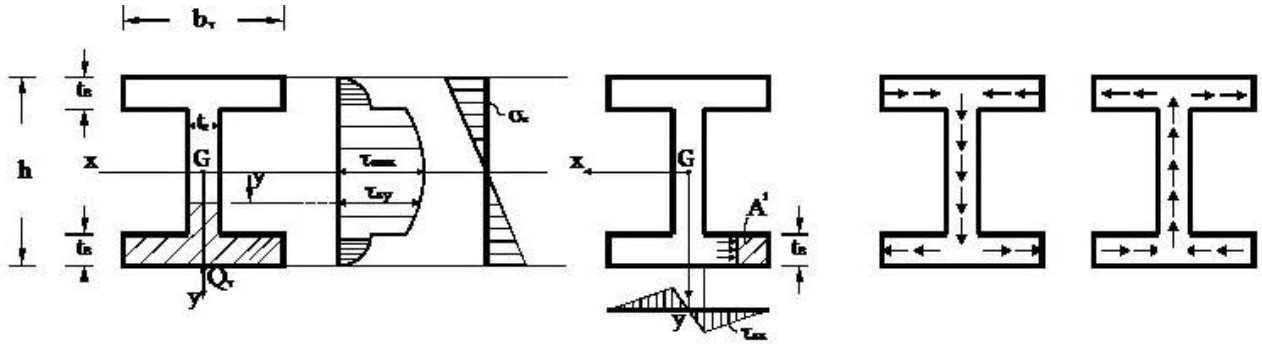
Above, for the I beam cross-section, in the case of positive shear force and positive bending, the directions of shear flux (flow) τt is shown.





I Profile:

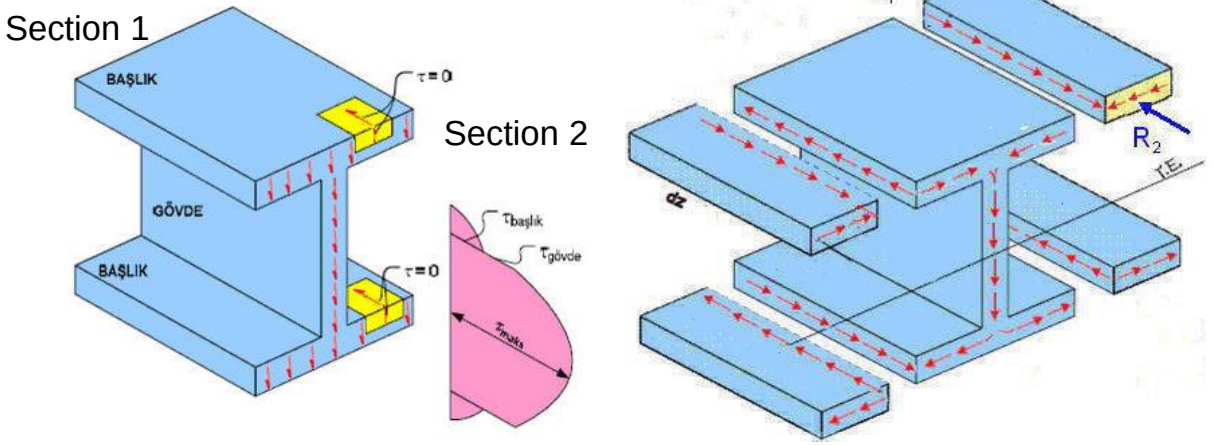




$$\tau_{zy} = \frac{Q_y . S_x}{b_{(y)} . I_x}$$

In calculating the shear stress component τ_{zx}, t_b must be considered instead of b(y) and A' must be considered instead of A.

Above, for the I beam cross-section, in the case of positive shear force and positive bending, the directions of shear flux τt is shown.



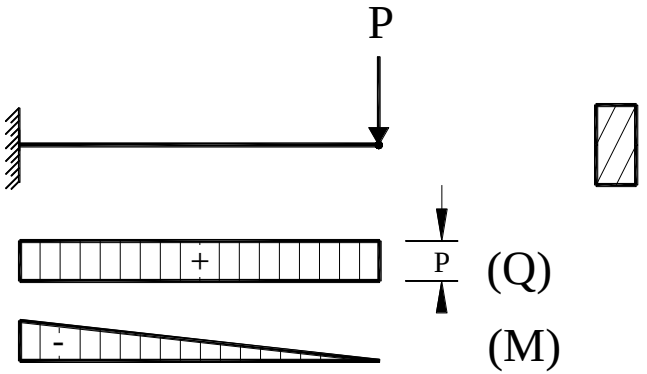
Cross-Section Dimensioning:

Cross-section dimensioning (sizing) in case of bending moment with shear, stresses coming from bending moment can be expressed as $\sigma_z = \frac{M_x}{I_x}.y$ and stresses coming from shear force can be expressed as $\tau_{zy} = \frac{Q_y.S_x}{b_{(y)}.I_x}$

Normal stresses's maximum value is on the farthest fiber from the neutral axis. However, shear stresses's maximum value is on the vicinity of the neutral axis. In addition, if compared, shear stresses's maximum values are smaller than normal stresses's maximum values.

We can see this in an example;

$$\sigma_{max} = \frac{M}{W} = \frac{Pl}{\frac{bh^2}{6}} = \frac{6Pl}{bh^2} \qquad \tau_{max} = 1,5\frac{Q}{A} = \frac{1,5P}{bh} \qquad \frac{\tau_{max}}{\sigma_{max}} = \frac{h}{4l}$$



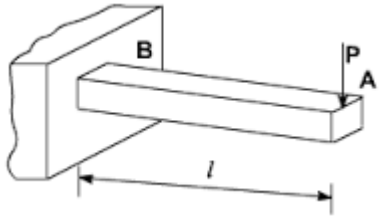
Because h/l value is generally small enough, the important stresses are σ normal stresses . Because of this, the dimensioning problem is based on the normal stresses σ .

Therefore sizing is performed according to the following formula $\frac{M}{W} \leq \sigma_{em}$

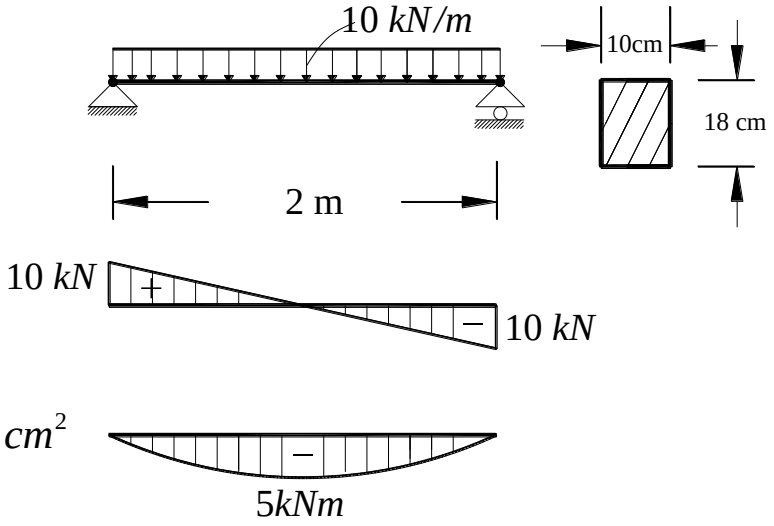
In addition investigation is also made in terms of shear stress.

Namely, it is verified that $\frac{k.Q}{A} \leq \tau_{em}$

If this is not not satisfied, the size is increased.



Problem: Check the beam cross-section for the loading condition given in the figure.

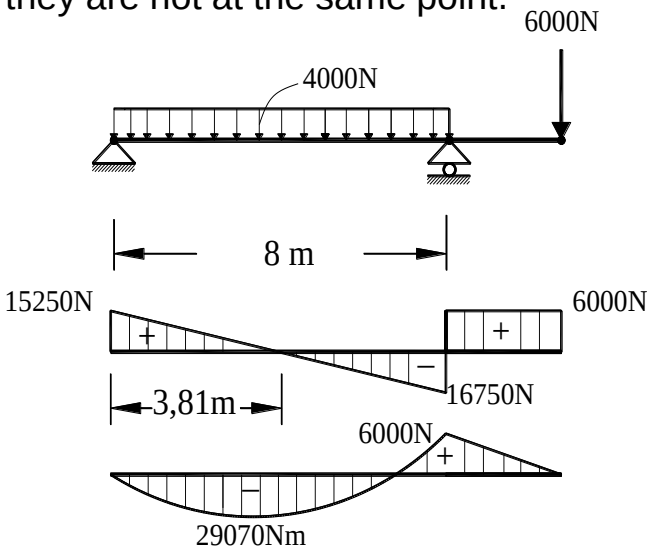


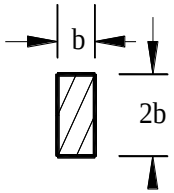
$\sigma_{em} = 1000 N/cm^2, \tau_{em} = 90 N/cm^2$ $\quad Q_{\max} = 10kN$; $M_{\max} = 5kNm$

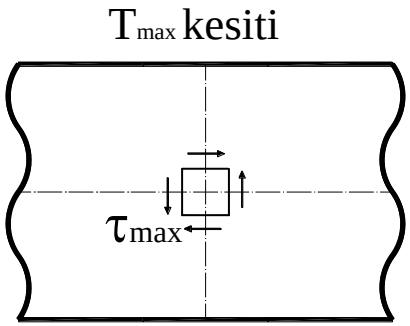
a) Investigation in terms of bending: $\sigma = \dfrac{5.10^5}{\dfrac{10.18^3}{12}}.9 = 930 < 1000 N/cm^2$

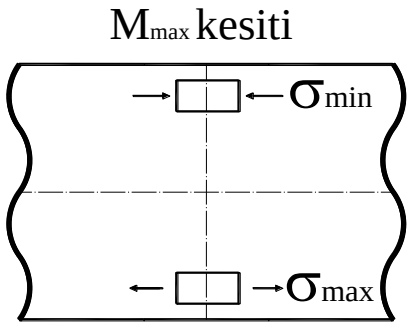
b) Investigation in terms of shear force: $\tau_{\max} = \dfrac{10.10^3.9.10.4,5}{10.\dfrac{10.18^3}{12}} = 83 < 90\ N/cm^2$

Maximum values of normal and shear stresses in a cross-section often are not in the same section and even they are not at the same point.





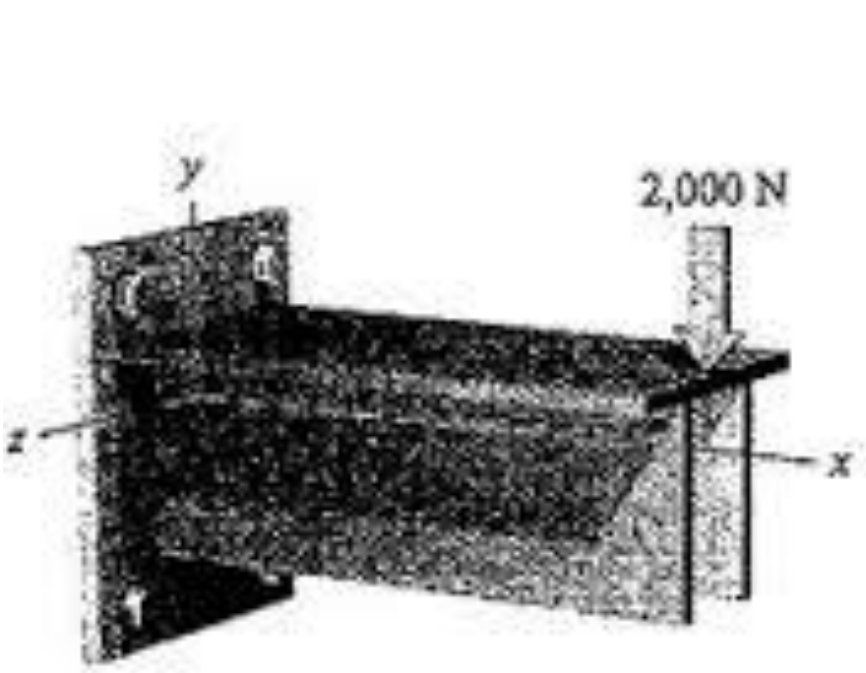


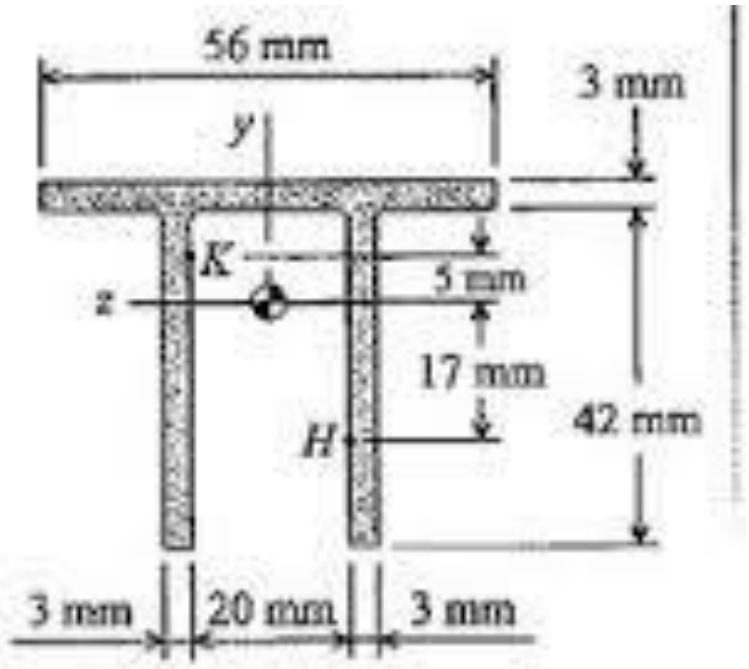


For simple bending, while using $\frac{M}{W}$ formula, bending moment was constant everywhere. Whereas, bending moment is variable in the case of bending with shear.

For this reason, it should be decided in advance that whether or not the cross-section is constant along the bar. If the cross-section is constant along the bar, it is clear that the maximum moment must be used for the dimensioning of the cross-section. For the case bending with shear, a cross-section dimensioned by bending moment is usually sufficient in terms of shear force also. Situations that are critical in terms of shear force are as follows:

1) Shear strength is very small element compared to compressive and tensile strength of the material.
2) If the ratio of height/span is large, shear stresses can be very high in magnitude.
3) In cases where the maximum shear stress is too larger than the average shear stress: I profile sections can be given as an example for this case. In some rarely encountered situations, considering a combination of the effect of normal stress and shear stress may be required. Again I profile sections can be given as an example for this situation. At the point where the web of the section is connected with the flange, normal stresses is very close to maximum normal streses σ_{max} and the shear stresses is very close to the maximum shear streses τ_{max} . In this situation, making separate investigations for the normal and shear stresses are false. In such cases, principal stresses are determined, one of the failure hypothesis is selected, and calculations are made according to it.





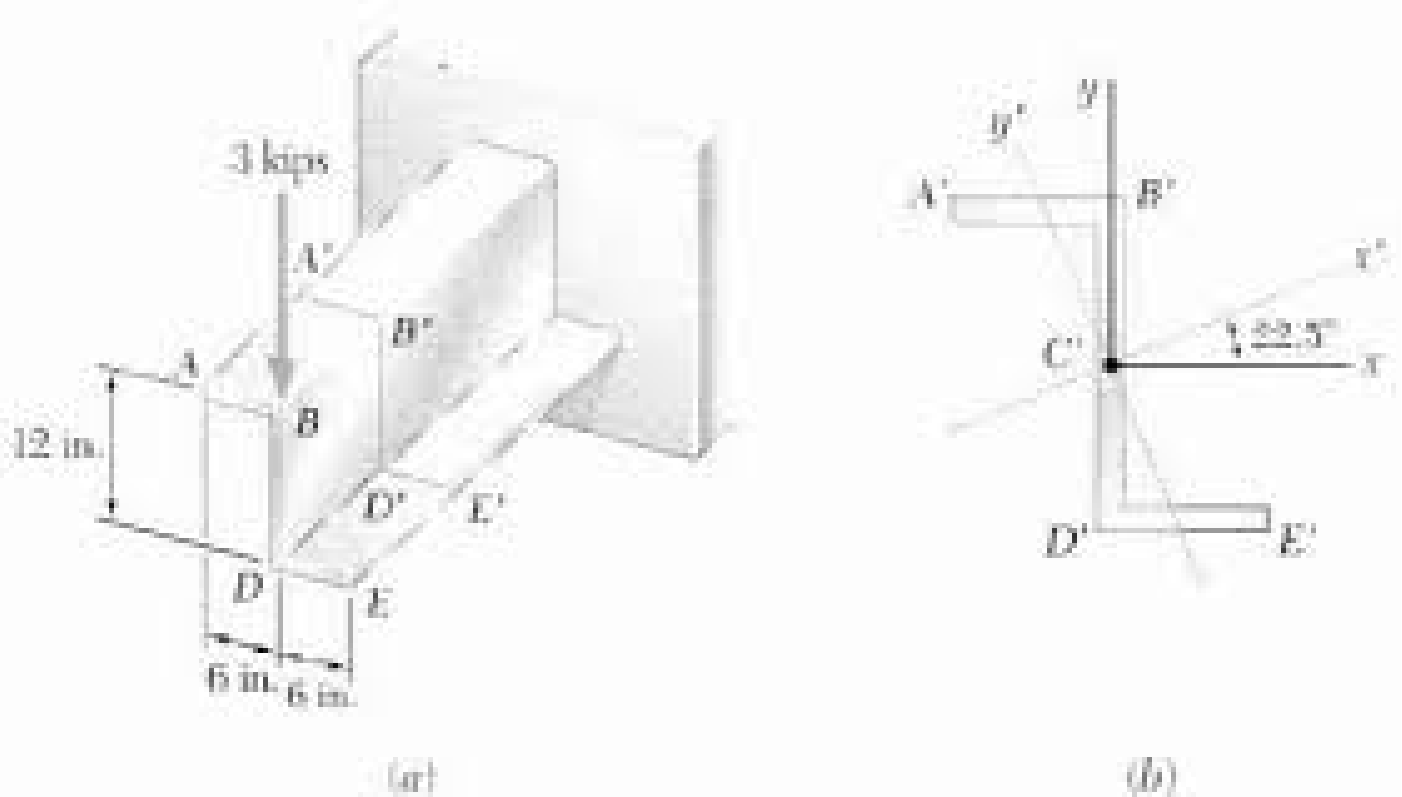
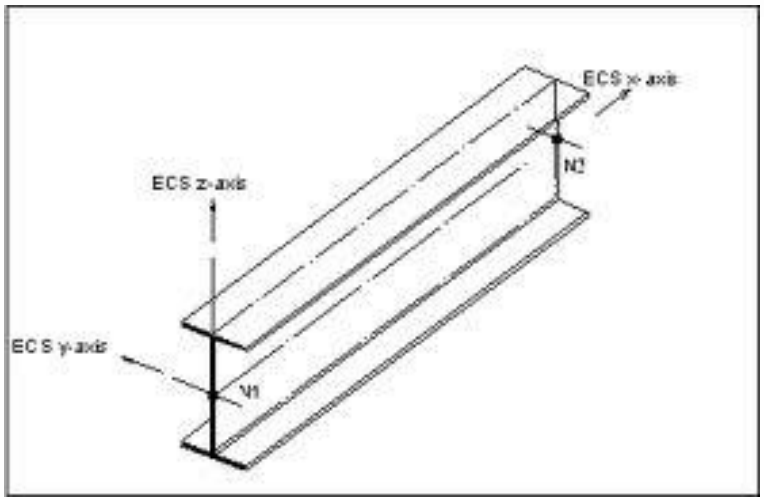
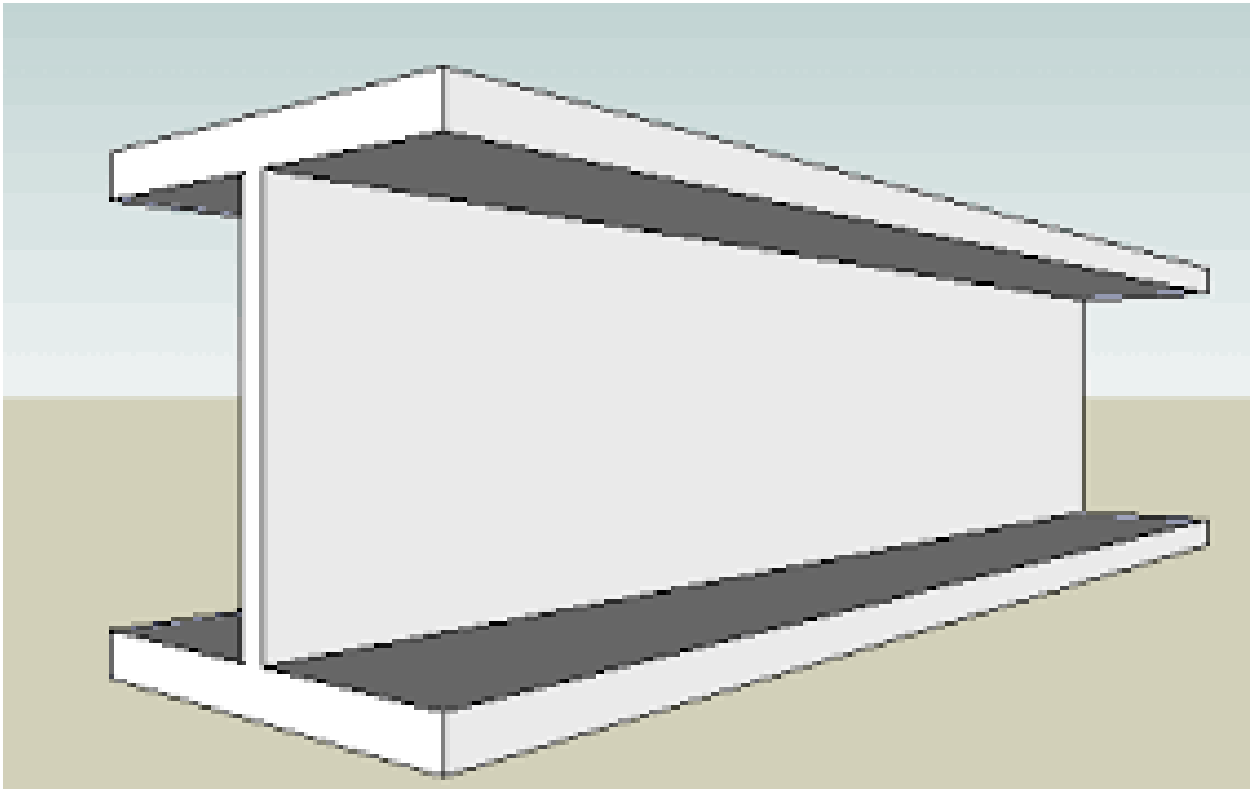
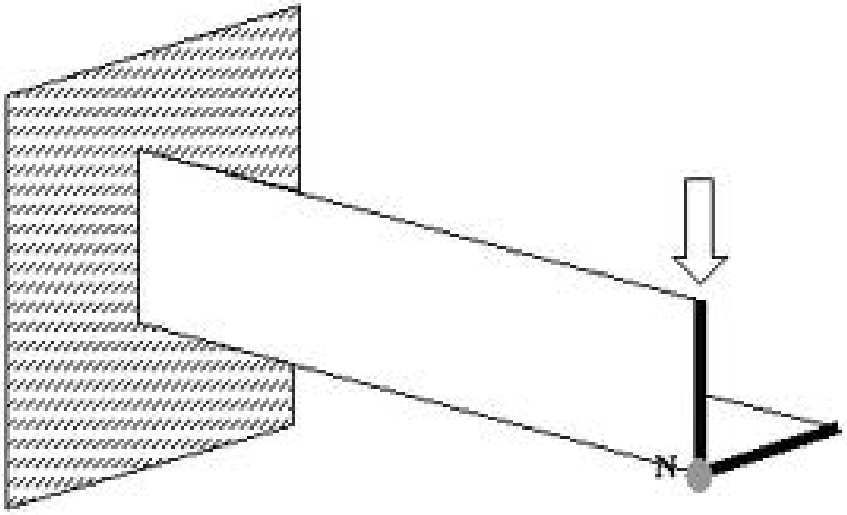

3 kips
A'
B'
A
B
12 in.
D'
E'
D
E
6 in.
6 in.
(a)
y
y'
A'
B'
x'
C'
22.5°
x
D'
E'
(b)

Fig. P6.85


ECS x- axis
N2
ECS z-axis
ECS y-axis
N1




N

$E = 10^{7}$ psi

$\mu = 0.3$

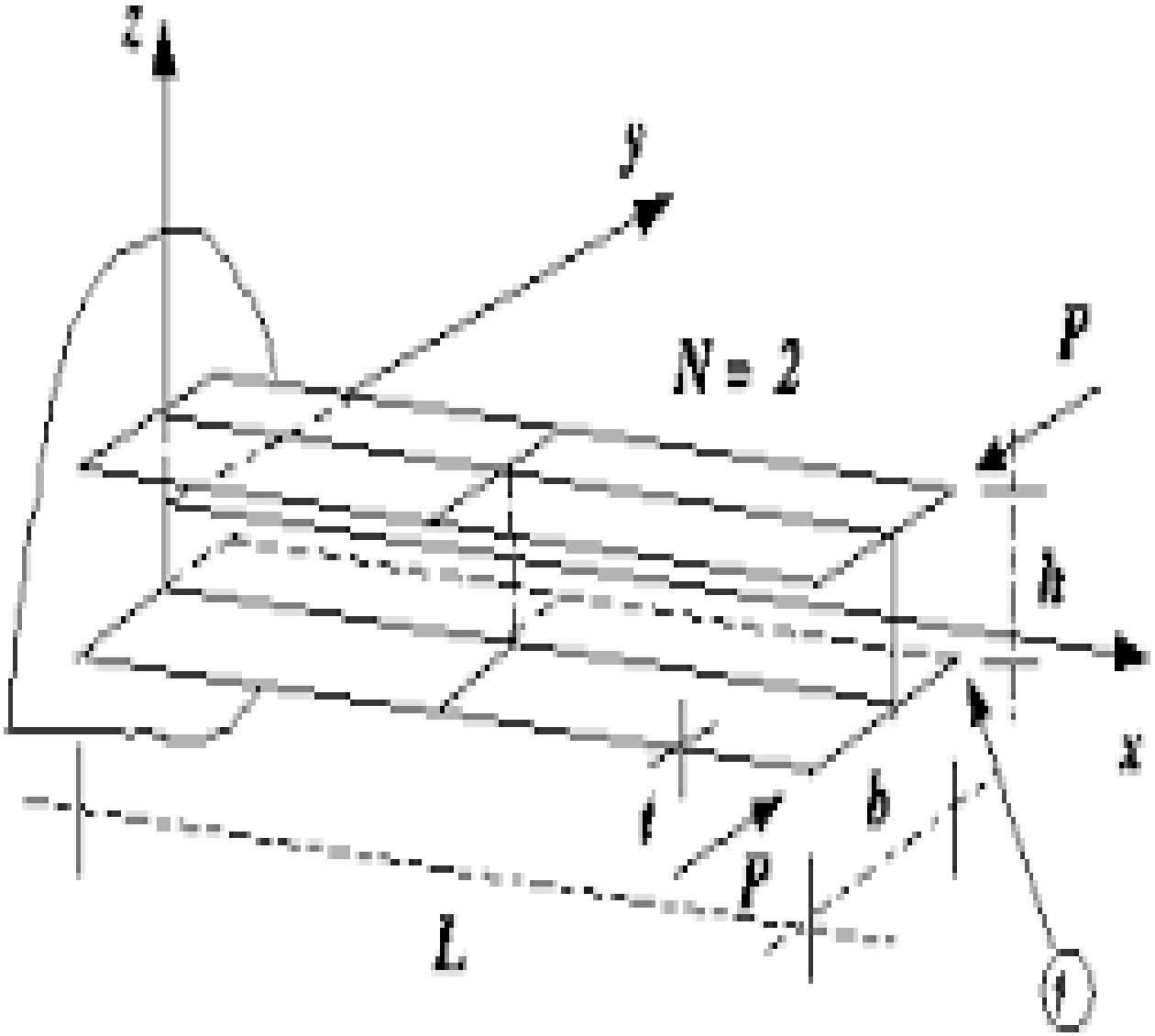
$t = 0.25''$

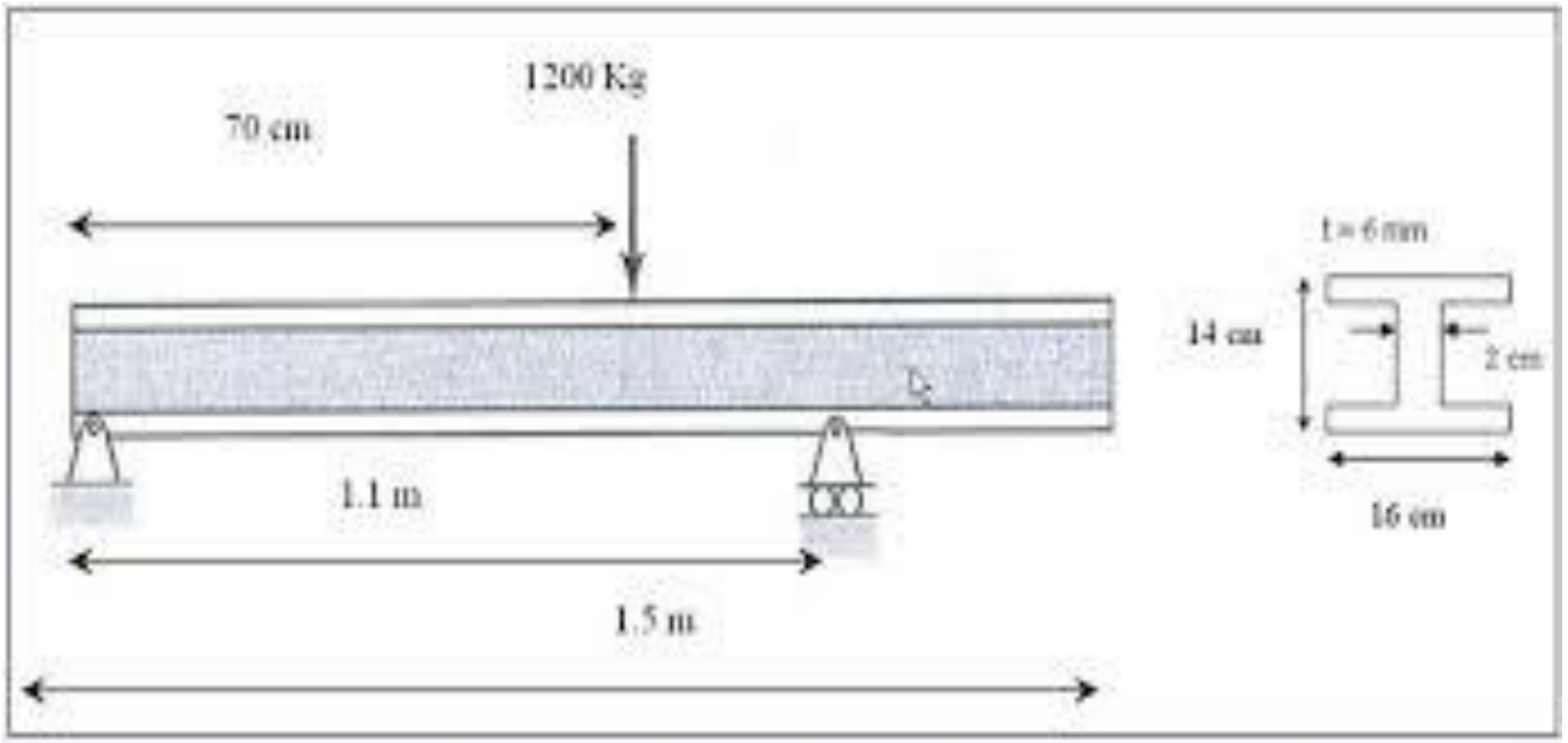
$L = 40''$

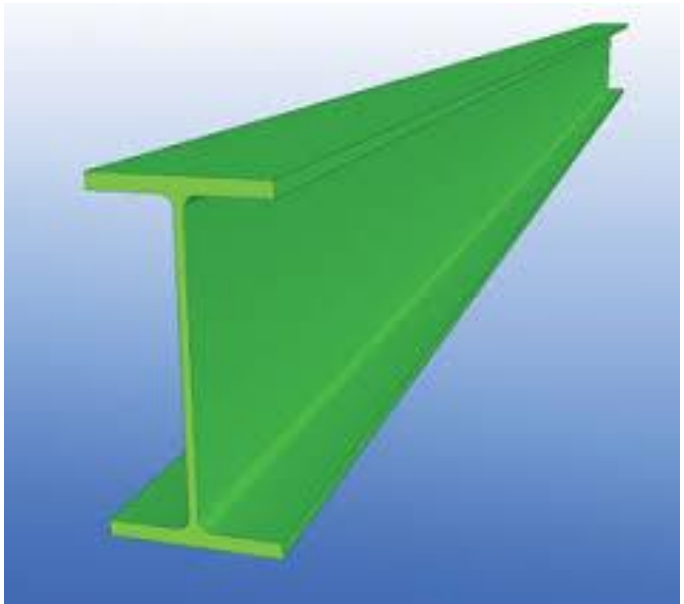
$b = 10''$

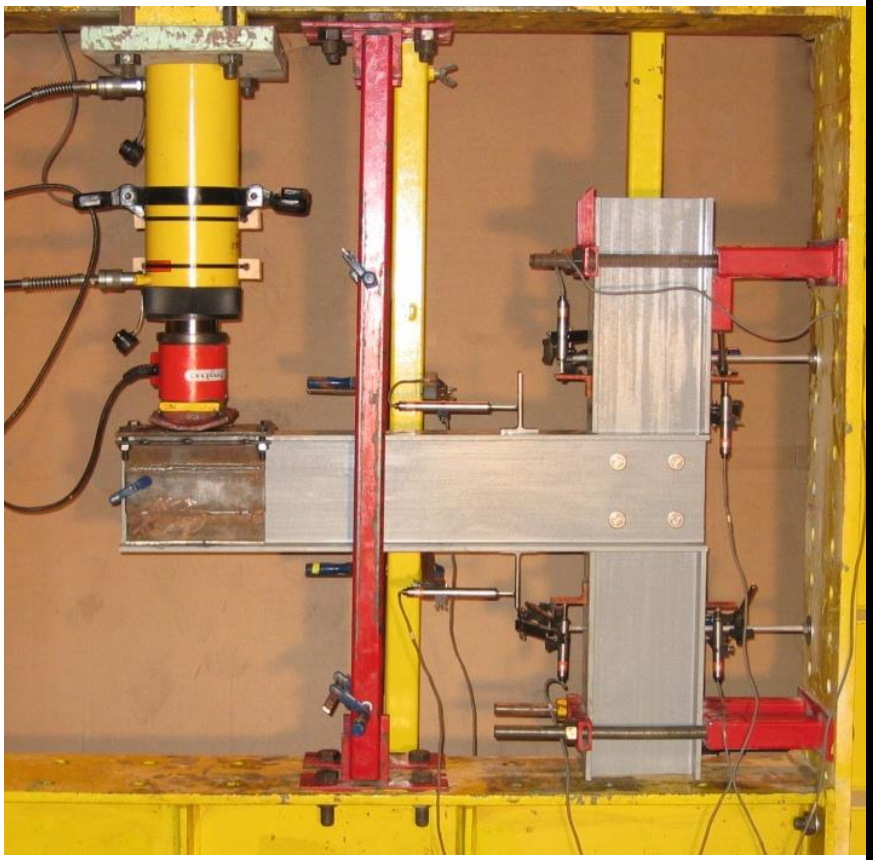
$h = 5''$

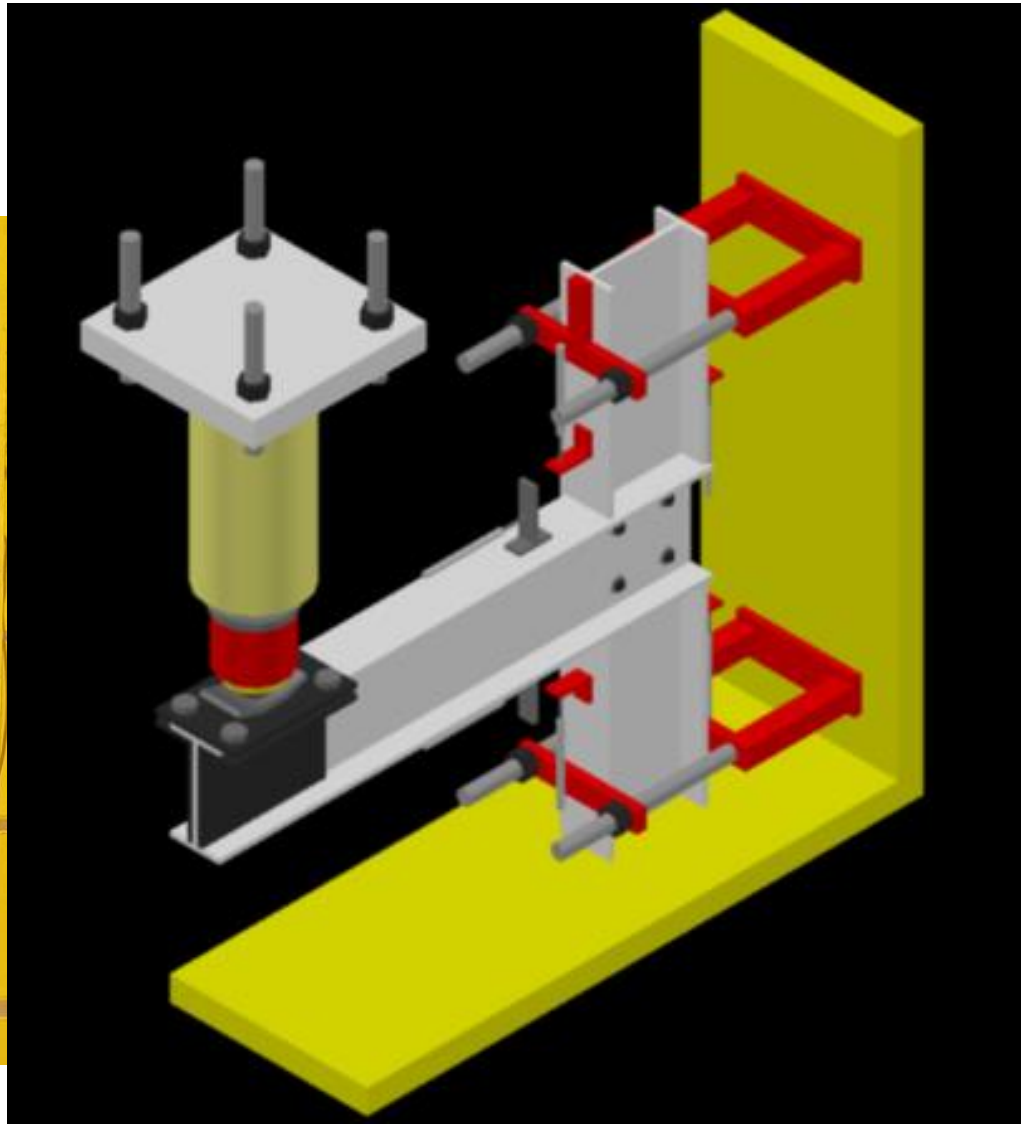
$P = 1600$ lb




1200 Kg
70 cm
1.1 m
1.5 m
t = 6 mm
14 cm
2 cm
16 cm

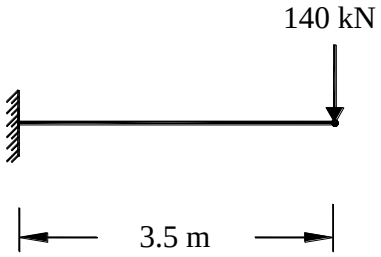






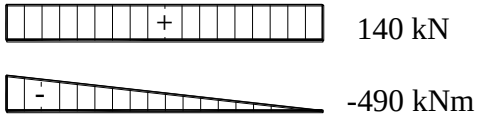
Problem: In the figure, loading condition and dimensions of a beam are given, check whether the cross-section is sufficient.

$$\sigma_{em} = 14000 N/cm^2, \tau_{em} = 8000 N/cm^2 \quad Q_{\max} = 140 kN \; ; \; M_{\max} = 490 kNm$$



$$Ix = \frac{1.18^3}{12} + 2.\left(\frac{9.2^3}{12} + 9.2.10^2\right) = 4098 cm^4 \; ; \; Wx = \frac{4098}{11} = 372.5 cm^3 \; ; \; \sigma_{\max} = 13151 N/cm^2$$

$$\sigma_{\max} = 13151 N/cm^2 < \sigma_{em} = 14000 N/cm^2$$

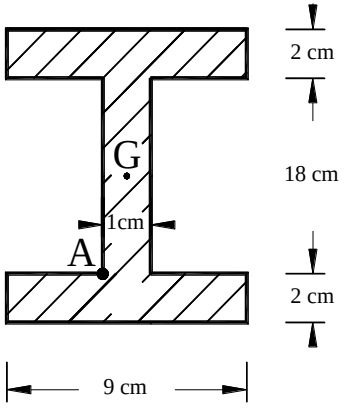


The maximum shear stress is on center of gravity G. Static moment of lower part of the G is

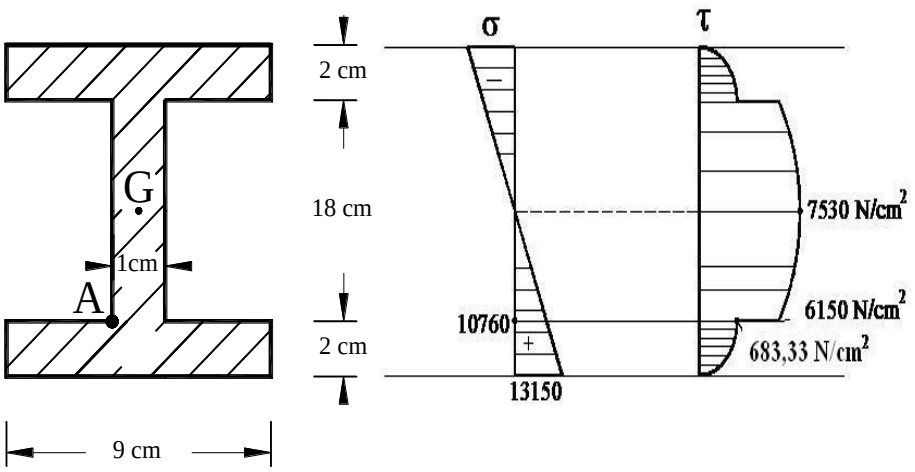
$$S_x = 9.2.10 + 9.1.4,5 = 220,5 cm^3 \quad \tau_{\max} = \frac{Q.S_x}{b.I_x} = \frac{140000.220,5}{1.4098} = 7530 < 8000 N/cm^2$$

To find the distribution of shear stress, shear stress at point A will be obtained:

$$S_x = 9.2.10 = 180 cm^3 \quad \tau_{Aüst} = \frac{140000.180}{1.4098} = 6150 N/cm^2 \quad \tau_{Aalt} = \frac{140000.180}{9.4098} = 683.33 N/cm^2$$



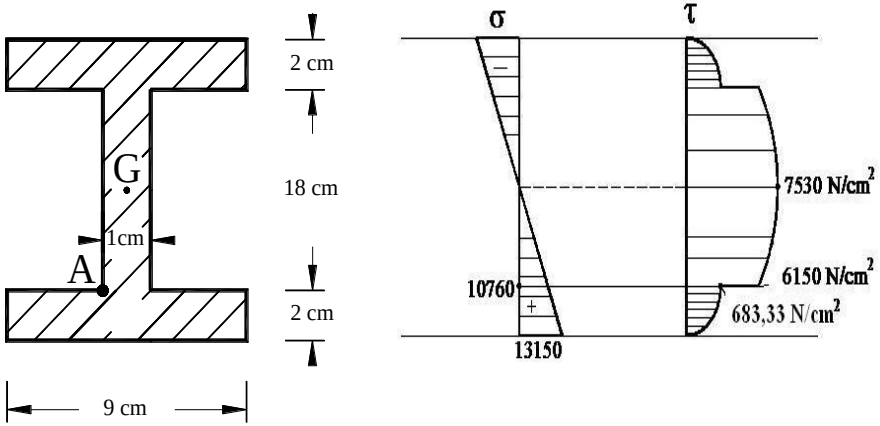
Cross-section



Although if considered separately the maximum shear and maximum normal stresses are not over the allowable stresses, in point A, σ and τ values are close to the maximum related stresses. This point may be more critical then the other points. The principal stresses in this point are

$$\sigma_{1,2} = \frac{10760}{2} \pm \sqrt{\left(\frac{10760}{2}\right)^2 + 6150^2} = 5830 \pm 8170 \rightarrow \sigma_1 = 13550 N/cm^2, \sigma_2 = -2790 N/cm^2$$

According to maximum shear stress hypothesis



$$\sigma_1 - \sigma_2 = \sigma_M \rightarrow 13550 - (-2790) = 16340 N/cm^2 > 14000$$

This results shows that the cross-section is not sufficient.

Distortional strain energy hypothesis:

$$\sigma_1^{\,2} + \sigma_2^{\,2} + \sigma_3^{\,2} - \sigma_1\sigma_2 - \sigma_1\sigma_3 - \sigma_2\sigma_3 \leq \sigma_M^{\,2}$$

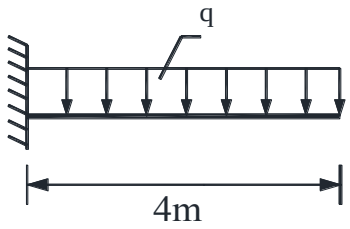
$13550^2 + (-2790)^2 - 13550(-2790) \not\leq 1400^2$ 15139>14000 unsafe.

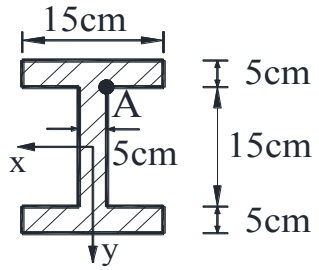
Problem: Find the maximum magnitute of the distributed load q which can be safely carried by the cantilever beam

a) with respect to normal stresses only,

b) with respect to shear stresses only,

c) with respect to the combined effect of normal and shear stresses which occur at neck point A by using maximum shear stress criteria. $\sigma_{safety} = 14000\, N/cm^2$, $\tau_{safety} = 8000\, N/cm^2$. **P.S.:** Cross section is symmetric with respect to x and y axes.





cross-section

Solution

$$I_x = \frac{15.25^3}{12} - \frac{10.15^3}{12} = 16718{,}7cm^4$$

$$a)\ M_x = -80000q\ (Ncm), \quad \sigma_{\max} = \frac{-80000q}{16718{,}7}(-12{,}5) \le 14000 \ \Rightarrow \ q \le 234{,}6\ N/cm$$

$$b)\ S_x^G = 5.15.10 + 5.7{,}5.7{,}5/2 = 890{,}6\, cm^3, \quad T_{\max} = 400q$$

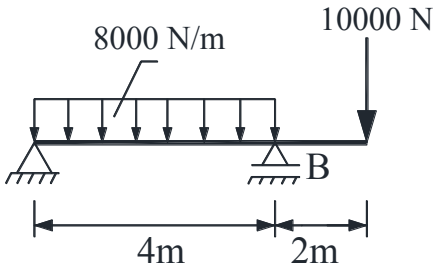
$$\tau_{\max} = \tau_G = \frac{400q.890{,}6}{5.16718{,}7} \le 8000 \quad \Rightarrow \quad q \le 1877{,}16\ N/cm$$

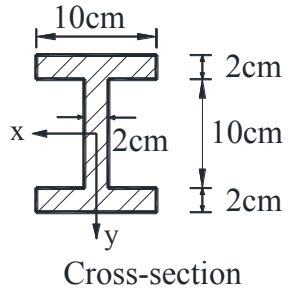
$$c)\ \sigma_z^A = \frac{-80000q}{16718{,}7}(-7{,}5) = 35{,}89q, \quad S_x^A = 15.5.10 = 750\, cm^3, \quad \tau^A = \frac{400q.750}{5.16718{,}7} = 3{,}59q$$

$$\sigma_{1,2} = 17{,}94q \pm 18{,}3q \quad \Rightarrow \quad \sigma_1 = 36{,}4q, \quad \sigma_2 = 0, \quad \sigma_3 = -0{,}441q$$

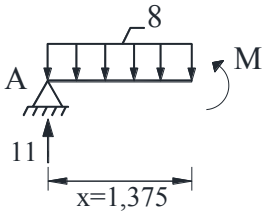
$$\sigma_1 - \sigma_3 = 36{,}4q - (-0{,}441q) = 36{,}841q \le 14000 \quad \Rightarrow \quad q \le 380\ N/cm$$

Problem: For the overhanging beam seen in the figure, maximum allowable stresses are $\tau_{safety} = 9000\ N/cm^2$, $\sigma_{safety} = 14000\ N/cm^2$. Check the safety of the beam with respect to shear and normal stresses. Draw the shear and normal stress distributions over the cross section for the critical locations.





Solution



$$\sum M_A = 0 \rightarrow B_y.4 - 8.4.2 - 10.6 = 0 \rightarrow B_y = 31kN \qquad A_y = 11kN$$

$$\sum M = 0 \rightarrow M + 8.\frac{1,375^2}{2} - 11.1,375 = 0 \rightarrow \quad M = 7,5625\,kNm$$

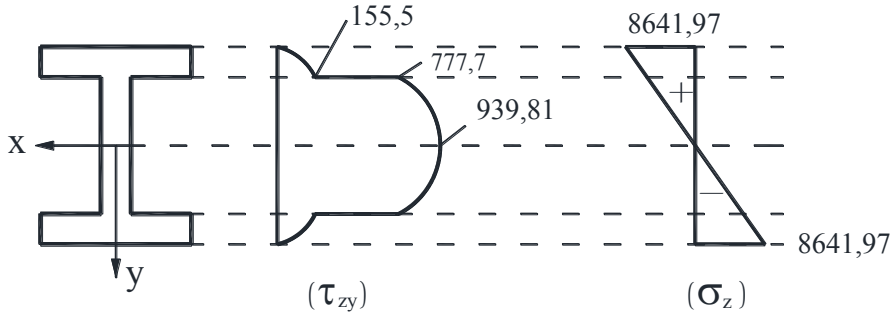
$$T_{max} = 21kN = 21000N, \ M_{max} = 20\,kNm = 2000000\,Ncm$$

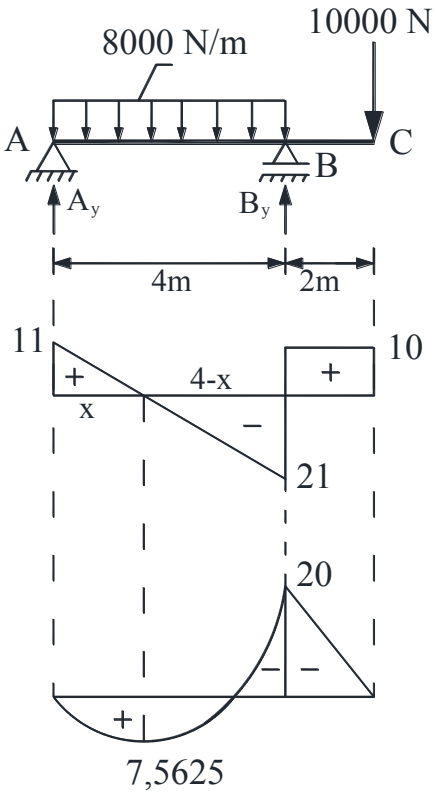
$$I_x = \frac{10.14^3}{12} - 2.\frac{4.10^3}{12} = 1620\,cm^4$$

$$(\sigma_z)_{\max} = \frac{2.10^6}{1620}.7 = 8641.97\,N/cm^2 \le \sigma_{safety} = 14000 \quad \text{(safe)}$$

$$\tau_{web} = \frac{21000.(10.2.6 + 5.2.5/2)}{1620.2} = 939,81\,N/cm^2 \le \tau_{safety} = 9000 \quad \text{(safe)}$$

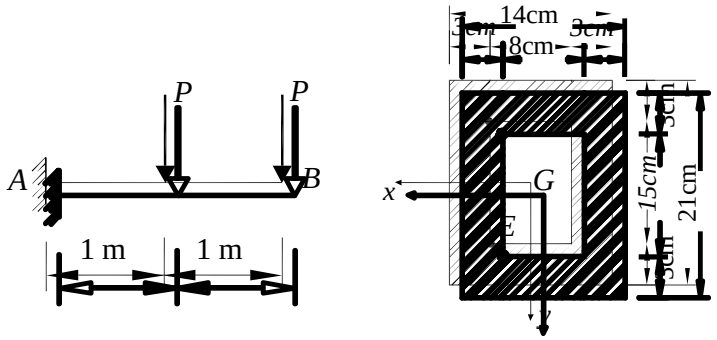
$$\tau_{flange}^{bottom} = \frac{21000.(10.2.6)}{1620.10} = 155,55\,N/cm^2, \quad \tau_{web}^{upper} = \frac{21000.(10.2.6)}{1620.2} = 777,77\,N/cm^2$$





Problem: The cantilever beam given below is under the affect of forces P. If the safety stress of the material is $\sigma_{safety} = 14000\ N/cm^2$,

a) Find the maximum value of the force P only considering the effect of bending moment.
b) Draw the diagrams of the normal and shear stresses at the most affected cross-section of the beam where force P is equal to the value found at part a.
c) Check the stress at point E using the max. shear stress hypothesis.



Solution:

1. The section under the most stress is section A.

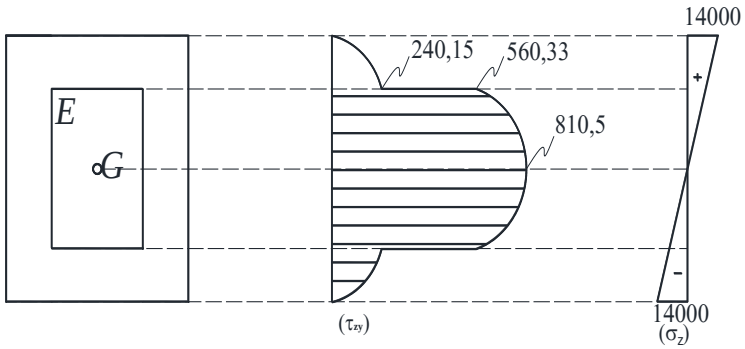
$$T_y = 2P,\ M_x = -3P,\ I_x = \frac{14.21^3}{12} - \frac{8.15^3}{12} = 8554,5\,cm^4$$

$$a)\ \sigma_{\max} = -\frac{300P}{8554,5}.(-10,5) = 0,368P \le \sigma_{em} = 14000 \Rightarrow P \le 38043\ N = 38,043\,kN$$

$$\tau_G = \frac{2.38043.(3.14.9 + 2.3.7,5.7,5/2)}{6.8554,5} \Rightarrow \tau_G = 810,5\,N/cm^2$$

$$b)\ (\tau_E)_{top} = \frac{2.38043.3.14.9}{14.8554,5} = 240,15\,N/cm^2,\quad \sigma_{\substack{\max\\ \min}} = \frac{-300.38043}{8554,5}(\mp 10,5) = \pm 14000$$

$$(\tau_E)_{bottom} = \frac{2.38043.3.14.9}{6.8554,5} = 560,33\,N/cm^2$$



$$\sigma_E = \frac{-300.38043}{8554,5}.7,5 \cong -10000\,N/cm^2,\quad \tau_E = 560,33\,N/cm^2$$

$$c)\ \sigma_{\substack{\max\\ \min}} = -\frac{10000}{2} \pm \sqrt{\left(\frac{10000}{2}\right)^2 + (560,33)^2} \Rightarrow \sigma_{\max} = 31,3\,N/cm^2,\ \sigma_{\min} = -10031,3\,N/cm^2$$

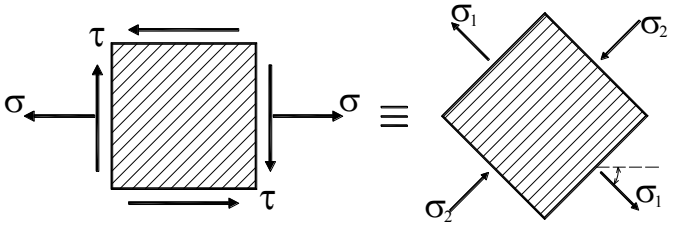
$$\sigma_{\max} - \sigma_{\min} \le \sigma_{safety} \Rightarrow 31,3 - (-10031,3) \le 14000 \Rightarrow 10062,6 \le 14000\ (Safe)$$

ORBITS OF THE PRINCIPAL STRESSES

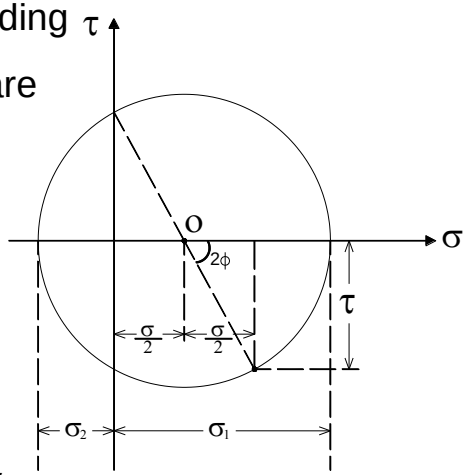
In the case bending with shear, the beam is subjected to two dimensional stress state due to the shear stresses. The figure (a) shows normal stress σ which comes from bending moment and shear stress τ which is due to the shear force. The principal stresses are found from MOHR circle

$$\sigma_1 = \frac{1}{2}\left[\sigma + \sqrt{\sigma^2 + 4\tau^2}\right] \quad \text{ve}$$

$$\sigma_2 = \frac{1}{2}\left[\sigma - \sqrt{\sigma^2 + 4\tau^2}\right]$$



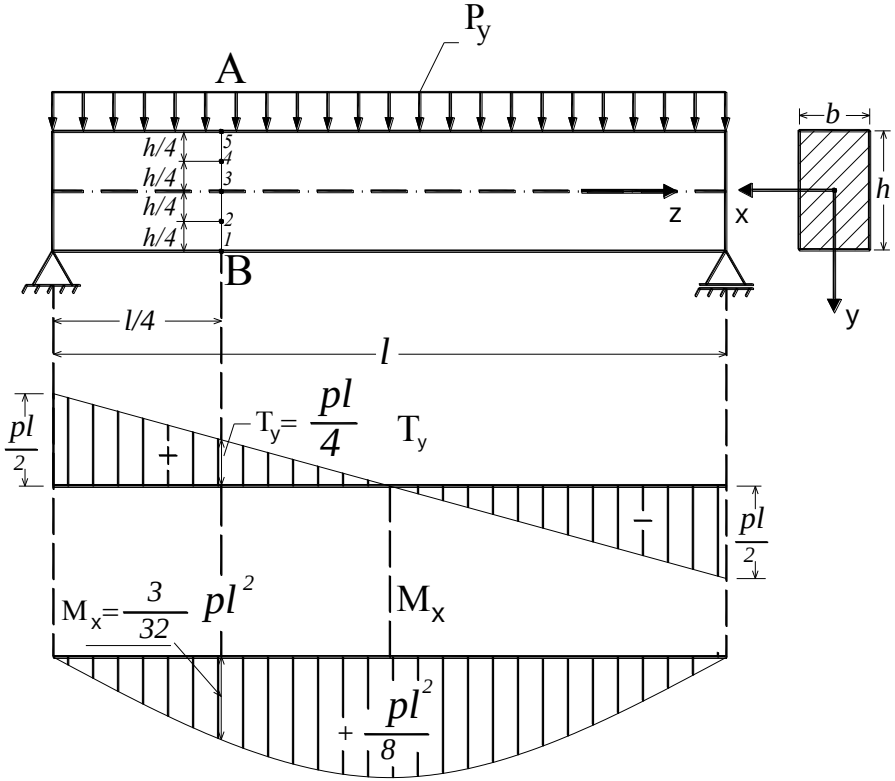
(a) (b)

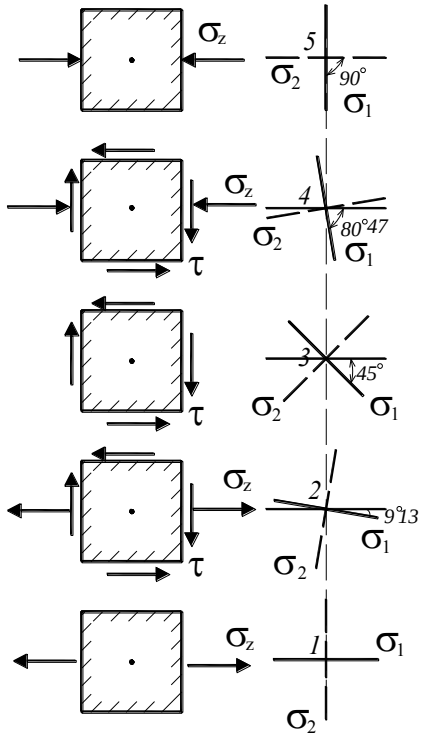


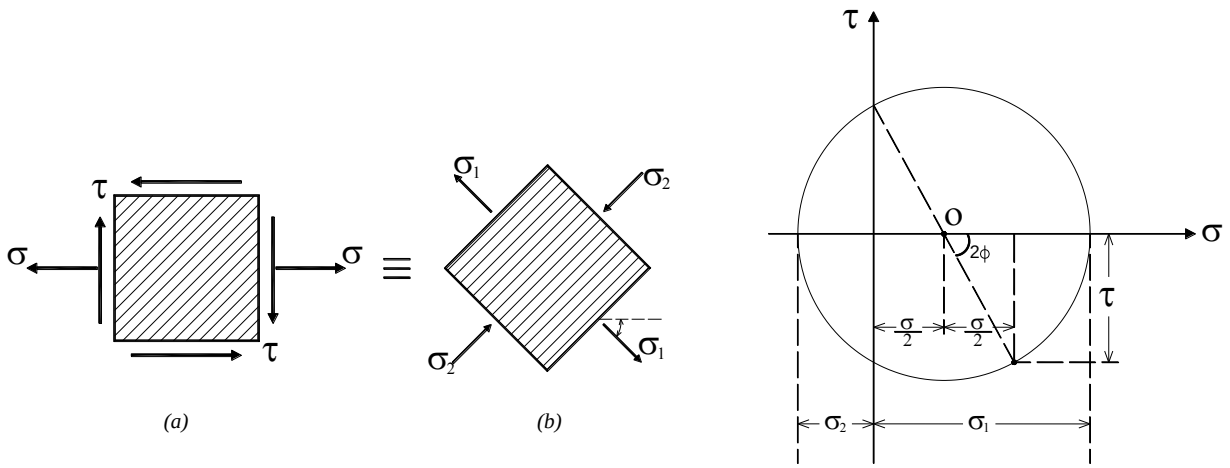
(c)

The angle between the axis of the bar and the tensile stress is φ

$$tg2\varphi = 2\tau / \sigma$$







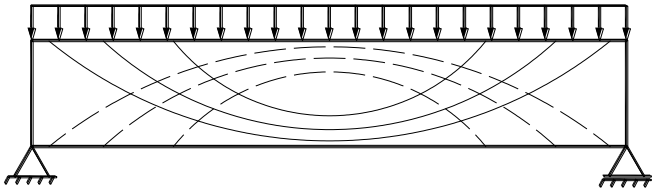
$$\sigma_{1,2} = \frac{\sigma_x + \sigma_y}{2} \pm \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2} \qquad (1)$$

$\sigma_x = \sigma,\ \sigma_y = 0, \tau_{xy} = \tau$ By considering these and using the formula (1)

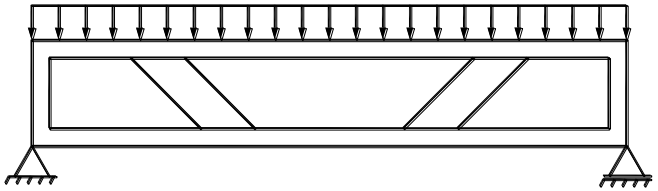
$$\sigma_1 = \frac{1}{2}\left[\sigma + \sqrt{\sigma^2 + 4\tau^2}\right] \quad \text{and}$$

$$\sigma_2 = \frac{1}{2}\left[\sigma - \sqrt{\sigma^2 + 4\tau^2}\right]$$

The expressions above give the principal stresses and its directions at any point of the beam. In figure (e), five elements and the directions of the principal stresses at the investigated section at $z = l/4$ is shown. Continuous lines are tensile pirincipal stresses σ_1, dashed lines are compressive pirincipal stresses σ_2 . Likewise, principal stress directions of all the other points on the beam are found: This means that the principal stress directions are known. Than, it is possible to find orbits of the principal stresses with graphical integration. The orbits are two sets of curved line families intersecting each other perpendicularly and the tangents to any curve at any point indicates the direction of principal stress for that curve at that point.

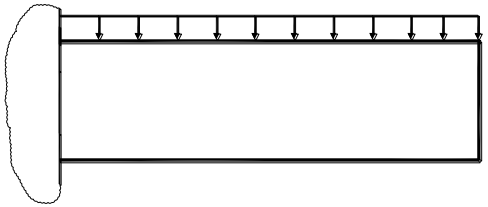


(f) Simply supported beam

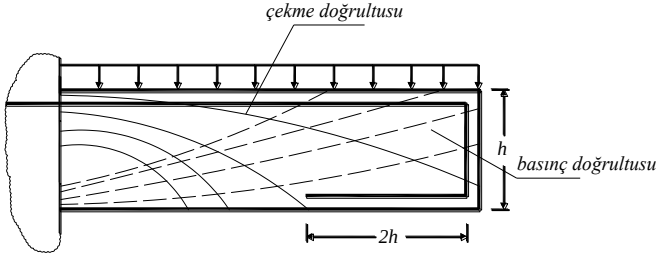


(g) Reinforcement placement for simply supported beam

If the beam is made with concrete which has very small resistance to tensile stresses, then there is need to use reinforcement to resist the tensile stresses and reinforcement forms need to follow orbits of tension stresses. Figures (f) and (g) show the orbits of stresses and the reinforcement placement of a simply supported beam. We place the upper reinforcement for montage requirements. To incease the comressive normal stress resistance of the concrete, we use stirrups. To place the stirrups we need upper reinforcement. The figures (h) and (k) show the orbits of stresses and reinforcement placement of a cantilever beam.

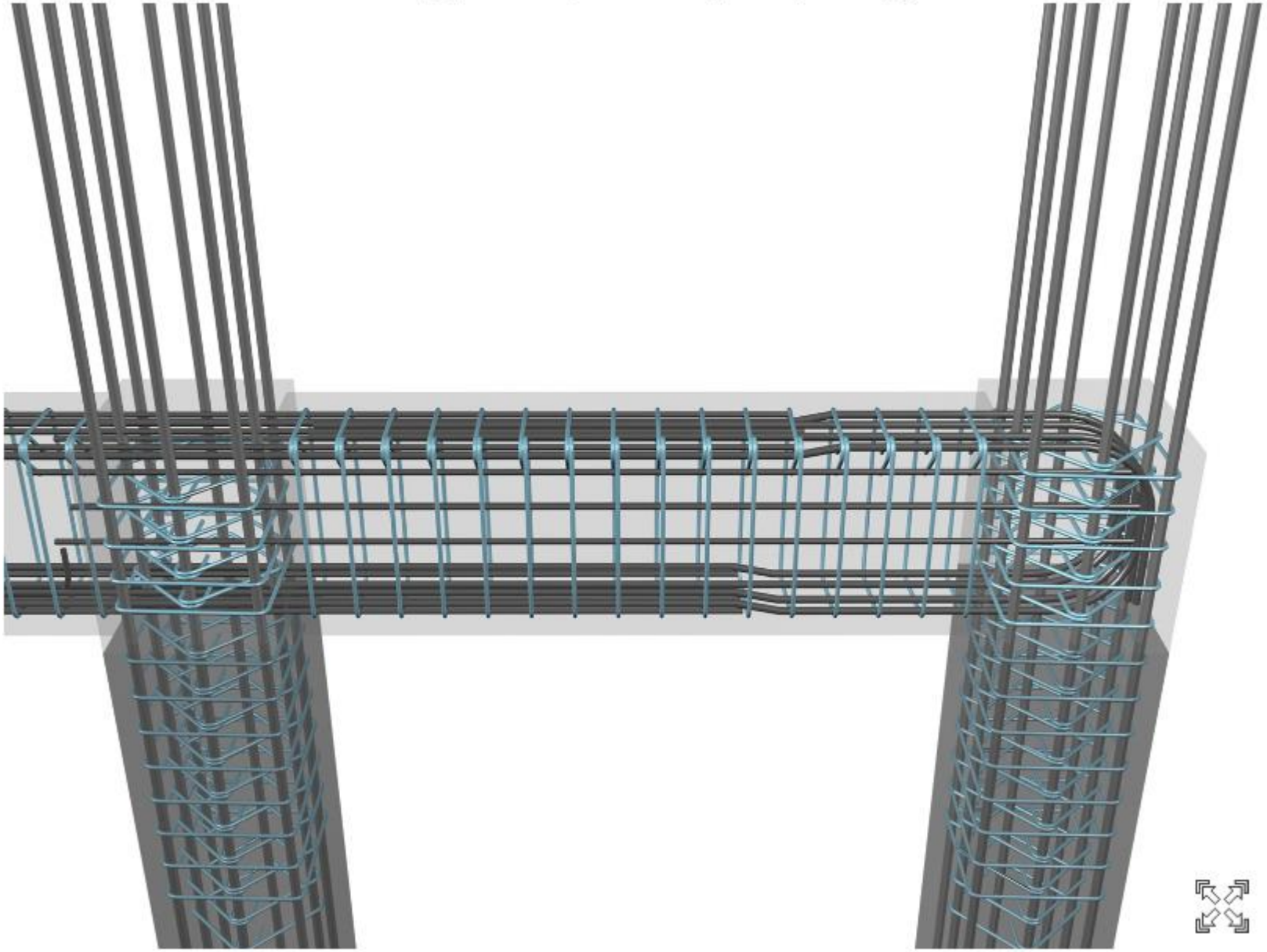


(h) Cantilever beam



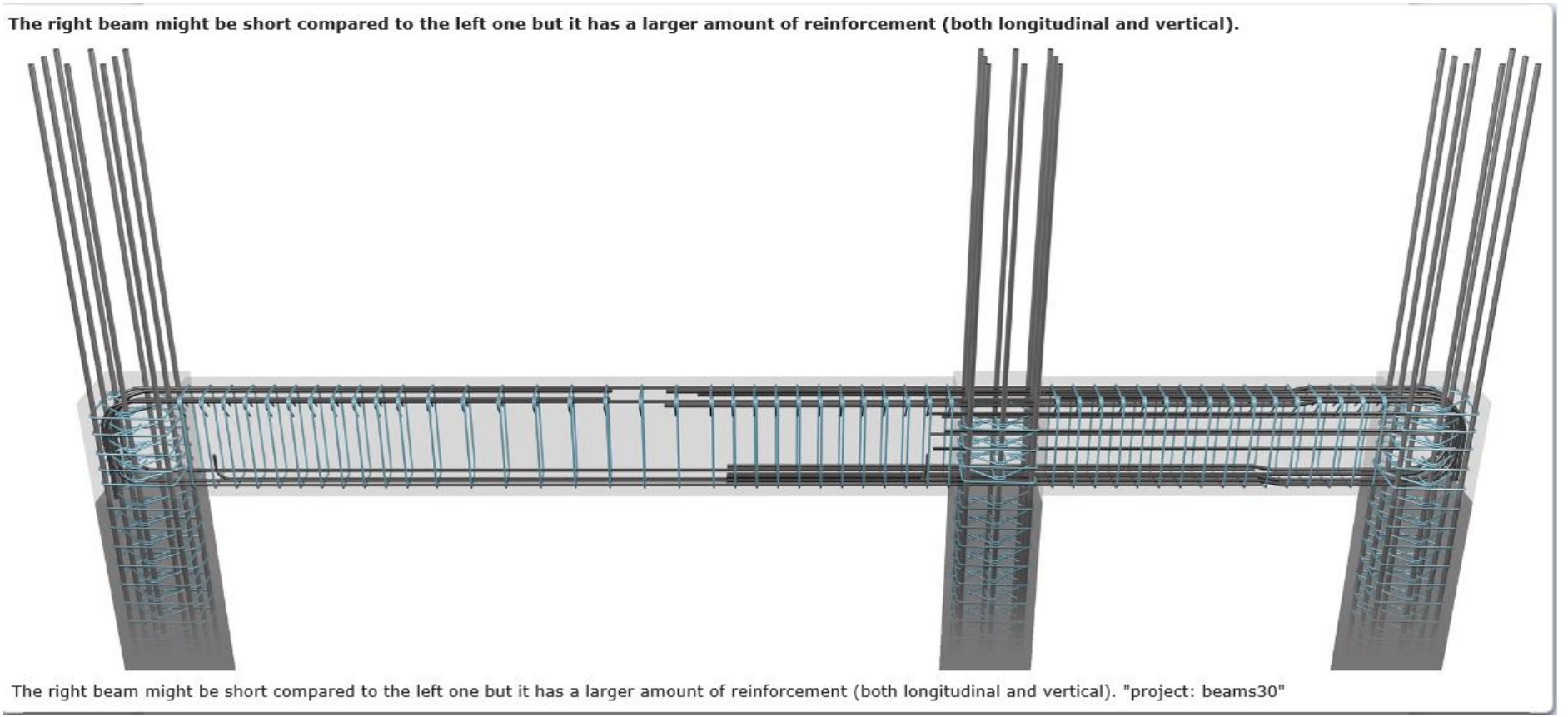
(k) Cantilever beam orbits of principal stress and reinforcement forms plan

The short column is reinforced with (a) narrow spaced strong stirrups and (b) with lateral rebars.

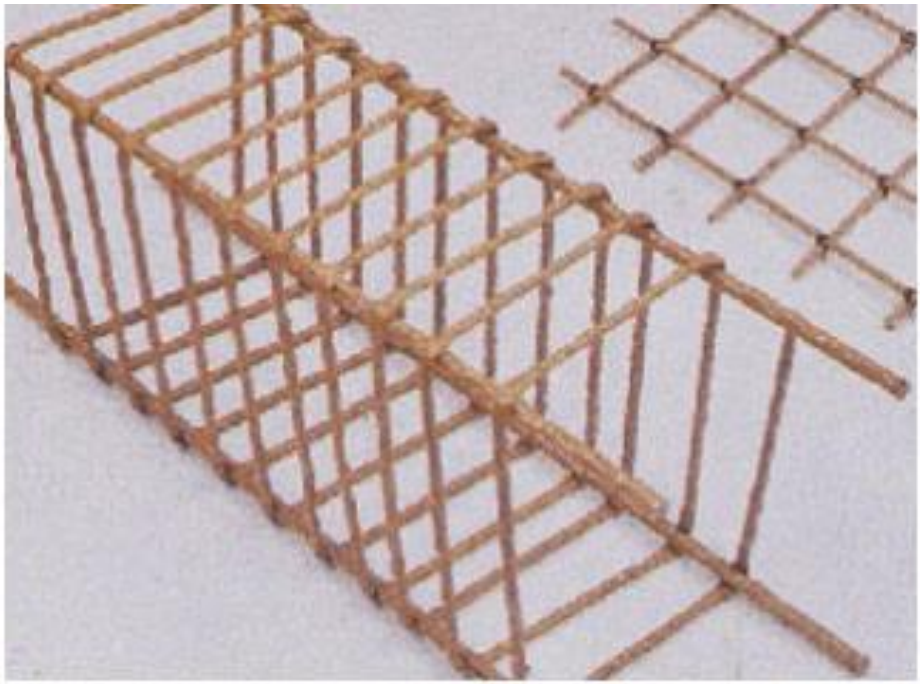


The short column is reinforced with (a) narrow spaced strong stirrups and (b) with lateral rebars.

The right beam might be short compared to the left one but it has a larger amount of reinforcement (both longitudinal and vertical).



The right beam might be short compared to the left one but it has a larger amount of reinforcement (both longitudinal and vertical). "project: beams30"



Stirrup: Etriye



Bent bar: Pilye

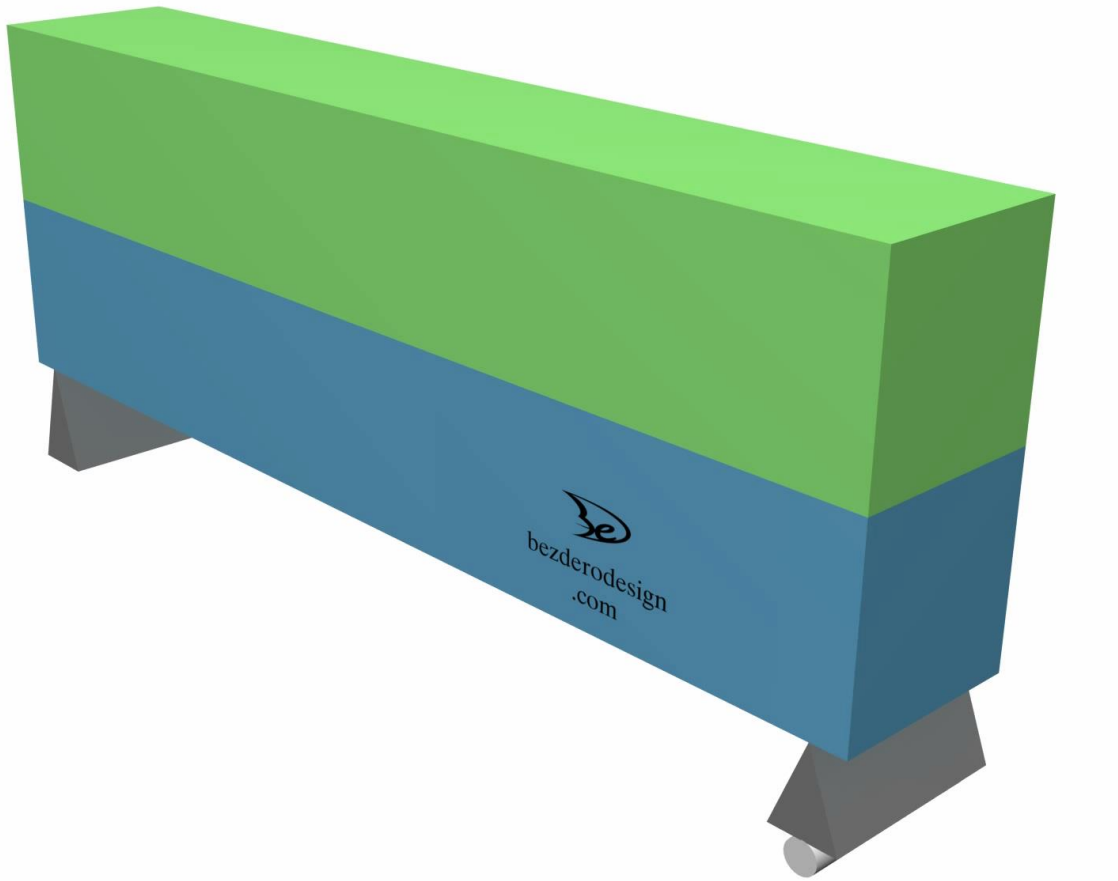
YILDIZ TECHNICAL UNIVERSITY
CIVIL ENGINEERING DEPARTMENT
MECHANICS DIVISION

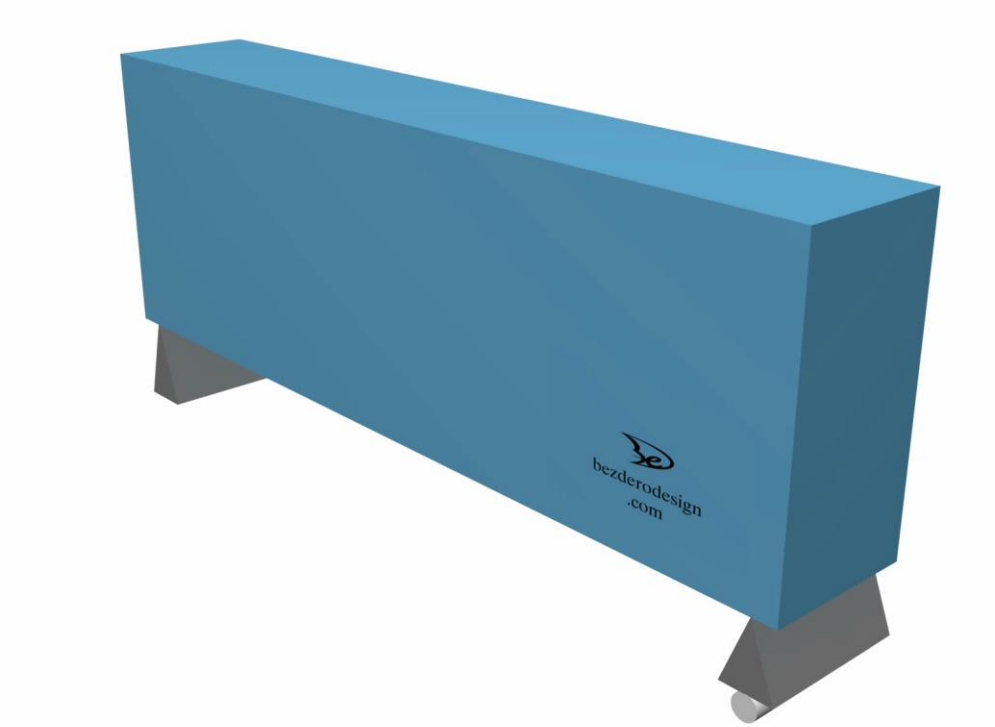
STRENGTH OF MATERIALS II

COMPOSITE BEAMS

1 Week, Third in Line

PROF. DR. TURGUT KOCATÜRK


bezderodesign
.com


bezderodesign
.com

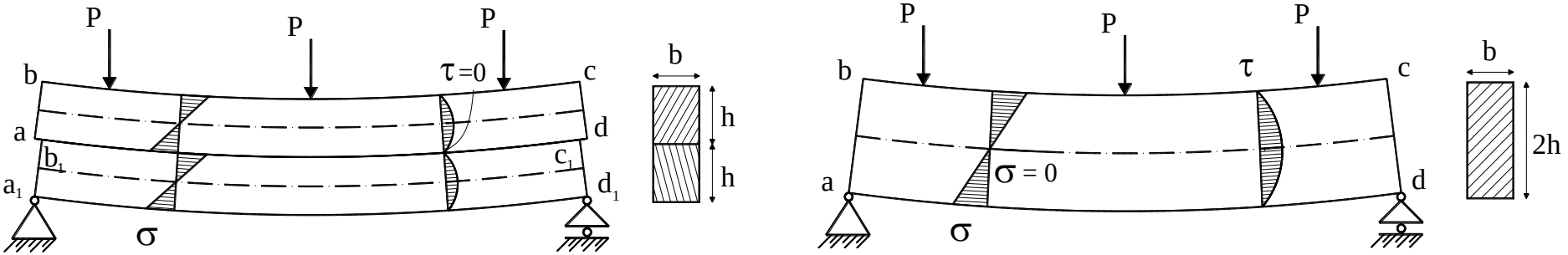
1- COMPOUND BEAMS

In order to increase the moment of resistance under the effect of bending, bars can be connected to each other by wedges, bolts, rivets or weld. The resulting beams are called compound beams. If these are under the effect of bending with shear, the connection elements almost always are under the effect of shear.

If there is no connection between the beams, the total moment of resistance of the system is

$$W_{total} = 2W_1 \qquad W_1 = bh^2/6$$

Relative movements will arise between ad and b_1c_1 edges.



If these beams are connected by using any connection elements to act (work) together, the total moment of resistance is: $W_{bileşik} = \frac{b(2h_1)^2}{6} = 4\frac{bh_1^2}{6} = 4W_1$ Compared to the first one, this system can carry two fold load.

The sign of stress σ will change only once and there will be no relative movement between the edges ad and b_1c_1

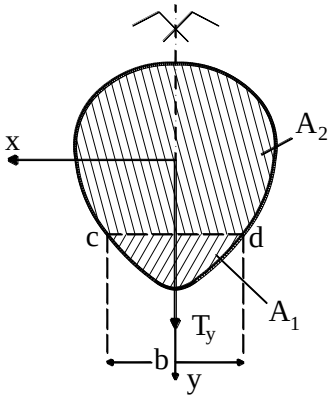
The connector for the compound beams can be glue, bolt, wedge, rivet or weld. Two beams work together with the help of connectors. The connector elements need to have enough strength to withstand the shear forces.

The shear force Z between the two sections must be known. The gravity center and area of the compound section is G and A respectively. Lets calculate the shear force Z on the combination plane $cd\,c'd'$ between the sections I and II which are a distance away from each other.

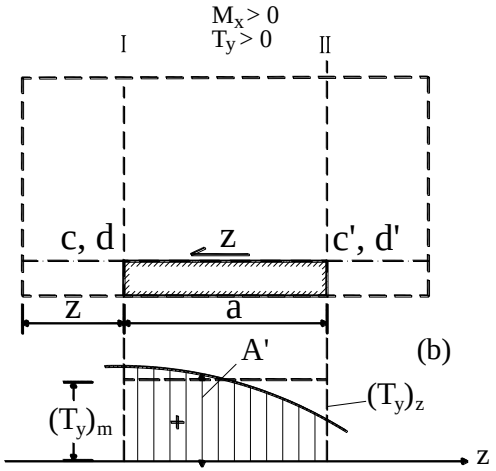
$$Z = \int_{z}^{z+a} \tau_{yz}.b.dz$$

$$\tau_{zy} = \frac{S_x.T_y}{b.I_x}$$

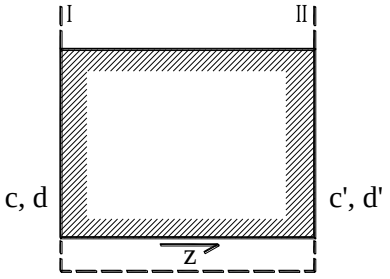
$$Z = \int_{z}^{z+a} \frac{S_x\,T_y}{b\,I_x}.b.dz = \frac{S_x}{I_x}\int_{z}^{z+a} T_y.dz = \frac{S_x}{I_x}.A'$$



(a)



(b)



(c)

S_x static moment of the added part A_1 with respect to the neutral axis x of the whole section

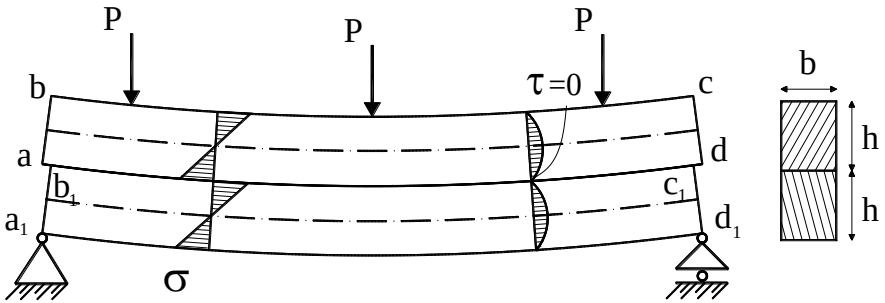
I_x the total moment of inertia of the whole section

A' The area of the shear force diagram between sections I and II

$\left(T_y\right)_m$ average shear stress. Thereby

$$A' = \int_{z}^{z+a} T_y.dz = \left(T_y\right)_m.a$$ and than $$Z = \frac{S_x.(T_y)_m}{I_x}.a = Z_1 a = q_m a \rightarrow Z_1 = q_m = \frac{S_x.(T_y)_m}{I_x}$$

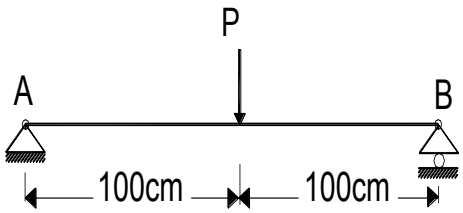
If Z_1 or q is defined as shear force for unit length (or shear flow) of the beam, then $Z = a\,Z_1$ ve $Z_1 = S_x.T_y/I_x$

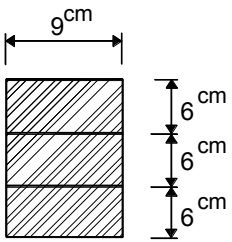


For beams of rectangular cross-section above $S_x = b \times h \times \frac{h}{2}$ $I_x = b(2h_1)^3 / 12 = 2b \times h^3 / 3$

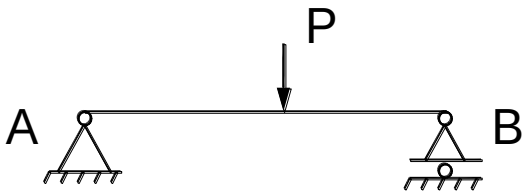
$Z = a\,Z_1$ ve $Z_1 = S_x . T_y / I_x$

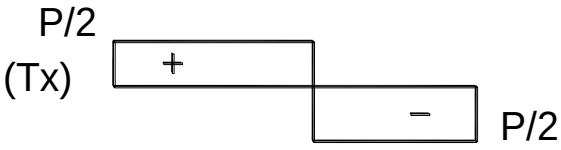
Problem: As seen in the figure, the beam with a span length of 2 meters is built from three equal timber parts (each have a rectangular cross section of 6cm*9cm) by means of gluing. If for the timber $\tau_{safety} = 90\,N/cm^2$ $\sigma_{safety} = 1300\,N/cm^2$ and for the glue $\tau_{safety_{glue}} = 40\,N/cm^2$ are given then find the greatest P load which can be carried safely.

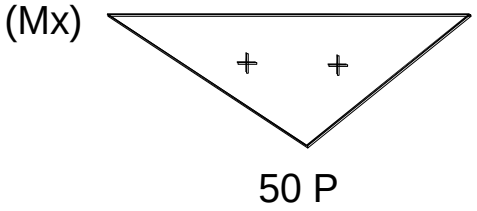




Solution:







With respect to bending:

$$\sigma_z = \frac{50P}{4374}.9 \le \sigma_{safety} = 1300 \rightarrow P \le 12636\,N$$

With respect to shear in timber:

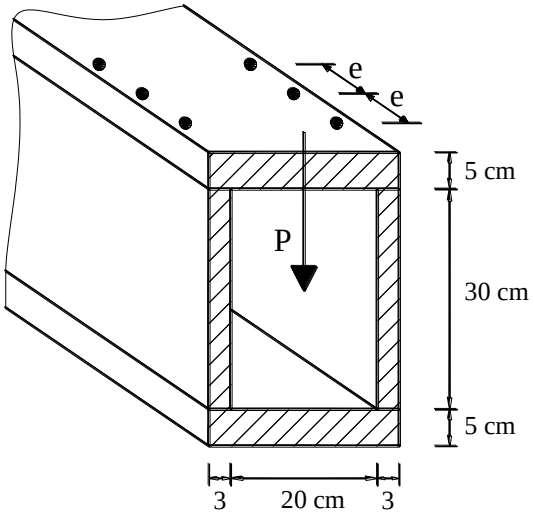
$$\tau_{zy} = \frac{\left(\frac{P}{2}\right).\left[(9.9).(4,5)\right]}{4374.9} \le \tau_{safety} = 90 \rightarrow P \le 19440\,N$$

With respect to shear in glue:

$$\tau_{glue} = \frac{\left(\frac{P}{2}\right).\left[(9.6).6\right]}{4374.9} \le \tau_{glue} = 40 \rightarrow P \le 9720\,N$$

$$\max P \le 9720\,N$$

Problem: Each screws shown in the figure can sustain 1 kN shear force. Taking into account bending with shear effects, determine the distance e between two contiguous (successive) screws if a load of P=6 kN is applied to the cross section.



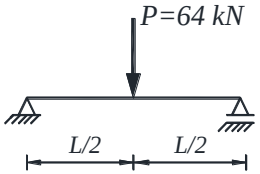
Solution:

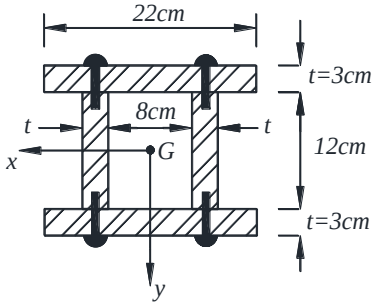
$$I_x = \frac{26.40^3}{12} - \frac{20.30^3}{12} = 93666,67\,cm^4$$

$$\tau = \frac{6}{I_x}\frac{26.5.17,5}{6} = 0,024\ kN/cm^2$$

$$0,024 \times e \times 6 = 2 \times 1 \ \rightarrow\ e = 13,89\,cm$$

Problem: *L*=5m long simply supported beam as shown in the figure is loaded at the middle point by a point load of *P*=64 *kN*. Cross-section of the beam is made with four horizontal timber assembled by screws. If each screw can carry *Q*=11 kN shear force, find the distance *e* between each screw. *t*=3 cm.





Solution:

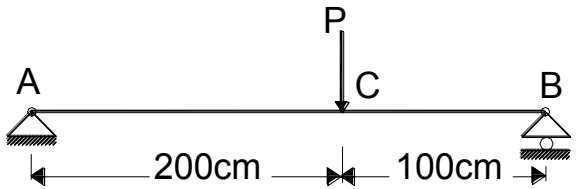
$$T_{\max} = P/2 = 32000\ N,$$

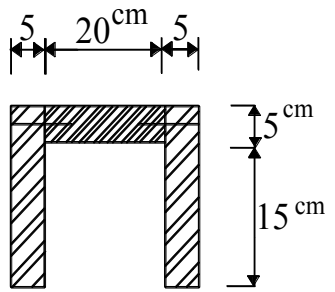
$$I_x = \frac{22.18^3}{12} - \frac{16.12^3}{12} = 8388\ cm^4$$

$$q = \frac{32000 \times (22 \times 3 \times 7{,}5)}{8388} = 1888{,}41\ N/cm \Rightarrow q\,\mathrm{e} \le 2\,\mathrm{Q} \Rightarrow 1888{,}41 \times \mathrm{e} \le 2 \times 1100 \Rightarrow \mathrm{e} \le 11{,}65\ \mathrm{cm}$$

Problem: The cross-section of the beam is made with three horizontal timbers assembled by screws as shown in the figure given in the right hand side. The allowable normal stress and the allowable shear stress of the material are $\sigma_{em} = 1050 N/cm^2$, $\tau_{em} = 200 N/cm^2$ respectively. Find the maximum value of load P safely carried by the simple beam,

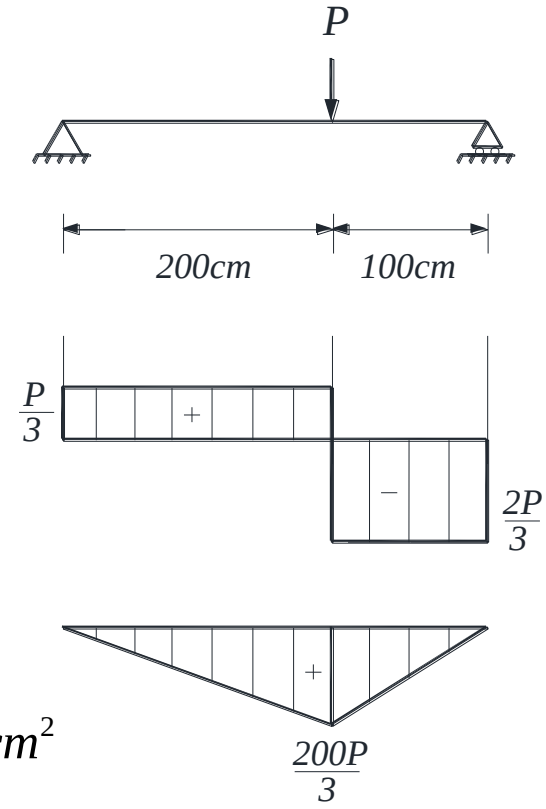
a) Draw the normal stress diagram for this obtained load P at the section at which the bending moment is maximum.
b) Draw the shear stress diagram for this obtained load P at the section at which the shear force is maximum.
c) If each screw can carry a shear force Q=800 N, find the distance between each screw.





$$y_G = \frac{2\cdot 20\cdot 5\cdot 10 + 20\cdot 5\cdot 17{,}5}{2\cdot 20\cdot 5 + 20\cdot 5} = 12{,}5cm$$

$$I_x = \frac{2\cdot 5\cdot 20^3}{12} + 2\cdot 20\cdot 5\cdot (2{,}5)^2 + \frac{20\cdot 5^3}{12} + 20\cdot 5\cdot 5^2 = 10625cm^4$$



Calculation in terms of bending:

$$\sigma = \frac{M_{\max}}{I_x}\cdot y = \frac{200P}{3\cdot 10625}\cdot 12{,}5 \le 1050 \;\rightarrow\; P \le 13387{,}5N$$

Maximum shear stress for this load is;

$$\tau_{\max} = \frac{T_{\max}\cdot S_x}{b\cdot I_x} = \frac{\frac{2}{3}\cdot 13387{,}5\cdot (20\cdot 5\cdot 5 + 2\cdot 7{,}5\cdot 5\cdot \frac{7{,}5}{2})}{2\cdot 5\cdot 10625} = 65{,}625N/cm^2$$

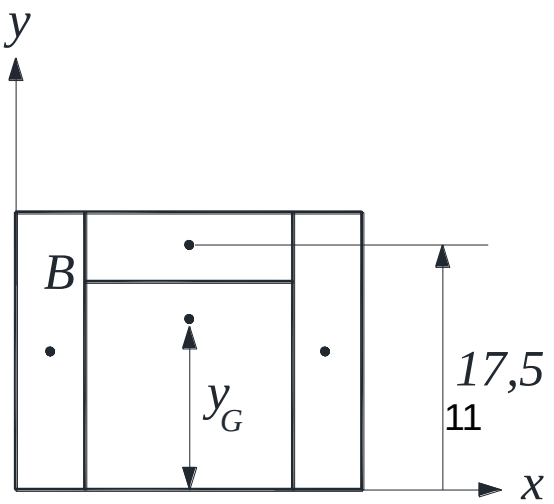
Lets obtain the shear stresses at the sections where there are screws for determining the spacing of the screws:

$$\tau_B = \frac{T_{\max}\cdot S_x}{b\cdot I_x} = \frac{\frac{2}{3}\cdot 13387{,}5\cdot (5\cdot 20\cdot 5)}{2\cdot 5\cdot 10625} = 42N/cm^2$$

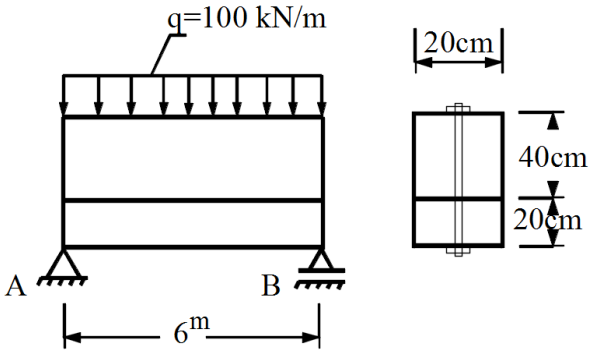
Spacing of the screws is;

$$\tau_B\cdot e\cdot 5 \le 800 \;\rightarrow\; e\cdot 42\cdot 5 \le 800 \;\rightarrow\; e \le 3{,}809cm$$

$$z_1 = \tau_{a\cdot b} \;\rightarrow\; z = z_1\cdot e \le 800 \;\rightarrow\; e \le 3{,}809cm$$



Problem: The cross-section of the beam is made with two horizontal timbers assembled by bolts as shown in the figure given in the right hand side. The allowable shear stress for bolts is $\tau_{em} = 11000\ N/cm^2$. Find the necessary number of bolts and the distance between bolts. Diameter of the bolts is 2cm.



Solution:

$$z = a\ z_1 = \frac{S_1 \cdot T_{\max}}{I_x} \cdot a$$

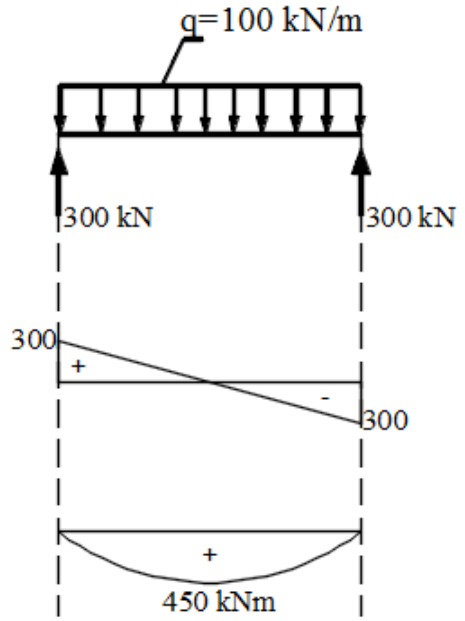
$$I_x = \frac{20 \cdot 60^3}{12} = 360000\,cm^4$$

$$S_x = 20 \cdot 20 \cdot 20 = 8000\,cm^3$$

$$z = a\ \ z_1 = \frac{S_x \cdot T_{\max}}{I_x} \cdot a \le \prod \frac{d^2}{4} \cdot \tau_{em}$$

$$\frac{8000 \cdot 300000}{360000} \cdot a \le \prod \frac{2^2}{4} \cdot 11000$$

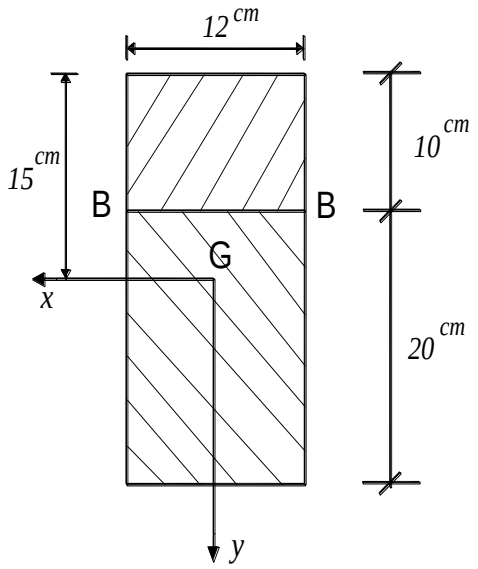
$$6666,67 \cdot a = 34557,52 \qquad a \le 5,18\ cm$$



It is difficult to drill a hole at 5.18cm spacing. Because, it is difficult to measure 5.18cm by a metre in the field or on site. Hence we round 5.18 down to 5.

$a = 5cm\ is\ chosen.$ $\qquad number\ of\ necessary\ bolts = \frac{600}{5} = 120$

Problem: Two wooden elements made of using same material are connected with glue as shown. The allowable shear stress for the material is $\tau_{em} = 200\ N/cm^2$, for the glue is $\tau_{em} = 160\ N/cm^2$, determine the shear force T_y safely carried by the cros-section.



Solution:

For the whole section: $I_x = \frac{1}{12}.12.(30)^3 = 27000\ cm^4$

Static moment of the area under the center of gravity with respect to the point G:

$S_x = 12.15.(7,5) = 1350\ \text{cm}^3$

$$\tau_{zy(\text{G})} = \frac{T_y.S_x}{b.I_x} = \frac{T_y.1350}{12.27000} = 200 \rightarrow T_y \leq 48000\ N$$

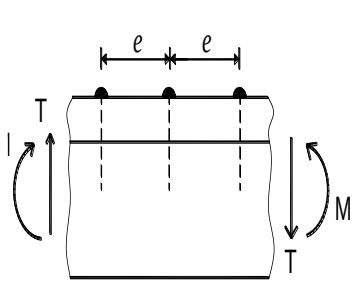
Assume that 2 elements work together, shear force acting on the connection surface B-B is obtained:

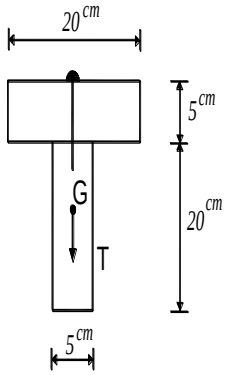
$S_{x(\text{B})} = 12.10.(15-5) = 1200\ \text{cm}^3$

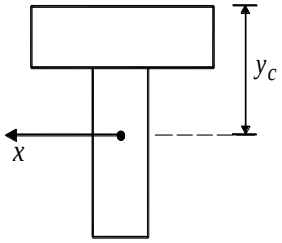
$$\tau_{zy(\text{B})} = \frac{T_y.S_x}{b.I_x} = \frac{T_y.1200}{12.27000} = 160 \rightarrow T_y \leq 43200\ N$$

Thus, the shear force carried by the compound section is found as $T_y \leq 43200\ N$.

Problem 2: A wooden beam carrying a bending moment and a shear force $T = 400\ N$ is made of using 2 20x5 cm planks connecting by nails as shown in the figure. Each nail connecting the planks can carry $Q = 800\ N$ shear force, find the distance between the nails in the direction of the beam so that the beam cross-section can work as a single piece?







Solution:

$$I_x = \frac{1}{12}.5.20^3 + 5.20.(15 - 8,75)^2 + \frac{1}{12}.20.5^3 + 20.5.(8,75 - 2,5)^2 = 11354,17\ cm^4$$

S_x (static moment of the upper plank wrt the x axis)

$$S_x = (20.5)(8,75 - 2,5) = 625\ \text{cm}^3$$

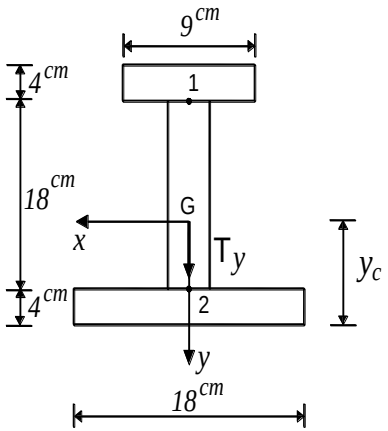
$$Z_1 = T\frac{S_x}{I_x} = 4000.\frac{625}{11354,17} \rightarrow Z_1 = 220,2\ N/cm$$

Each nail has to carry the shear force acting on the distance e

$$Z_1.e = Z \rightarrow e = \frac{Z}{Z_1}$$

Spacing of the nails $e = \frac{800}{220,2} = 3,63\ cm \approx 3,5\ cm$

Problem: The three piece cross-section shown is under the effect of T_y shear force. The pieces are connected by glue. The allowable shear stress of the glue is $150\ N/cm^2$, and the allowable shear stress of the material is $200\ N/cm^2$,



a) Find the shear force T_y safely carried by the cross-section.

b) Draw the shear stress diagram according to the shear force found a part a.

Solution:

a) $$y_c = \frac{18.4.2 + 4.18.(9+4) + 9.4.(2+18+4)}{18.4 + 4.18 + 9.4} = 10,8\ cm$$

Moment of inertia wrt the x axis given in the section is

$$I_x = \frac{1}{12}.18.4^3 + 18.4.(10,8-2)^2 + \frac{1}{12}.4.18^3 + 4.18.(13-10,8)^2 + \frac{1}{12}.9.4^3 + 9.4.(24-10,8)^2 = 1428$$

Now, let's find the shear force taking the max. shear stresses on the glue at the section **1** and **2** as $150\ N/cm^2$:

$$S_{x(1)} = 9.4.(24-10,8) = 475,2\ \text{cm}^3$$

$$\tau_1 = \frac{T_y.S_{x1}}{b_1.I_y} = \frac{T_y}{4}.\frac{475,2}{14284,8} = 150 \rightarrow T_y \leq 18036,4\ N$$

$$S_{x(2)} = 18.4.(10,8-2) = 633,6\ \text{cm}^3$$

$$\tau_2 = \frac{T_y.S_{x2}}{b_2.I_y} = \frac{T_y}{4}.\frac{633,6}{14284,8} = 150 \rightarrow T_y \leq 13527,3\ N$$

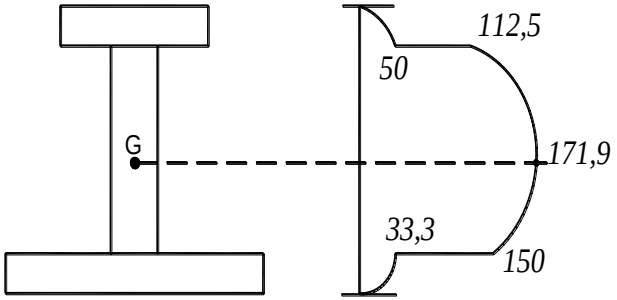
The max. shear stress will occur at the center of gravity G.

$$S_{x(\mathrm{G})} = 18.4.(10,8-2) + 4.(10,8-4).\left(\frac{10,8-4}{2}\right) = 726,08\ cm^3$$

$$\tau_G = \frac{T_y.S_{xG}}{b.I_x} = \frac{T_y.(726,08)}{4.(14284,8)} = 200 \rightarrow T_y = 15739,1\ N$$

The min. value of these forces found will be considered. $T_y \leq 13527,3\ N$

b) Bu değer kesme kuvveti olarak alınırsa,



$$\tau_{yG} = \frac{(13527,3).(726,08)}{4.(14284,8)} = 171,9\ N/cm^2$$

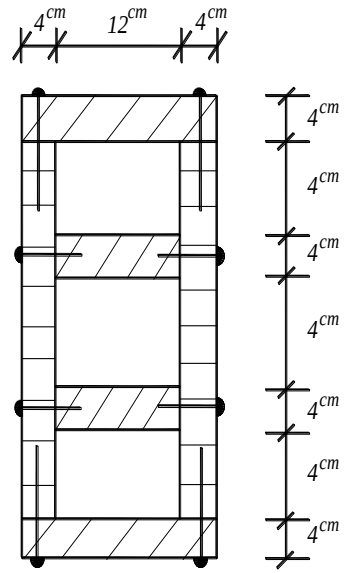
$$\tau_{y1(top)} = \frac{(13527,3).(475,2)}{9.(14284,8)} = 50\ N/cm^2$$

$$\tau_{y1(bottom)} = \frac{(13527,3).(475,2)}{4.(14284,8)} = 112,5\ N/cm^2$$

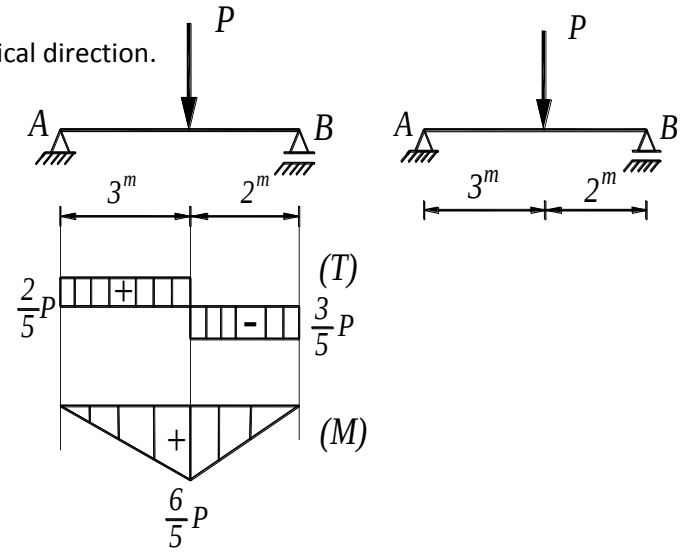
$$\tau_{y2(top)} = \frac{(13527,3).(633,6)}{4.(14284,8)} = 150\ N/cm^2$$

$$\tau_{y2(bottom)} = \frac{(13527,3).(633,6)}{18.(14284,8)} = 33,3\ N/cm^2$$

Problem: The beam shown is under the effect of load P. The beam is constructed by $4\,cm$ thick planks and nails that can carry $800\,N$ shear force. The allowable normal stress of the wood is $\sigma_{güv} = 1050\ N/cm^2$;



a) Find the load P safely carried ny the beam,
b) Determine the spacing of the nails in the horizpntal and vertical direction.



Solution:

a) From the shear force and bending moment diagrams

$T_{\max} = 0,6P$, $M_{\max} = 1,2P$ is found.

Moment of inertia of the section:

$$I_x = \frac{1}{12}.20.28^3 - \frac{1}{12}.12.4^3.3 - 12.4.8^2.2 = 30250,67\ cm^4$$

According to the bending moment;

$$\sigma = \frac{M_{\max}}{I_x}.y = \frac{(1,2).10^2.P}{30250,67}.14 \le 1050 \rightarrow P \le 18906,7\ N$$

b) Shear stress at the section A;

$$S_x = 20.4.12 = 960\ cm^3 \quad \tau_A = \frac{T_{\max}.S_x}{b.I_x} = \frac{(0,6).(18906,7).960}{8.(30250,67)} = 45\ N/cm^2$$

Spacing of the nails in the vertical direction is e_1, one nail will transfer the force acting on the area of $e_1.4$.

$$\tau_A.e_1.4 \le Q_1 = 800 \rightarrow 45.e_1.4 \le 800 \rightarrow e_1 \le 4,44\ cm$$

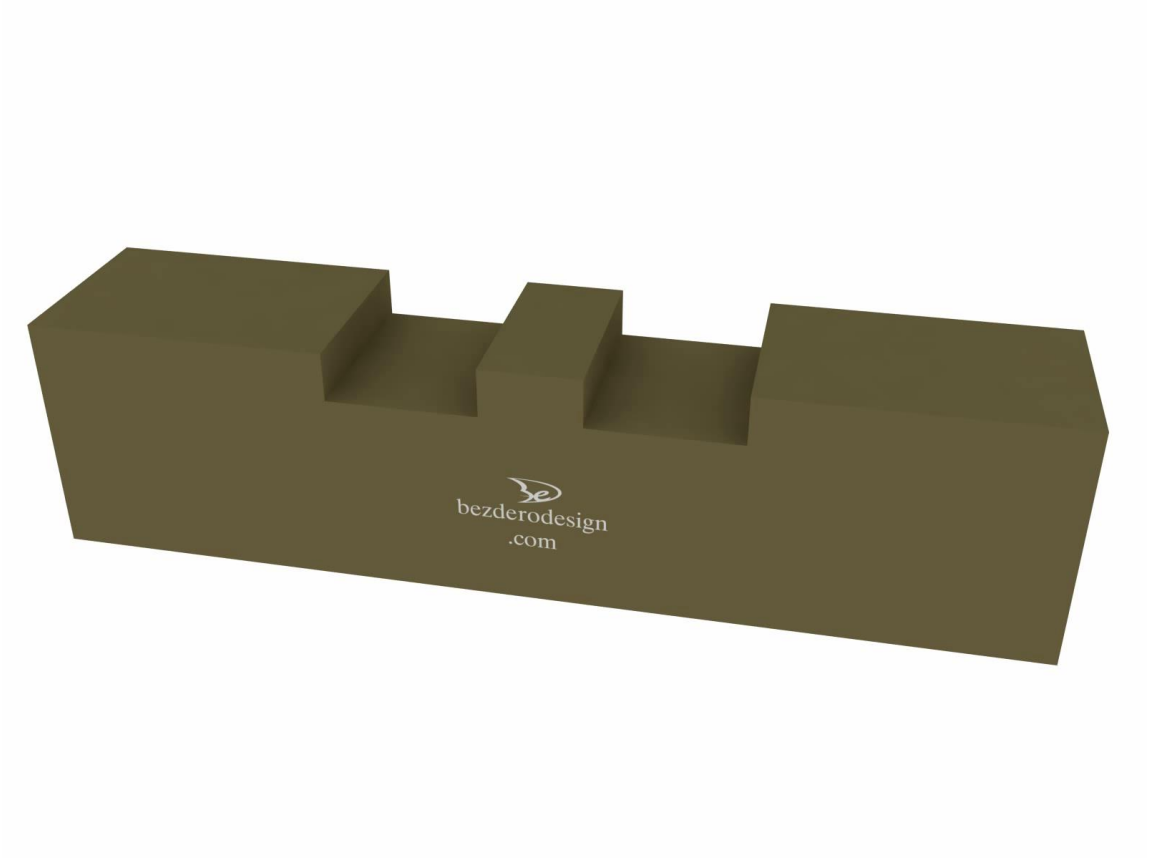
For the nails in the horizontal direction, let's calculate the shear stress on the section where the nails are working:

$$S_x = \frac{12}{2}.4.4 = 96\ cm^3$$

$$\tau_B = \frac{T_{\max}.S_x}{b.I_x} = \frac{(0,6).(18906,7).96}{4.(30250,67)} = 9\ N/cm^2$$

Spacing between the horizontal screw e_2 is;

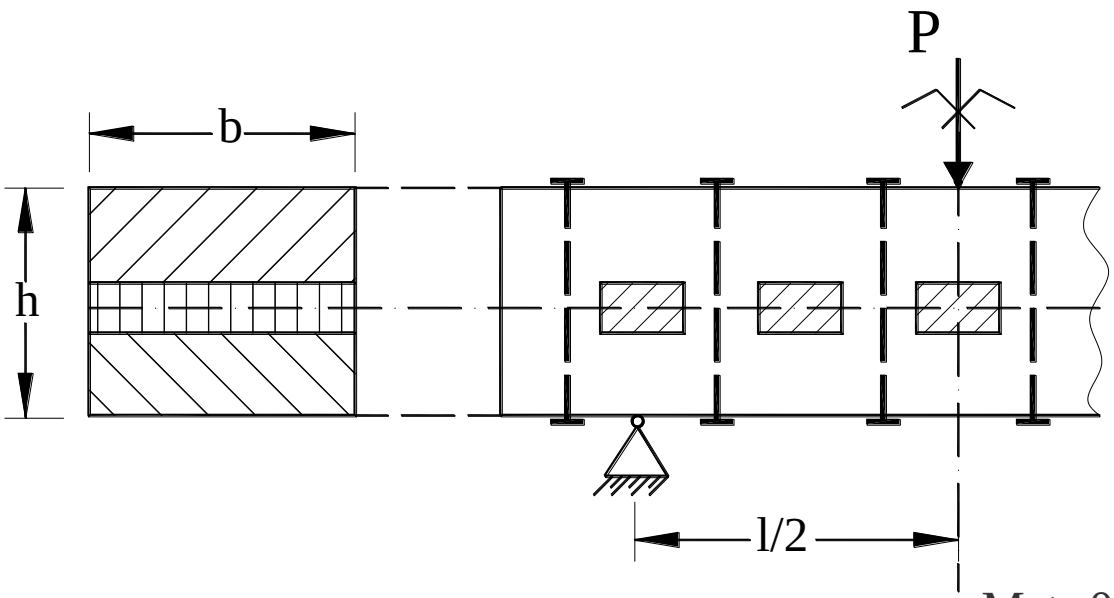
$\tau_B.e_2.4 \le 800 \rightarrow e_2 \le 22,2\ cm$ is obtained.

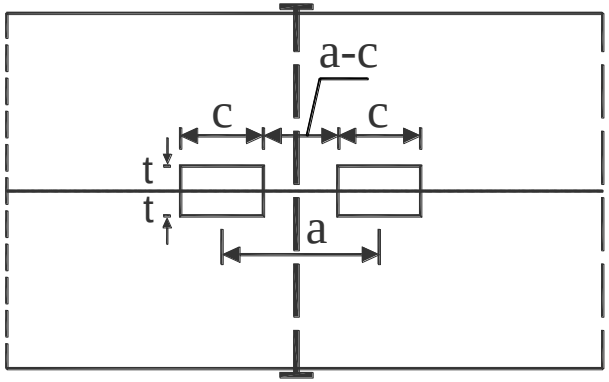

bezderodesign
.com

Wedged Beams

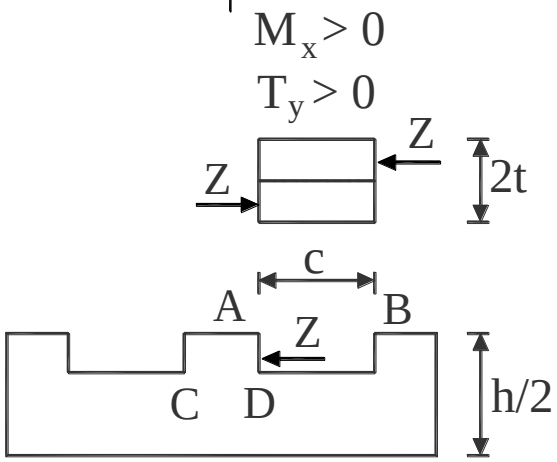
In wood structures, wedges are used for combining the two parts and obtaining buil-up beams.

c : wedge length, $2t$: wedge height, b : the width of wedge and socket, a : wedge intervals





(a)



(b)



1) To avoid shear in wedges

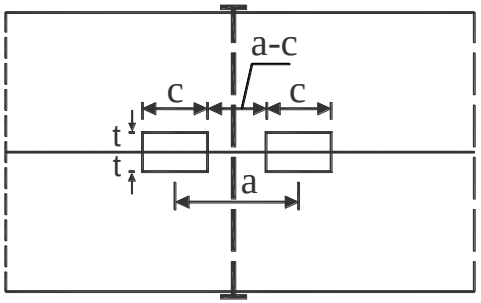
$$Z \le cb\tau'_{em}$$ τ'_{em} The allowable shear stress for the fiber direction of oak-wedge.

2) Normal stresses between wedge and beam musn't exceed the allowable normal stresses

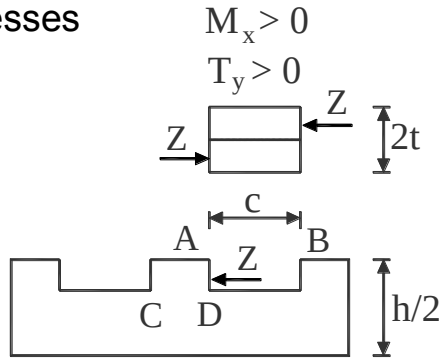
$$Z \le t.b.\sigma_{bem}$$

3) There must be no shearing in the beam:

$$Z \le b(a-c)\tau_{em}$$



(a)



(b)

In a well designed combination, if possible, the strenths (1), (2) and (3) must be equal to each other.

Half-wedge height is taken as

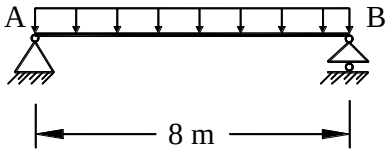
$$t \cong h/20$$

If the wedge distances are held constant, by staying on the safe side, it can be assumed that

$$\left(T_y\right)_m = T_{\max}$$

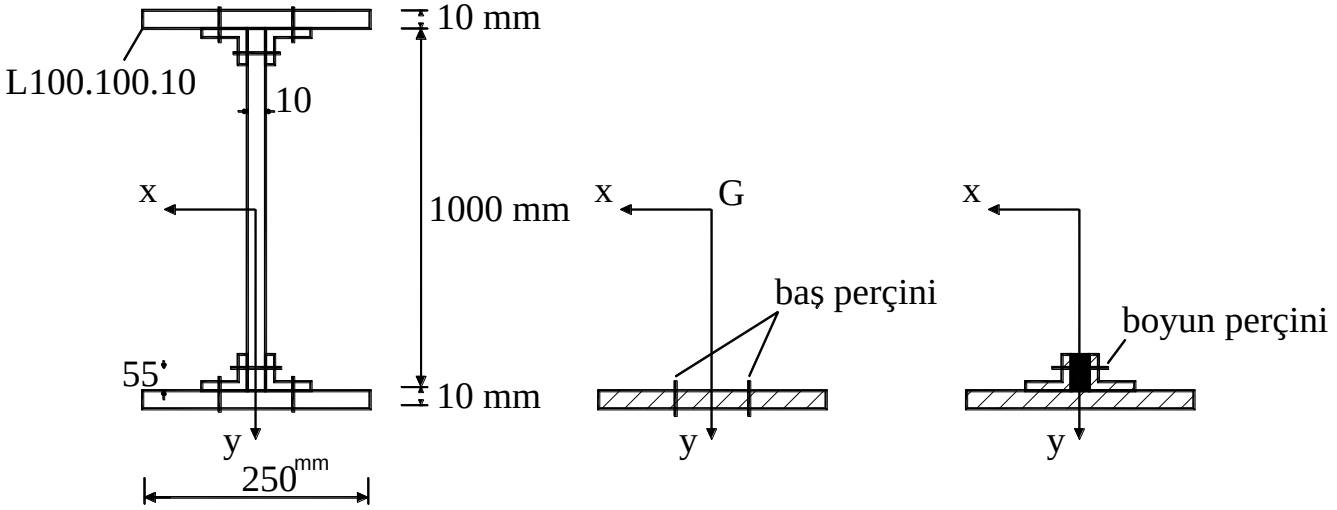
Useful moment of resistance of wooden beams combined by wedges is 85% of that of the ideal beam. In the displacement calculations, it is necessary to decrease the moment of inertia by 40%.

Example: $\sigma_{em} = 14000\ \text{N/cm}^2,\ \tau_{em} = 7000 N/cm^2,\quad \tau'_{em} = 0,8.\sigma_{em}$



$\sigma_l = 28000\ N/cm^2$ Ensure that safety is satisfied for bolted built-up beam.

Bolt diameter is taken as 20 mm, determine the bolt spacing.



Moment of inertia of the section without holes is

$$I_x' = \frac{1x100^3}{12} + 4\left[177 + 19,2x47,2^2\right] + 2.\frac{25.1^3}{12} + 25.1.(50,5)^2 \quad Ix' = 381900cm^4$$

If the moment of inertia of the bolt holes of diameter $d = 20mm$ in the flange is subtracted from I'_x

$Ix = 381900cm - 40000 = 341900cm^4$ Than moment of resistance of the section is obtained as follows:

$W_x = 341900/51 = 6704cm^3$ ve $\sigma_{\max} = \frac{Pl^2}{8}.\frac{1}{Wx} = \frac{1000.800^2}{8.6704} = 11933 < 14000N/cm^2$

The largest shear stress in the web is $\tau_{\max} = \frac{P.l.S_x}{2.t.I_x} = \frac{1000.800.3924,98}{2.1.341900} = 4592 < 7000N/cm^2$

$$Z = a.Z_1 = \frac{S_x (T_y)_m}{I_x}.a \quad ve \quad Z \le 2\frac{\pi d^2}{4}.\tau_{em} \quad den \quad a \le \frac{\tau_{em}.2\pi d^2/4}{(T_y)_m}.\frac{I_x}{S_x}$$

$$Z = a.Z_1 = \frac{S_x (T_y)_m}{I_x}.a \quad \text{ve} \quad \text{Z} \le \text{t.d.}\sigma_l \quad den \quad \text{a} \le \frac{\text{t.d.}\sigma_l}{\left(T_y\right)_m}.\frac{I_x}{S_x}$$

Here S_x is the static moment of the half section with respect to the axis X .

I_x is the moment of inertia of the whole section.

S_x and I_x are calculated by removing the related quantities of bolt holes. Later bolt calculation will be performed by choosing the bolt diameter as $d = 2cm$ and by taking $T_{\max} = Pl/2 = 400kN$

from $a \le \frac{t.d.\sigma_l}{T_{\max}}.\frac{I_x}{S_x}$ $a = 17,3cm$ and from $a \le \frac{\tau_{em}.2.\pi.d^2/4}{T_{\max}}.\frac{I_x}{S_x}$ $a = 21,8cm$

is found. Even the small value $17,3cm$ for spacing is higher than $a_{\max} = 6d$

In this case $a = 6d = 12cm$ is taken. The same space could be enough for the head bolts.

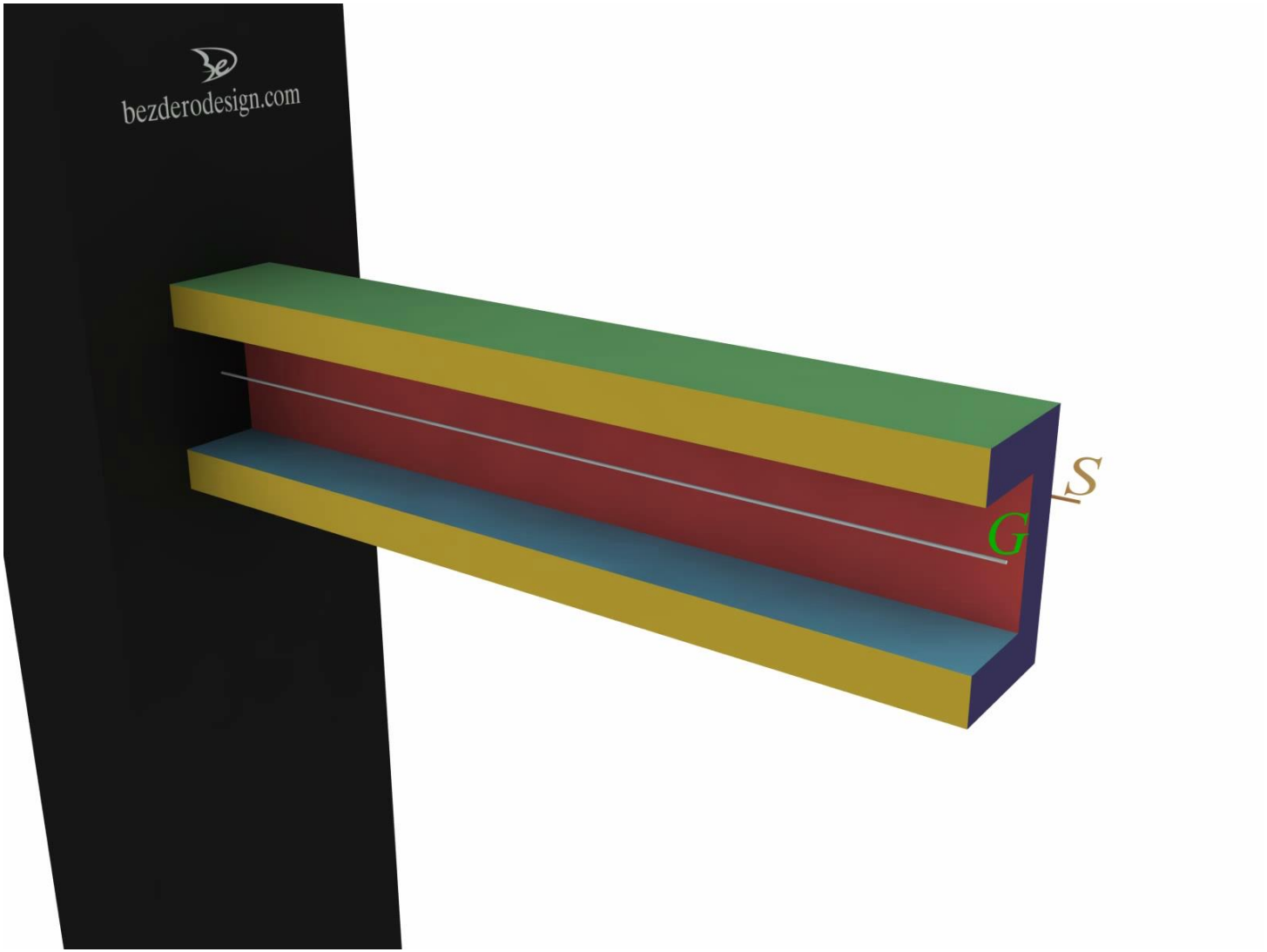
YILDIZ TECHNICAL UNIVERSITY
CIVIL ENGINEERING DEPARTMENT
MECHANICS DIVISION

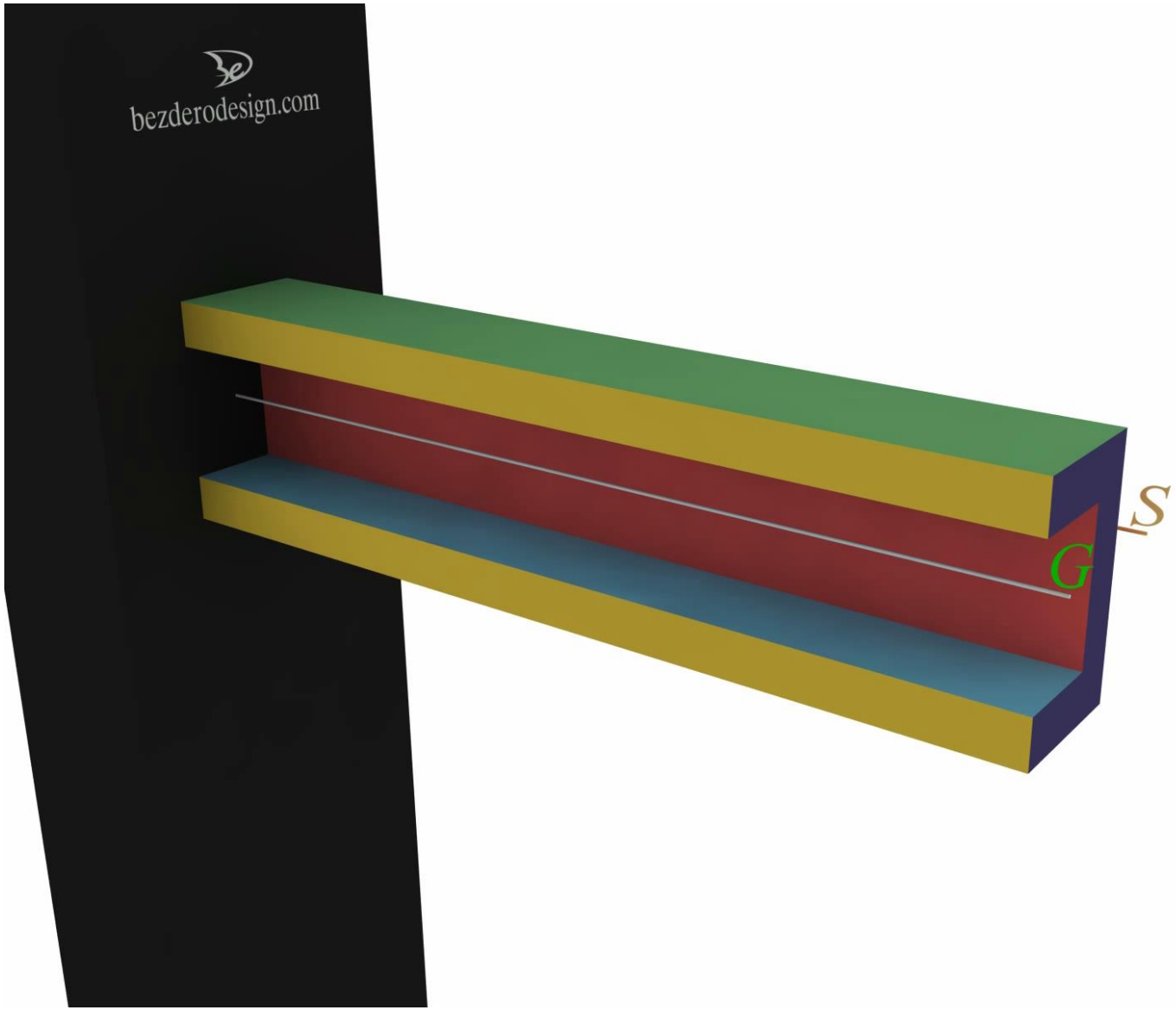
STRENGTH OF MATERIALS II

DEFORMATION IN BENDING WITH SHEAR

1 Week, Fourth in Line

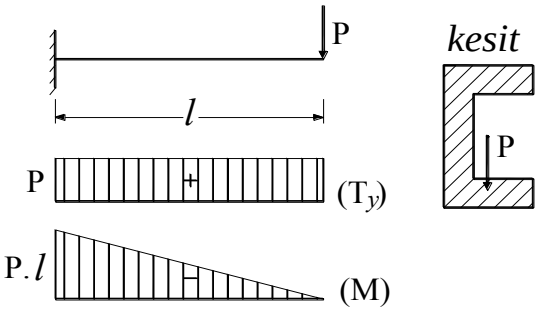
PROF. DR. TURGUT KOCATÜRK

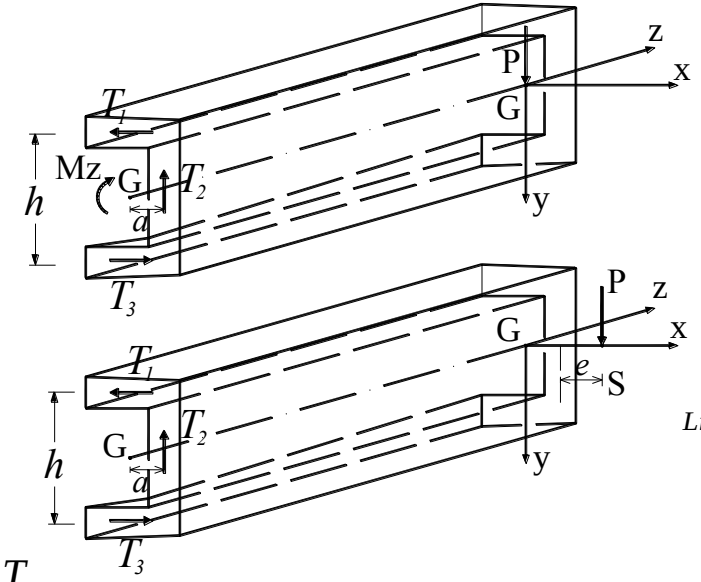

bezderodesign.com
S
G

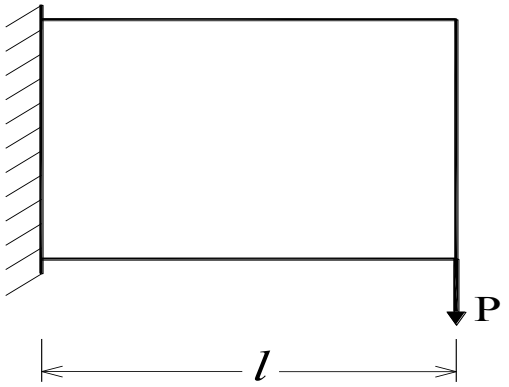

bezderodesign.com
S
G

When the cross section is not symmetrical with respect to the line of the forces:

If the line of the forces passes from the gravity center of the cross section, then bending, shear and torsion are all existent (together) in the cross section. If there is only bending with shear when the shear forces pass from the point S instead of the point the G, then the point S is called as the shear center.







$$\sum F_x = 0 \rightarrow -T_1 + T_3 = 0 \rightarrow T_1 = T_3$$

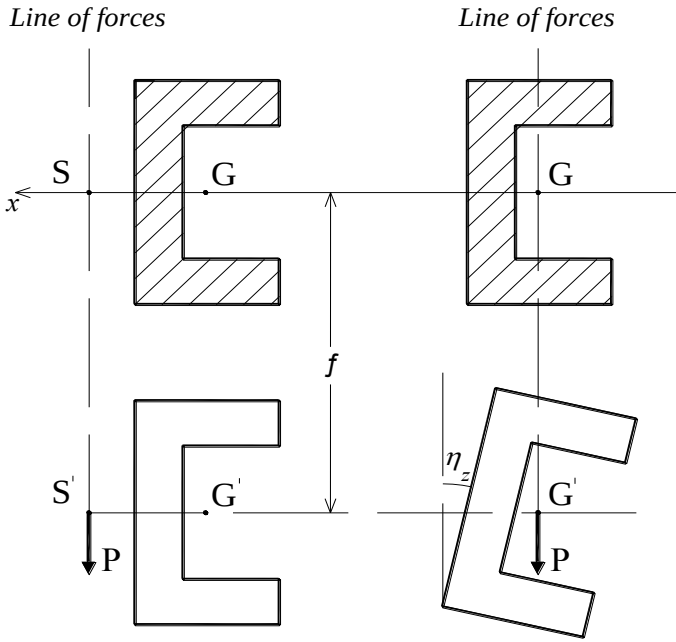
$$\sum F_y = 0 \rightarrow T_2 - P = 0 \rightarrow T_2 = P = T_y$$

$$\sum M_G = 0 = M_z - T_1.h - T_y.a = 0$$

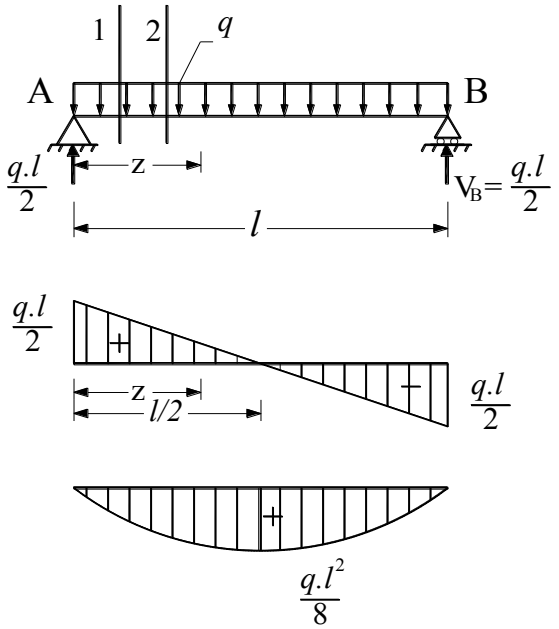
$$M_z = T_1.h + T_y.a$$

Where is the place of P for $M_z = 0$?

$$\sum M_S = 0 = -T_1.h + T_y.e = 0 \rightarrow e = \frac{T_1}{T_y}.h$$



Let's find the shear center of an U section. First the distribution of the shear stresses on the upper and lower arms will be calculated.



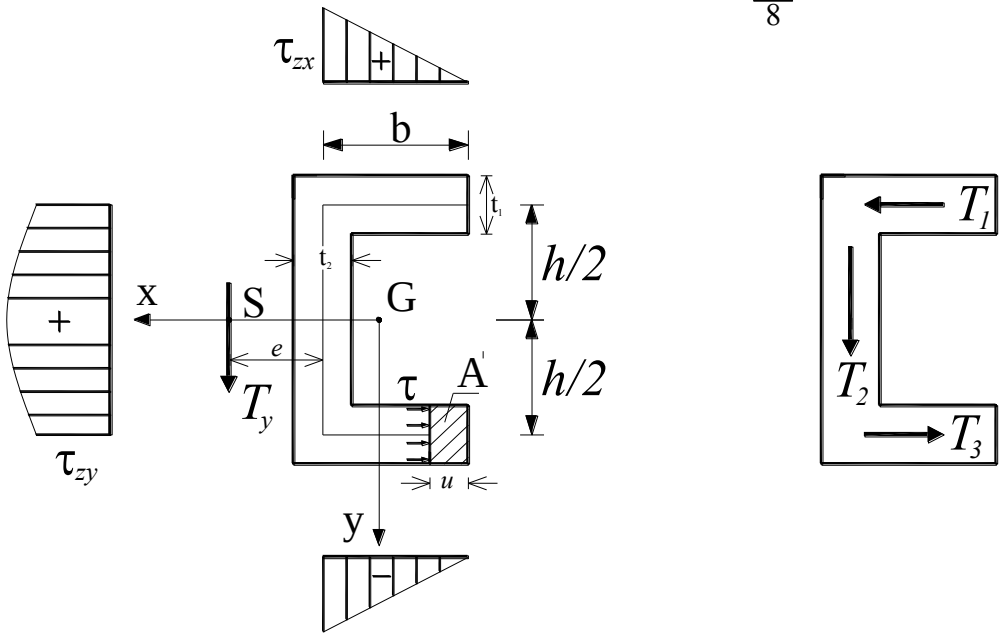
S'_x shows the static moment of the shaded area A' .

$$\tau_{zx} = \frac{T_y.S'_x}{t_1.I_x} = \frac{T_y.u.t_1.h/2}{t_1 I_x} = \frac{T_y.h}{2.I_x}.u$$

If $t = t_1$ then the shear force of the upper arm is

$$T_3 = \int_0^b \tau_{zx}.t.du = \frac{T_y.h.t.b^2}{4I_x}$$

Now the point S from which the resultant of the resultant forces T_1 , T_2 and T_3 of the arms pass will be found.

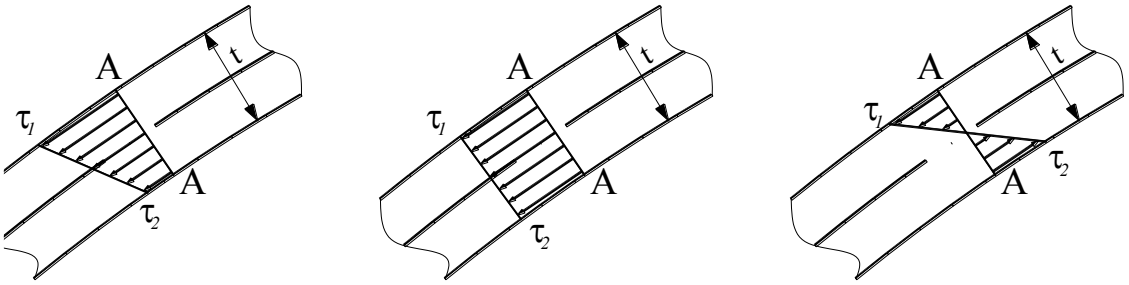


The static moment about the point S is

$$T_2.e - T_1.h = 0 \rightarrow e = h.\frac{T_1}{T_2} = \frac{h^2.b^2.t}{4I_x}$$ is found.

Shear Center of Open Sections

The shear center is also defined as the point from which the resultant of the shear stresses of the section passes in the case of bending with shear force when there is no torsion. In this case, the difference between the distribution of shear stresses in bending without torsion and in bending with torsion must be explained.

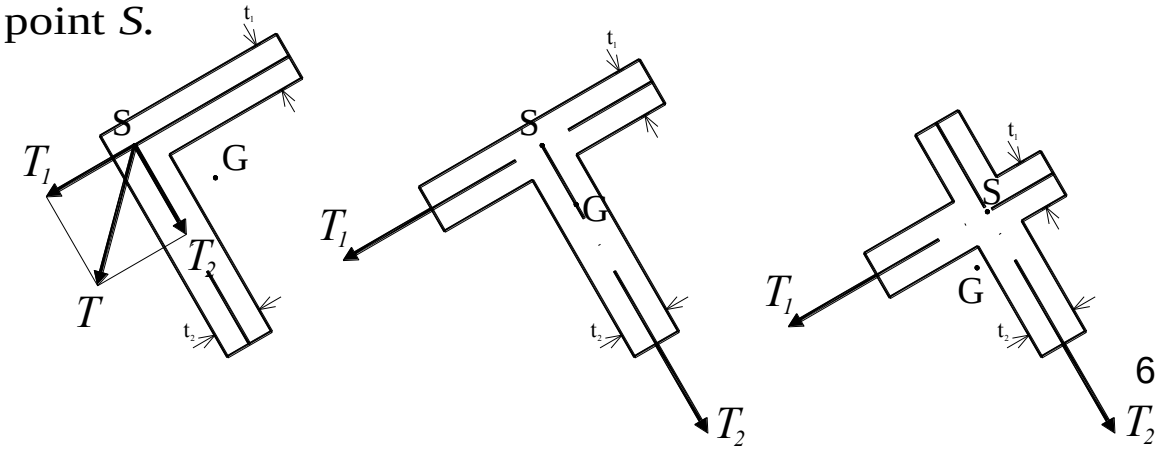


$\tau_1 \neq \tau_2$ $\tau_1 = \tau_2$

Bending with torsion *Bending without torsion* *Only torsion*

(a) *(c)*

The shear center of the cross sections formed by two thin arms is the point of intersection o the axes of the arms. Because the resultants of the shear stresses T_1 and T_2 pass from the related axes of the arms. Therefore the shear center S is the intersection point of these axes. When two thin rectangular sections intersect, then the line of action of the resultant forces of all the section must pass fromthe intersection point of the axes of the rectangular section and this point is the shear center. Because the line of action of resultants of the shear stresse of each section passes from the point S.

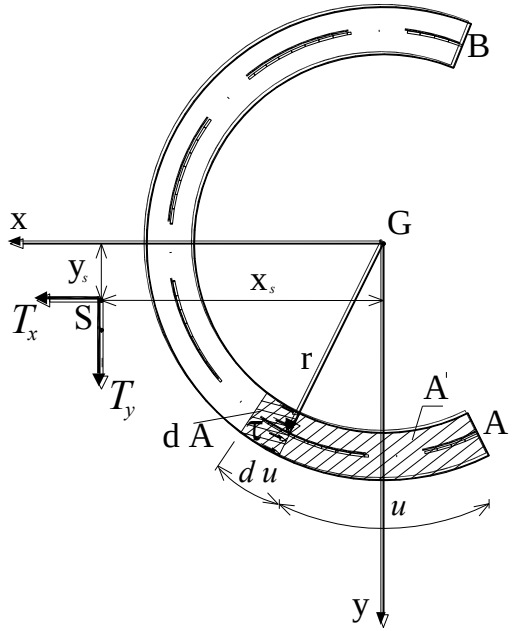


Lets consider a general open section. The gravity center of this section is *G.* The origin of the coordinate axes is located at the point G and the axes of these coordinate axes are principal axes. Namely product of inertia is zero $I_{xy} = 0$ The components of the shear forces in the directions of coordinate axes are T_x and T_y . First lets calculate the the shear flow (flux) produced by T_y

$\tau = \dfrac{T_y . S_x}{t . I_x}$ S_x is the first moment of area of the shaded area A' with respect to the axis x.

It is more suitable to use shear flow τt instead of shear stress τ

$$\tau t = \frac{T_y . S_x}{I_x}$$

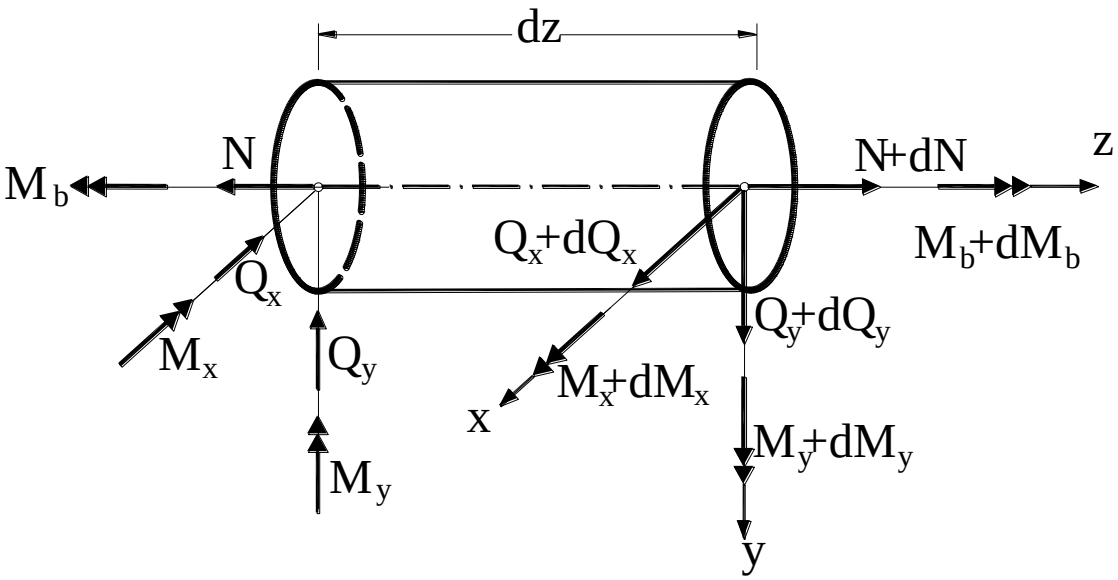


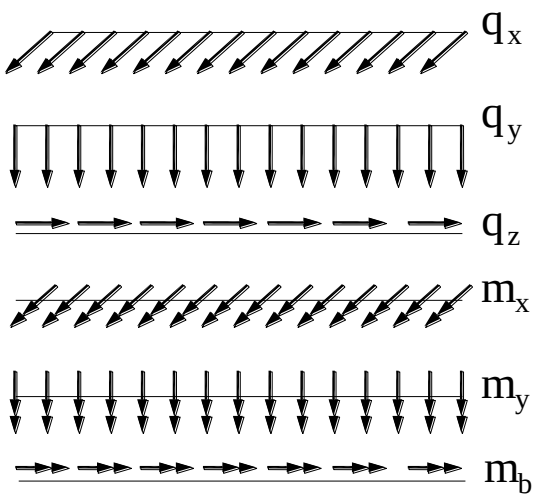
The moment of the resultant according to the point *G* is equal to the moment of components according to the same point.

$$T_y . x_s = \int_0^c r(\tau t)du = \int_0^c r(\frac{T_y S_x}{I_x})du \rightarrow \quad x_s = \frac{1}{I_x}\int_0^c rS_x du$$

By using the similiar way of calculation

$$y_s = -\frac{1}{I_y}\int_0^c rS_y du$$ This minus sign is because of $dMy / dz = -T_x$





$$\sum M_x = 0 \rightarrow -M_x + m_x\,dz - Q_y\,dz + M_x + dM_x = 0 \rightarrow \frac{dM_x}{dz} = Q_y - m_x \qquad \textbf{(6.4)}$$

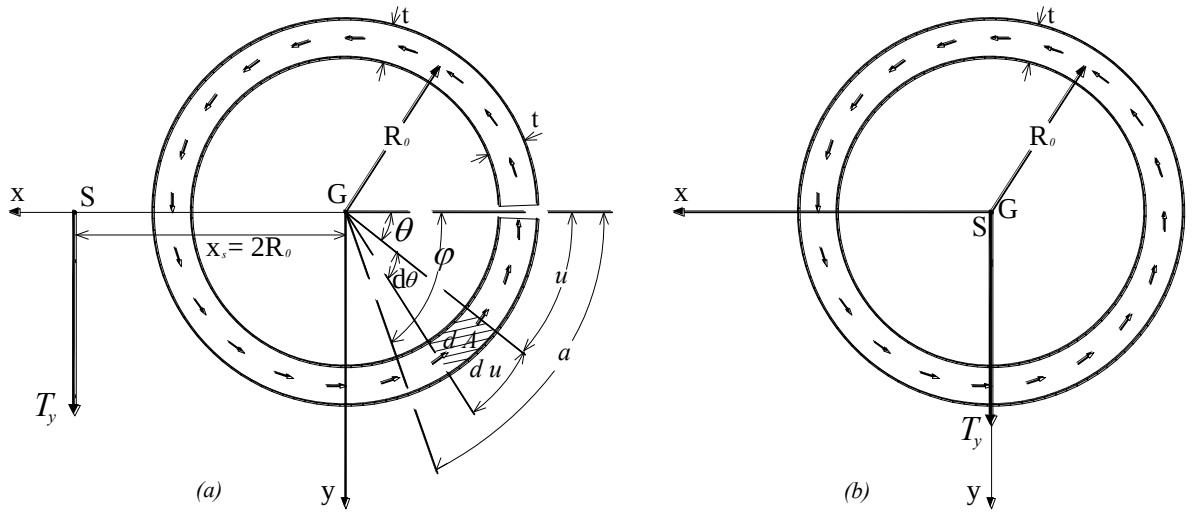
$$\sum M_y = 0 \rightarrow -M_y + m_y\,dz + Q_x\,dz + M_y + dM_y = 0 \rightarrow \frac{dM_y}{dz} = -Q_x - m_y \qquad \textbf{(6.5)}$$

$$\sum M_z = 0 \rightarrow -M_b + m_b\,dz + M_b + dM_b = 0 \rightarrow \frac{dM_b}{dz} = -m_b \qquad \textbf{(6.6)}$$

As it is noticed from Eqs. (6.1-6.6), among $Q_x, Q_y, N \text{ and } M_x, M_y, M_b$, only $N \text{ and } M_b$ are independent of the others.

$$\tau_{zy} = \frac{dM_x}{dz}\cdot\frac{S_x}{b.I_x} \;;\; \frac{\mathrm{dM}_x}{\mathrm{dz}} = Q_y \rightarrow \tau_{yz} = \frac{Q_y S_x}{bI_x} \qquad \tau_{zx} = \frac{dM_y}{dz}\cdot\frac{S_y}{b.I_y} \;;\; \frac{\mathrm{dM}_y}{\mathrm{dz}} = -Q_x \rightarrow \tau_{zx} = -\frac{Q_x S_y}{bI_y}$$

EXAMPLE: Lets find the shear center of the cut thin annulus:



$$S_x = \int_0^a t\,du\,R_o \sin\theta = \int_0^\varphi tR_0\,d\theta R_o \sin\theta = tR_0^2 \int_0^\varphi \sin\theta d\theta = tR_0^2(1-\cos\varphi)\ ,\quad r = R_0\ ,\quad c = 2\pi R_o \quad (Whole\ length)$$

$$I_o = R_o^2\,2\pi R_0 t = 2\pi R_0^3 t \rightarrow I_o = I_x + I_y = 2I_x = 2I_y \rightarrow I_x = \pi R_0^3 t$$

$$x_s = \frac{1}{\pi R_0^3 t}.\int_0^c R_0.t.R_0^2(1-\cos\varphi)du$$

$$x_s = \frac{1}{I_x}\int_0^c rS_x du \quad \text{den} \qquad x_s = \frac{1}{\pi R_0^3 t}.\int_0^{2\pi} t.R_0^3(1-\cos\varphi).R_o d\varphi$$

$$x_s = \frac{tR_0^4}{tR_0^3\pi}.\int_0^{2\pi}(1-\cos\varphi)d\varphi = \frac{R_0}{\pi}(2\pi - \sin 2\pi) = 2R_o$$

In open thin sections, shear center can be found by using a different view. This is the relation between the displacements of each part in case of bending without torsion. For example, lets consider an *I* section with different flanges:

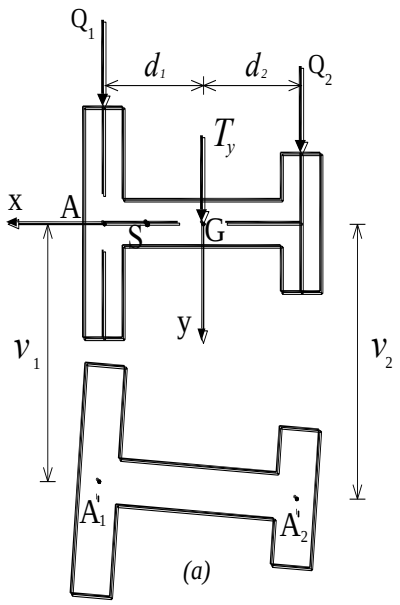
$$\frac{1}{\rho_x}=\frac{M_x}{E\,I_x}=\pm\frac{\mathbf{d}^2v/\mathbf{d}z^2}{\left[1+\left(\mathbf{d}v/\mathbf{d}z\right)^2\right]^{3/2}}$$ When the slope $\mathrm{d}v/\mathrm{d}z$ is small then it can be assumed that $\frac{\mathrm{d}^2v}{\mathrm{d}z^2}=\frac{M_x}{E\,I_x}$

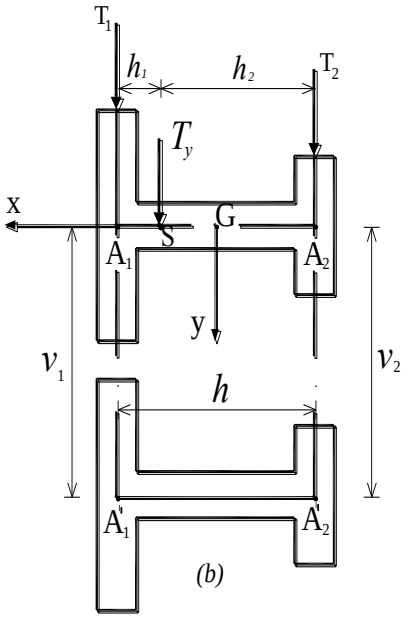
This section is symmetrical with respect to the x axis. Because the web of the section is thin and perpendicular to the direction of the force, the force web can carry can be ignored. Therefore it can be assumed that the shear force is carried by the two flanges.

If the displacements of the flanges are equal to each other, $v_1=v_2$, then the curvature of the two flanges must be equal to each other.

Namely $\frac{M_1}{E\,I_{x1}}=\frac{M_2}{E\,I_{x2}}$ By considering $dM_x/dz=T_y$

then $\frac{T_1}{I_{x1}}=\frac{T_2}{I_{x2}}\rightarrow\frac{T_1}{T_2}=\frac{I_{x1}}{I_{x2}}$



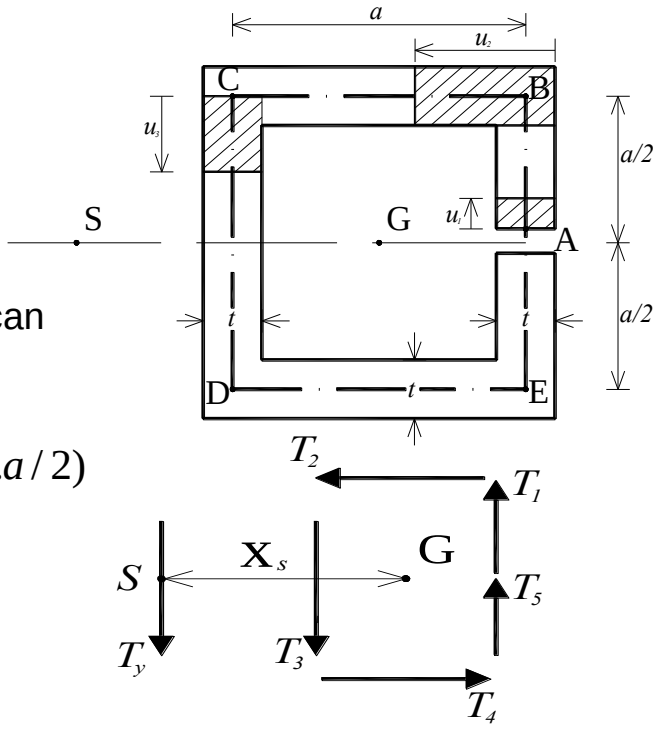


Here I_{x1} and I_{x2} are the second moment of areas with respect to the axis x

By considering $h=h_1+h_2$

$$T_1\,h_1=T_2\,h_2\rightarrow\frac{h_1}{h_2}=\frac{T_2}{T_1}=\frac{I_{x2}}{I_{x1}}\quad or\quad h_1=h\frac{I_{x2}}{I_{x1}+I_{x2}}\quad and\quad h_2=h\frac{I_{x1}}{I_{x1}+I_{x2}}$$

PROBLEM: Find the shear center of the given section.



$$\tau(u_1)=\frac{T_y}{t.I_x}\left(t.\frac{u_1^2}{2}\right)=\frac{T_y.u_1^2}{2I_x} \qquad T_1=\int_0^{a/2}\frac{T_y.u_1^2.t}{2I_x}.du_1=\frac{T_y.t.a^3}{48I_x}$$

$$I_x=\frac{2.t.a^3}{12}+2.t.a.\left(\frac{a}{2}\right)^2=\frac{2}{3}t.a^3 \quad 2\times\frac{a.t^3}{12}$$ is very small end therefore can be omitted.

$$\tau(u_2)=\frac{T_y S_x}{t\,I_x}=\frac{T_y}{t\,I_x}\left[\frac{a}{4}\frac{t\,a}{2}+(u_2.t.a/2)\right]= \qquad \frac{T_y.a^2}{8I_x} \qquad +\frac{T_y}{I_x.t}(u_2.t.a/2)$$

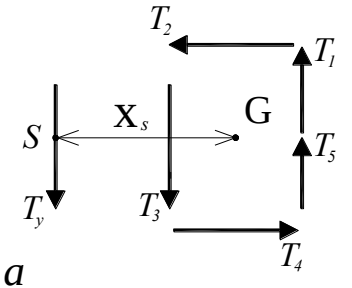
At the node B, because $\tau_{(u_1)}=\tau_{(u_2)}$ this expression can be written.

$$\tau(u_2)=\tau(u_1)_B+\frac{T_y}{I_x.t}(u_2.t.a/2)$$

$$\tau(u_2)=\frac{T_y}{8I_x}(a^2+4u_2a) \;\rightarrow\; T_2=\int_0^a\tau(u_2).t.du_2=\frac{3}{8}\frac{T_y.t.a^3}{I_x} \qquad \tau(u_3)=\tau(u_2)_c+\frac{T_y.(u_3.t)(a-u_3)/2}{I_x.t}=\frac{5T_y.a^2}{8I_x}+\frac{T_y(u_3)(a-u_3)}{I_x.t}$$

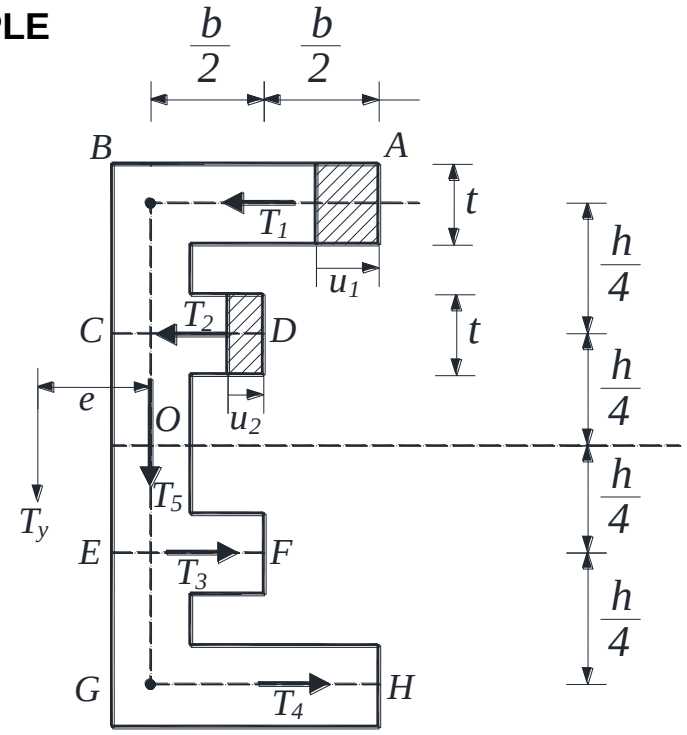
$$\tau(u_3)=\frac{T_y}{8I_x}(5a^2+4u_3a-4u_3^2) \;\rightarrow\; T_3=\int_0^a\tau(u_3).t.du_3 \qquad T_3=\frac{17}{24}.\frac{T_y}{I_x}a^3t \qquad |T_5|=|T_1| \;\; ; \;\; |T_4|=|T_2|$$

$$\sum M_G=T_y.x_s=2.T_1.\frac{a}{2}+T_2.a+T_3.\frac{a}{2} \qquad T_y x_s=2.\frac{T_y.t.a^3}{48I_x}.\frac{a}{2}+\frac{3}{8}\frac{T_y.t.a^3}{I_x}.a+\frac{17}{24}.\frac{T_y}{I_x}.a^3.t.\frac{a}{2}$$



$$x_s=\frac{1}{I_x}\left(\frac{2ta^4}{96}+\frac{3ta^4}{8}+\frac{17ta^4}{48}\right)=\frac{ta^4}{I_x}\left(\frac{2}{96}+\frac{36}{96}+\frac{34}{96}\right) \qquad x_s=\frac{ta^4}{I_x}.\left(\frac{72}{96}\right)=\frac{3}{4}\frac{ta^4}{I_x}=\frac{3}{4}.\frac{ta^4}{\frac{2}{3}ta^2}=\frac{9}{8}a$$

EXAMPLE



Find the shear center of the cross-section given.

SOLUTION

$$|T_1| = |T_4| \quad ; \quad |T_2| = |T_3|$$

$$\tau_{(u_1)} = \frac{T_y (u_1 \cdot t \cdot h/2)}{t \cdot I_x} = \frac{T_y \cdot h \cdot u_1}{2 \cdot I_x}$$

$$T_1 = \int_0^b \tau_{(u_1)} t \, du = \frac{T_y \cdot h \cdot t \cdot b^2}{4 \cdot I_x}$$

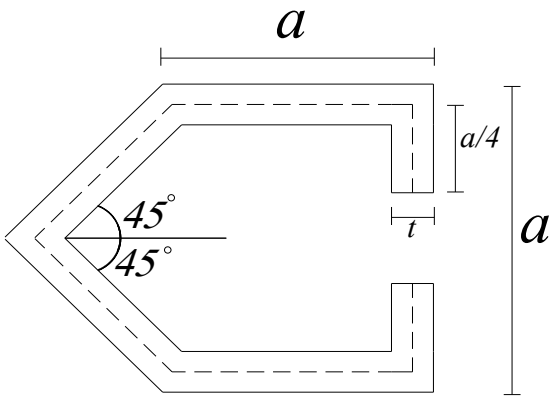
$$\tau_{(u_2)} = \frac{T_y (u_2 \cdot t \cdot h/4)}{t \cdot I_x} = \frac{T_y \cdot h \cdot u_2}{4 \cdot I_x}$$

$$T_2 = \int_0^{b/2} \tau_{(u_2)} t \, du_2 = \frac{T_y \cdot h \cdot t \cdot b^2}{32 \cdot I_x}$$

$$\sum M_0 = T_y \cdot e = (T_1 \cdot h) + (T_2 \cdot h/2)$$

$$e = \frac{17}{64} \cdot \frac{b^2 h^2 t}{I_x}$$

Problem: Obtain the shear center of the cross-section given on the right hand side in terms of I_x , a and t. t is thickness and it is constant everywhere.



Solution:

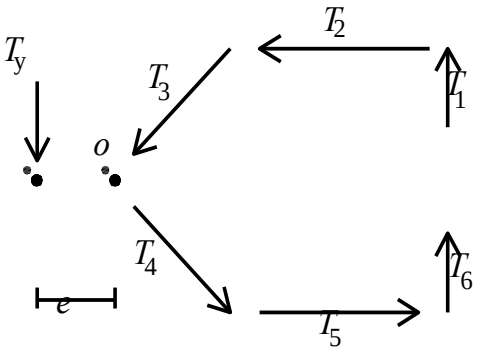
$$\tau_1(u_1) = \frac{T_y}{t.I_x}.t.u_1\left(\frac{a}{4}+\frac{u_1}{2}\right) \qquad \tau_1(u_1) = \frac{T_y.u_1}{2I_x}\left(\frac{a}{2}+u_1\right)$$

$$T_1 = \int_0^{a/4} \tau_1(u_1).t.du_1 = \frac{T_y.t}{I_x}\left|\frac{a.u_1^2}{8}+\frac{u_1^3}{6}\right|_0^{a/4} = \frac{T_y.t.a^3}{96.I_x}$$

First solution way:

$$\tau_2(u_2) = \tau_1\left(u_1 = \frac{a}{4}\right) + \frac{T_y}{I_x.t}u_2.t.\frac{a}{2}$$

$$\tau_2(u_2) = \frac{3.T_y.a^2}{32.I_x} + \frac{T_y.a.u_2}{2.I_x} \qquad \tau_2(u_2) = \frac{T_y.a}{2I_x}.\left[\frac{3a}{16}+u_2\right]$$



$T_1 = T_6$
$T_2 = T_5$
$T_3 = T_4$

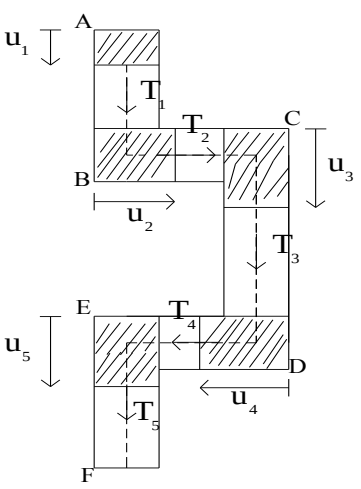
Second solution way:

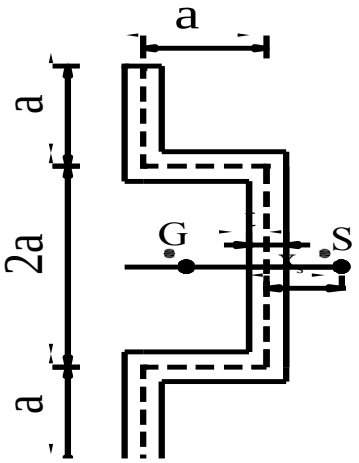
$$\tau_2(u_2) = \frac{T_y}{t.I_x}.\left[t.\frac{a}{4}.\frac{3a}{8}+u_2.t.\frac{a}{2}\right] \qquad \tau_2(u_2) = \frac{T_y.t}{t.I_x}.\left[\frac{3a^2}{32}+u_2.\frac{a}{2}\right] \qquad \tau_2(u_2) = \frac{T_y}{2I_x}.a\left[\frac{3a}{16}+u_2\right]$$

$$T_2 = \int_0^a \tau_2(u_2).t.du_2 = \frac{T_y.a.t}{2.I_x}\left|\frac{3a.u_2}{16}+\frac{u_2^2}{2}\right|_0^a = \frac{11.T_y.a^3.t}{32.I_x} \qquad \Sigma M_o = T_y.e = 2\left[T_1.(a+\frac{a}{2})+T_2.(\frac{a}{2})\right]$$

$$T_y.e = 2\left[\frac{T_y.t.a^3}{96.I_x}.(\frac{3a}{2})+\frac{11.T_y.a^3.t}{32.I_x}.(\frac{a}{2})\right] \rightarrow \boxed{e = \frac{3.t.a^4}{8.I_x}}$$

Problem: Obtain the distance of the shear center to the right edge of the cross-section given on the right hand side in terms of I_x , a and t. t is thickness and it is constant everywhere.





Solution:

$$\tau(u_1)=\frac{T_y}{t.I_x}\left[t.u_1.(2a-\frac{u_1}{2})\right]=\frac{T_y}{I_x}(2au_1-\frac{u_1^2}{2})$$

$$T_1=\int_0^a\frac{T_y}{I_x}(2au_1-\frac{u_1^2}{2})tdu_1=\frac{T_y.t}{I_x}(2a.\frac{u_1^2}{2}-\frac{u_1^3}{6})\Big|_0^a \qquad T_1=\frac{T_y.t}{I_x}.\frac{5a^3}{6}$$

$$\tau_1(a)=\frac{3T_ya^2}{2I_x} \qquad \tau(u_2)=\frac{3T_ya^2}{2.I_x}+\frac{T_y}{I_x.t}(u_2.t.a)=\frac{3T_ya^2}{2I_x}+\frac{T_y}{I_x}u_2.a$$

$$\tau(u_2)=\frac{3T_ya^2}{2.I_x}+\frac{T_y.a}{I_x}u_2=\frac{T_ya}{I_x}(\frac{3}{2}.a+u_2) \qquad T_2=\frac{T_y.a}{I_x}\int_0^a(\frac{3}{2}a+u_2)tdu_2=\frac{T_y.a.t}{I_x}(\frac{3}{2}a.u_2+\frac{u_2^2}{2})\Big|_0^a \qquad T_2=\frac{T_y.a.t}{I_x}(\frac{3a^2}{2}+\frac{a^2}{2})=\frac{2T_y.a^3.t}{I_x}$$

$$(\tau_2)_C=\frac{T_y}{I_x}.\frac{5}{2}a^2 \qquad \tau(u_3)=(\tau_2)_C+\frac{T_y\left[u_3.t.(a-u_3/2\right]}{t.I_x} \qquad \tau(u_3)=\frac{T_y}{I_x}\left[\frac{5}{2}a^2+au_3-\frac{u_3^2}{2}\right]$$

$$T_3=\int_0^{2a}\frac{T_y}{I_x}\left[\frac{5}{2}a^2+au_3-\frac{u_3^2}{2}\right]tdu_3=\frac{T_y.t}{I_x}\left[\frac{5}{2}a^2u_3+a\frac{u_3^2}{2}-\frac{u_3^3}{6}\right]\Big|_0^{2a}= \qquad T_3=\frac{T_y.t}{I_x}\left[\frac{5}{2}a^2.2a+a\frac{(2a)^2}{2}-\frac{(2a)^3}{6}\right]=\frac{T_y.t}{I_x}.\frac{17a^3}{3}$$

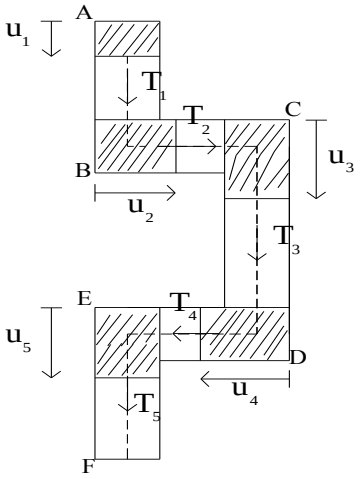
$$(\tau_3)_D=\frac{T_y}{I_x}\left[\frac{5}{2}a^2+a.2a-\frac{(2a)^2}{2}\right]=\frac{T_y}{I_x}.\frac{5}{2}a^2$$

$$\tau_4=(\tau_3)_D+\frac{T_y.u_4.t.a}{t.I_x}=\frac{T_y}{I_x}\left[\frac{5}{2}a^2+(-a).u_4\right] \qquad T_4=\int_0^a\frac{T_y.a}{I_x}\left[\frac{5}{2}a-u_4\right]tdu_4=\frac{T_y.a.t}{I_x}\left[\frac{5}{2}au_4-\frac{u_4^2}{2}\right]\Big|_0^a=\frac{2T_y.a.t}{I_x}$$

$$(\tau_4)_E = \frac{T_y}{I_x}.\frac{5}{2}a^2 - \frac{T_y}{I_x}.a^2 = \frac{3}{2}\frac{T_y}{I_x}.a^2$$

$$\tau_5 = \frac{3}{2}.\frac{T_y}{I_x}.a^2 + \frac{T_y}{I_x}\left[-au_5 - \frac{u_5^2}{2}\right]$$

$$T_5 = \frac{T_y}{I_x}\int_0^a (\frac{3}{2}a^2 - au_5 - \frac{u_5^2}{2})tdu_5 = \frac{T_y.t}{I_x}(\frac{3}{2}a^3 - \frac{1}{2}a^3 - \frac{1}{6}a^3) \qquad T_5 = \frac{T_y.t}{I_x}.\frac{5a^3}{6}$$

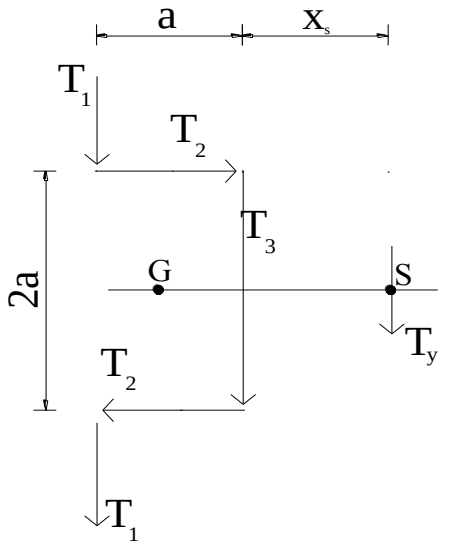


$$2T_1.(a + x_s) + T_3.x_s - T_2.2a = 0$$

$$2T_1a + 2T_1x_s + T_3x_s - 2T_2a = 0$$

$$x_s(2T_1 + T_3) + 2T_1a - 2T_2a = 0$$

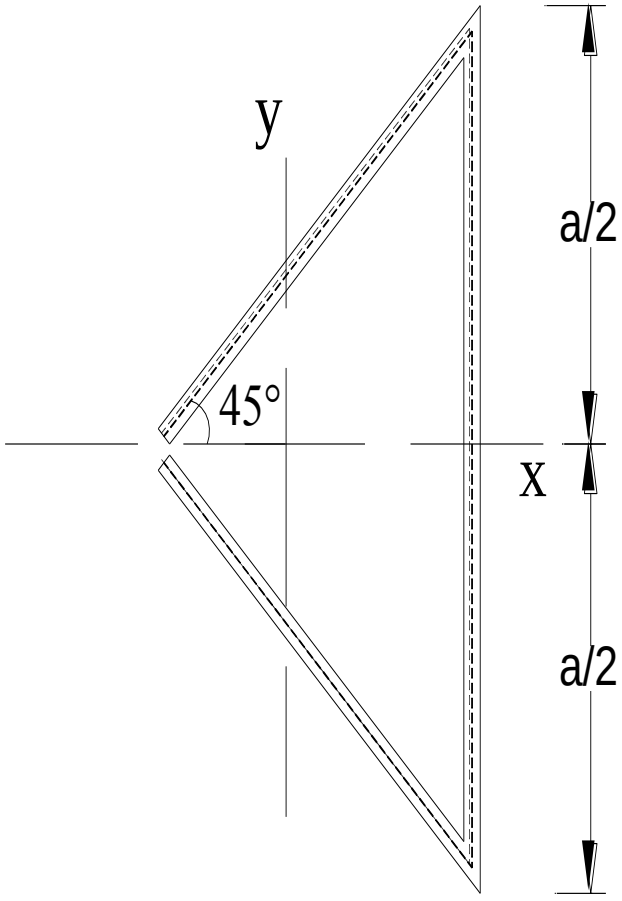
$$x_s = \frac{2a(T_2 - T_1)}{2T_1 + T_3}$$



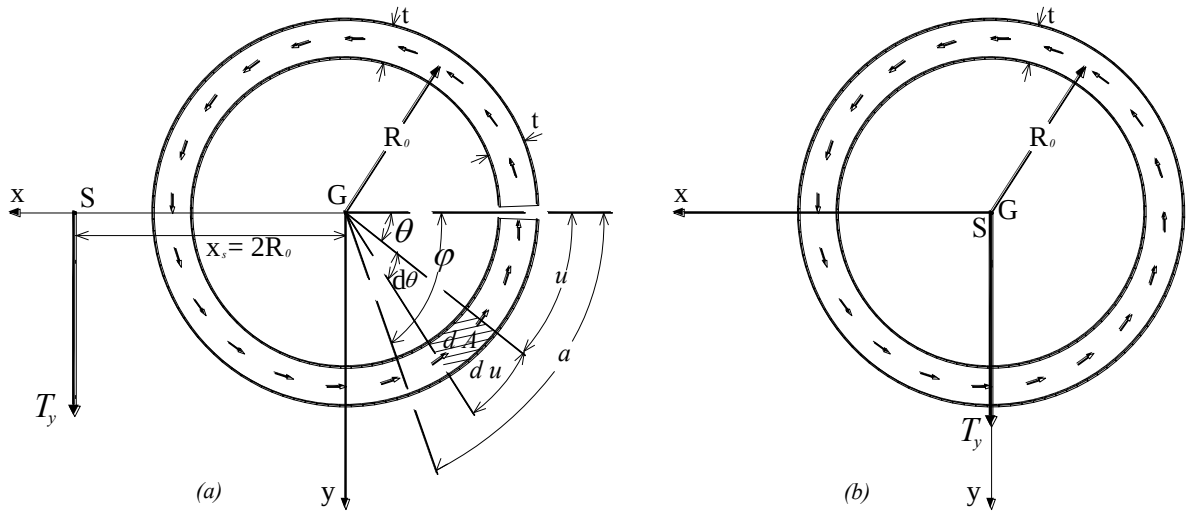
$$x_s = \frac{2a\left[\frac{2T_y a^3.t}{I_x} - \frac{5T_y a^3 t}{6I_x}\right]}{\frac{T_y.a^3.t}{I_x}\left(\frac{10}{6} + \frac{17}{3}\right)} = \frac{2a.\frac{(12-5)}{6}}{(\frac{10}{6} + \frac{34}{6})}$$

$$x_s = \frac{\frac{14}{6}a}{\frac{44}{6}} = \frac{14}{6}.\frac{6}{44}a = \frac{14}{44}a = \frac{7}{22}a$$

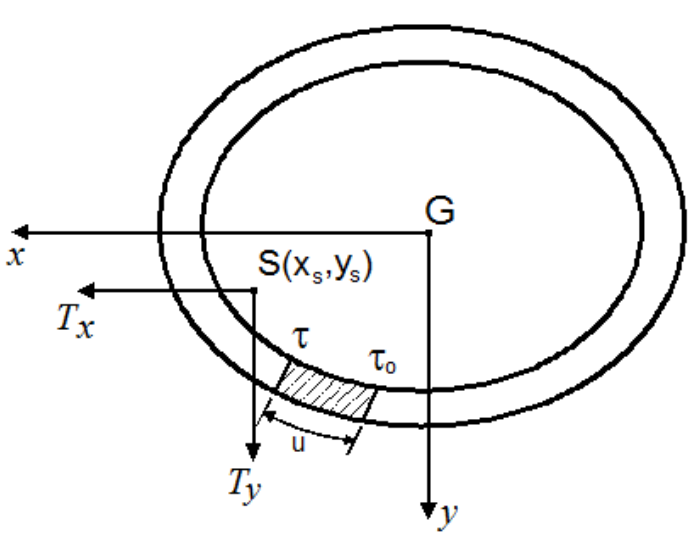
Problem: The cross-section given by the figüre is symmetrical with respect to given horizontal axis. Without calculation, define the shear center of the cross section. Explain the reason of your answer.



SHEAR CENTER IN CLOSED TUBE CROSS SECTİONS



SHEAR CENTER IN CLOSED TUBE CROSS SECTİONS



In general, the shear center of the closed sections is obtained in the same manner as open sections.

Lets place the shear forces T_x, T_y on the shear center S. Now we want to find the coordinates x_s, y_s of the shear center by using the equations

$$T_y x_s = M_1(G) \qquad (1) \qquad T_x y_s = M_2(G) \qquad (2)$$

$$T_y . x_s = \int_0^c r(\tau t)du = \int_0^c r(\frac{T_y S_x}{I_x})du \rightarrow \; x_s = \frac{1}{I_x}\int_0^c rS_x du$$

$$T_x . y_s = \int_0^c r(\tau t)du = \int_0^c r(\frac{T_x S_y}{I_y})du \rightarrow \; y_s = -\frac{1}{I_y}\int_0^c rS_y du$$

The calculations performed until now is the same as in the open sections. But in calculating the shear force on the section at u which is seen in Fig. 12.67, the following equation can not be used:

$$\tau = T_y S_x / (t I_x)$$

Because the shear stress on the section at u=0 is not zero. As it is known, the shear stress is zero for the section at u=0 in open sections, but shear stress is not zero on the section at u=0 for closed sections. Therefore the shear flux for any closed section can be calculated by using the following equations:

$$\tau t = \tau_0 t_0 + T_y S_x / I_x \qquad q = q_0 + T_y S_x / I_x \qquad (3)$$

Here τ_o or q_o values are for the starting section (origin section, onset section) and they must be calculated beforehand (in advance).

Birim dönme açısı enerji esasından hareketle hesaplanacaktır. Bunun için boyu birim olan bir eleman göz önüne alınsın, Şekil12.14. M_b momentleri etkisinde bu elemanın iki ucunun birbirine göre dönmesi ω dır. Dış kuvvetlerin işi

$M_b\,\omega/2$ olur.

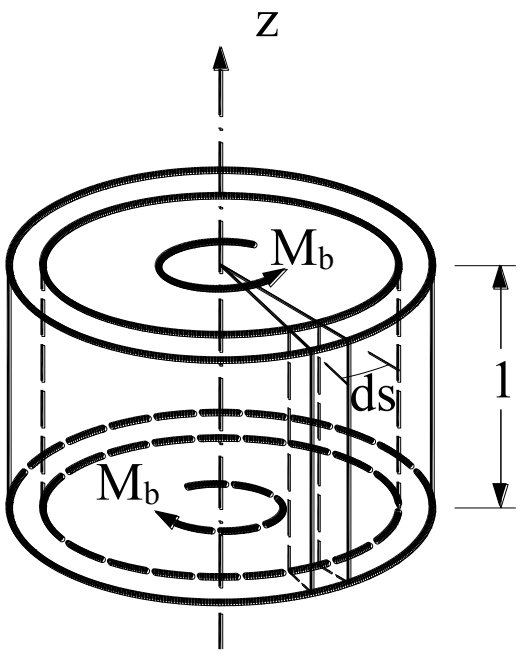
Şekil değiştirme sırasında elemanda depo edilen enerji ise

$$V=\int_{(V)}\frac{\tau^2}{2G}\,dv=\oint\frac{q^2}{2G\,t^2}\,t\,ds=\oint\frac{q^2}{2G\,t}\,ds \qquad (12.40)$$

olarak elde edilir. Dış kuvvetlerin işinin depo edilen enerjiye eşit olması gerektiğinden birim dönme açısı aşağıdaki şekilde elde edilir:

$$\omega=\frac{M_b}{4G\,A^2}\oint\frac{ds}{t}=\frac{1}{2G\,A}\oint\frac{q\,ds}{t} \qquad (12.41)$$

$$\omega=\frac{1}{2G\,A}\oint\frac{q\,ds}{t}$$



Şekil 12.14

Let's take into consideration the closed section in Fig. 12.67. In order to calculate the shear flux q_o, a closed tube of unit length is considered. For bending with shear only, namely if there is no torsion, the rotation angle of this tube of unit length must be zero:

$$\omega = \frac{1}{2G\,A}\oint \frac{q\,du}{t} = 0$$

From the above equation, we arrive at the following expression:

$$\oint \frac{q\,du}{t} = 0$$

In the case of bending without torsion, shear flux must satisfy the condition given above. If the expression given before for the shear flux are substituted in the equation given above, we get the following equation

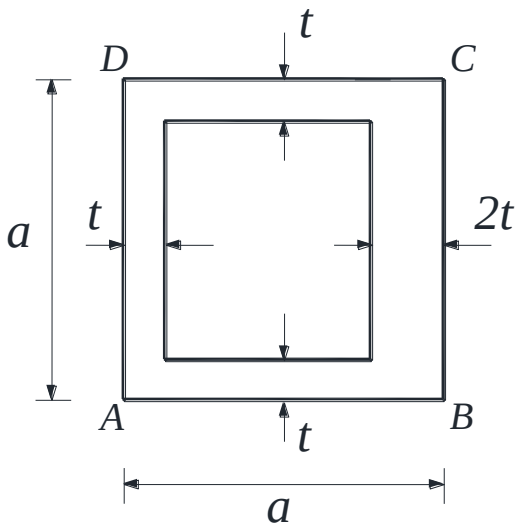
$$q = q_0 + \frac{T_y\,S_x}{I_x} \qquad (3)$$

$$\oint \left(q_0 + \frac{T_y S_x}{I_x} \right) \frac{du}{t} = 0 \;\rightarrow\; q_0 = -\frac{T_y}{I_x}\,\frac{\oint S_x\,du/t}{\oint du/t}$$

After obtaining q_o, by replacing it into equation (3) and later on substituting Eq. (3) into Eq. (1) an (2), the shear center are obtained.

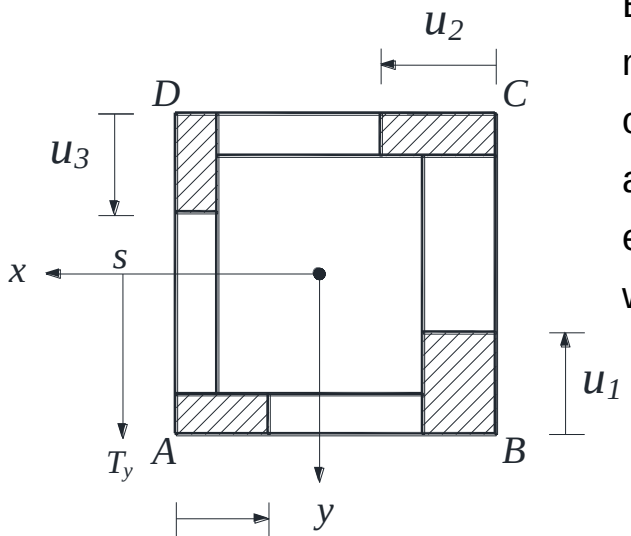
Example Problem: Let's find the shear center of a squarish tube of closed section with the edge length a. Its three arm's thickness is t, and the thickness of the fourth arm is 2t

Solution: The set of xy axes are passed from the geometric center of the cross section which are parallel to the sides of the figure. Because the x axis is the symmetry axis, the shear center is on the x axis. By neglecting the terms with t^3, the second moment of inertia about the x axis is obtained as follows:



$$I_x = \frac{ta^3}{12} + \frac{2ta^3}{12} + 2at\left(\frac{1}{2}a\right)^2 = \frac{3}{4}a^3t$$

For performing the necessary calculations on the cross section, let's choose the point B as the starting point. It is necessary to calculate q_o at the point B. Let's choose the calculation route as BCDAB. For calculating the static moments for each arm, the starting points and the u_i variables are shown on the figure. The static moments for each arm is the static moment of the area between the starting point and the considered section. Because the expressions of these static moments are different on each arm, these are written separately below:



$$S_x^1 = 2u_1 t\left(\frac{1}{2}a - \frac{1}{2}u_1\right); \quad S_x^2 = -\frac{1}{2}au_2 t$$

$$S_x^3 = -\frac{1}{2}a^2 t - u_3 t\left(\frac{1}{2}a - \frac{1}{2}u_3\right); \qquad S_x^4 = -\frac{1}{2}a^2 t - u_4 t\left(\frac{1}{2}a - \frac{1}{2}u_4\right)$$

The necessary integrals for calculating q_o and the value of q_o is given below:

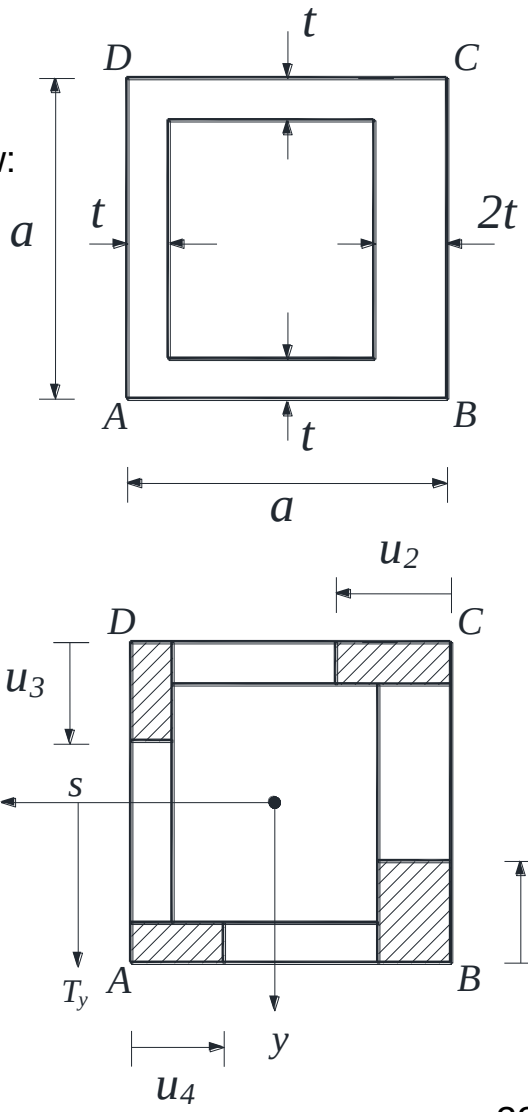
$$\oint S_x \frac{du}{t} = \int_0^a 2u_1\left(\frac{1}{2}a - \frac{1}{2}u_1\right)u_1 + \int_0^a -\frac{1}{2}au_2 du_2$$

$$+\int_0^a \left[-\frac{1}{2}a^2 - u_3\left(\frac{1}{2}a - \frac{1}{2}u_3\right)\right]du^3 + \int_0^a \left[-\frac{1}{2}a^2 - u_4\left(\frac{1}{2}a - \frac{1}{2}u_4\right)\right]du^4$$

$$\oint S_x \frac{du}{t} = \frac{1}{12}a^3 - \frac{1}{4}a^3 - \frac{7}{12}a^3 - \frac{1}{4}a^3 = -a^3$$

$$\oint \frac{du}{t} = \int_0^a \frac{du_1}{2t} + \int_0^a \frac{du_2}{t} + \int_0^a \frac{du_3}{t} + \int_0^a \frac{du_4}{t} = \frac{7a}{2}$$

$$q_0 = -\frac{T_y}{I_x}\cdot\frac{-a^3}{7a/2t} = \frac{2T_y a^2 t}{7I_x} = \frac{8}{21}\frac{T_y}{a}$$



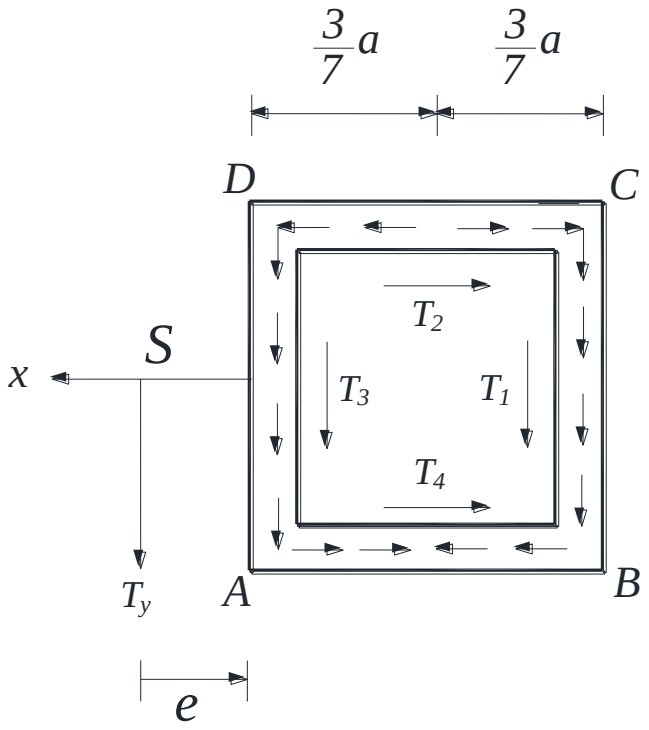
The shear flux on each arm can be obtained as follows by using the equation $q=q_o+T_yS_x/I_x$

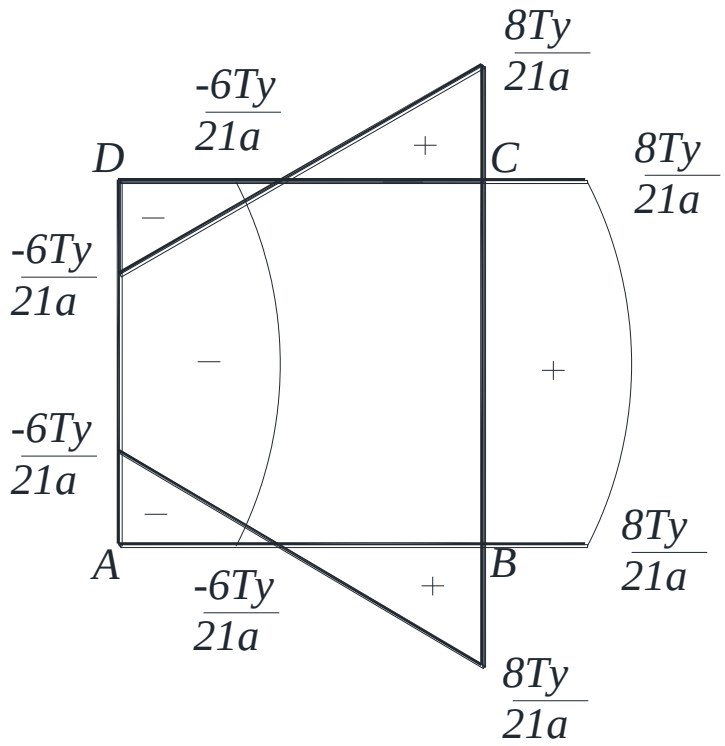
$$q_1=\frac{8}{21}\frac{T_y}{a}+\frac{4}{3}\frac{T_y}{a}\left(\frac{u_1}{a}-\frac{u_1^2}{a^2}\right)=\frac{T_y}{a}\left(\frac{8}{21}+\frac{4}{3}\frac{u_1}{a}-\frac{4}{3}\frac{u_1^2}{a^2}\right)$$

$$q_2=\frac{8}{21}\frac{T_y}{a}-\frac{2}{3}\frac{T_y}{a}\frac{u_2}{a}=\frac{T_y}{a}\left(\frac{8}{21}-\frac{2}{3}\frac{u_2}{a}\right)$$

$$q_3=\frac{8}{21}\frac{T_y}{a}-\frac{2}{3}\frac{T_y}{a}\left(1+\frac{u_3}{a}-\frac{u_3^2}{a^2}\right)=\frac{T_y}{a}\left(-\frac{6}{21}-\frac{2}{3}\frac{u_3}{a}+\frac{2}{3}\frac{u_3^2}{a^2}\right)$$

$$q_4=\frac{8}{21}\frac{T_y}{a}+\frac{2}{3}\frac{T_y}{a}\left(-1+\frac{u_4}{a}\right)=\frac{T_y}{a}\left(-\frac{6}{21}+\frac{2}{3}\frac{u_4}{a}\right)$$





By integrating the shear flux on each arm, the shear forces of the arms can be obtained as follows:

$$T_1 = \int_0^a q_1 du_1 = \frac{38}{63} T_y; \qquad T_2 = \int_0^a q_2 du_2 = \frac{1}{21} T_y$$

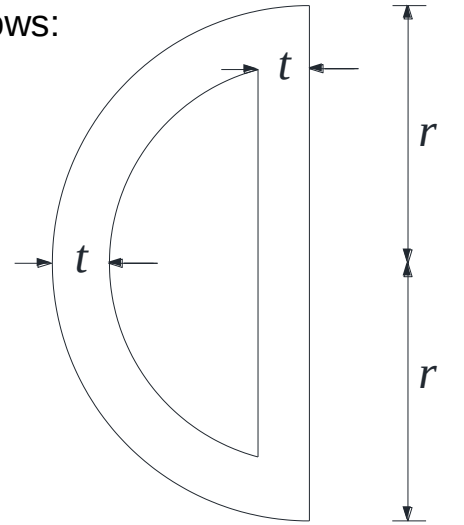
$$T_3 = \int_0^a q_3 du_3 = -\frac{25}{63} T_y; \qquad T_4 = \int_0^a q_4 du_4 = \frac{1}{21} T_y$$

The graphics of the shear flux and the shear forces are given in Fig. 12.70b. As seen in the Figure, the equilibrium equations are satisfied and the distance e for the shear center is found as folows:

$$T_y.e = \frac{38}{63} T_y a + \frac{1}{21} T_y a \qquad \rightarrow \qquad e = \frac{41}{63} a$$

Example Problem: Let's find the shear center of a semi circle tube of closed section with the radius r. Its thickness is t

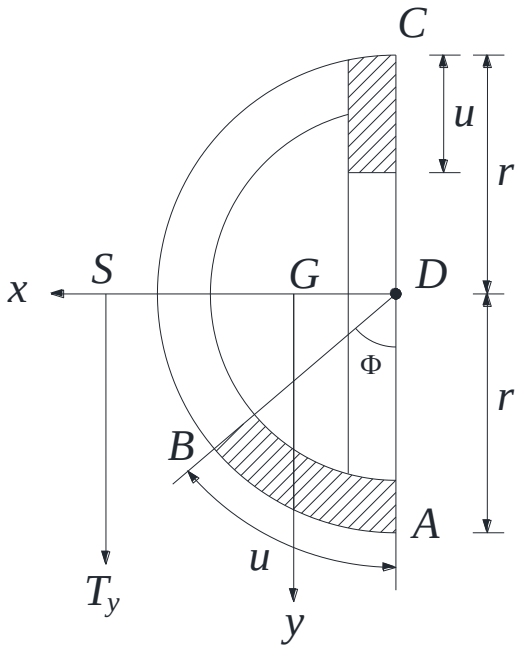
Solution: The set of xy axes are passed from the geometric center of the cross section.Because the x axis is the symmetry axis, the shear center is on the x axis. The second moment of inertia about the x axis is obtained as follows:



$$I_x = \frac{1}{12} t (2r)^3 + \int_0^{\pi} r^2 \cos^2 \varphi r t d\varphi$$

$$I_x = \frac{2}{3} r^3 t + \frac{\pi}{2} r^3 t = 2,24 r^3 t$$

For performing the necessary calculations on the cross section, let's choose the point A as the starting point. It is necessary to calculate q_o at the point A. Let's choose the calculation route as ABCA. The expressions of the static moments between ABC and CA are different from each other. These static moments are as follows:



$$S_x^1 = \int r \cos\alpha dF = \int_0^{\varphi} r^2 t \cos\alpha d\alpha = r^2 t \sin\varphi$$

$$S_x^2 = S_x^1(\pi) - ut(r - u/2) = -ut(r - u/2)$$

The necessary integrals for calculating q_o and the value of q_o is given below:

$$\oint S_x \frac{du}{t} = \int_o^{\pi} r^2 \sin\varphi d\varphi + \int_0^{2a} -ut(r - u/2) du = 2r^3 - \frac{2}{3} r^2 = \frac{4}{3} r^2$$

$$\oint \frac{du}{t} = \frac{\pi r + 2r}{t} = 5,14 \frac{r}{t}$$

$$q_0 = -\frac{T_y}{2,24 r^3 t} \cdot \frac{4r^3/3}{5,14 r/t} = -\frac{T_y}{8,64 r}$$

The shear flux on the circular arm and the vertical arm can be obtained as follows by using the equation $q=q_o+T_yS_x/I_x$

$$q_1 = -\frac{T_y}{8,64r} + \frac{T_y r^2 t}{2,24r^3 t}\sin\varphi = \frac{T_y}{2,24r}\left(\sin\varphi - 0,259\right)$$

$$q_2 = -\frac{T_y}{8,64r} - \frac{T_y ut\left(r-u/2\right)}{2,24r^3 t} = \frac{T_y}{2,24r}\left(\frac{u^2}{2r^2} - \frac{u}{r} - 0,259\right)$$

The zero shear flux is obtained at the angle

$$\sin\varphi_0 = 0.259 \rightarrow \varphi_0 = 15^o$$

The directions of shear fluxes are seen on the figure. The shear center can be found by taking moment about the point D as follows:

$$T_y.e = \int q_1 r du = \int_0^{\pi} q_1 r^2 d\varphi = \frac{T_y.\mathrm{r}}{2,24}\int_0^{\pi}\left(\sin\varphi - 0,259\right)d\varphi$$

$$e = 0,53a$$

TORSION AND SİMPLE STATES OF STRENGTH OF MATERIALS

In the simple states of strength of materials, the distributed forces on the section were tranferred to the gravity center of the cross section. If the gravity center and shear center coincide, then there is no need to make some changes in the definitions and investigations. If these two centers are different, there is need to make changes. Because it was seen before that the shear forces passing from the gravity center G causes torsion effects on the section.

Let the stress resultant at the gravity center be the vectors $\vec{R}$ and $\vec{M}$. The vectors can be expressed in terms of their components along the coordinate axes x, y, z as folows:

$$\vec{R} = T_x \vec{i} + T_y \vec{j} + N\vec{k} \qquad and \qquad \vec{M} = M_x \vec{i} + M_y \vec{j} + M_z \vec{k}$$

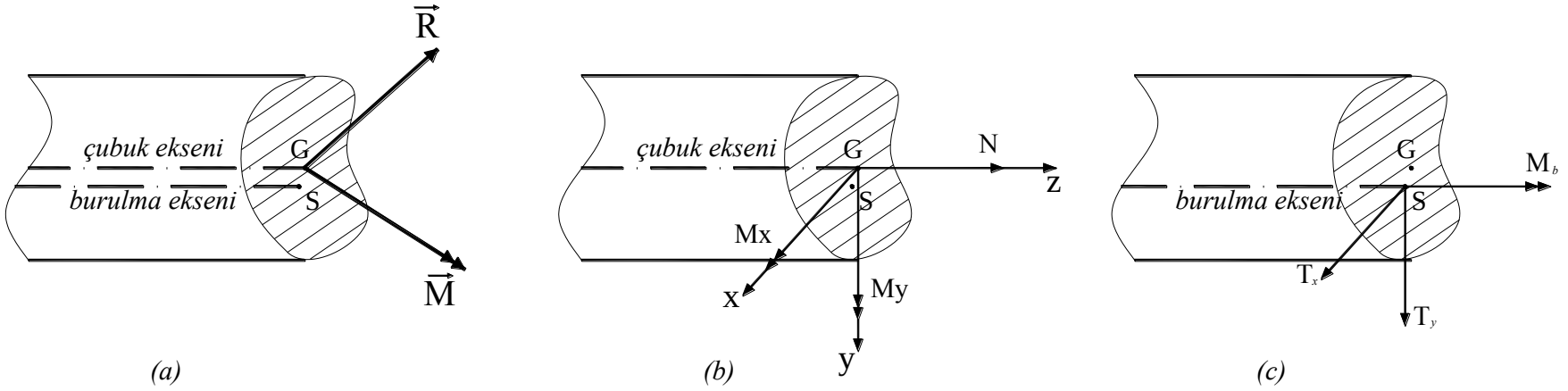
N : Axial normal force which is the same with the axial normal force in the case of simple axial normal force case of the strength of materials which passes from the gravity center G of the cross section

M_x ve M_y : Bending moments which is the same with the bending moments in the case of simple bending moment case of the strength of materials which act at the gravity center G of the cross section

T_x and T_y which pass from the gravity center of the cross section are not the same with the shear forces in the case of simple shaer force case of the strength of materials which act at the gravity center G of the cross section. Because if these shear forces passes from the gravity center, they cause rotation in addition to translation in the considered element.

M_z is not the torsion alone, because the shear forces must be transferred to the shear center S, and in this tranferring proces some additional torsions arise and these must be added to M_z. After adding these additional torsions to M_z, we obtain the torsion M_b.

$$M_b = M_z + T_x y_s - T_y x_s$$



If G and S coincides, then $x_s=y_s=0$ and therefore $M_b = M_z$

According to this explanation, the lever arm of the transversal force is taken as dS instead of dG.

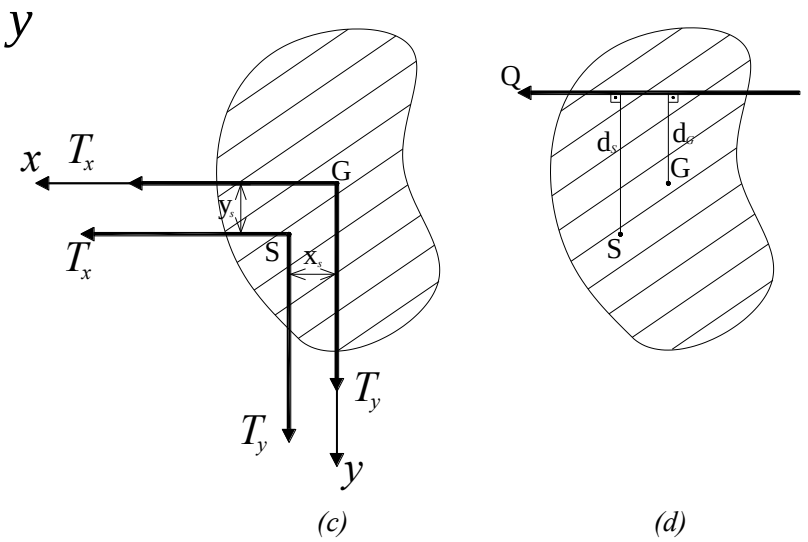
The obtained results can be summarized as follows:

N : Axial normal force which passes from G den geçen x, y

Bending moments M_x *and* M_y

are the moments which are about the axes passing from the gravity center G of the cross section.

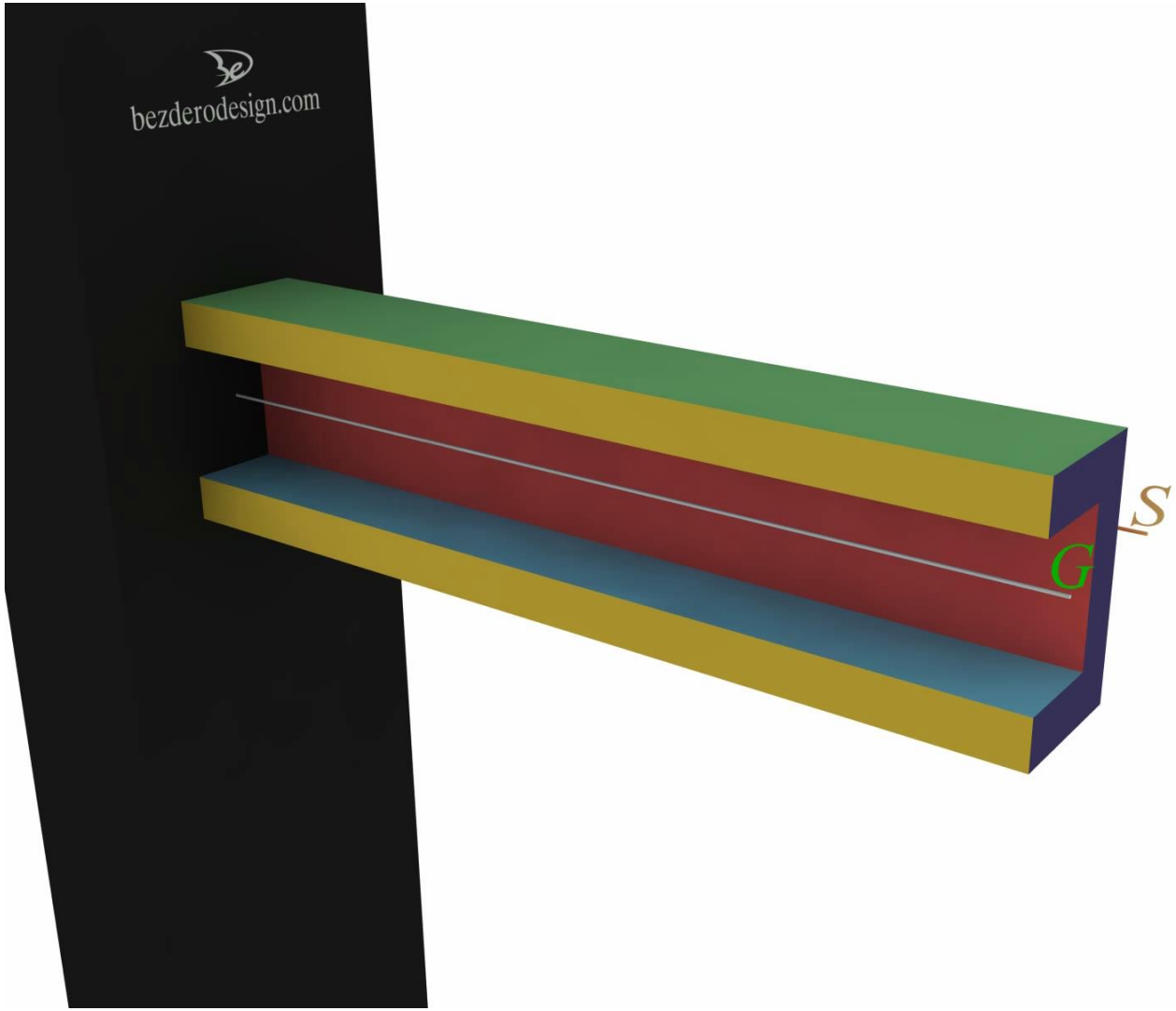
T_x *and* T_y are the shear forces which pass from the shear center S and are in the plane of cross section



M_b : Torsion about the torsion axis which passes from the point S.

ELASTIC CURVE
2 Weeks, Fifth in Line

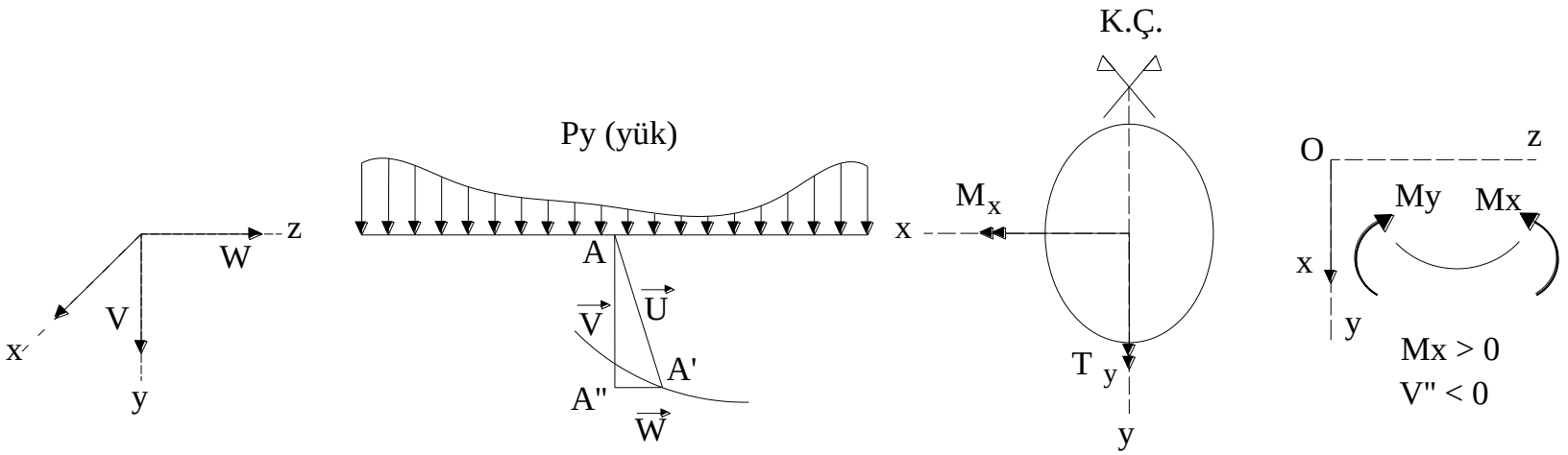
Prof. Dr. Turgut KOCATÜRK


bezderodesign.com
S
G

$u_z = w = 0$

WHAT IS ELASTIC CURVE?

Bar axis which is straight before bending becomes curve after bending. This curve is called as elastic curve.

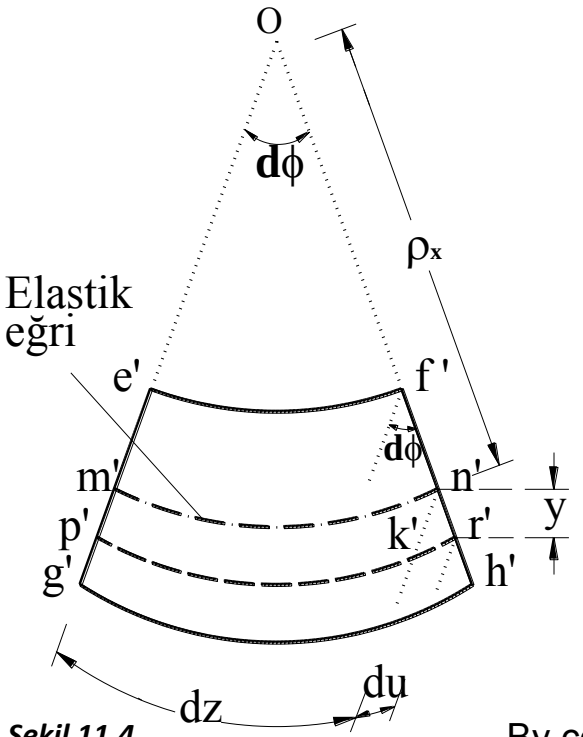


Straight bending occurs when the applied force plane coincides with the bending plane.

$\vec{u}$ displacement vector $u_y = v$

$u_z = w$

When working with the first-order theory, the displacement w is always one degree smaller than the displacement v in the bar. So it will be asumed that u_z =w=0. And also it will be assumed that u_y=v and v will be parallel to y axis. In this case, the only unknown is the function v(z). In order to obtain this function which is called elastic curve function, there are various ways. But all the methods are based on diferantial equation of the function of the elastic curve.



Şekil 11.4

1.2 Differantial Equation

If the fiber *m-n* is accepted on the neutral axis, then *mn=m'n'=dz*. Let's calculate the change in the displacement on the fiber *p-r* which is y distance away from *m-n* fiber. From Figure 11.4, it can be seen that *k'r'=p'r'-p'k'=du*. Hence, it can be written *p'k'=m'n'=dz*.

$$\varepsilon_z = \frac{p'r' - p'k'}{p'k'} = \frac{du}{dz} \quad \textit{Also;} \quad du = y\,d\varphi\,, \quad dz = \rho_x\,d\varphi \quad (11.2)$$

From here, $\frac{du}{dz} = \frac{y}{\rho_x}$ (11.3)

From the equations (11.1) and (11.3), $\varepsilon_z = \frac{y}{\rho_x}$ (11.4)

By considering the Hook's Law, it can be found that $\sigma_z = E\varepsilon_z = \frac{E}{\rho_x} y$ (11.5)

It is known that $M_x = \int_A y\sigma_z dA$ (11.6) Substituting the eqaution (11.5) into the equation (11.6),

It can be written that. $\frac{E}{\rho_x}\int_A y^2 dA = \frac{E}{\rho_x} I_x = M_x$ *Therefore;* $\frac{1}{\rho_x} = \frac{M_x}{E\,I_x}$ *(11.9)* *By substituting (11.9) into (11.5)*

we obtain $\sigma_z = \frac{E}{\rho_x} y = \frac{M_x}{I_x} y$ (11.10) From the differential geometry, $\frac{1}{\rho x} = \pm \frac{d^2v/dz^2}{(1+(dv/dz)^2)^{3/2}}$

$$\frac{M_x}{E\,I_x} = \frac{1}{\rho x} = \pm \frac{d^2v/dz^2}{(1+(dv/dz)^2)^{3/2}}$$

The term dv/dz is very small in flat curves. Therefore the square of this term can be neglected compared to 1 in the denominator, and hence the equation is linearized.

$$\frac{M_x}{EI_x} = -\frac{d^2v}{dz^2}$$

The important point in the elastic curve problems is to find a relationship between the dispalcement $v(z)$ and the load p_y. If the relationship is to be independent from the boundary conditions, the equation needs to be a diffrential equation defined by the derivatives. This equation can be determined by using the moment-curvature relationship;

The curvature term on the right side of the equation is defined by $v(z)$

$$\frac{M_x}{EI_x} = -\frac{d^2v}{dz^2} \rightarrow \frac{1}{EI_x}\frac{dM_x}{dz} = \frac{1}{EI_x}Q_y = -\frac{d^3v}{dz^3} \quad \frac{1}{EI_x}\frac{dQ_y}{dz} = \frac{1}{EI_x}\left(-p_y\right) = -\frac{d^4v}{dz^4} \rightarrow \quad \frac{d^4v}{dz^4} = \frac{p_y}{EI_x}$$

The differential equation is always valid in both cases of statically determinate and statically indeterminate systems. When the differential equation is lineer, the superposition rule is acceptable. For example, the displacements because of the loads p_1+p_2 will be equal to the summation of the v_1+v_2. Here v_1, v_2 are the displacements because of P_1 and P_2, respectively.

$$EI_x\frac{d^4v_1}{dz^4} = p_1 \qquad EI_x\frac{d^4v_2}{dz^4} = p_2 \quad \text{Hence,} \quad EI_x\frac{d^4(v_1+v_2)}{dz^4} = p_1 + p_2$$

Derivatives of displacements are proportional to the geometrical and mechanical quantities of the bar.

$v(z)$:Displacements

$v'(z)$: Slopes or section rotations

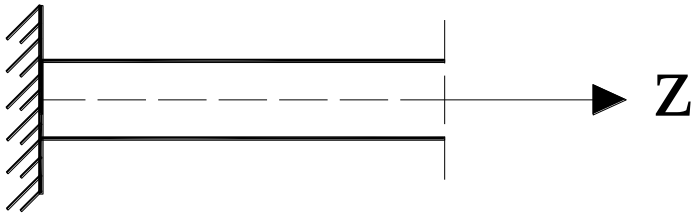
$-EI_x v''(z) = M_x$: Bending moment

$-EI_x v''' = Q_y$:Shear force

$EI_x v^{iv}(z) = p_y$:Magnitude of the uniform load

1.2.1 Boundary Conditions

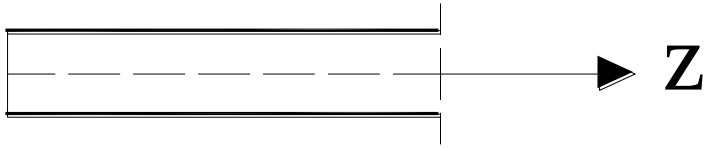
1.2.1.1 Fixed Support



For $z = 0, \quad v = 0$ and $v' = 0$

v and v' are called as the geometric boundary condirions. Even if the support is not constrained along the z axis, the conditions will not change.

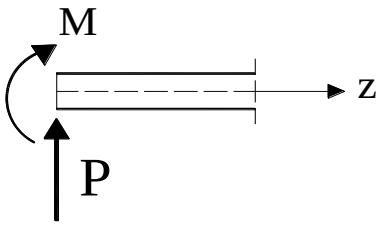
1.2.1.2 Free End

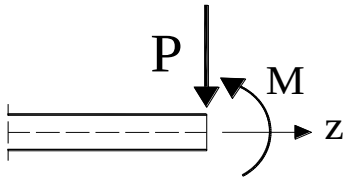


For $z = 0, \quad - EI_x v'' = 0$ and $- EI_x v''' = 0$

These boundary conditions are related to the force and the moment. Therefore, they are defined as dynamic boundary conditions.

Left end





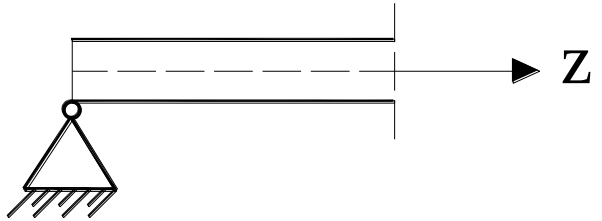
Right end

For $z = 0, \quad - EI_x v'' = M$ and $- EI_x v''' = P$

For $z = l, \quad - EI_x v'' = M$ and $- EI_x v''' = P$

1.2.1.3 Pin Support

For $z = 0, \quad v = 0$ and $v'' = 0$



1.4 Solution Methods

1.4.1 Integration Method:

If the system is determinate and hence the moment function is known in the beginning, the elastic curve function of a beam can be derived by using the equation (2c) and the geometric boundary conditions.

$$-E I_x v''(z) = M_x \qquad (2c)$$

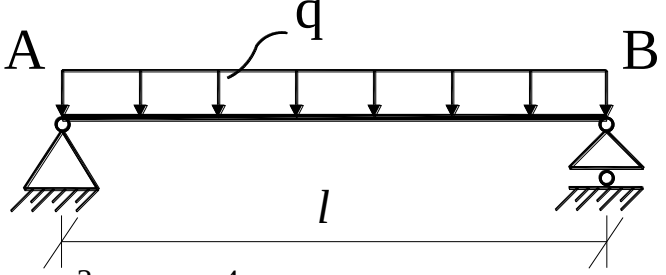
In general, even if the system is determinate or indeterminate, the elastic curve function can be detemined by using the equation (2e), the geometric and the dynamic boundary conditions.

$$E I_x v^{iv}(z) = p_y \qquad (2e)$$

The application of the method will be described in the upcoming problems.

Problem 1.1

Determine the elastic curve of the uniformly loaded beam shown and calculate the maximum displacement by using fourth order differential equation.



$$EIv^{iv} = q \rightarrow EIv''' = c_1 + qz \rightarrow EIv'' = c_2 + c_1 z + q\frac{z^2}{2}$$

$$EIv' = c_3 + c_2 z + c_1\frac{z^2}{2} + q\frac{z^3}{6} \rightarrow EIv = c_4 + c_3 z + c_2\frac{z^2}{2} + c_1\frac{z^3}{6} + q\frac{z^4}{24}$$

Boundary conditions:

$$z = 0 \rightarrow v = 0 \rightarrow EIv = c_4 = 0 \rightarrow (1)$$

$$z = 0 \rightarrow v'' = 0 \rightarrow EIv'' = c_2 = 0 \rightarrow (2)$$

$$z = l \rightarrow v = 0 \rightarrow c_3 l + c_1\frac{l^3}{6} + q\frac{l^4}{24} = 0 \rightarrow (3)$$

$$z = l \rightarrow v'' = 0 \rightarrow EIv'' = c_1 l + q\frac{l^2}{2} = 0 \Rightarrow c_1 = -q\frac{l}{2} \rightarrow (4)$$

Using the equation (3),

$$\left(-q\frac{l}{2}\right).\frac{l^3}{6} + c_3 l + q\frac{l^4}{24} = 0 \Rightarrow -q\frac{l^4}{12} + c_3 l + q\frac{l^4}{24} = 0 \rightarrow c_3 = q\frac{l^3}{24} \Rightarrow EI\,v = q\frac{l^3}{24}z - q\frac{l}{12}z^3 + q\frac{z^4}{24}$$

The elastic curve is symmetric, so the maximum displacement will be at the middle of the beam.

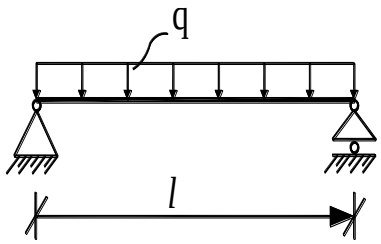
$$z = \frac{l}{2} \rightarrow EI\,v = EI\,v_{\max} = \left[q\frac{l^3}{24}\left(\frac{l}{2}\right) - q\frac{l}{12}\left(\frac{l}{2}\right)^3 + q\frac{(l/2)^4}{24}\right] = \frac{5ql^4}{384} \rightarrow v_{\max} = \frac{5ql^4}{384EI}$$

$$EIv'' = -q\frac{l}{2}z + q\frac{z^2}{2} \rightarrow M = -EIv'' = q\frac{l}{2}z - q\frac{z^2}{2} \quad M(l/2) = q\frac{l}{2}\frac{l}{2} - q\frac{(l/2)^2}{2} = \frac{ql^2}{4} - \frac{ql^2}{8}$$

$$M(l/2) = \frac{ql^2}{4} - \frac{ql^2}{8} = \frac{ql^2}{8}$$

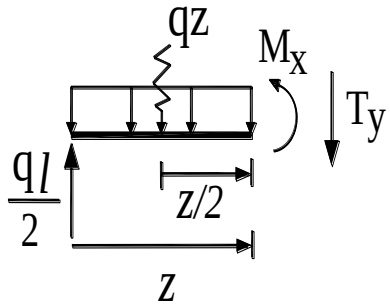
Problem 1.1

Determine the elastic curve of the uniformly loaded beam shown and calculate the maximum displacement by using second order differential equation.



$$\sum M = 0 \rightarrow \frac{ql}{2}.z - qz.\frac{z}{2} - M_x = 0$$

$$M_x = \frac{ql}{2}z - \frac{qz^2}{4}$$



$$-EIv'' = M_x = \frac{ql}{2}z - \frac{qz^2}{2}$$

$$-EIv' = \frac{qlz^2}{4} - \frac{qz^3}{6} + C_1$$

$$-EIv = \frac{qlz^3}{12} - \frac{qz^4}{24} + C_1 z + C_2$$

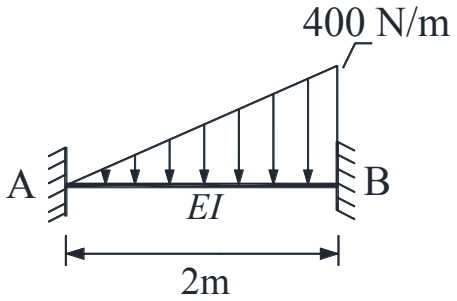
$$z = 0 \rightarrow v = 0 = \frac{ql.0^3}{12} - \frac{q.0^4}{24} + C_1.0 + C_2 \rightarrow C_2 = 0$$

$$z = l \rightarrow v = 0 = \frac{ql.l^3}{12} - \frac{q.l^4}{24} + C_1 l = 0$$

$$C_1 = \frac{ql^3}{24} - \frac{l^3}{12} = -\frac{ql^3}{24}$$

$$EIv = -\frac{qlz^3}{12} + \frac{qz^4}{24} + \frac{ql^3}{24}z$$

Problem: Obtain the elastic curve for the double cantilever beam seen in the figure. Determine the maximum displacement using the elastic curve expression you derived.



Solution:

$$EIv^{IV} = \frac{400}{2}z = 200z \qquad EIv^{III} = 200\frac{z^2}{2} + c_1 \qquad EIv^{II} = 200\frac{z^3}{6} + c_1 z + c_2$$

$$EIv^{I} = 200\frac{z^4}{24} + c_1\frac{z^2}{2} + c_2 z + c_3 \qquad EIv = 200\frac{z^5}{120} + c_1\frac{z^3}{6} + c_2\frac{z^2}{2} + c_3 z + c_4$$

$$x = 0,\ v = 0 \rightarrow\ c_4 = 0 \qquad EIv = 200\frac{0}{120} + c_1\frac{0}{6} + c_2\frac{0}{2} + c_3 0 + c_4$$

$$x = 0,\ v^I = 0 \rightarrow\ c_3 = 0 \qquad EIv^I = 200\frac{0}{24} + c_1\frac{0}{2} + c_2 0 + c_3$$

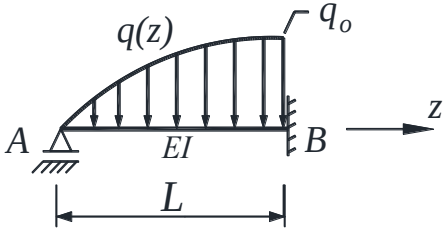
$$\left.\begin{array}{l} x = 2m,\ v = 0 \rightarrow EIv = 200\dfrac{2^5}{120} + c_1\dfrac{2^3}{6} + c_2\dfrac{2^2}{2} = 1,33c_1 + 2c_2 = -53,33 \quad (1) \\ x = 2m,\ v' = 0 \rightarrow EIv' = 200\dfrac{2^4}{24} + c_1\dfrac{2^2}{2} + c_2 2 = 2c_1 + 2c_2 = -133,33 \quad (2) \end{array}\right\} \quad \begin{array}{l} c_1 = -119,4 \\ c_2 = 52,735 \end{array}$$

$$v = \frac{1}{EI}\left(\frac{200}{120}z^5 - \frac{119,4}{6}z^3 + \frac{52,735}{2}z^2\right)$$

$$v \text{ is } v_{max} \text{ when } v' = 0 \rightarrow\ 8,35z^4 - 59,7z^2 + 52,735z = 0 \quad \begin{cases} z = 0 \\ z = -3,03 \\ z = 2 \\ z = 1,041 \end{cases}$$

$z = 1,041$ is selected. $\qquad v_{max} = v\big|_{z=1,041} = \dfrac{7,04}{EI}$

Problem: $q(z)=q_0\left(\frac{2z}{L}-\frac{z^2}{L^2}\right)$ parabolic load is acting on the indeterminate beam with a span length of *L*. Find the elastic curve of the beam using the integration method, rotation of point A, and support reactions.

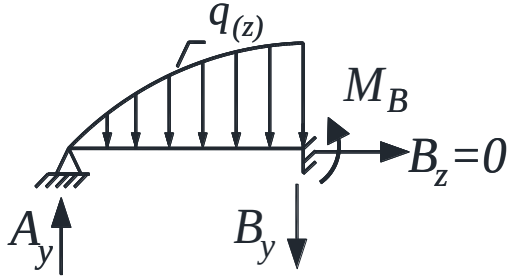


Çözüm:

$$EIv^{(iv)}=q(z)=\frac{2q_0z}{L}-\frac{q_0z^2}{L^2},\qquad EIv'''=\frac{q_0z^2}{L}-\frac{q_0z^3}{3L^2}+C_1,$$

$$EIv''=\frac{q_0z^3}{3L}-\frac{q_0z^4}{12L^2}+C_1z+C_2\qquad EIv'=\frac{q_0z^4}{12L}-\frac{q_0z^5}{60L^2}+C_1\frac{z^2}{2}+C_2z+C_3,$$

$$EIv=\frac{q_0z^5}{60L}-\frac{q_0z^6}{360L^2}+C_1\frac{z^3}{6}+C_2\frac{z^2}{2}+C_3z+C_4$$



Boundary Conditions

$$at\ z=0,\ z=0,\ v=0\Rightarrow C_4=0,\quad and\quad v''=0\Rightarrow C_2=0,$$

$$at\ z=L,\ v=0\Rightarrow\frac{q_0L^5}{60L}-\frac{q_0L^6}{360L^2}+C_1\frac{L^3}{6}+C_3L=0$$

$$at\ z=L,\ v'=0\Rightarrow\ \frac{q_0L^4}{12L}-\frac{q_0L^5}{60L^2}+C_1\frac{L^2}{2}+C_3=0$$

$$C_1=-\frac{19}{120}q_0L,\ C_3=\frac{1}{80}q_0L^3$$

$$\Omega_A=v'(z=0)=\frac{q_0L^3}{80EI}$$

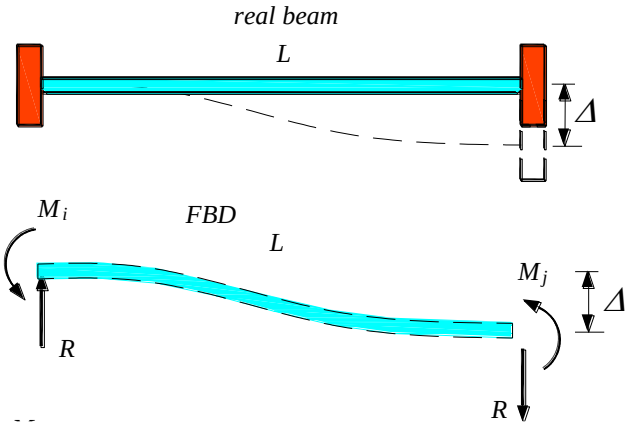
$$A_y=-EIv'''(z=0)=\frac{19}{120}q_0L$$

$$B_y=-EIv'''(z=L)=-\frac{61}{120}q_0L$$

$$M_B=-EIv''(z=L)=-\frac{11}{120}q_0L^2$$

Problem 1.1

Determine the elastic curve and support reactions of the beam due to the unit support displacement of the support B by using fourth order differential equation.



$$EIv^{iv} = q \rightarrow EIv''' = c_1 + qz \rightarrow EIv'' = c_2 + c_1 z + q\frac{z^2}{2}$$

$$EIv' = c_3 + c_2 z + c_1\frac{z^2}{2} + q\frac{z^3}{6} \rightarrow EIv = c_4 + c_3 z + c_2\frac{z^2}{2} + c_1\frac{z^3}{6} + q\frac{z^4}{24}$$

Boundary conditions:

$$z = 0 \rightarrow v = 0 \rightarrow EIv = c_4 = 0 \qquad (1)$$

$$z = 0 \rightarrow v' = 0 \rightarrow EIv' = c_3 = 0 \qquad (2)$$

$$z = l \rightarrow v = 1 \rightarrow EIy(l) = EI = c_1\frac{l^3}{6} + c_2\frac{l^2}{2} = EI \qquad (3)$$

$$z = l \rightarrow v' = 0 \rightarrow EIv' = c_1\frac{l^2}{2} + c_2 l = 0 \Rightarrow c_2 = -\frac{c_1 l}{2} \qquad (4)$$

Using the equation (3),

$$z = l \rightarrow v = 1 \rightarrow EIy(l) = EI = c_1\frac{l^3}{6} - \frac{c_1 l}{2} \times \frac{l^2}{2} = c_1\frac{2l^3}{12} - \frac{3c_1 l^3}{12} = -\frac{c_1 l^3}{12} = EI \rightarrow c_1 = -\frac{12EI}{l^3}$$

$$c_2 = -\frac{c_1 l}{2} = \frac{12EI}{l^3} \times \frac{l}{2} = \frac{6EI}{l^2}$$

The elastic curve is as follows:

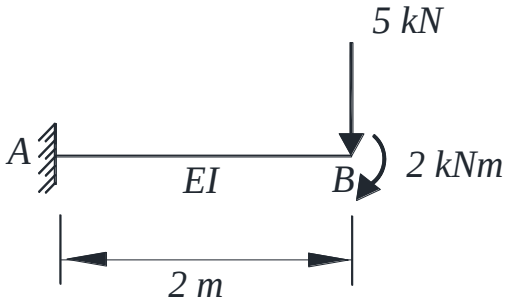
$$EIv = c_1\frac{z^3}{6} + c_2\frac{z^2}{2} = -\frac{12EIz^3}{6l^3} + \frac{6EIz^2}{2l^2} = -\frac{2EIz^3}{l^3} + \frac{3EIz^2}{l^2}$$

$$M = -EIv'' = \frac{12EIz}{l^3} - \frac{6EI}{l^2} \qquad at\ z = 0 \rightarrow M = -EIv'' = -\frac{6EI}{l^2}$$

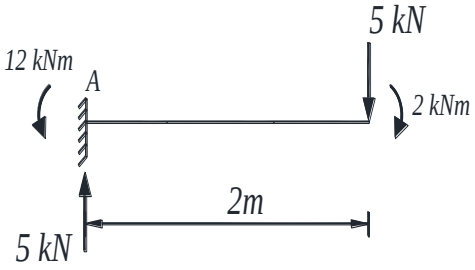
$$at\ z = l \rightarrow M = -EIv'' = \frac{12EIl}{l^3} - \frac{6EI}{l^2} = \frac{6EI}{l^2}$$

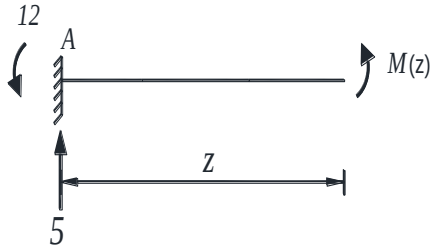
$$\sum M = 0 = \frac{-2 \times 6EI}{l^2} + R_A \times l = 0 \rightarrow R_A = \frac{12EI}{l^3} = -R_B$$

Problem: Find the elastic curve equation of the cantilever beam shown below by using the integration method. Calculate the deflection at point B.



$$M(z)+12-5z=0 \Rightarrow M(z)=5z-12$$





$$-EIv'' = 5z-12 \Rightarrow -EIv' = \frac{5z^2}{2}-12z+C_1$$

$$-EIv = \frac{5z^3}{6}-6z^2+C_1z+C_2$$

Boundary Conditions:

$$z=0,\ v(0)=0 \Rightarrow C_2=0$$

$$z=0,\ v'(0)=0 \Rightarrow C_1=0$$

$$v(z)=\frac{1}{EI}\left[6z^2-\frac{5}{6}z^3\right], \quad v(B)=v(2)=\frac{1}{EI}\left[62^2-\frac{5}{6}2^3\right]=\frac{52}{3EI}$$

In open thin sections, shear center can be found by using a different viewpoint (point of view). This is the relation between the displacements of each part in case of bending without torsion. For example, lets consider an *I* section with different flanges:

$$\frac{1}{\rho_x}=\frac{M_x}{E\,I_x}=\pm\frac{\mathbf{d}^2v/\mathbf{d}z^2}{\left[1+\left(\mathbf{d}v/\mathbf{d}z\right)^2\right]^{3/2}}$$ When the slope $\mathrm{d}v/\mathrm{d}z$ is small then it can be assumed that $\frac{\mathrm{d}^2v}{\mathrm{d}z^2}=\frac{M_x}{E\,I_x}$

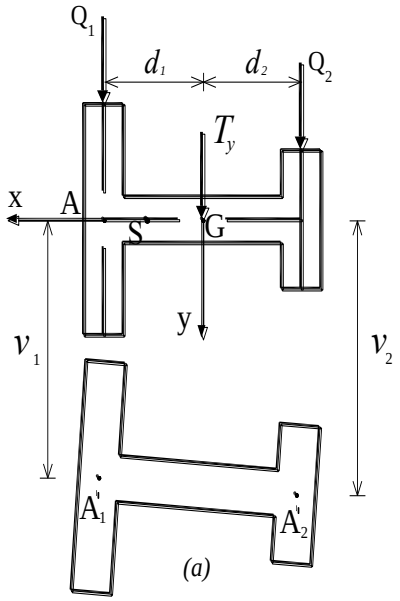
This section is symmetrical with respect to the x axis. Because the web of the section is thin and perpendicular to the direction of the force, the force web can carry can be ignored. Therefore it can be assumed that the shear force is carried by the two flanges.

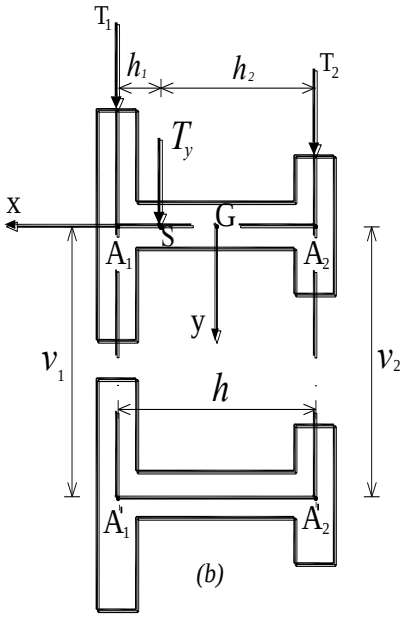
If the displacements of the flanges are equal to each other, $v_1=v_2$, then the curvature of the two flanges must be equal to each other.

Namely $\frac{M_1}{E\,I_{x1}}=\frac{M_2}{E\,I_{x2}}$ By considering $dM_x/dz=T_y$

then $\frac{T_1}{I_{x1}}=\frac{T_2}{I_{x2}}\rightarrow\frac{T_1}{T_2}=\frac{I_{x1}}{I_{x2}}$

Here I_{x1} and I_{x2} are the second moment of areas with respect to the axis x





By considering $h=h_1+h_2$

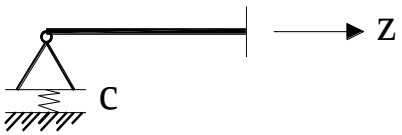
$$T_1h_1=T_2h_2\rightarrow\frac{h_1}{h_2}=\frac{T_2}{T_1}=\frac{I_{x2}}{I_{x1}}\rightarrow h_2=h_1\frac{I_{x1}}{I_{x2}}\qquad h=h_1+h_1\frac{I_{x1}}{I_{x2}}=h_1+\frac{h_1I_{x2}+h_1I_{x1}}{I_{x2}}\rightarrow$$

$$h_1=h\frac{I_{x2}}{I_{x1}+I_{x2}}\qquad and\qquad h_2=h\frac{I_{x1}}{I_{x1}+I_{x2}}$$

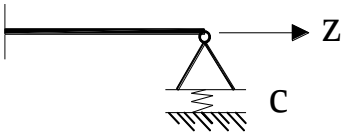
Other Boundary Conditions

1.2.1.4 Pin support with elastic displacement

There is a support reaction proportional to the displacement which is opposite in direction to the displacement of the support.



Left end



Right end

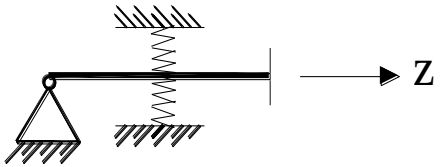
At Left End for $z = 0,\ -E I_x\ v''' = cv$ and $v'' = 0$

At Right End for $z = l,\ \ E I_x\ v''' = cv$ and $v'' = 0$

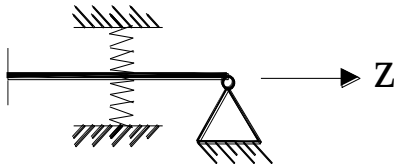
c is a proportion factor and it is always positive and is called as spring constant (c= N/cm).

1.2.1.5 Pin support with elastic rotation

There is a couple occured at the support with respect to the rotation angle.



Left end



Right end

At left end for $z = 0, \quad v = 0 \quad \text{and } E I_x \, v'' = m v'$

At right end for $z = l, \quad v = 0 \quad \text{and } -E I_x \, v'' = m v'$

m is a proportion constant, its value is always positive and its unit is Ncm. For left end, if v' is positive, then the bending moment is negative. If v' is positive for right end, then bending moment is positive.

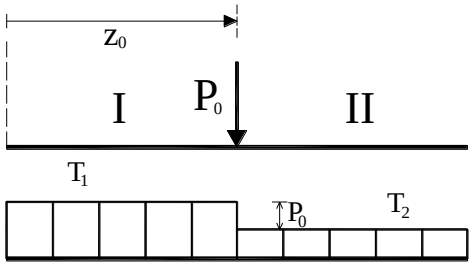
1.2.1.6 Support with elastic displacement and rotation

At left end for $z = 0, \; -E I_x \, v''' = cv \text{ and } E I_x \, v'' = m v'$

At right end for $z = l, \quad E I_x \, v''' = cv \;\; and \;\; -E I_x \, v'' = m v'$

1.3 Discontunity on Elastic Curve

If there is a single force, or a force couple or a hinge on the bar, the related derivative of the elastic curve equation has discontunities.



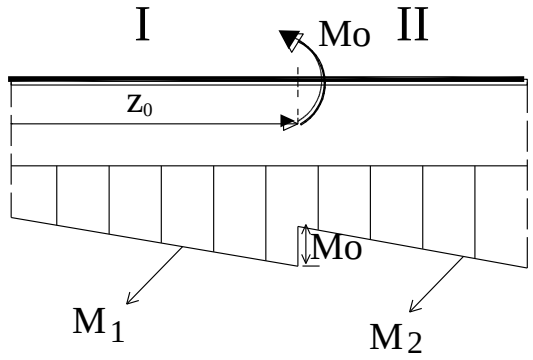
When $z = z_o$,

$$T_2 - T_1 = -P_o$$

$$(-EI_x\, v_2''') - (-EI_x\, v_1''') = -P_o$$

$$-EI_x\, v_1''' - (-EI_x\, v_2''') = P_o$$

Hence, $$v_2''' - v_1''' = \frac{P_o}{EI_x}$$



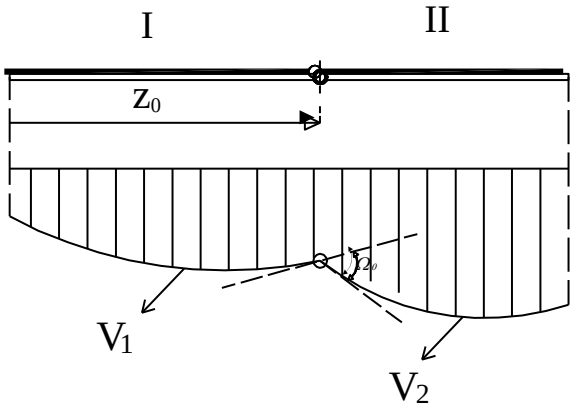
When $z = z_o$,

$$M_2 - M_1 = -M_o$$

$$(-EI_x\, v_2'') - (-EI_x\, v_1'') = -M_o$$

$$-EI_x\, v_1'' - (-EI_x\, v_2'') = M_o$$

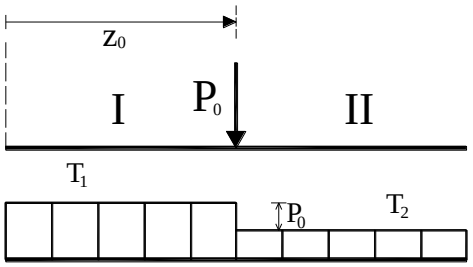
Hence, $$v_2'' - v_1'' = \frac{M_o}{EI_x}$$



When $z = z_o$, $v_2' - v_1' = \Omega_0$

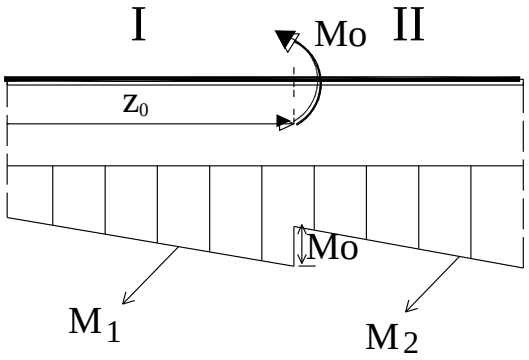
$$M_x(z_o) = 0$$

When discontinuties exist, the following relations are valid at these points for various discontinuties.



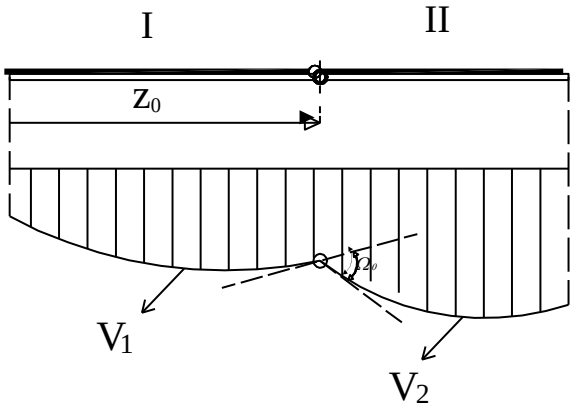
For $Z = Z_o$,

$$v_1 = v_2 \; ; \quad v_1' = v_2' \; ; \quad v_1'' = v_2'' \; ; \quad v_2''' - v_1''' = P_o / \left(EI_x \right)$$



For $Z = Z_o$,

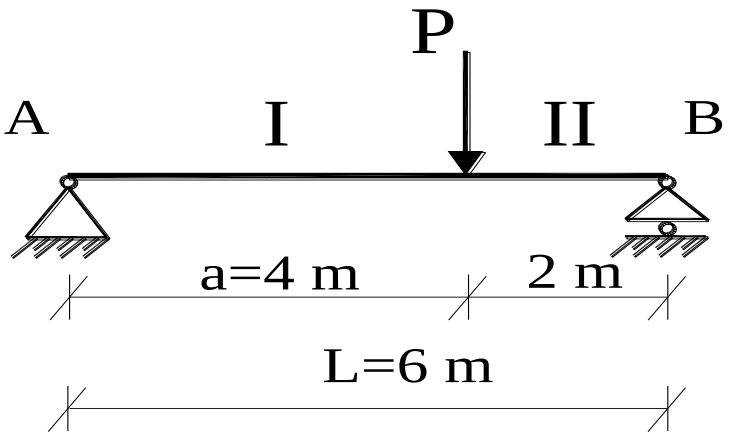
$$v_1 = v_2 \; ; \quad v_1' = v_2' \; ; \quad v_2'' - v_1'' = M_o / \left(EI_x \right) \; ; \quad v_1''' = v_2'''$$



For $Z = Z_o$,

$$v_1 = v_2 \; ; \quad v_2' - v_1' = \Omega_o \; ; \quad v_1'' = v_2'' = 0 \; ; \quad v_1''' = v_2'''$$

Problem 1.2: Find the elastic curve of the beam shown.



Solution:

$$EIv_1^{iv} = 0 \rightarrow EIv_1^{iii} = c_1 \rightarrow EIv_1^{ii} = c_1 x + c_2$$

$$EIv_1^{i} = c_1 \frac{x^2}{2} + c_2 x + c_3 \rightarrow EIv_1 = c_1 \frac{x^3}{6} + c_2 \frac{x^2}{2} + c_3 x + c_4$$

Similarly, $$EIv_{II} = D_1 \frac{x^3}{6} + D_2 \frac{x^2}{2} + D_3 x + D_4$$

Boundary conditions: $$x = 0 \rightarrow v = 0 \rightarrow c_4 = 0$$

$$x = 0 \rightarrow v^{ii} = 0 \rightarrow c_2 = 0$$

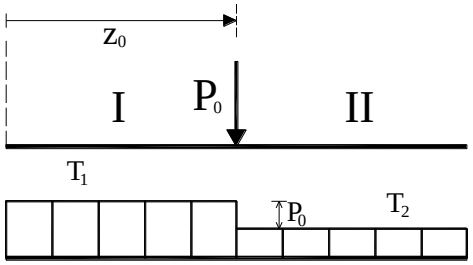
Continuity conditions:

$$x = a \rightarrow v_I = v_{II} \rightarrow c_1 \frac{a^3}{6} + c_3 a = D_1 \frac{a^3}{6} + D_2 \frac{a^2}{2} + D_3 a + D_4 \rightarrow (1)$$

$$v_I^{i} = v_{II}^{i} \rightarrow c_1 \frac{a^2}{2} + c_3 = D_1 \frac{a^2}{2} + D_2 a + D_3 \rightarrow (2)$$

$$v_I^{ii} = v_{II}^{ii} \rightarrow c_1 a = D_1 a + D_2 \rightarrow (3)$$

$$v_{II}^{iii} - v_I^{iii} = \frac{P}{EI_x} \rightarrow D_1 - c_1 = \frac{P}{EI_x} \rightarrow (4)$$



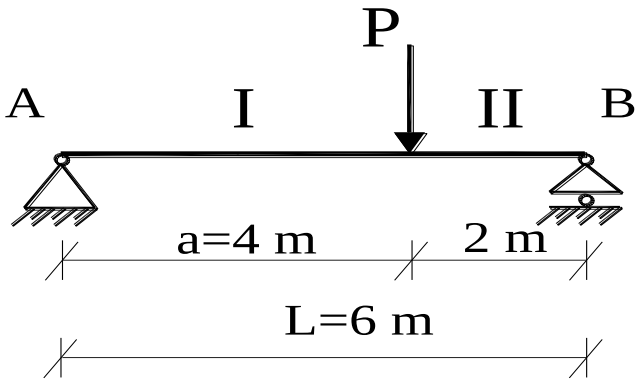
Boundary conditions:

$$x = L \rightarrow v_{II} = 0 \rightarrow D_1 \frac{L^3}{6} + D_2 \frac{L^2}{2} + D_3 L + D_4 = 0 \rightarrow (5)$$

$$v_{II}^{ii} = 0 \rightarrow D_1 L + D_2 = 0 \rightarrow (6)$$

$$v_1 = v_2 \; ; \quad v_1' = v_2' \; ;$$

$$v_1'' = v_2'' \; ; \quad v_2''' - v_1''' = P_o / (EI_x)$$



$$x = a \rightarrow v_I = v_{II} \rightarrow c_1 \frac{a^3}{6} + c_3 a = D_1 \frac{a^3}{6} + D_2 \frac{a^2}{2} + D_3 a + D_4 \rightarrow (1)$$

$$v_I' = v_{II}' \rightarrow c_1 \frac{a^2}{2} + c_3 = D_1 \frac{a^2}{2} + D_2 a + D_3 \rightarrow (2)$$

$$v_I'' = v_{II}'' \rightarrow c_1 a = D_1 a + D_2 \rightarrow (3)$$

$$v_{II}''' - v_I''' = \frac{P}{EI_x} \rightarrow D_1 - c_1 = \frac{P}{EI_x} \rightarrow (4)$$

$$x = L \rightarrow v_{II} = 0 \rightarrow D_1 \frac{L^3}{6} + D_2 \frac{L^2}{2} + D_3 L + D_4 = 0 \rightarrow (5)$$

$$v_{II}'' = 0 \rightarrow D_1 L + D_2 = 0 \rightarrow (6)$$

$$\begin{bmatrix} 4^3/6 & 4 & -4^3/6 & -4^2/2 & -4 & -1 \\ 4^2/2 & 1 & -4^2/2 & -4 & -1 & 0 \\ 4 & 0 & -4 & -1 & 0 & 0 \\ -1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 6^3/6 & 6^2/2 & 6 & 1 \\ 0 & 0 & 6 & 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} c_1 \\ c_3 \\ D_1 \\ D_2 \\ D_3 \\ D_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ P/EI \\ 0 \\ 0 \end{bmatrix} \rightarrow (7)$$

By solving the system of equations we found the constants:

$$c_1 = -0.333 \frac{P}{EI}; c_3 = 1.7778 \frac{P}{EI}$$

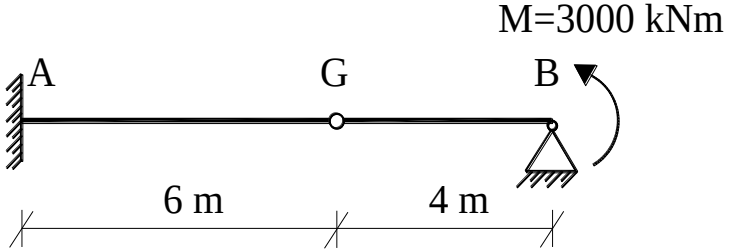
$$D_1 = 0.667 \frac{P}{EI}; D_2 = -4 \frac{P}{EI}; D_3 = 9.778 \frac{P}{EI}; D_4 = -10.667 \frac{P}{EI}$$

$$v_I = -0.333 \frac{P}{EI} \frac{x^3}{6} + 1.7778 \frac{P}{EI} x$$

$$x = 4 \rightarrow v_I = 3.5556 \frac{P}{EI}$$

Problem 1.3

Calculate the vertical displacement at point G on the Gerber beam by using the integration method.



Solution:

$$E\,I\,v_1^{iv} = q = 0 \qquad E\,I\,v_1''' = C_1 \qquad E\,I\,v_1'' = C_1\,z + C_2 \qquad E\,I\,v_1' = C_1\frac{z^2}{2} + C_2\,z + C_3 \qquad E\,I\,v_1 = C_1\frac{z^3}{6} + C_2\frac{z^2}{2} + C_3\,z + C_4$$

Similarly, $E\,I\,v_2 = D_1\frac{z^3}{6} + D_2\frac{z^2}{2} + D_3\,z + D_4$

boundary condition at z=0, $v_1 = 0 \rightarrow C_4 = 0 \qquad v_1' = 0 \rightarrow C_3 = 0$

boundary condition at z=L, $E\,I\,v_2 = 0 \rightarrow D_1\frac{L^3}{6} + D_2\frac{L^2}{2} + D_3.L + D_4 = 0 \qquad -E\,I\,v_{II}'' = 3000 = M = -D_1\,l - D_2$

$$-EI_x\,v_I'' = 0 = 6C_1 + C_2 \;=\; 0$$

Hinge conditions:

$$-EI_x\,v_{II}'' = 0 = 6D_1 + D_2 \;=\; 0$$

$$v_I = v_{II} \rightarrow C_1\frac{6^3}{6} + C_2\frac{6^2}{2} \;=\; D_1\frac{6^3}{6} + D_2\frac{6^2}{2} + D_3\,6 + D_4$$

$$v_I''' = v_{II}''' \rightarrow E\,I\,v_I''' = E\,I\,v_{II}''' \rightarrow C_1 = D_1$$

$$0 + 0 + 166{,}67\,D_1 \;+\; 50\,D_2 \;+\; 10\,D_3 \;+ D_4 \;=\; 0$$

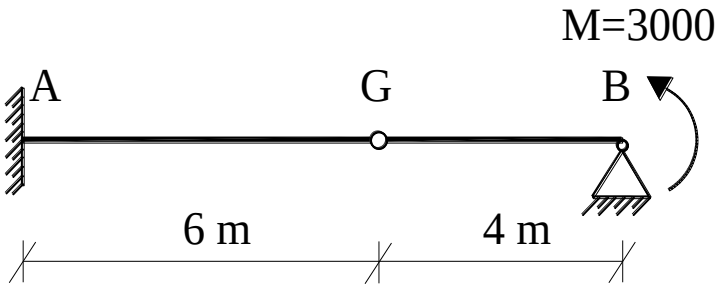
$$0 + 0 + 10\,D_1 + \;D_2 + 0 + 0 \;=\; -3000$$

$$0 + 0 + 6D_1 + D_2 + 0 + 0 = 0$$

$$6C_1 + C_2 + 0 + 0 + 0 + 0 = 0$$

$$C_1 + 0 - D_1 + 0 \;+\; 0 \;+\; 0 \;=\; 0$$

By solving the set of equations;



$$C_1 = -750, \quad C_2 = 4500, \quad D_1 = -750, \quad D_2 = 4500, \quad D_3 = -25000, \quad D_4 = 150000$$

$$v_1(z) = \frac{1}{EI}\left(-750\ \frac{z^3}{6} + 4500\ \frac{z^2}{2}\right)$$

$$v_2(z) = \frac{1}{EI}\left(-750\ \frac{z^3}{6} + 4500\ \frac{z^2}{2} - 25000\ z + 150000\right)$$

$$\mathrm{v_I}(6) = \mathrm{v_{II}}(6) = \frac{54000}{E\,I} \quad \mathrm{m}$$

1.4.2. Mohr method/moment-area method

Moment graph of the beam must be known for using this method. This method calculates the elastic curve by using a mechanical analogy. This analogy has two group of variations.

First group: From the equilibrium conditions on a beam, the following equations can be written between distributed force=q, shear force= T and bending moment=M:

$$\frac{dT}{dz} = -q \qquad \frac{dM}{dz} = T \qquad \frac{dT}{dz} = \frac{d^2M}{dz^2} = -q$$

Second group: A beam has also the following relations from definition and geometric basis between

$$curvature = \frac{1}{\rho x} \qquad slope = \theta \quad and \ displacement \quad v$$

$$\frac{d\theta}{dz} = \frac{d^2v}{dz^2} = -\frac{1}{\rho_x} = -\frac{M_x}{EI_x}$$

$$\frac{dT}{dz} = -q \rightarrow \frac{d\theta}{dz} = -\frac{M_x}{EI_x} \qquad \frac{dM}{dz} = T \rightarrow \frac{dv}{dz} = \theta \qquad \frac{d^2M}{dz^2} = -q \rightarrow \frac{d^2v}{dz^2} = -\frac{M_x}{EI_x}$$

It can be seen from the comparison of the sets of equations that there are mathematical analogy pairs between the six quantities:

Force	$\Leftrightarrow$	Curvature	$q \Leftrightarrow -\frac{1}{\rho_x} = -\frac{M_x}{EI_x}$
Shear force	$\Leftrightarrow$	Slope	$T \Leftrightarrow \theta$
Bending moment	$\Leftrightarrow$	Displacement	$M_x \Leftrightarrow v$

By assuming the curvature of the real beam as a force affecting on the conjugate beam, the shear force diagram of the conjugate beam gives the rotation of the real beam, the moment diagram of the conjugate beam gives the displacement diagram of the real beam.

Table 1.1 a) Boundary conditions for real and conjugate systems

Force	⇔	Curvature
Shear force	⇔	Slope
Bending moment	⇔	Displacement

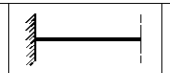
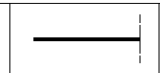
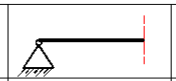
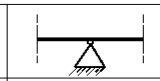
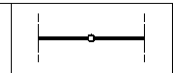
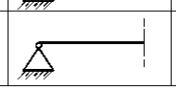
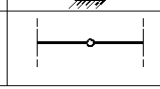
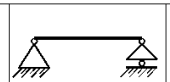
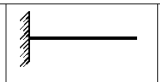
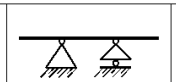
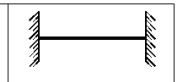
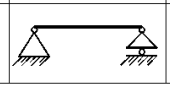
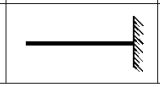
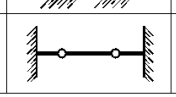
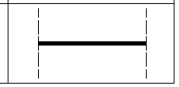
REAL SYSTEM					
CONJUGATE SYSTEM					

Table 1.1 b) Boundary conditions for real and conjugate systems

REAL SYSTEM				
CONJUGATE SYSTEM				
	a	b	c	d

Based on this basic idea, Mohr's method help us find the slopes and displacements of the beam by means of shear force and bending moment diagrams of another beam loaded by fictitious curvature load. This fictitious beam is called conjugate beam system of the real beam system.

The conjugate beam end conditions may not be the same as the end conditions of the actual system. Appropriate end conditions must be determined by considering the analogies between the above quantities.

Deflection and slope are zero at the fixed support on real beam; conjugate beam moment and shear forces must be zero in this section. Hence, fixed support on real system is replaced with free end on conjugate system.

By utilizing the reciprocity in analogy, free end of the actual beam system is replaced with fixed end of the conjugated beam system.

If the support is an immovable hinge in the vertical direction on the real system, then although $v = 0$, the rotation angle θ is not zero at this support. So, on the conjugate system, this support have to be a hinge without deflection, because such a support has zero moment (means zero displacement in the real system) and the shear force is not zero (means rotation angle is different than zero).

An intermediate support on real system corresponds to a hinged joint on the conjugate system.

A hinged joint on real system corresponds to an intermediate support without deflection on the conjugate system.

After this preliminary information, the solution procedure of the method to be followed is given below:

1) First the moment diagram Mx is drawn for the real system. Then the values of moment diagram are divided by $E I_x$ to obtain the curvature diagram.

2) Conjugate of the real system is obtained. The curvature diagram of the real systems is considered as the load diagram for the conjugate system.

3) The shear force diagram of the conjugate system gives the slope diagram of the real system.

4) Bending moment diagram of the conjugate system gives the displacements of the actual system.

This method is particularly useful for obtaining rotations and displacements of the beams loaded with concentrated forces and moments.

1.4.2. Mohr method/moment-area method

Mechanical analogy:

$$\frac{dT}{dz} = -q \qquad \frac{dM}{dz} = T \qquad \frac{dT}{dz} = \frac{d^2M}{dz^2} = -q$$

$$curvature = \frac{1}{\rho x} \qquad slope = \theta \qquad and\ \ displacement \quad v$$

$$\frac{d\theta}{dz} = \frac{d^2v}{dz^2} = -\frac{1}{\rho_x} = -\frac{M_x}{EI_x}$$

$$\frac{dT}{dz} = -q \rightarrow \frac{d\theta}{dz} = -\frac{M_x}{EI_x} \qquad \frac{dM}{dz} = T \rightarrow \frac{dv}{dz} = \theta \qquad \frac{d^2M}{dz^2} = -q \rightarrow \frac{d^2v}{dz^2} = -\frac{M_x}{EI_x}$$

By assuming the curvature of the real beam as a force affecting on the conjugate beam, the shear force diagram of the conjugate beam gives the rotation of the real beam, the moment diagram of the conjugate beam gives the displacement diagram of the real beam.

Force $\Leftrightarrow$ Curvature $\quad q \Leftrightarrow -\frac{1}{\rho_x} = -\frac{M_x}{EI_x}$

Shear force $\Leftrightarrow$ Slope $\quad T \Leftrightarrow \theta$

Bending moment $\Leftrightarrow$ Displacement $\quad M_x \Leftrightarrow v$

Table 1.1 a) Boundary conditions for real and conjugate systems

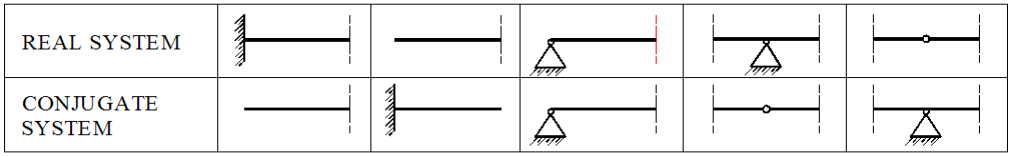
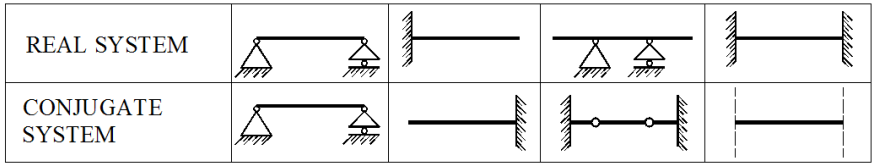
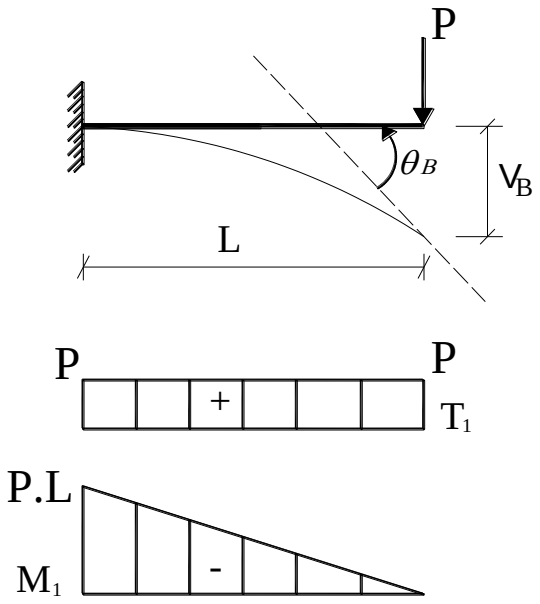


Table 1.1 b) Boundary conditions for real and conjugate systems

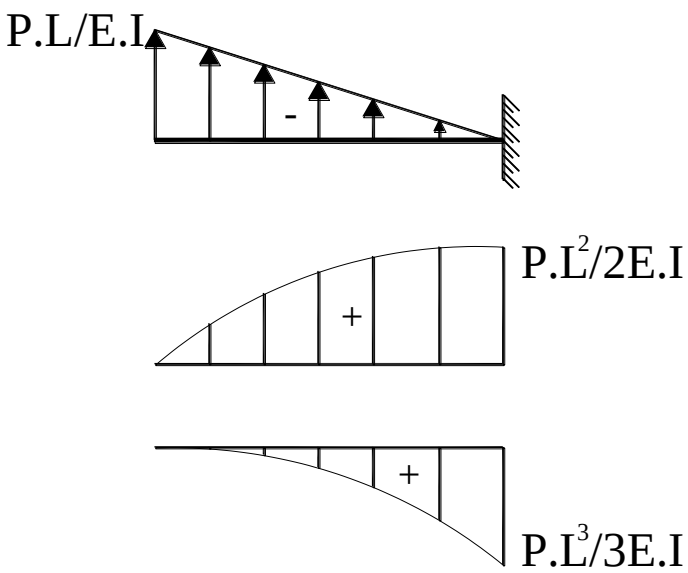


Problem 1.6: In the figure shown below, a cantilever beam is loaded with a concentrated force P. Find the elastic curve of the beam, calculate the angle of rotation and displacement at the free end of the beam.

Obtained quantities for real system:



Obtained quantities for conjugate system:



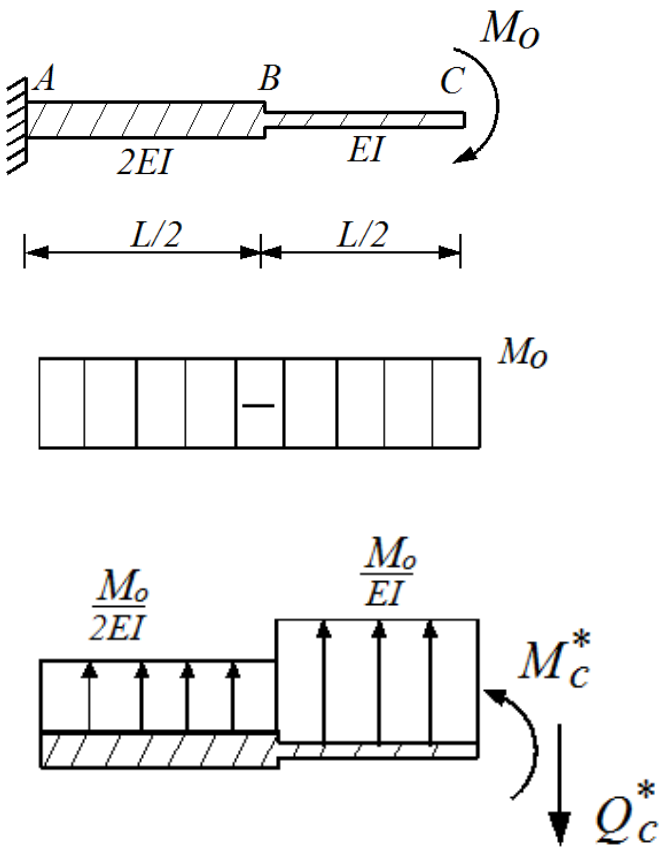
$$q \Leftrightarrow -\frac{1}{\rho_x} = -\frac{M_x}{EI_x}$$

$$T \Leftrightarrow \theta$$

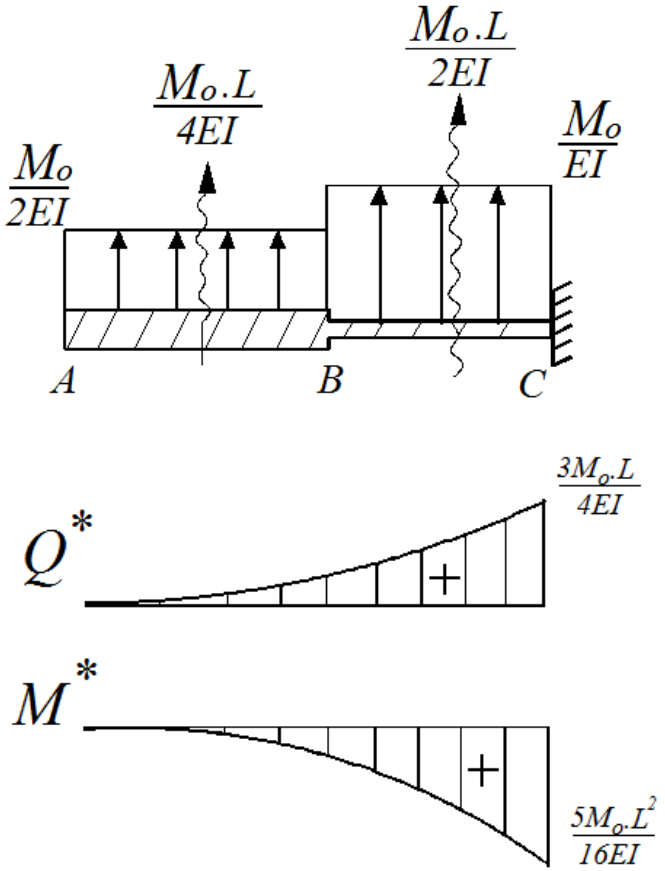
$$M_x \Leftrightarrow v$$

Problem : In the figure shown below, a cantilever beam of variable cross-section is loaded with a concentrated moment Mo. Find the elastic curve of the beam, calculate the angle of rotation and displacement at the free end of the beam.

Obtained quantities for real system:



Obtained quantities for conjugate system:



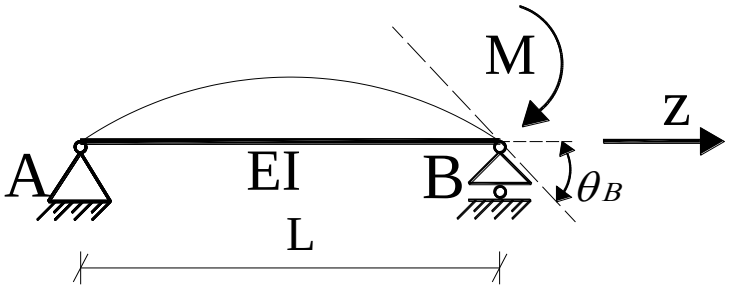
$$+\uparrow \sum F_y = 0 = \frac{M_0}{2EI}.\frac{L}{2} + \frac{M_0}{EI}.\frac{L}{2} - Q_c^* \rightarrow Q_c^* = \frac{3M_0L}{4EI}$$

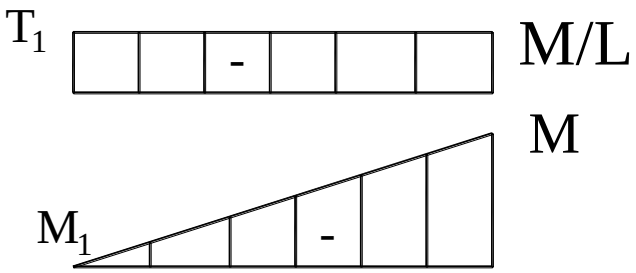
$$\sum M = 0 = \frac{M_0L}{4EI}.\frac{3L}{4} + \frac{M_0L}{2EI}.\frac{L}{4} - M_c^*$$

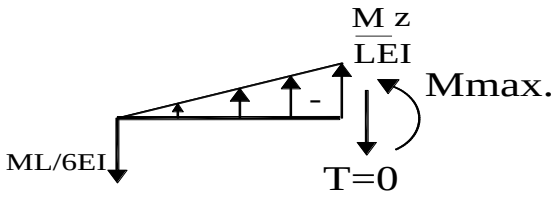
$$M_c^* = \frac{3M_0L^2}{16EI} + \frac{M_0L^2}{8EI} = \frac{5M_0L^2}{16EI}$$

Problem 1.7: The simple beam given below is loaded with a concentrated force at the point B. Find the rotation angle of the end point B of the beam.

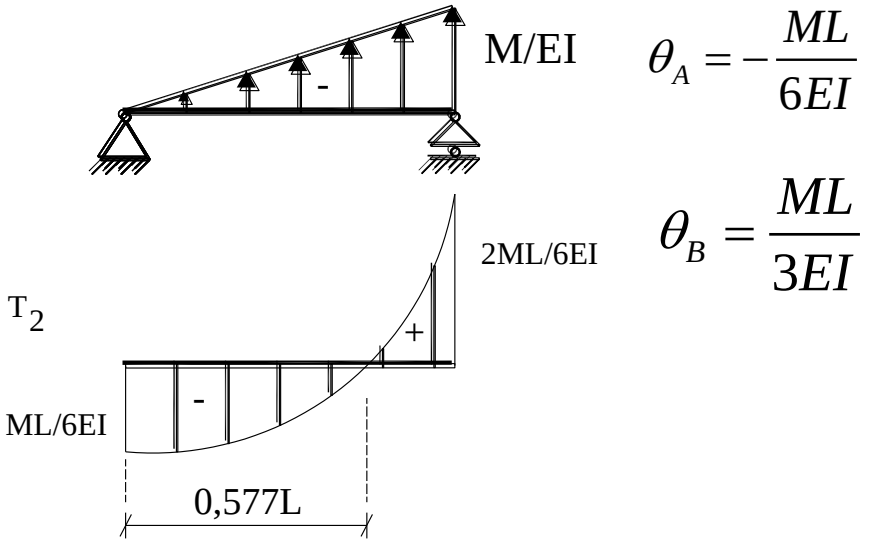
Obtained quantities for real system:







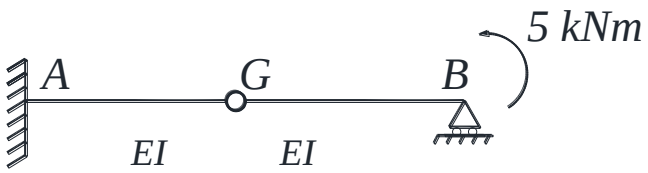
Obtained quantities for conjugate system:



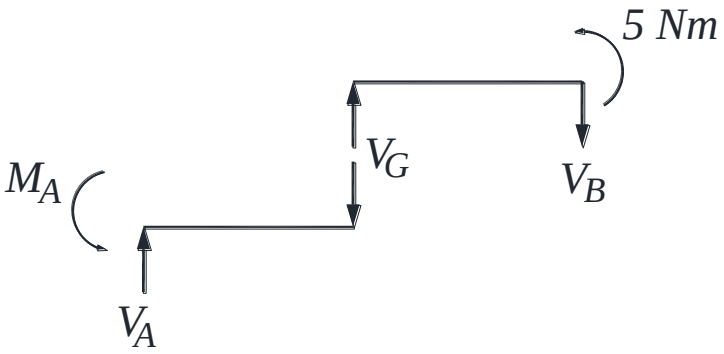
$$\frac{M.Z}{LEI}.\frac{Z}{2}-\frac{ML}{6EI}=0 \qquad Z=\sqrt{\frac{2}{6}}L$$

$$\frac{ML}{6EI}.Z-\frac{MZ}{LEI}.\frac{Z}{2}.\frac{Z}{3}+M_{max}=0 \qquad M_{max}=v_{max}=-\frac{1}{9}.\sqrt{\frac{2}{6}}.\frac{M.L^2}{EI}$$

EXAMPLE:



Calculate the vertical displacement δ_G at point G on the Gerber beam, the rotations at the right and left side of point G, and the relative rotation at point G. AG=GB=3m

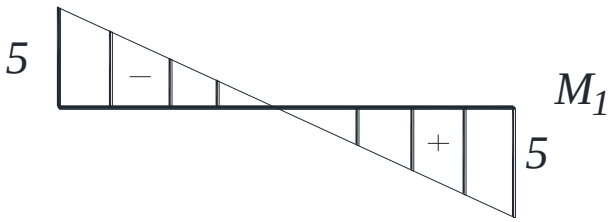


$$V_G + V_B = 0$$

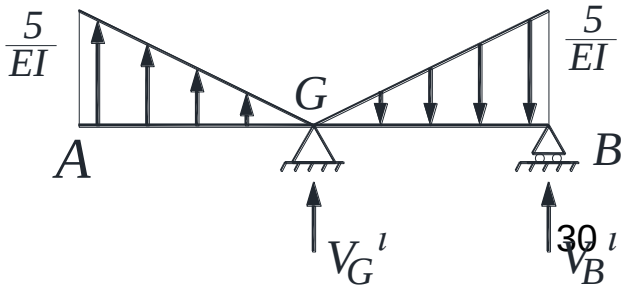
$$\sum M_B = 0 \quad \rightarrow \quad V_G \cdot 3 - 5 = 0 \quad \rightarrow \quad V_G = 5/3 kN$$

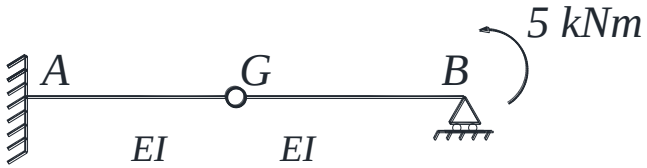
$$V_A = V_G = 1.67 kN$$

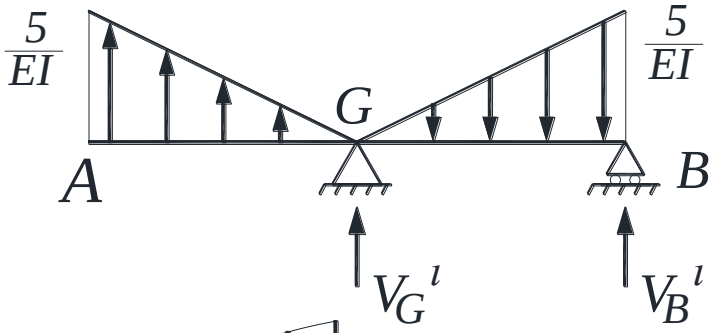
$$\sum M_A = 0 \quad \rightarrow \quad -M_A + 1{,}67 \cdot 3 = 0 \quad \rightarrow \quad M_A = 5 kNm$$



Fictitious Loading







$$\sum M_B = 0$$

$$\frac{-7,5}{EI}\cdot 1 + 3\cdot V_G{}' + \frac{7,5}{EI}\cdot 5 = 0 \quad \rightarrow \quad V_G{}' = \frac{-10}{EI}$$

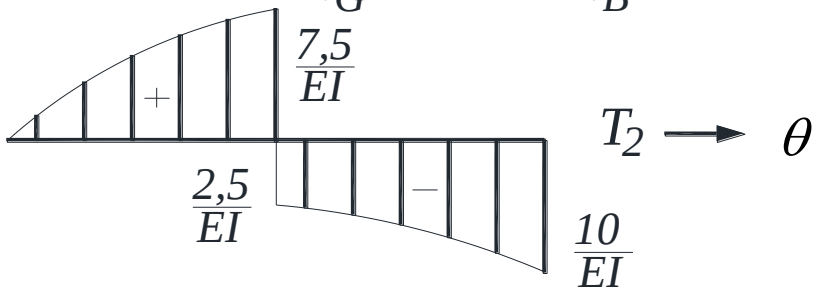
$$V_B{}' = \frac{10}{EI}$$

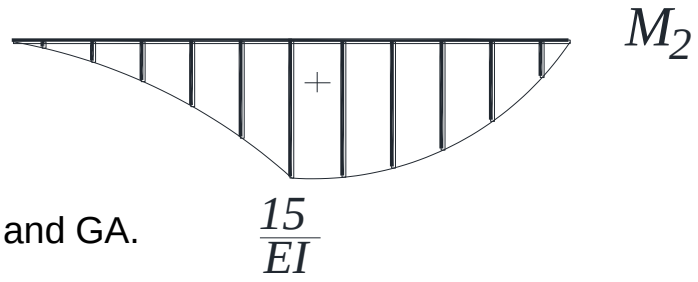
$$\theta_{GA} = 7,5 / EI$$

$$\theta_{GB} = 2,5 / EI$$

$$\theta_G = 10 / EI$$

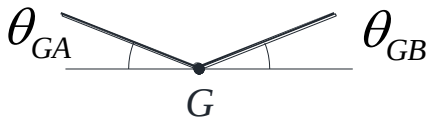


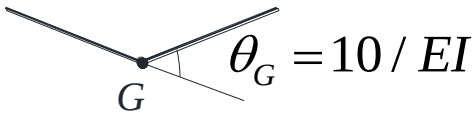




θ_G , is the relative rotation at point G between the parts GB and GA.

$$\delta_G = 15 / EI$$





Problem: Find the deflection at point A of the beam given in Figure 1 by using the MOHR method. $\sum M_B = 0 \rightarrow C_y = 3333,33\,kN\,(\downarrow)$, $\sum F_y = 0 \rightarrow B_y = 8333,33\,kN\,(\uparrow)$ $M_B = -5000 \times 2 = -10000\,kNm$

$$\sum M_C^* = 0 \rightarrow \frac{10000}{EI} \times \frac{3}{2} \times \frac{2 \times 3}{3} - B_y^* \times 3 = 0$$

$$B_y^* = \frac{10000}{EI}\,kN$$

$$\sum M_A^* = 0 \rightarrow M_A^* - \frac{10000}{EI} \times \frac{2}{2} \times \frac{2 \times 2}{3} - \frac{10000}{EI} \times 2 = 0$$

$$M_A^* = v_A = \frac{100000}{3EI} = \frac{33333,3}{EI}$$

Second solution way

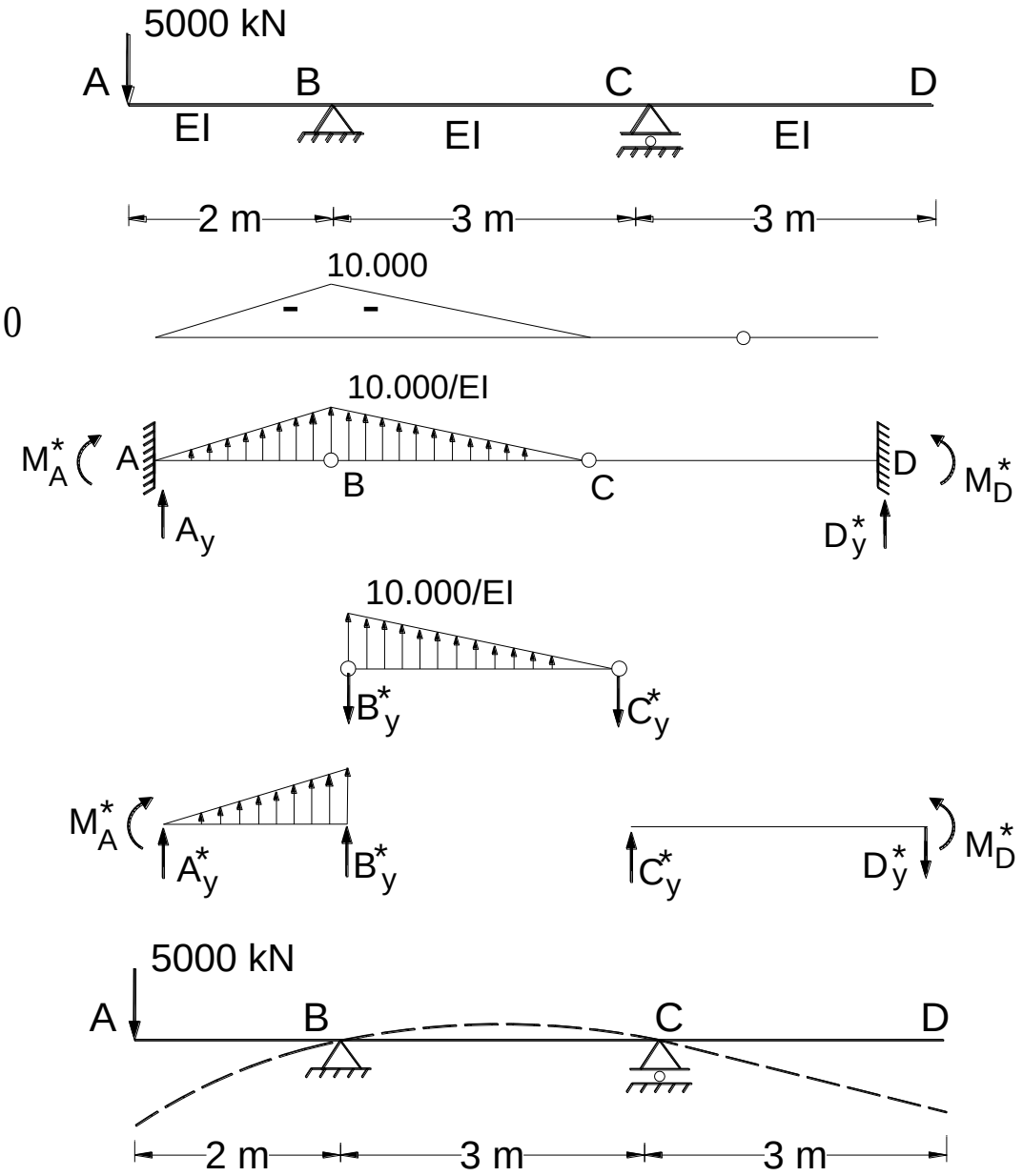
$$\sum M_{C_{right}}^* = 0 \rightarrow D_y^*.3 + M_D^* = 0,$$

$$\sum M_{B_{right}}^* = 0 \rightarrow D_y^*.6 + M_D^* + \frac{1}{2}.3.\frac{10000}{EI} = 0$$

$$D_y^* = -\frac{5000}{EI}$$

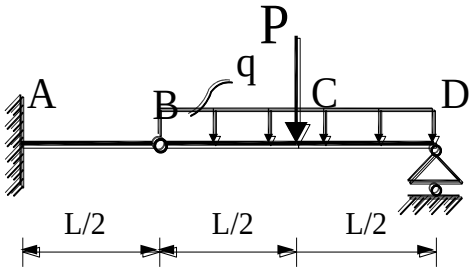
$$\sum F_y^* = 0 \rightarrow A_y^* = -\frac{20000}{EI}, \sum M_{B_{left}}^* = 0$$

$$v_A = M_A^* = \frac{33333,3}{EI}$$



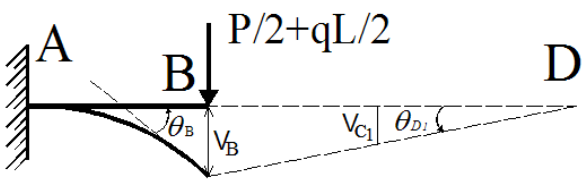
Problem 1.12: For the beam under given loading which is seen in the figure , calculate the following quantities by using the function of elastic curve.

$$\theta_{Bleft},\ V_B, V_C \ and\ \theta_D$$

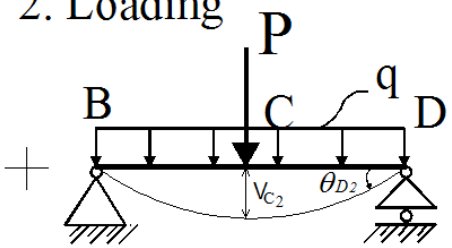


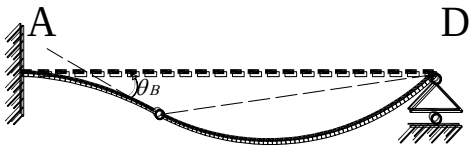
Answer:

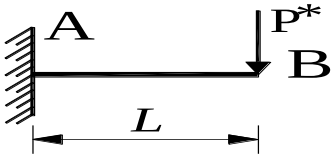
1. Loading



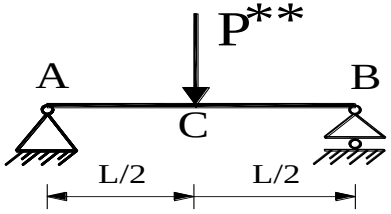
2. Loading



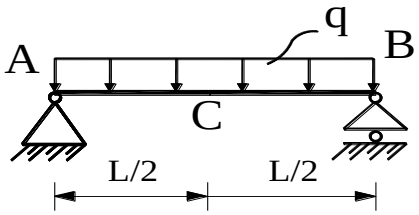




$$\theta_B = \frac{P^* L^2}{2EI}\ ;\quad V_B = \frac{P^* L^3}{3EI}$$

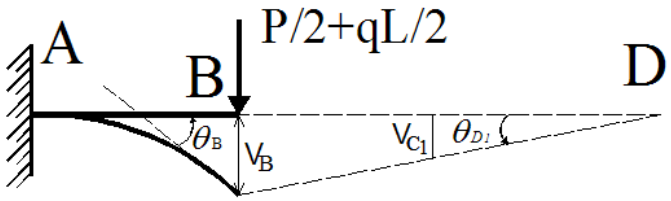


$$V_C = \frac{P^{**} l^3}{48EI} \qquad \theta_A = -\theta_B = \frac{P^{**} l^2}{16EI}$$

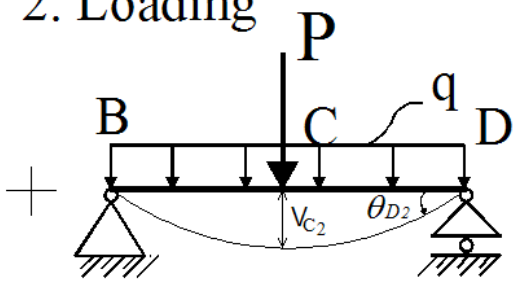


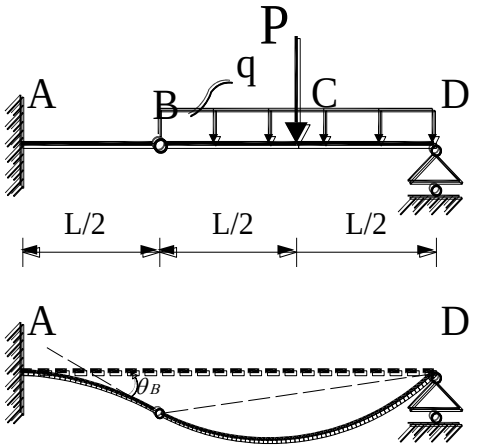
$$V_C = \frac{5ql^4}{384EI} \qquad \theta_A = -\theta_B = \frac{ql^3}{24EI}$$

1. Loading



2. Loading

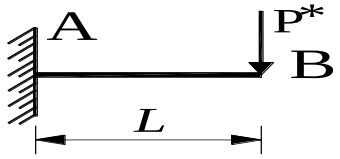




First loading;

$$\theta_B = \frac{\left(\frac{P}{2}+\frac{qL}{2}\right)\cdot\left(\frac{L}{2}\right)^2}{2EI} = \frac{(P+qL)L^2}{16EI} \qquad V_B = \frac{\left(\frac{P}{2}+\frac{qL}{2}\right)\cdot\left(\frac{L}{2}\right)^3}{3EI} = \frac{(P+qL)L^3}{48EI}$$

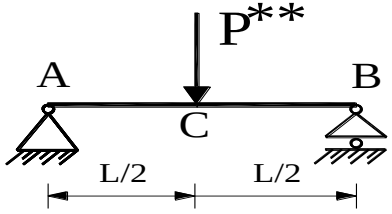
$$V_{c_1} = \frac{V_B}{2} = \frac{(P+ql)l^3}{96EI}; \quad \theta_{D_1} = -\frac{V_B}{l} = -\frac{(P+ql)l^2}{48EI}$$



$$\theta_B = \frac{P^* L^2}{2EI}; \quad V_B = \frac{P^* L^3}{3EI}$$

Second loading;

$$V_{C_2} = \frac{(5ql+8P)l^3}{384EI} \quad ; \theta_{D_2} = -\frac{(3P+2ql)l^2}{48EI}$$

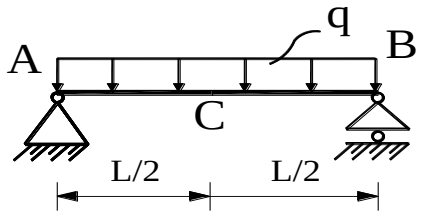


$$V_C = \frac{P^{**} l^3}{48EI} \qquad \theta_A = -\theta_B = \frac{P^{**} l^2}{16EI}$$

Total loading;

$$V_C = V_{C_1} + V_{C_2} = \frac{(P+ql)l^3}{96EI} + \frac{(5ql+8P)l^3}{384EI} \rightarrow V_C = \frac{(9ql+12P)l^3}{384EI}$$

$$\theta_D = -\frac{(P+ql)l^2}{48EI} + \frac{-(3P+2ql)l^2}{48EI} \rightarrow \theta_D = \frac{-(4P+3ql)l^2}{48EI}$$



$$V_C = \frac{5ql^4}{384EI} \qquad \theta_A = -\theta_B = \frac{ql^3}{24EI}$$

4. Method of Superposition

1. selecting the reaction redundants
2. establish the force-displacement relations
3. consistence of deformation (compatibility equation)

consider a propped cantilever beam

(i) select R_B as the redundant, then

$$R_A = qL - R_B \qquad M_A = \frac{qL^2}{2} - R_B L$$

force-displacement relation

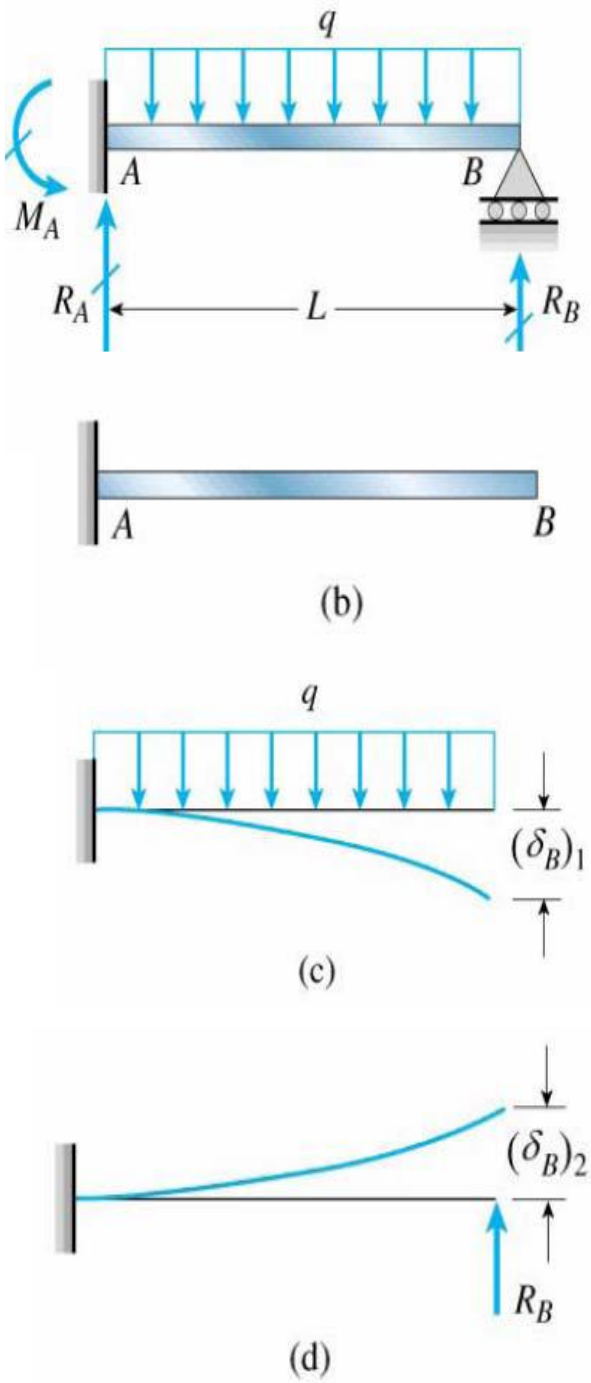
$$(\delta_B)_1 = \frac{qL^4}{8EI} \qquad (\delta_B)_2 = -\frac{R_B L^3}{3EI}$$

compatibility equation

$$\delta_B = (\delta_B)_1 + (\delta_B)_2 = \frac{qL^4}{8EI} - \frac{R_B L^3}{EI} = 0$$

$$\frac{qL^4}{8EI} = \frac{R_B L^3}{3EI}$$

$$R_B = \frac{3qL}{8} \quad \Rightarrow \quad R_A = \frac{5qL}{8} \qquad M_A = \frac{qL^2}{8}$$



(ii) select the moment M_A as the redundant

$$\sum M_B = 0 = -M_A - \frac{qL^2}{2} + R_A L$$

$$R_A = \frac{qL}{2} + \frac{M_A}{L} \qquad R_B = \frac{qL}{2} - \frac{M_A}{L}$$

force-displacement relation

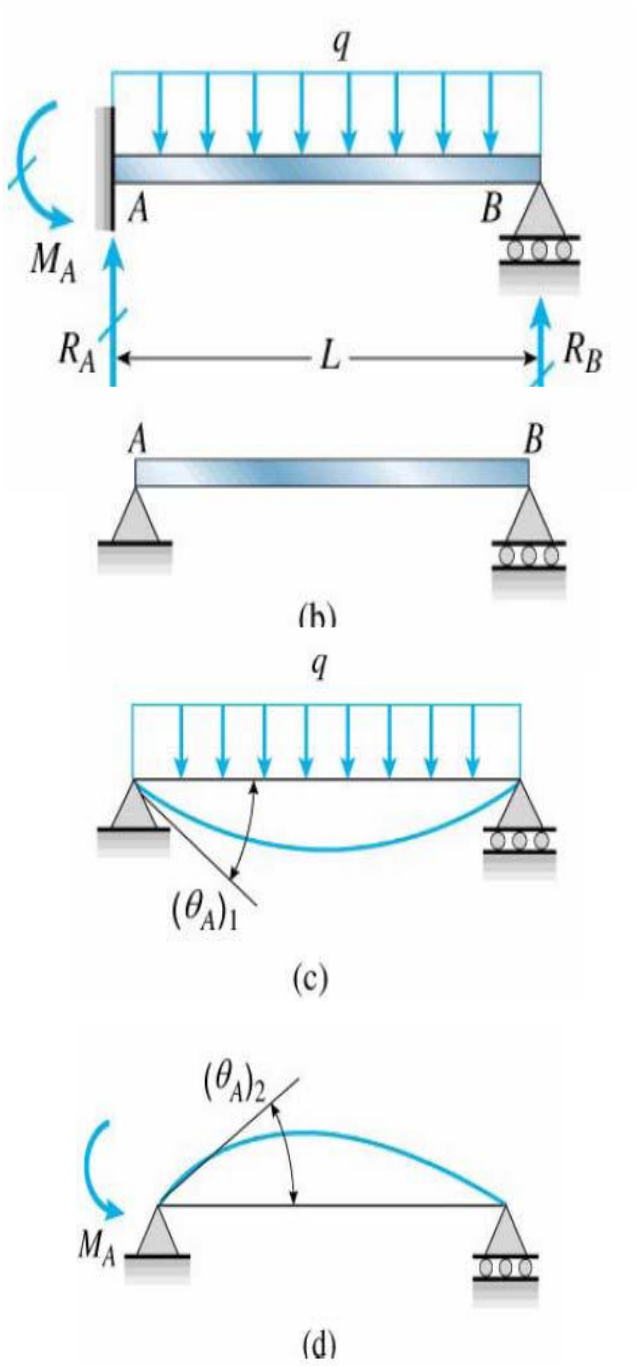
$$(\theta_A)_1 = \frac{qL^3}{24EI} \qquad (\theta_A)_2 = -\frac{M_A L}{3EI}$$

compatibility equation

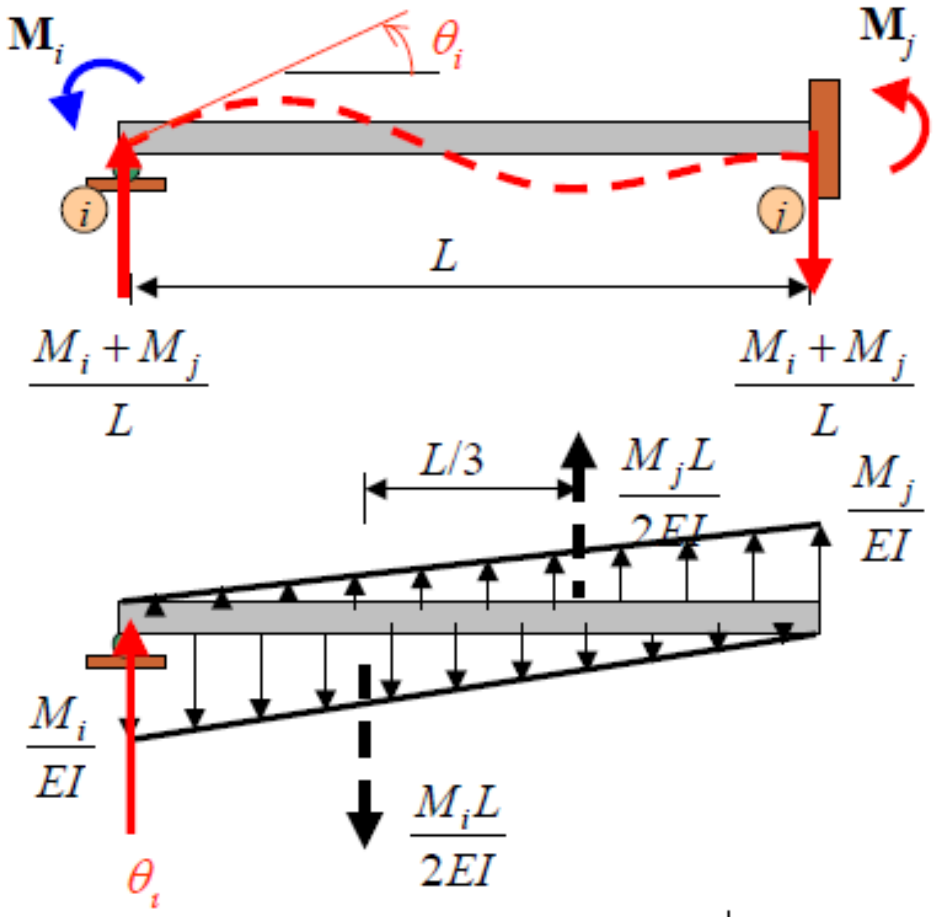
$$(\theta_A)_1 + (\theta_A)_2 = \frac{qL^3}{24EI} - \frac{M_A L}{3EI} = 0$$

thus $M_A = qL^2/8$

and $R_A = 5qL/8 \qquad R_B = 3qL/8$



Stiffness Coefficients Derivation: Fixed-End Support



Real beam

Conjugate beam

$$+\circlearrowleft \Sigma M'_i = 0: \quad -(\frac{M_i L}{2EI})(\frac{L}{3}) + (\frac{M_j L}{2EI})(\frac{2L}{3}) = 0$$

$$M_i = 2M_j \quad ---(1)$$

$$+\uparrow \quad \Sigma F_y = 0: \quad \theta_i - (\frac{M_i L}{2EI}) + (\frac{M_j L}{2EI}) = 0 \quad ---(2)$$

From (1) *and* (2);

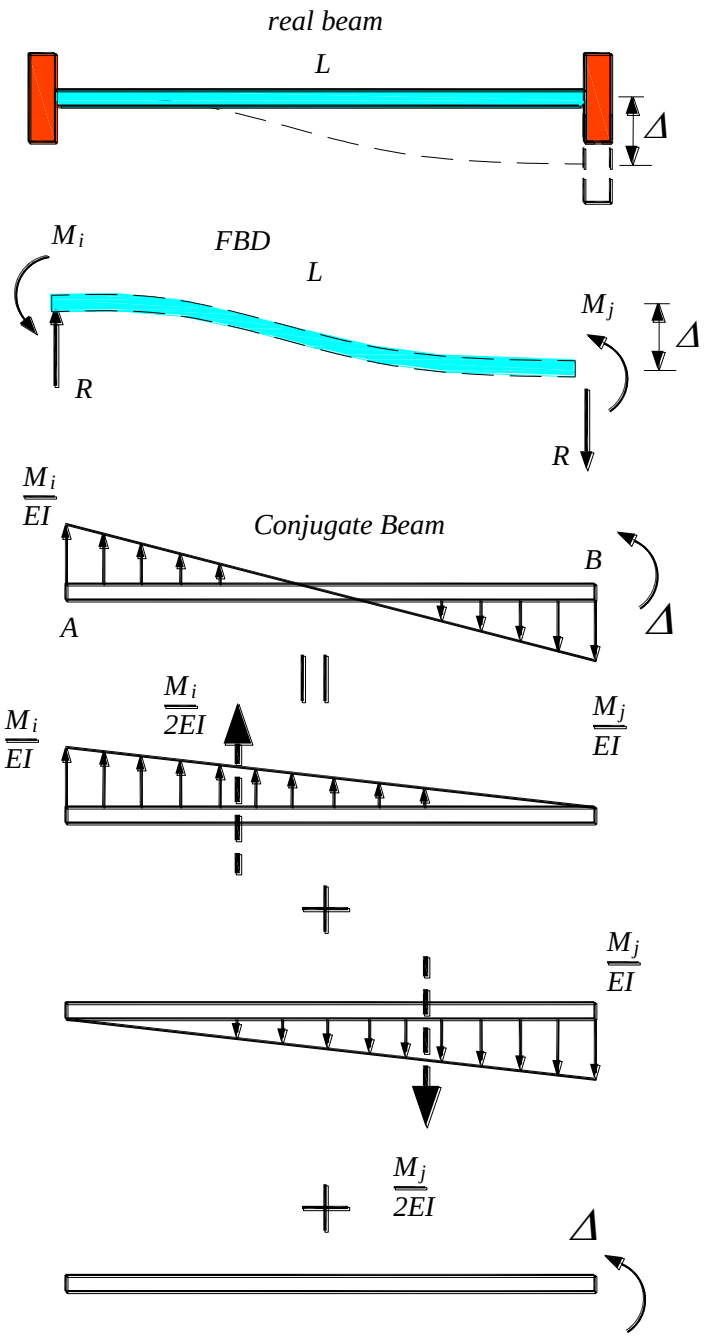
$$M_i = (\frac{4EI}{L})\theta_i$$

$$M_j = (\frac{2EI}{L})\theta_i$$

$$\sum F_y^* = 0 \rightarrow \frac{M_i}{EI}\frac{L}{2} - \frac{M_j}{EI}\frac{L}{2} = 0 \rightarrow M_i = M_j = M$$

$$\text{+} \quad \Sigma M_B = 0: \quad -\Delta - (\frac{ML}{2EI})(\frac{L}{3}) + (\frac{ML}{2EI})(\frac{2L}{3}) = 0,$$

$$M = \frac{6EI\Delta}{L^2} \qquad 18$$



Real beam

$$\sum M = 0 \rightarrow M_i + M_j - RL = 0 \rightarrow M_i = RL - M_j$$

Conjugate beam

$$\sum M_B^* = 0 \rightarrow \frac{(RL - M_j)L}{2EI} \times \frac{2L}{3} - \frac{M_j L}{2EI} \times \frac{L}{3} - \Delta = 0$$

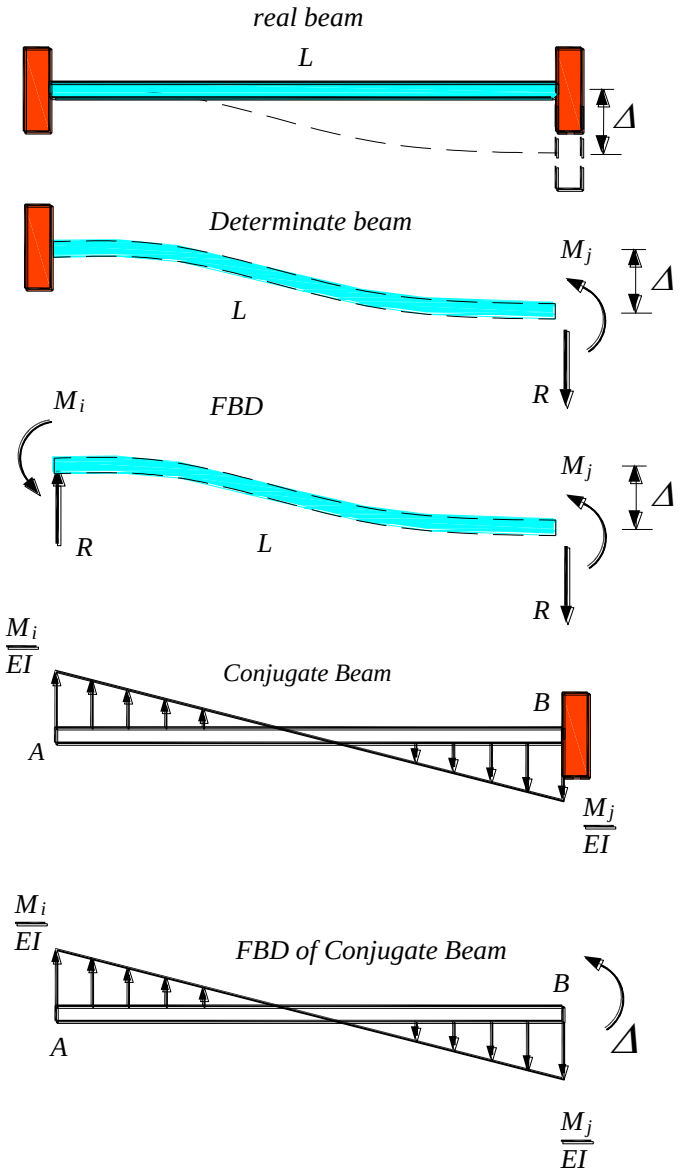
$$\frac{RL^3}{3EI} - \frac{M_j L^2}{3EI} - \frac{M_j L^2}{6EI} = \Delta \rightarrow \frac{RL^3}{3EI} - \frac{M_j L^2}{2EI} = \Delta$$

$$\theta_B = \frac{(RL - M_j)L}{2EI} - \frac{M_j L}{2EI} = \frac{RL^2}{2EI} - \frac{M_j L}{EI} = 0 \rightarrow R = \frac{2M_j}{L}$$

$$M_i = \frac{2M_j}{L} \times L - M_j \rightarrow M_i = M_j$$

$$\frac{2M_j}{L}\frac{L^3}{3EI} - \frac{M_j L^2}{2EI} = \Delta \rightarrow \frac{4M_j L^2}{6EI} - \frac{3M_j L^2}{6EI} = \Delta \rightarrow \frac{M_j L^2}{6EI} = \Delta$$

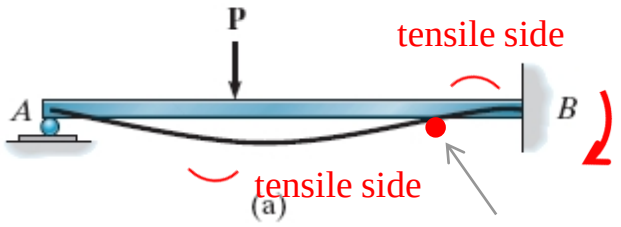
$$M_j = \frac{6EI}{L^2}\Delta$$

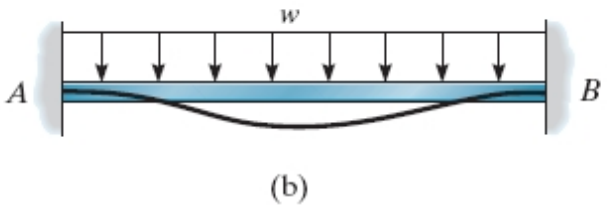


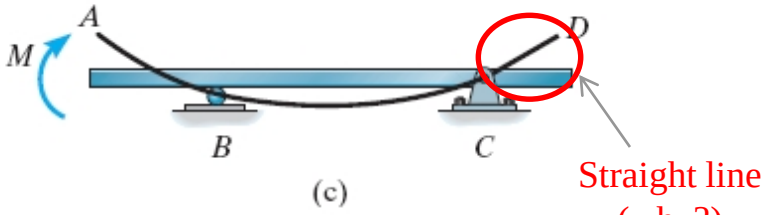
Example 8.1

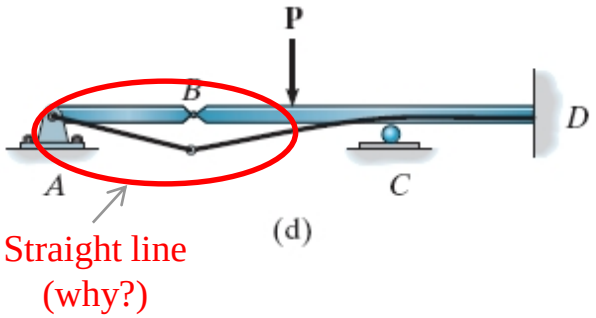
Draw the deflected shape of each of the beams.

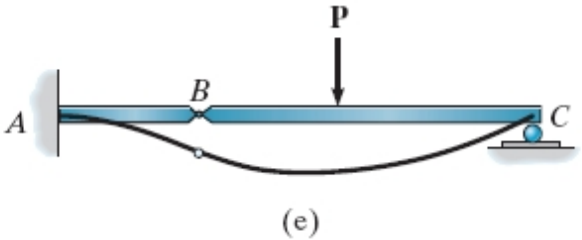
Need to show 1st order (slope) and 2nd order (curvature) information

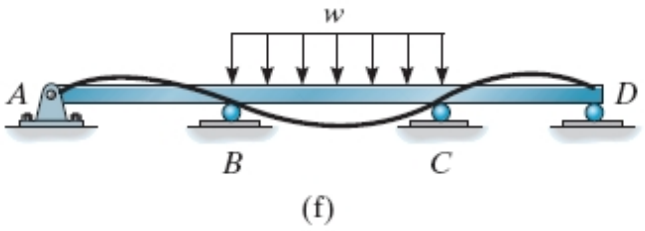






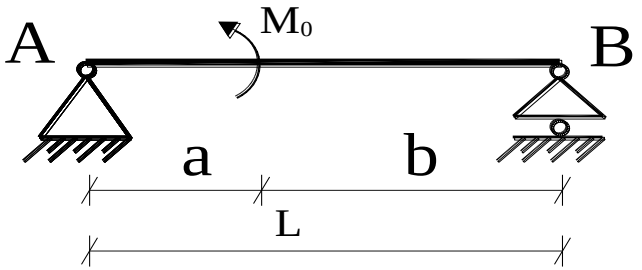






Problem 1.11: A simple beam is loaded with a concentrated moment M_0.

Find the rotation angles of the supports θ_A, θ_B



$$\sum M_B = 0 \qquad R_A.L - \frac{M_0.a}{LEI}\frac{a}{2}\left(b+\frac{a}{3}\right) + \frac{M_0.b}{LEI}.\frac{b}{2}.\frac{2b}{3} = 0$$

$$R_A = \frac{1}{2}.\frac{M_0}{L^2EI}.\left(\frac{a^3}{3} + \frac{3ba^2}{3} - \frac{2b^3}{3}\right)$$

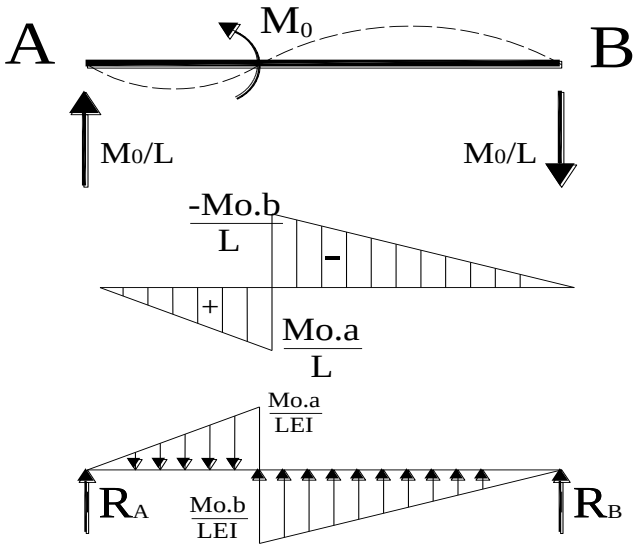
$$\theta_A = R_A = \frac{M_0}{6L^2EI}.\left(a^3 + 3a^2b - 2b^3\right)$$

$$\sum M_A = 0 \qquad -\frac{M_0 a}{LEI}\frac{2}{2}\frac{2a}{3} + \frac{M_0 b}{LEI}\frac{b}{2}\left(a+\frac{b}{3}\right) + R_B.L = 0$$

$$R_B = \frac{M_0}{2L^2EI}.\left(\frac{2a^3}{3} - \frac{3ab^2}{3} - \frac{b^3}{3}\right)$$

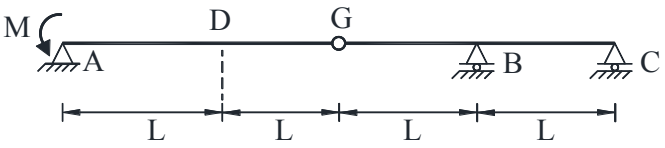
$$R_B = \frac{M_0}{6L^2EI}.\left(2a^3 - 3ab^2 - b^3\right)$$

$$\theta_B = -R_B = \frac{M_0 a}{6L^2EI}.\left(-2a^3 + 3ab^2 + b^3\right)$$



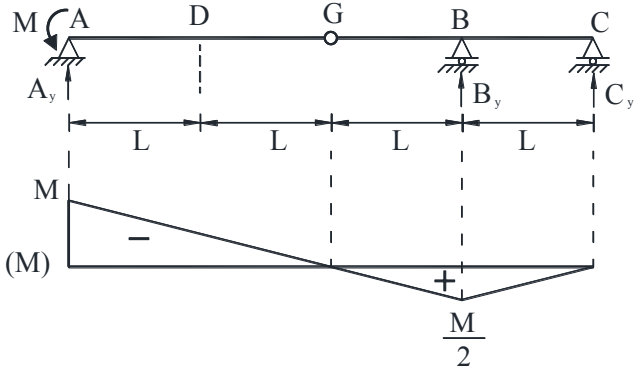
$$\theta_A = R_A \qquad \theta_B = T_B = -R_B$$

Problem: Find the displacement and the rotation at D point of the beam given in the figure by employing Mohr's method. (*EI* is constant along the beam).



Solution

Actual system

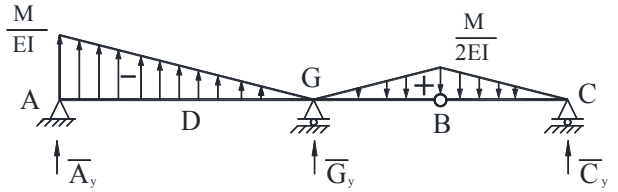


$$\sum (M_G)_{sol} = 0 \rightarrow M - A_y.2L = 0 \rightarrow A_y = \frac{M}{2L}$$

$$\sum M_C = 0 \rightarrow M - A_y.4L - B_y.L = 0 \rightarrow B_y = -\frac{M}{L}$$

$$\sum F_y = 0 \rightarrow A_y + B_y + C_y = 0 \rightarrow C_y = \frac{M}{2L}$$

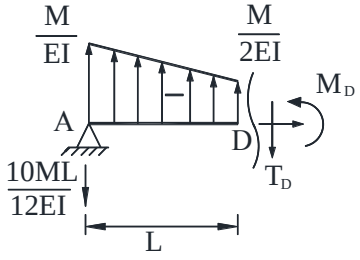
Conjugate system



For conjugate system:

$$\sum \left(\overline{M}_B\right)_{sağ} = 0 \rightarrow \overline{C}_y.L - \frac{1}{2}L\frac{M}{2EI}\frac{L}{3} = 0 \rightarrow \overline{C}_y = \frac{ML}{12EI}$$

$$\sum \overline{M}_G = 0 \rightarrow \overline{C}_y.2L - \frac{1}{2}2L.\frac{M}{2EI} - \overline{A}_y.2L - \frac{1}{2}2L.\frac{M}{EI}.\frac{2L}{3} = 0 \quad \rightarrow \overline{A}_y = -\frac{10ML}{12EI}$$



$$\sum F_y = 0 \rightarrow \frac{10ML}{12EI} - \frac{L}{2}\left(\frac{M}{EI} + \frac{M}{2EI}\right) + T_D = 0 \rightarrow \quad T_D = -\frac{ML}{12EI} = \theta_D$$

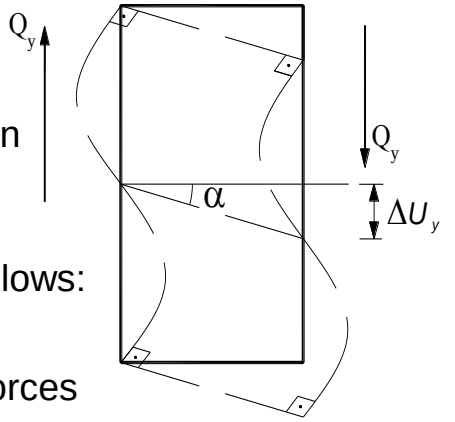
$$\sum M_D = 0 \rightarrow M_D + \frac{10ML}{12EI}L - \frac{1}{2}\left(L\frac{M}{2EI}\frac{2L}{3}\right) - \frac{M}{2EI}L\frac{L}{2} = 0 \rightarrow \quad M_D = -\frac{5ML^2}{12EI} = v_D$$

1- DEFORMATION IN BENDING WITH SHEAR

Because the shear stresses have no effect on the change of lenghts of fibers,

there will be no effect of the shear force Q_y on the curvature caused by bending.

Therefore curvature in bending with shear can be considered as equal to the curvature in bending only. Namely,



$$\frac{1}{\rho_x} = \omega_x = \frac{M_x}{EI_x} \quad (1)$$ The translation because of shear force can be calculated as follows:

$$(\Delta U_y)_{kesmeden} = k'_y \frac{Q_y}{G.A}.\Delta z \quad (2)$$ By using the energy equivalence of internal and external forces

The coefficient k'_y can be obtained. The strain energy in a beam element of length Δz because of shear only is

$$\Delta V = \Delta z \int_A \frac{\tau^2}{2G} dA \quad (3)$$

This is the internal strain energy due to shear stresses in a beam of length Δz .

The work of external forces is $$\Delta W = \frac{1}{2} Q_y.\Delta U_y = \frac{1}{2} Q_y \left(k'_y \frac{Q_y}{G.A}.\Delta z \right) = k'_y \frac{Q_y^{\,2}}{2G.A}.\Delta z \quad (4)$$

By using (3) and (4) $$\Delta W = \Delta V \rightarrow k'_y \frac{Q_y^{\,2}}{2G.A}.\Delta z = \Delta z \int_A \frac{\tau^2}{2G} dA \rightarrow k'_y = \frac{A}{Q_y^{\,2}} \int (\tau_{zx}^2 + \tau_{zy}^2) dA$$

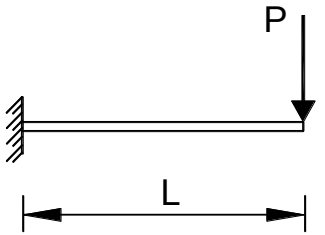
For a rectangular cross section, by taking $\tau_{zx} = 0$ $$k'_y = \frac{A}{Q_y^{\,2}} \int_A \left(\frac{Q_y.S_x}{b.I_x} \right)^2 dA = 1,2 \quad (6)$$

Cross Section	Circle	Rectangle	Annular Cross Section
k'	1,18	1,2	2

KESME KUVVETİNİN ELASTİK EĞRİYE ETKİSİ

Daha önceki incelemelerde elastik eğri bulunurken sadece eğilme momentinin etkisi göz önüne alındı. Kesme kuvveti ise daha önce bulunan elastik eğriye ek terimler getirir. Bu terimlerin katkısı eğilme momenti terimleri kadar büyük değilse de bazı hallerde göz önüne almak gerekir.

Elastik eğriyi, $\nu = \nu_M + \nu_T$ şeklinde eğilme momenti ile kesme kuvvetinin etkilerini gösteren iki terimin toplamında ibaret olarak göz önüne alalım. Kesme kuvvetinin etkisini gösteren ν_T teriminin türevi Bölüm 12 de verilen (12.15) denkleminden aşağıdaki şekilde elde edilir.



$$\Delta \nu_T = k_y^{'} \frac{T_y}{GF} \Delta z \quad \rightarrow \quad \frac{d\nu_T}{dz} = k_y^{'} \frac{T_y}{GF} \qquad (13.31)$$

Yukarıda bulunan sonucu şekil 13.59 de yükleme durumu verilen dikdörtgen kesitli konsol kirişe uygulayalım. Konsol kirişin kesme kuvveti diyagramı $T_y = P$ olduğundan bu değer (13.31) denkleminde yerine konulduktan sonra denklem 0 dan L ye kadar integre edildiğinde $z = 0$ da $\nu_T = 0$ olduğu göz önüne alınarak $\nu_T(\mathrm{L}) = \mathrm{k}_y^{'} \dfrac{PL}{(\mathrm{GF})}$ elde edilir. Konsol kiriş ucunun toplam çökmesi ise moment etkisi de göz önüne alınarak aşağıda verilen şekildedir.

$$\delta = \frac{PL^3}{3EI_x} + k_y^{'} \frac{PL}{GF} = \frac{PL^3}{3EI_x}\left[1 + \frac{3k_y^{'} EI_x}{GFL^2}\right] \qquad 13.32)$$

Konsol kiriş kesiti dikdörtgen olduğundan $k_y^{'} = 6/5$ olup $E/G = 2(1+\upsilon) = 2,6$ alındığında

$$\delta = \frac{PL^3}{3EI_x}\left[1 + \frac{3}{10}\frac{h^2 E}{L^2 G}\right] = \frac{PL^3}{3EI_x}\left[1 + 0,78\frac{h^2}{L^2}\right]$$

elde edilir. Bulunan sonuçta görüldüğü gibi kesme kuvvetinin etkisi h/L ye bağlı olmakla birlikte bu örnekte azdır

Kesme kuvvetinin etkisini farklı şekilde hesaba katmak için (13.31) ifadesinin bir kez daha türevi alındıktan sonra, $\nu = \nu_M + \nu_T$ bağıntısı göz önüne alınarak toplam elastik eğrinin ikinci mertebe diferansiyel denklemi, aşağıda verilen şekilde bulunur.

$$\frac{d^2\nu_T}{dz^2} = \frac{k_y^{'}}{GF}\frac{dT_y}{dz} = -\frac{k_y^{'}p}{GF}; \qquad \frac{d^2\nu}{dz^2} = -\left[\frac{M_x}{EI_x} + \frac{k_y^{'}p}{GF}\right] \tag{3.34}$$

Elastik eğri eşlenik kiriş yöntemi ile bulunmak istenirse yukarıda verilen ifadeye göre eğrilik diyagramına $k_y^{'}p/(\mathrm{GF})$ terimini eklemek gerekir.

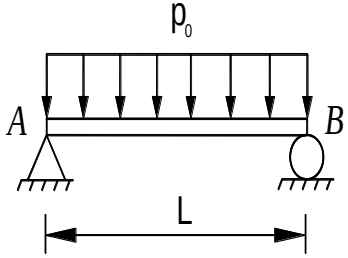
Kesme kuvvetinin elastik eğriye etkisi;

a) yüksekliğin açıklığa göre büyük olan kirişlerde,

b) putrel gibi ince kalınlıklı kirişlerde,

c) tekil yüklerin bulunması halinde önem kazanmaktadır.

Örnek Problem 13.19

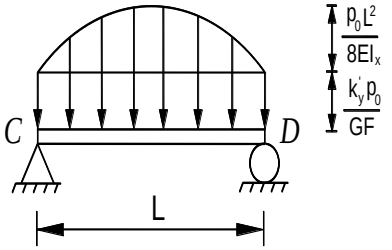


Şekil 13.60 da görülen basit kirişin düzgün yayılı yük ile yüklenmesi halinde kesme kuvvetinin etkisini de göz önüne alarak kirişin orta noktasının çökmesini ve kesme kuvvetinin etkisinin yüzdesini bulunuz.

Kirişin kesiti dikdörtgen olup $\upsilon = 0,3$ ve $h/L = 0,1$ dir.

Çözüm:

Şekil 13.61 de eşlenik kirişe yüklenecek yükler görülmektedir. Kirişin orta noktasının çökmesi



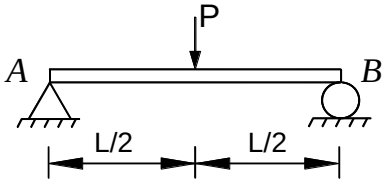
$$f = \frac{5p_0L^4}{384EI_x} + \frac{k_y' p_0 L^2}{8GF} = \frac{5p_0L^4}{384EI_x}\left[1 + \frac{384k_y' EI_x}{40GFL^2}\right]$$

şeklinde bulunur. Kesit dikdörtgen olduğundan $k_y' = 6/5$ olup ve $E/G = 2(1+\upsilon) = 2,6$ olduğundan aşağıdaki sonuç bulunur.

$$f = \frac{5p_0L^4}{384EI_x}\left[1 + 2,496\frac{h^2}{L^2}\right] = \frac{5p_0L^4}{384EI_x}\left[1 + 0,025\right]$$

Yukarıda görüldüğü gibi kesme kuvvetinin etkisi % 2,5 dir.

Örnek Problem 13.20:



Şekil 13.62 de görülen basit kirişin orta noktasında P kuvveti ile yüklenmesi halinde kesme kuvvetinin etkisini de göz önüne alarak orta noktasının çökmesini ve kesme kuvvetinin etkisinin yüzdesini;

a) Kesiti putrel ($k_y^{'} \cong 2,5; I_x/F \cong h^2/6$) $h/L = 1/10$ ve $\upsilon = 0,3$ alarak,

b) Kesiti dikdörtgen $h/L = 1/10$ ve $\upsilon = 0,3$ alarak bulunuz,

c) Eşlenik kiriş yöntemi ile orta noktanın çökmesini bulunuz.

Çözüm:

Kesme kuvvetinin kirişin orta noktasının çökmesine olan etkisi, (13.31) formülü 0 dan $L/2$ ye kadar integre edildiğinde, kesme kuvvetinin $P/2$ olduğu göz önüne alınarak, $k_y^{'} PL/(4\mathrm{GF})$ olarak bulunur. Momentin etkisi ise $PL^3/(48\,\mathrm{EI}_x)$.

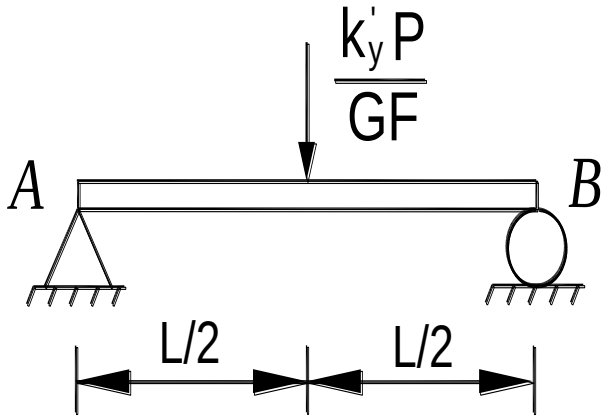
a) Putrel için orta noktanın çökmesi, $k_y^{'} \cong 2,5;\ E/G = 2,6;\ I_x/F \cong h^2/6\ ve\ h/L = 0,1$ alındığında

$$f = \frac{PL^3}{48EI_x} + \frac{k_y^{'} PL}{4GF} = \frac{PL^3}{48EI_x}\left(1 + \frac{48k_y^{'} EI_x}{4GFL^2}\right) = (1 + 0,13)$$

olarak bulunur. Yukarıdaki sonuçta görüldüğü gibi kesme kuvvetinin etkisi % 13 civarındadır.

b) Kesit dikdörtgen alındığında $k_y^{'} = 1,2;\ E/G = 2,6;\ I_x/F = h^2/12\ ve\ h/L = 0,1$ olduğundan kesme kuvvetinin etkisi % 3 civarında elde edilir.

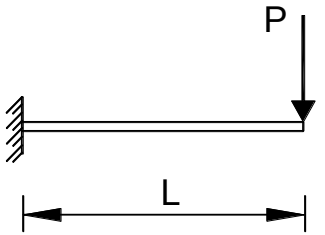
c) Tekil kuvvet olması halinde; tekil kuvveti çok küçük e aralığına etki eden yayılı yük olarak düşünüp eğrilik diyagramına ek olarak $k_y^{'}\left(P/e\right)/\left(GF\right)$ terimi gelecektir. Limite gidildiğinde eşlenik sisteme yük olarak gelecek ek terim $k_y^{'}P/\left(GF\right)$ dir. Şekil 13.63 de verilen basit kirişin orta noktasının çökmesini eşlenik kiriş yöntemi ile hesaplayalım. Şekil 13.63 de eşlenil kiriş ve bu kirişe yüklenen kesme kuvvetine ait yük görülmektedir (eğrilik diyagramı görülmemektedir). Bu yük altında eşlenik kirişin orta noktasındaki moment, yani kesme kuvvetinden dolayı çökme $k_y^{'}P/\left(4GF\right)$ dir. Bu terime ek olarak eğilmeden dolayı çökme terimi olan $PL^3/\left(48EI_x\right)$ gelecektir.



KESME KUVVETİNİN ELASTİK EĞRİYE ETKİSİ

Daha önceki incelemelerde elastik eğri bulunurken sadece eğilme momentinin etkisi göz önüne alındı. Kesme kuvveti ise daha önce bulunan elastik eğriye ek terimler getirir. Bu terimlerin katkısı eğilme momenti terimleri kadar büyük değilse de bazı hallerde göz önüne almak gerekir.

Elastik eğriyi, $\nu = \nu_M + \nu_T$ şeklinde eğilme momenti ile kesme kuvvetinin etkilerini gösteren iki terimin toplamında ibaret olarak göz önüne alalım. Kesme kuvvetinin etkisini gösteren ν_T teriminin türevi Bölüm 12 de verilen (12.15) denkleminden aşağıdaki şekilde elde edilir.



$$\Delta \nu_T = k_y^{'} \frac{T_y}{GF} \Delta z \quad \rightarrow \quad \frac{d\nu_T}{dz} = k_y^{'} \frac{T_y}{GF} \qquad (13.31)$$

Yukarıda bulunan sonucu şekil 13.59 de yükleme durumu verilen dikdörtgen kesitli konsol kirişe uygulayalım. Konsol kirişin kesme kuvveti diyagramı $T_y = P$ olduğundan bu değer (13.31) denkleminde yerine konulduktan sonra denklem 0 dan L ye kadar integre edildiğinde $z = 0$ da $\nu_T = 0$ olduğu göz önüne alınarak $\nu_T(\mathrm{L}) = \mathrm{k}_y^{'} \dfrac{PL}{(\mathrm{GF})}$ elde edilir. Konsol kiriş ucunun toplam çökmesi ise moment etkisi de göz önüne alınarak aşağıda verilen şekildedir.

$$\delta = \frac{PL^3}{3EI_x} + k_y^{'} \frac{PL}{GF} = \frac{PL^3}{3EI_x}\left[1 + \frac{3k_y^{'} EI_x}{GFL^2}\right] \qquad 13.32)$$

Konsol kiriş kesiti dikdörtgen olduğundan $k_y^{'} = 6/5$ olup $E/G = 2(1+\upsilon) = 2,6$ alındığında

$$\delta = \frac{PL^3}{3EI_x}\left[1 + \frac{3}{10}\frac{h^2 E}{L^2 G}\right] = \frac{PL^3}{3EI_x}\left[1 + 0,78\frac{h^2}{L^2}\right]$$

elde edilir. Bulunan sonuçta görüldüğü gibi kesme kuvvetinin etkisi h/L ye bağlı olmakla birlikte bu örnekte azdır

Kesme kuvvetinin etkisini farklı şekilde hesaba katmak için (13.31) ifadesinin bir kez daha türevi alındıktan sonra, $\nu = \nu_M + \nu_T$ bağıntısı göz önüne alınarak toplam elastik eğrinin ikinci mertebe diferansiyel denklemi, aşağıda verilen şekilde bulunur.

$$\frac{d^2\nu_T}{dz^2} = \frac{k_y^{'}}{GF}\frac{dT_y}{dz} = -\frac{k_y^{'}p}{GF}; \qquad \frac{d^2\nu}{dz^2} = -\left[\frac{M_x}{EI_x} + \frac{k_y^{'}p}{GF}\right] \tag{3.34}$$

Elastik eğri eşlenik kiriş yöntemi ile bulunmak istenirse yukarıda verilen ifadeye göre eğrilik diyagramına $k_y^{'}p/(\mathrm{GF})$ terimini eklemek gerekir.

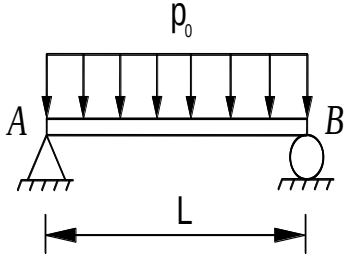
Kesme kuvvetinin elastik eğriye etkisi;

a) yüksekliğin açıklığa göre büyük olan kirişlerde,

b) putrel gibi ince kalınlıklı kirişlerde,

c) tekil yüklerin bulunması halinde önem kazanmaktadır.

Örnek Problem 13.19

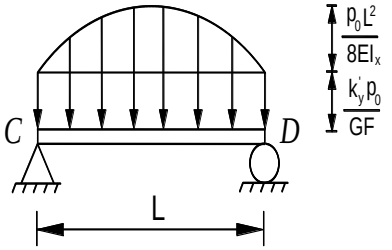


Şekil 13.60 da görülen basit kirişin düzgün yayılı yük ile yüklenmesi halinde kesme kuvvetinin etkisini de göz önüne alarak kirişin orta noktasının çökmesini ve kesme kuvvetinin etkisinin yüzdesini bulunuz.

Kirişin kesiti dikdörtgen olup $\upsilon = 0,3$ ve $h/L = 0,1$ dir.

Çözüm:

Şekil 13.61 de eşlenik kirişe yüklenecek yükler görülmektedir. Kirişin orta noktasının çökmesi



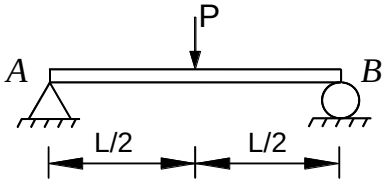
$$f = \frac{5p_0L^4}{384EI_x} + \frac{k_y' p_0 L^2}{8GF} = \frac{5p_0L^4}{384EI_x}\left[1 + \frac{384k_y' EI_x}{40GFL^2}\right]$$

şeklinde bulunur. Kesit dikdörtgen olduğundan $k_y' = 6/5$ olup ve $E/G = 2(1+\upsilon) = 2,6$ olduğundan aşağıdaki sonuç bulunur.

$$f = \frac{5p_0L^4}{384EI_x}\left[1 + 2,496\frac{h^2}{L^2}\right] = \frac{5p_0L^4}{384EI_x}\left[1 + 0,025\right]$$

Yukarıda görüldüğü gibi kesme kuvvetinin etkisi % 2,5 dir.

Örnek Problem 13.20:



Şekil 13.62 de görülen basit kirişin orta noktasında P kuvveti ile yüklenmesi halinde kesme kuvvetinin etkisini de göz önüne alarak orta noktasının çökmesini ve kesme kuvvetinin etkisinin yüzdesini;

a) Kesiti putrel ($k_y^{'} \cong 2,5; I_x/F \cong h^2/6$) $h/L = 1/10$ ve $\upsilon = 0,3$ alarak,

b) Kesiti dikdörtgen $h/L = 1/10$ ve $\upsilon = 0,3$ alarak bulunuz,

c) Eşlenik kiriş yöntemi ile orta noktanın çökmesini bulunuz.

Çözüm:

Kesme kuvvetinin kirişin orta noktasının çökmesine olan etkisi, (13.31) formülü 0 dan $L/2$ ye kadar integre edildiğinde, kesme kuvvetinin $P/2$ olduğu göz önüne alınarak, $k_y^{'} PL/(4\mathrm{GF})$ olarak bulunur. Momentin etkisi ise $PL^3/(48\,\mathrm{EI}_x)$.

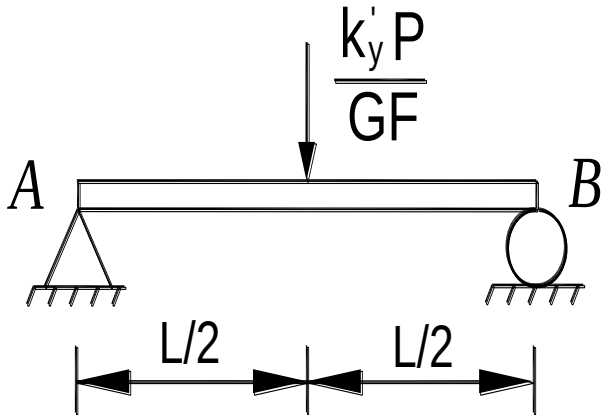
a) Putrel için orta noktanın çökmesi, $k_y^{'} \cong 2,5;\ E/G = 2,6;\ I_x/F \cong h^2/6\ ve\ h/L = 0,1$ alındığında

$$f = \frac{PL^3}{48EI_x} + \frac{k_y^{'} PL}{4GF} = \frac{PL^3}{48EI_x}\left(1 + \frac{48k_y^{'} EI_x}{4GFL^2}\right) = (1 + 0,13)$$

olarak bulunur. Yukarıdaki sonuçta görüldüğü gibi kesme kuvvetinin etkisi % 13 civarındadır.

b) Kesit dikdörtgen alındığında $k_y^{'} = 1,2;\ E/G = 2,6;\ I_x/F = h^2/12\ ve\ h/L = 0,1$ olduğundan kesme kuvvetinin etkisi % 3 civarında elde edilir.

c) Tekil kuvvet olması halinde; tekil kuvveti çok küçük e aralığına etki eden yayılı yük olarak düşünüp eğrilik diyagramına ek olarak $k_y^{'}\left(P/e\right)/\left(GF\right)$ terimi gelecektir. Limite gidildiğinde eşlenik sisteme yük olarak gelecek ek terim $k_y^{'}P/\left(GF\right)$ dir. Şekil 13.63 de verilen basit kirişin orta noktasının çökmesini eşlenik kiriş yöntemi ile hesaplayalım. Şekil 13.63 de eşlenil kiriş ve bu kirişe yüklenen kesme kuvvetine ait yük görülmektedir (eğrilik diyagramı görülmemektedir). Bu yük altında eşlenik kirişin orta noktasındaki moment, yani kesme kuvvetinden dolayı çökme $k_y^{'}P/\left(4GF\right)$ dir. Bu terime ek olarak eğilmeden dolayı çökme terimi olan $PL^3/\left(48EI_x\right)$ gelecektir.



YILDIZ TECHNICAL UNIVERSITY
CIVIL ENGINEERING DEPARTMENT
MECHANICS DIVISION

STRENGTH OF MATERIALS II

SIXTH CHAPTER

COMBİNED LOADING

BENDING WITH TORSION

1.0 Week, Sixth in Line

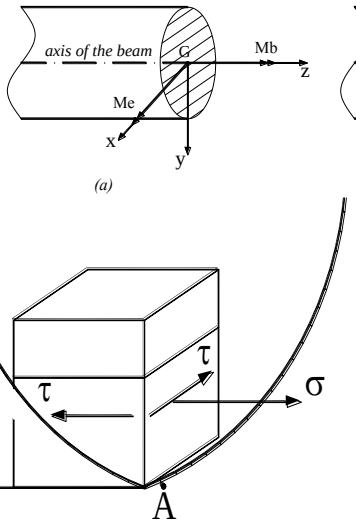
PROF. DR. TURGUT KOCATÜRK

1- BENDING WITH TORSION

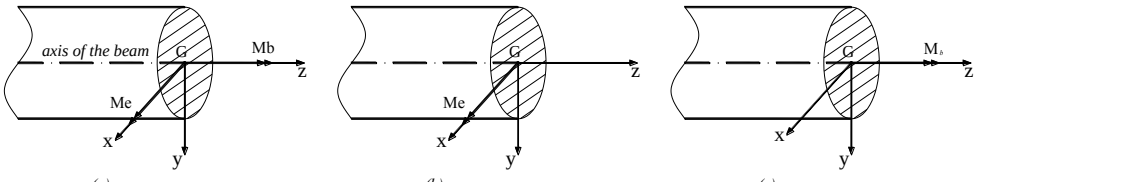
If the bending is existent with torsion, this situation is called bending with torsion. The combined state of strength of materials can be obtained by superposing the simple states of strength of materials.

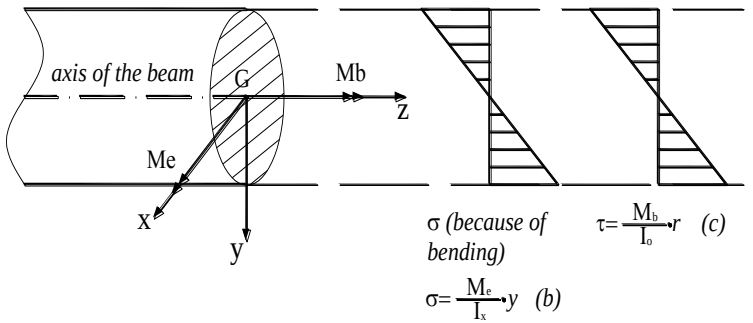
We can't superpose σ and τ directly.

Therefore we should investigate the combined effects of these stresses. The cross section is a circle with the diameter D. Lets consider an infinitesimal element on which both σ and τ have their maximum values.



Both σ and τ are maximum at the point A

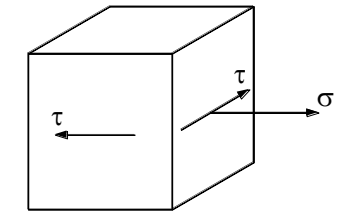




$$\sigma_{\max,e} = \frac{M_e}{I_x}.y_{\max} = \frac{M_e}{\frac{\pi D^4}{64}}.\frac{D}{2} = \frac{32M_e}{\pi D^3}$$

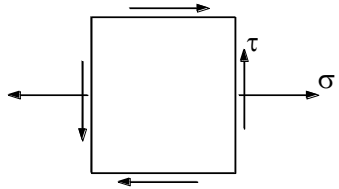
$$\tau_{\max,b} = \frac{M_b}{I_o}.r_{\max} = \frac{M_b}{\frac{\pi D^4}{32}}.\frac{D}{2} = \frac{16M_b}{\pi D^3} \qquad (1)$$

We must find the principal stresses to use failure criteria. These principal stresses are given below:



plane stress state

(a)



Look of fig.3a from upper side

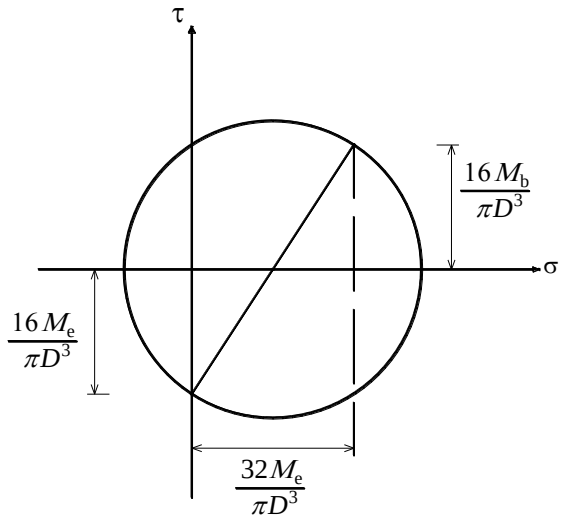
(b)

$$\sigma_{1,2} = \frac{16M_e}{\pi D^3} \pm \frac{16}{\pi D^3}\sqrt{M_e^2 + M_b^2} \rightarrow \sigma_{1,2} = \frac{16}{\pi D^3}\left[M_e \pm \sqrt{M_e^2 + M_b^2}\right] \qquad (2)$$

$$\tau_{\max} = \sqrt{\left(\frac{1}{2}.\frac{32M_e}{\pi D^3}\right)^2 + \left(\frac{16M_b}{\pi D^3}\right)^2} \rightarrow \tau_{\max} = \frac{16}{\pi D^3}\sqrt{M_e^2 + M_b^2} \qquad (3)$$

$$\sigma_{\max,e} = \frac{M_e}{I_x}.y_{\max} = \frac{M_e}{\frac{\pi D^4}{64}}.\frac{D}{2} = \frac{32M_e}{\pi D^3}$$

$$\tau_{\max,b} = \frac{M_b}{I_o}.r_{\max} = \frac{M_b}{\frac{\pi D^4}{32}}.\frac{D}{2} = \frac{16M_b}{\pi D^3} \qquad (1)$$



Dimensioning (Design): In the case of two and three dimensional stress states, we must use failure criteria for dimensioning th elements. We must investigate whether the considered element can withstand the combined effect of the stresses. Lets use the maximum shear stress yield criterion.

$$\sigma_1 = \frac{16}{\pi D^3}\left[M_e + \sqrt{M_e^2 + M_b^2}\right]$$

$$\sigma_2 = 0$$

$$\sigma_3 = \frac{16}{\pi D^3}\left[M_e - \sqrt{M_e^2 + M_b^2}\right]$$

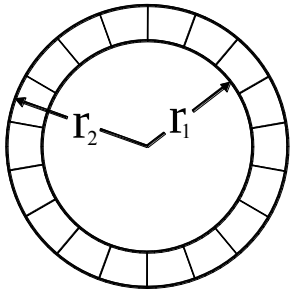
$$\sigma_{\max} - \sigma_{\min} \le \sigma_{em} \;\rightarrow\; 2.\frac{16\sqrt{M_e^2 + M_b^2}}{\pi D^3} \le \sigma_{em} \qquad (4)$$

These formulas can be expressed in a concise form by making the following fictitious moment definition: $M_f = \sqrt{M_e^2 + M_b^2}$

$$\frac{M_f}{W_e} = \frac{32M_f}{\pi D^3} \le \sigma_{em} \qquad (5)$$

Hollow circle

$$I_x = \frac{\pi\left(r_2^4 - r_1^4\right)}{4} \; ; \quad I_o = \frac{\pi\left(r_2^4 - r_1^4\right)}{2}$$



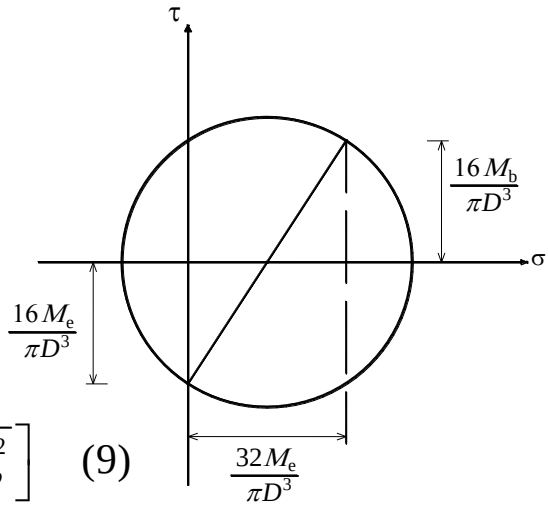
$$\sigma = \frac{4M_e}{\pi\left(r_2^4 - r_1^4\right)}.r_2 \qquad (6)$$

The maximum normal stress when there is only bending moment affecting on the section.

$$\tau = \frac{2M_b}{\pi\left(r_2^4 - r_1^4\right)}.r_2 \qquad (7)$$

The maximum shear stress when there is only torsion affecting on the section.

By using the formulas (6) and (7), we obtain the maximum shear stress and maximum normal stresses for the combined stress state, as follows:



$$\tau_{\max} = \sqrt{\left[\frac{4M_e r_2}{2\pi\left(r_2^4 - r_1^4\right)}\right]^2 + \left[\frac{2M_b r_2}{\pi\left(r_2^4 - r_1^4\right)}\right]^2} \qquad (8)$$

$$\sigma_{1,2} = \frac{2r_2}{\pi\left(r_2^4 - r_1^4\right)}\left[M_e \pm \sqrt{M_e^2 + M_b^2}\right] \qquad (9)$$

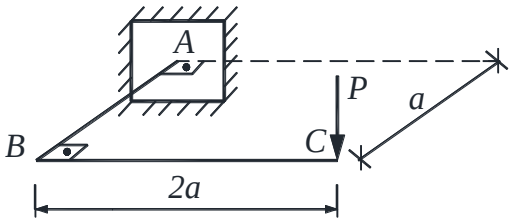
$$\sigma_{\max} - \sigma_{\min} \le \sigma_{allow} \rightarrow \frac{2r_2}{\pi\left(r_2^4 - r_1^4\right)}\left\{\left[M_e + \sqrt{M_e^2 + M_b^2}\right] - \left[M_e - \sqrt{M_e^2 + M_b^2}\right]\right\} = \frac{2r_2}{\pi\left(r_2^4 - r_1^4\right)} 2\sqrt{M_e^2 + M_b^2} \le \sigma_{allow}$$

We define a fictitious moment: $M_f = \sqrt{M_e^2 + M_b^2}$

$$\frac{M_f}{W_e} = \frac{4r_2 M_f}{\pi\left(r_2^4 - r_1^4\right)} = \frac{32D_2 M_f}{\pi\left(D_2^4 - D_1^4\right)} \le \sigma_{allow} \qquad (10)$$

By using one of the expressions (5) or (10) with respect to the geometrical property of the section (full section or hollow section), we can find the necessary diameter D of the section. When using the formula (10), the ratio D_1/D_2 must be given.

Problem: 2D planar ABC bar given with the dimensions and loading case is a full circle having a diameter of d=10 cm. If P=10 kN and σ_{safety} =10000 N/cm^2, find the max. value of the length a according to the max. shear stress hypothesis?



Solution:

$$\frac{M_f}{W_e} = \frac{32M_f}{\pi D^3} \le \sigma_{em} \qquad (5)$$

$$M_b = 2Pa$$

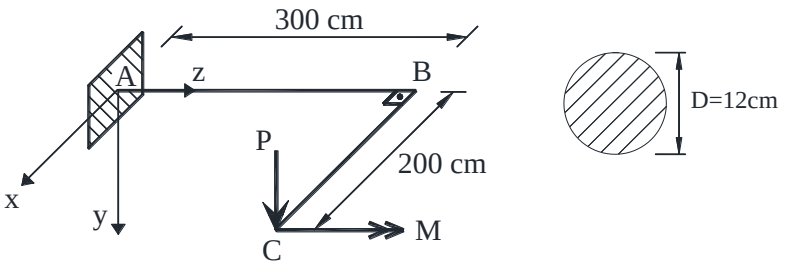
$$M_e = Pa$$

$$M_f = \sqrt{M_e^2 + M_b^2} \ (\text{Acc. to the max. shear stress hypothesis}) \Rightarrow M_f = \sqrt{(Pa)^2 + (2Pa)^2} = 2,236Pa$$

$$\frac{M_f}{W_e} = \frac{32M_f}{\pi D^3} \le \sigma_{em} \qquad (5)$$

$$\frac{32}{\pi d^3} M_f \le \sigma_{safety} \Rightarrow \frac{32}{\pi 10^3}.2,236.10000.a \le 10000 \Rightarrow a \le 43,906\ cm$$

Problem: If $M = 2\times10^6$ Ncm determine the maximum allowable P load by considering maximum shear stress criterion (ignore the contribution of shear force). $\sigma_{safety} = 2\times10^4\ \mathrm{N/cm^2}$



$$\frac{M_f}{W_e} = \frac{32M_f}{\pi D^3} \le \sigma_{em} \qquad (5)$$

Solution:

$$I_o = \frac{\pi.12^4}{32} = 2034,72cm^4, \quad I_x = \frac{\pi.12^4}{64} = 1017,36\,cm^4$$

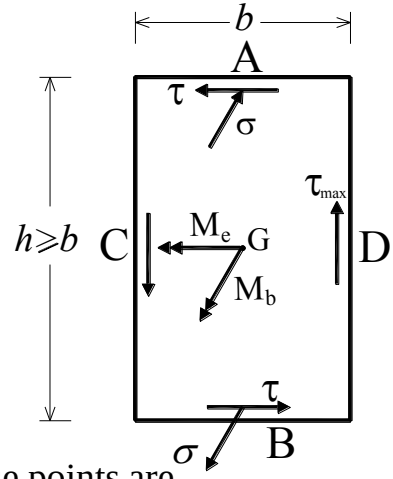
$$M_z = 200P + 2.10^6\ (Ncm) \rightarrow \quad \tau = \frac{200P + 2.10^6}{I_o}.6 = 0,59P + 5897,6$$

$$M_x = -300P\ (Ncm) \rightarrow\ \sigma_z = \frac{-300P}{I_x}(-6) = 1,77P$$

$$\sigma_{1,2} = \frac{\sigma}{2} \pm \sqrt{\left(\frac{\sigma}{2}\right)^2 + \tau^2} \rightarrow \sigma_{\max} - \sigma_{\min} \le \sigma_{safety} \rightarrow\ 1,13P^2 + 6959,17P + 34781685,76 \le 10^8$$

$$P = 5118.12\,N$$

The state when the section is rectangular (Summary)



If $M_e > \lambda M_b$ then the most critical points are the points A and B.

$$\sigma = M_e / W_e \;\; ; \;\; \tau = M_b / W_b' \qquad W_e = bh^2/6 \;\; ; \;\; W_b' = \eta_3 h b^2$$

By means of σ and τ, we can obtain the maximum normal stresses of the combined state as $\sigma_{\max}$ and $\sigma_{\min}$ and then we use the maximum shear stress yield criterion for dimensioning as follows:

$$\sigma_{\max} - \sigma_{\min} = \frac{M_f}{W_e} = \frac{6M_f}{bh^2} \le \sigma_{em} \;\; ; \;\; M_f = \sqrt{M_e^2 + (\alpha M_b)^2} \;\; , \;\; \alpha = \frac{2W_e}{W_b'} = \frac{1}{3\eta_3} \cdot \frac{h}{b}$$

If $M_e \le \lambda M_b$, then the most critical points are the points C and D. The simple shear stresses at these points are

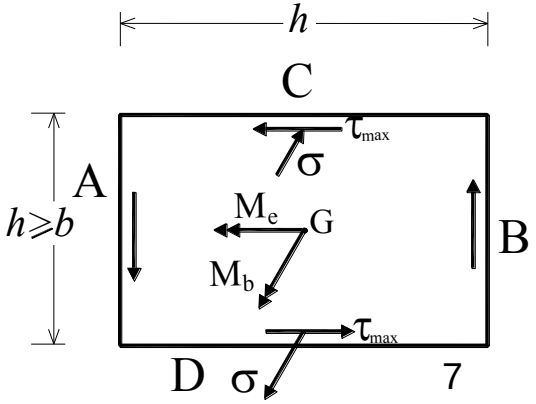
$$\sigma = 0 \;; \quad \tau = M_b / W_b \qquad W_b = \eta_1 h b^2$$

$$\tau_{\max} = \frac{M_b}{W_b} = \frac{M_b}{\eta_1 h b^2}$$

According to the maximum shear stress yield criterion, we can write the following yield criterion:

$$\sigma_{\max} - \sigma_{\min} = 2\tau_{\max} \le \sigma_{em} \qquad \frac{M_f}{W_e} = 2\tau_{\max} = \frac{2M_b}{\eta_1 h b^2} \le \sigma_{em} \;\; ; \;\; M_f = \frac{1}{3\eta_1}\left(\frac{h}{b}\right) M_b$$

If the neutral axis in bending is parallel to the longer edge, then the most critical points are always the points C and D. In this case, the normal and shear stresses at these points are



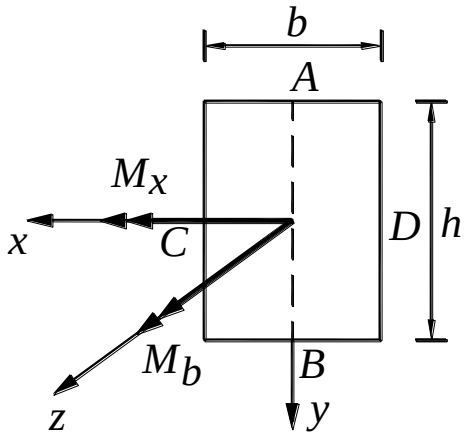
$$\sigma = \frac{M_e}{W_e} \quad and \quad \tau_{\max} = \frac{M_b}{W_b} \;\; ; \;\; W_e = \frac{hb^2}{6} \;\; ; \;\; W_b = \eta_1 h b^2$$

After obtaining the principal normal stresses, by using the maximum shear stress yield criterion, we obtain the following comparison equation:

$$\frac{M_f}{W_e} \le \sigma_{em} \;\; ; \;\; M_f = \sqrt{M_e^2 + \left(\beta M_b^2\right)} \;\; ; \;\; \beta = \frac{2W_e}{W_b} = \frac{1}{3\eta_1}$$

h/b	1,0	1,5	2,0	3,0	4,0	6,0	8,0	10,0
η_1	0,208	0,231	0,246	0,267	0,282	0,299	0,307	0,313
η_3	0,208	0,269	0,31	0,355	0,378	0,402	0,413	0,421
α	1,6	1,857	2,148	2,817	3,524	4,975	6,455	7,917
λ	0	1,108	1,65	2,468	3,149	4,475	5,804	7,131
β	1,6	1,441	1,354	1,247	1,181	1,114	1,085	1,064

Problem: Find the dimensions of the bar with a rectangular cross-section such that h=3b using the max shear stress hypothesis. The reaction moments are M_x=42x10^5 Nmm, M_b=15x10^5 Nmm, and the allowable normal stress is $\sigma_{all} = 120\ MPa$. **NOTE:** 1 MPa = 1 N/mm^2.



h/b	1,0	1,5	2,0	3,0
η_1	0,208	0,231	0,246	0,267
η_3	0,208	0,269	0,31	0,355
α	1,6	1,857	2,148	2,817
λ	0	1,108	1,65	2,468
β	1,6	1,441	1,354	1,247

Where h/b=3, $\lambda = 2,468,\ M_e > \lambda M_b \rightarrow 42\times10^5 \geq 2.468\times15\times10^5 = 37.02\times10^5$. In this case A and B are the points under the greatest stresses. They are the most critical points.

$$\sigma = \frac{M_e}{w_e} = \frac{42.10^5}{b(3b)^2/6} = \frac{28.10^5}{b^3} \qquad \tau = \frac{M_b}{w_{b'}} = \frac{15.10^5}{0,355b^2.3b} = \frac{14,08.10^5}{b^3}$$

When h/b=3, $\eta_3 = 0,355,\ w_{b'} = \eta_3 b^2 h$

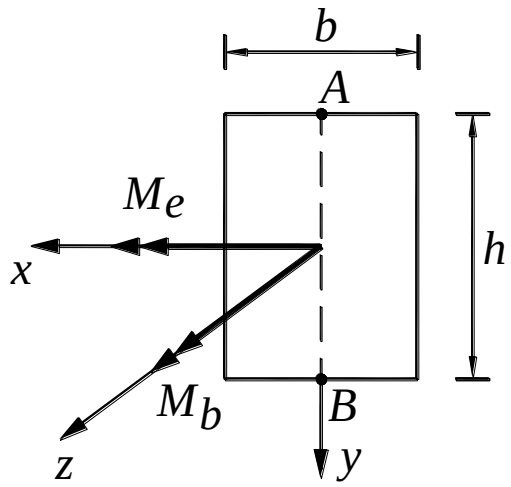
$$\sigma_{\max,\min} = \frac{\sigma}{2} \pm \sqrt{\left(\frac{\sigma}{2}\right)^2 + \tau^2} \qquad \sigma_{\max} - \sigma_{\min} = \sqrt{\sigma^2 + 4\tau^2} \leq \sigma_{all}$$

$$\sqrt{\left(\frac{28.10^5}{b^3}\right)^2 + 4.\left(\frac{14,08.10^5}{b^3}\right)^2} = \frac{39,71.10^5}{b^3} \leq 120 \quad \rightarrow b \geq 32,10\ mm \quad h \geq 96,3\ mm$$

Problem 1: For a rectangular cross section of h/b = 2, M_b=100 kNcm and M_e=300 kNcm. Under the conditions that neutral axis is parallel to the small edge and σ_{em}=14000 N/cm^2, find b and h.

Solution: For h/b=2 we obtain λ=1,65 from the table. In this case the condition $M_e \geq \lambda M_b$ ($300 \geq 1{,}65.100$) is satisfied. The most critical points of the section for this case are the points A ve B noktaları olur. The normal and shear stresses at these points are;

$$\sigma = \frac{M_e}{W_e} = \frac{M_e}{\frac{1}{2}bh^2} \qquad \tau = \frac{M_b}{W_b'} = \frac{M_b}{\eta_3 hb^2} = \frac{M_b}{0,31hb^2}$$



By using the maximum shear stress hypthesis (Tresca Hypthesis) ;

$$\sigma_{\max} - \sigma_{\min} = 2\sqrt{(\frac{M_e}{2W_e})^2 + (\frac{M_b}{W_b'})^2} \leq \sigma_{em} = \sigma_m$$

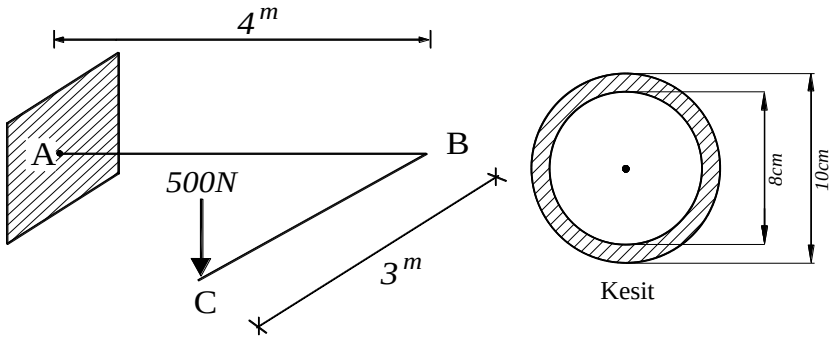
$$2\sqrt{(\frac{300000}{2.\frac{1}{6}bh^2})^2 + (\frac{100000}{0,31hb^2})^2} \leq 14000 \text{ N/cm}^2$$

$$h = 2b \qquad 2\sqrt{(\frac{225000}{b^3})^2 + (\frac{161290,32}{b^3})^2} \leq 14000$$

$$\frac{276838,52}{b^3} \leq 7000 \rightarrow \frac{h}{2} = b \geq 3,41 \text{ cm}$$

h/b	1,0	1,5	2,0
η_1	0,208	0,231	0,246
η_3	0,208	0,269	0,31
α	1,6	1,857	2,148
λ	0	1,108	1,65
β	1,6	1,441	1,354

Poblem 3: The bar ABC is in a horizontal plane. Its dimensions and loading condition is given in the figure. The cross section is an annulus and its inner diameter is 8cm, outer diameter is 10cm. The support A is fixed. Obtain the stresses at point A and check out whether the section is safe or not. $\sigma_{em} = 10000\ N/\text{cm}^2$.



Solution:

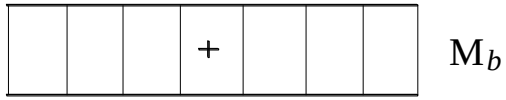
$$\sigma_{\max e} = \frac{M_e}{I_x}.\frac{d}{2} = \frac{200000}{\frac{\pi}{64}\left(10^4 - 8^4\right)}.5 = 3450,51\ N/cm^2$$

$$\tau_{\max b} = \frac{M_b}{I_o}.\frac{d}{2} = \frac{150000}{\frac{\pi}{32}\left(10^4 - 8^4\right)}.5 = 1293,94\ N/cm^2$$

2000 Nm=200000 Ncm



1500 Nm=150000 Ncm

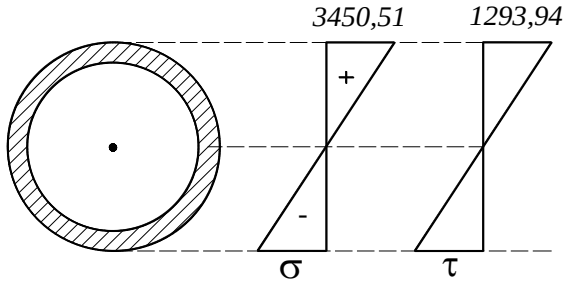


According to the maximum shear stress criterion (Tresca stress criterion):

$$\sigma_m = \frac{32 D_2 M_f}{\pi\left(D_2^4 - D_1^4\right)} \le \sigma_{em}$$

$$M_f = \sqrt{(200000)^2 + (150000)^2} = 250000\ Ncm$$

$$\frac{32.10.250000}{\pi\left(10^4 - 8^4\right)} = 4313,14 \le 10000\ N/cm^2$$ Okay.



Problem 2: The section of the cantilever beam AB given in the figure is an annulus and the inner diameter of the annulus is d, the outer diameter of the annulus is 2d. For $\sigma_{em} = 14\,kN/cm^2$, determine the dimension d for the cross section to carry the given load safely by using

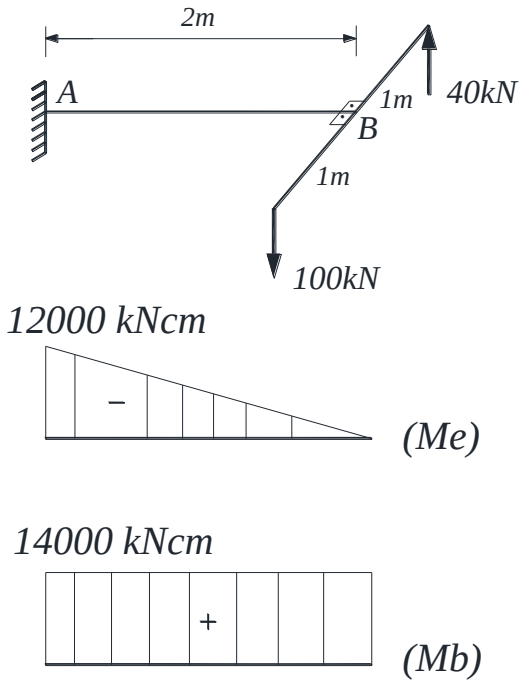
a) Maximum shear stress hypothesis (Tresca hypothesis);
b) and distortion energy (shear stress energy) hypothesis.

$$\sigma = \frac{M_e}{I_x}\cdot d = \frac{12000}{I_0}\cdot 2d \;;\; \tau = \frac{M_b}{I_0}\cdot d = \frac{14000}{I_0}\cdot d$$

$$\sigma_{1,2} = \frac{\sigma}{2} \pm \sqrt{(\frac{\sigma}{2})^2 + \tau^2} = \frac{12000}{2I_0}\cdot 2d \pm \sqrt{(\frac{12000}{2I_0}\cdot 2d)^2 + (\frac{14000}{I_0}\cdot d)^2}$$

$$\sigma_1 = \frac{d}{I_0}(12000 + 18439,09) = 30439,09\cdot\frac{d}{I_0} \;;\; I_0 = \frac{\pi}{32}\left[(2d)^4 - (d)^4\right]$$

$$\sigma_2 = \frac{d}{I_0}(12000 - 18439,09) = -6439,09\cdot\frac{d}{I_0} \;;\; I_0 = \frac{15\pi}{32}d^4$$



a) According to maximum shear stress hypothesis (Tresca hypothesis);

$$\tau_{\max} = \frac{\sigma_{\max} - \sigma_{\min}}{2} = \frac{30439,09 - (-6439,09)}{2(\frac{15\pi}{32} \cdot d^4)} \cdot d = \frac{12521,27}{d^3} \le \tau_{em} = \frac{\sigma_{em}}{2} = 7 \quad d \ge 12,14cm$$

b) According to distortion (shear stress) energy hypothesis;

$$\sigma_1^{\,2} + \sigma_2^{\,2} - \sigma_1 \cdot \sigma_2 = (30439,09 \cdot \frac{d}{I_0})^2 + (-6439,09 \cdot \frac{d}{I_0})^2 - \left[\left(30439,09 \frac{d}{I_0} \right) \left(-6439,09 \cdot \frac{d}{I_0} \right) \right] =$$

$$1,164 \cdot 10^9 \cdot \frac{d^2}{(\frac{15\pi}{32} d^4)^2} = 536,75 \cdot 10^6 \cdot \frac{1}{d^6} \le \sigma_{em}^{\,2} = 196 \qquad d \ge 11,83cm$$

$d \ge 12,14cm$ *must be chosen.*

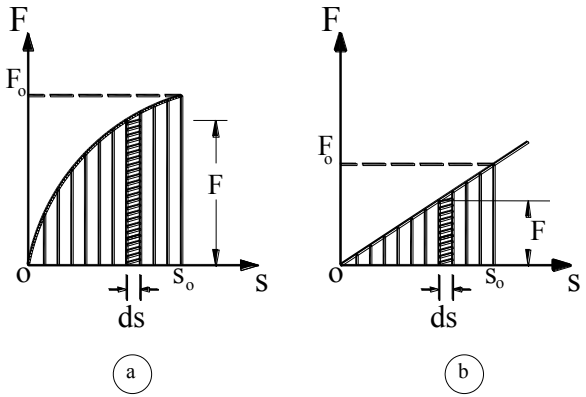
ENERGY METHODS
Two Weeks, Seventh in line

Prof. Dr. Turgut KOCATÜRK

Elastic Energy

Some definitions which will be used in the investigations are given below:

- **Work of external forces:** Natural solid bodies change their shapes when they are loaded and the application points of the forces moves and therefore they **produce** work. This work is shown by $\boldsymbol{U_d}$ and called as the work of the external forces.



$$U_d = \int_0^{s_o} F(s)\,ds$$

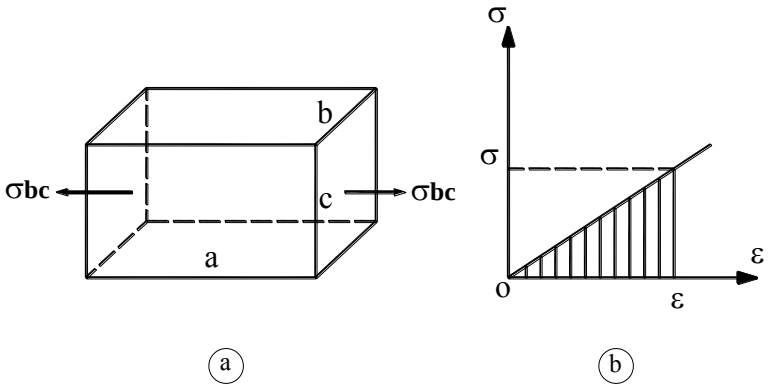
F: Force

s: displacement which is in the direction of the force

Ud: The work done in between 0-s_0

- **Internal Energy:** The work done by the external forces are used in the deformation within the solid body. This energy is called as deformation energy and shown by $\boldsymbol{U_i}$.

Strain energy due to normal stresses:

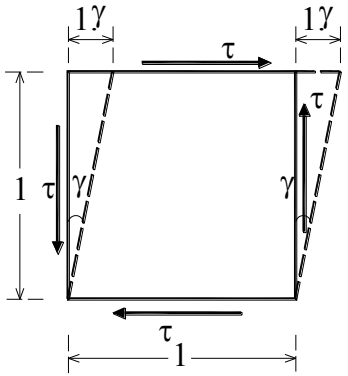


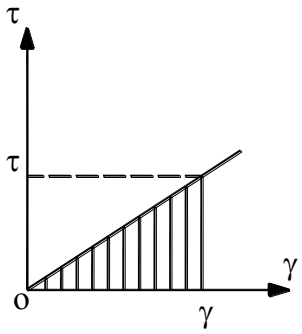
$$U_i = \int_V u_i\,dV$$

$$U_i = \frac{1}{2}(\sigma bc)(\varepsilon a) = \frac{1}{2}abc(\sigma\varepsilon)$$

$$V = abc \qquad u_i = \frac{U_i}{V} = \frac{1}{2}\sigma\varepsilon \qquad u_i = \frac{\sigma^2}{2E} = \frac{E\varepsilon^2}{2}$$

Strain energy due to shear stresses:

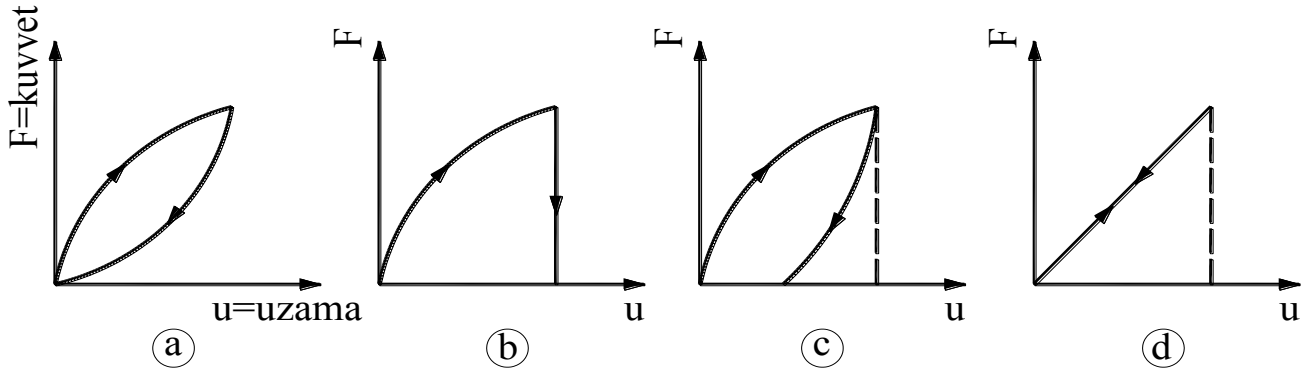




$$u_i = \frac{1}{2}(1.1).\tau.(1.\gamma) = \frac{\tau\gamma}{2}$$

$$u_i = \frac{\tau^2}{2G} = \frac{G\gamma^2}{2}$$

- **Conservation of Mechanical Energy:** In the cases without energy loss it is necessary that *Ud = Ui* . In the exact elastic body, when the loads are removed, if there is no mechanical energy loss, the body returns to its initial state and the energy which is stored in the body come off. The deformation energy depends only on the quantities which define the deformed state of the body and generally called as elastic potential energy.



Tam elastik cisim Tam plastik cisim Elasto-plastik cisim Doğrusal elastik cisim

- **Static Loading:** While calculating deformation, it is assumed that both forces and displacements increase from zero gradually (statical loading). External and internal forces are in equilibrium all the time and hence dynamical phenomena are not allowed.

Three Dimensional Stress State:

$$\varepsilon_x = \frac{1}{E}\left[\sigma_x - \nu\left(\sigma_y + \sigma_z\right)\right] \qquad \varepsilon_y = \frac{1}{E}\left[\sigma_y - \nu\left(\sigma_x + \sigma_z\right)\right] \qquad \varepsilon_z = \frac{1}{E}\left[\sigma_z - \nu\left(\sigma_x + \sigma_y\right)\right]$$

$$u_i = \frac{1}{2}(\varepsilon_1\sigma_1 + \varepsilon_2\sigma_2 + \varepsilon_3\sigma_3) \qquad u_i = \frac{1}{2E}\left[\sigma_1^2 + \sigma_2^2 + \sigma_3^2 - 2\nu(\sigma_1\sigma_2 + \sigma_2\sigma_3 + \sigma_3\sigma_1)\right]$$

$$u_i = G\left[\varepsilon_1^2 + \varepsilon_2^2 + \varepsilon_3^2 + \left(\frac{\nu}{1-2\nu}\right)(\varepsilon_1 + \varepsilon_2 + \varepsilon_3)^2\right]$$

$$u_i = \frac{1}{2}[\varepsilon_x\sigma_x + \varepsilon_y\sigma_y + \varepsilon_z\sigma_z + \tau_{xy}\gamma_{xy} + \tau_{xz}\gamma_{xz} + \tau_{yz}\gamma_{yz}]$$

$$u_i = \frac{1}{2E}\left[\sigma_x^2 + \sigma_y^2 + \sigma_z^2 - 2\nu(\sigma_x\sigma_y + \sigma_x\sigma_z + \sigma_y\sigma_z\right] + \frac{1}{2G}(\tau_{xy}^2 + \tau_{xz}^2 + \tau_{yz}^2)$$

$$u_i = G\left[\varepsilon_x^2 + \varepsilon_y^2 + \varepsilon_z^2 + \left(\frac{\nu}{1-2\nu}\right)(\varepsilon_x + \varepsilon_y + \varepsilon_z)^2\right] + \frac{G}{2}(\gamma_{xy}^2 + \gamma_{xz}^2 + \gamma_{yz}^2)$$

$$\frac{\partial u_i}{\partial \sigma_x} = \varepsilon_x \qquad \frac{\partial u_i}{\partial \varepsilon_x} = \sigma_x$$

$$\frac{\partial u_i}{\partial \varepsilon_x} = \frac{\partial\left\{G\left[\varepsilon_x^2 + \varepsilon_y^2 + \varepsilon_z^2 + \left(\frac{\nu}{1-2\nu}\right)(\varepsilon_x + \varepsilon_y + \varepsilon_z)^2\right] + \frac{G}{2}(\gamma_{xy}^2 + \gamma_{xz}^2 + \gamma_{yz}^2)\right\}}{\partial \varepsilon_x}$$

$$\left.\begin{aligned}
\sigma_x &= \frac{E\nu}{(1+\nu)(1-2\nu)}\left(\varepsilon_x + \varepsilon_y + \varepsilon_z\right) + 2G\varepsilon_x \\
\sigma_y &= \frac{E\nu}{(1+\nu)(1-2\nu)}\left(\varepsilon_x + \varepsilon_y + \varepsilon_z\right) + 2G\varepsilon_y \\
\sigma_z &= \frac{E\nu}{(1+\nu)(1-2\nu)}\left(\varepsilon_x + \varepsilon_y + \varepsilon_z\right) + 2G\varepsilon_z
\end{aligned}\right\}$$

$$\sigma_x = G\left[2\varepsilon_x + \left(\frac{\nu}{1-2\nu}\right)2(\varepsilon_x + \varepsilon_y + \varepsilon_z)\right] = 2G\varepsilon_x + \left(\frac{\nu}{1-2\nu}\right)\left(\frac{E}{2(1+\nu)}\right)2(\varepsilon_x + \varepsilon_y + \varepsilon_z) = \frac{E\nu}{(1+\nu)(1-2\nu)}\left(\varepsilon_x + \varepsilon_y + \varepsilon_z\right) + 2G\varepsilon_y$$

Dilatational and Distortional Strain Energies:

$$\begin{pmatrix} \sigma_1 & 0 & 0 \\ 0 & \sigma_2 & 0 \\ 0 & 0 & \sigma_3 \end{pmatrix} = \begin{pmatrix} \sigma_m & 0 & 0 \\ 0 & \sigma_m & 0 \\ 0 & 0 & \sigma_m \end{pmatrix} + \begin{pmatrix} \sigma_1 - \sigma_m & 0 & 0 \\ 0 & \sigma_2 - \sigma_m & 0 \\ 0 & 0 & \sigma_3 - \sigma_m \end{pmatrix} \qquad \sigma_m = \frac{\sigma_1 + \sigma_2 + \sigma_3}{3}$$

Lets calculate the energy densities for the first and second components of the right side of the above expression. Energy related with pure volume change is shown by $\boldsymbol{u_v}$ and energy related with pure shape change (distortion) is shown by $\boldsymbol{u_g}$. In this case

$$u_i = u_v + u_g$$

The general expression in terms of principal stresses are as follows:

$$u_i = \frac{1}{2E}\left[\sigma_1^2 + \sigma_2^2 + \sigma_3^2 - 2\nu(\sigma_1\sigma_2 + \sigma_2\sigma_3 + \sigma_3\sigma_1)\right]$$

$$u_v = \frac{1}{2E}(3\sigma_m^2 - 2\nu 3\sigma_m^2) = \frac{3(1-2\nu)}{2E}\sigma_m^2 = \frac{(1-2\nu)}{6E}(\sigma_1 + \sigma_2 + \sigma_3)^2$$

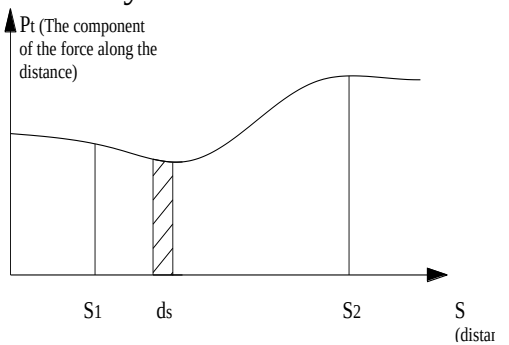
$$u_g = \frac{1}{6G}\left[\sigma_1^2 + \sigma_2^2 + \sigma_3^2 - (\sigma_1\sigma_2 + \sigma_2\sigma_3 + \sigma_1\sigma_3)\right]$$

$$u_g = \frac{1}{12G}\left[(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2\right]$$

Obtaining the elastic strain energy : The external forces do work during the deformation of the body. This work is stored in the body and can be taken back.

$$W_{1\to 2} = \int_{S_1}^{S_2} P_t \, ds$$

$Work = \text{distance} \times \text{the component of the force along the distance}$

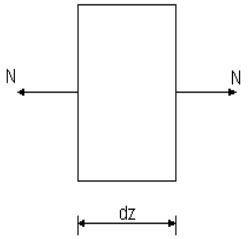


Energy density of the internal forces can be expressed as follows:

$$u_i = \frac{1}{2}\left(\sigma_x \varepsilon_x + \sigma_y \varepsilon_y + \sigma_z \varepsilon_z + \tau_{xy}\gamma_{xy} + \tau_{xz}\gamma_{xz} + \tau_{yz}\gamma_{yz}\right)$$

Here the stresses $\sigma_x, \sigma_y, \sigma_z, \tau_{xy}, \tau_{xz}, \tau_{yz}$ are because of the stress resultants $(N,\ T,\ M,\ M_b)$

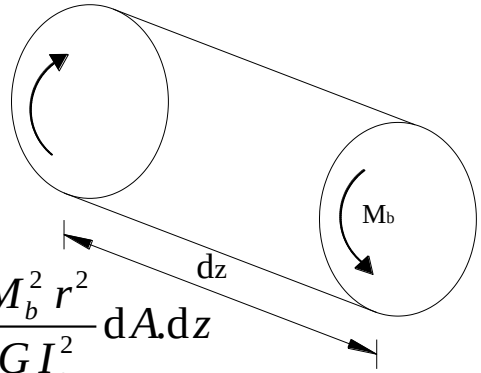
Potential Energy due to normal force: $\sigma_z = \frac{N}{A}$ $\varepsilon_z = \frac{N}{E\,A}$



$$u_i = \frac{1}{2}\sigma_z \varepsilon_z = \frac{1}{2}\frac{N}{A}\frac{N}{E\,A} = \frac{N^2}{2EA^2} \qquad U_i = \int_0^l \frac{N^2}{2EA}\,dz$$

Potential Energy due to torsion: $u_i = \frac{1}{2}\left(\sigma_x \varepsilon_x + \sigma_y \varepsilon_y + \sigma_z \varepsilon_z + \tau_{xy}\gamma_{xy} + \tau_{xz}\gamma_{xz} + \tau_{yz}\gamma_{yz}\right)$

If the cause of the shear stress is torsion, then the energy density (energy for unit volume) is as follows:



$$\tau = \frac{M_b}{I_o} r \qquad u_i = \frac{1}{2}\tau\gamma = \frac{1}{2}\left(\frac{M_b}{I_o} r\right)\left(\frac{M_b}{G\,I_o} r\right) = \frac{1}{2}\frac{M_b^2 r^2}{G\,I_o^2}$$

The energy in the volume dV is $dU_i = \frac{1}{2}\tau\gamma dV = \frac{1}{2}\frac{M_b^2 r^2}{G\,I_o^2} dx.dy.dz = \frac{1}{2}\frac{M_b^2 r^2}{G\,I_o^2} dA.dz$

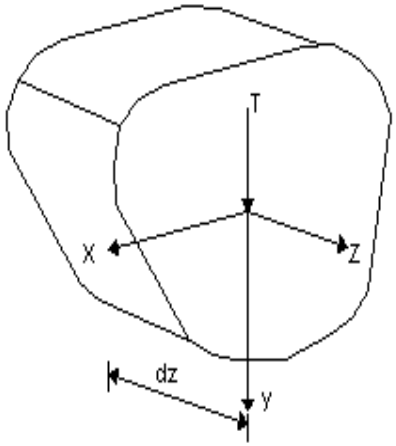
The energy in the beam of length dz is $\frac{1}{2}\frac{M_b^2}{G\,I_o^2}\int_A r^2 dA.dz = \frac{1}{2}\frac{M_b^2}{G\,I_o}.dz$

The energy in the whole beam, whose length is *l*, is $U_i = \int_0^l \frac{1}{2}\frac{M_b^2}{G\,I_o}.dz$

Potential Energy due to Shear Force: $$u_i = \frac{1}{2}\left(\sigma_x \varepsilon_x + \sigma_y \varepsilon_y + \sigma_z \varepsilon_z + \tau_{xy}\gamma_{xy} + \tau_{xz}\gamma_{xz} + \tau_{yz}\gamma_{yz}\right)$$

If the cause of the shear stresses in the above expression is shear force, then the energy density for such stresses is given as follows:

$$\tau = \left(T\,S_x\right)/\left(bI_x\right)$$



Energy density: $$u_i = \frac{1}{2}\tau\gamma = \frac{1}{2}\frac{T\,S_x}{bI_x}\frac{T\,S_x}{GbI_x} = \frac{1}{2}\frac{T^2\,S_x^2}{Gb^2I_x^2} \qquad \gamma = \tau/G$$

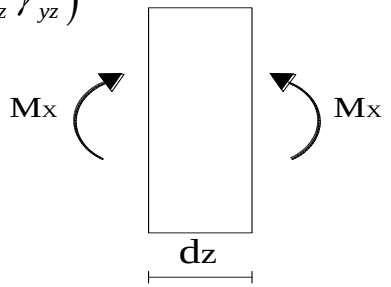
Energy dU stored in the volume dV: $$\mathrm{d}U_i = \frac{1}{2}\frac{T^2\,S_x^2}{G\,b^2\,I_x^2}\,\mathrm{d}V$$

Energy stored in the beam of length dz: $$\int_A \frac{1}{2}\frac{T^2\,S_x^2}{Gb^2I_x^2}\,\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z = \frac{T^2}{2G\,I_x^2}\int_A \frac{S_x^2}{b^2}\,\mathrm{d}A\,\mathrm{d}z \qquad k' = \frac{A}{I_x^2}\int_A \frac{S_x^2}{b^2}\,\mathrm{d}x\,\mathrm{d}y$$

Energy stored in the beam of length *l*: $$U_i = \int_0^l \frac{k'T^2}{2GA}\mathrm{d}z$$

Potential Energy due to Bending Moment: $$u_i = \frac{1}{2}\left(\sigma_x \varepsilon_x + \sigma_y \varepsilon_y + \sigma_z \varepsilon_z + \tau_{xy}\gamma_{xy} + \tau_{xz}\gamma_{xz} + \tau_{yz}\gamma_{yz}\right)$$

If the normal stresses are because of the bending moments, then the internal energy can be found as follows:



$$\sigma_z = \frac{M_x}{I_x}y \qquad \mathrm{d}U_i = M_x^2/(2.E.I_x^2).y^2\mathrm{d}V = M_x^2/(2.EI_x^2).y^2.\mathrm{d}A.\mathrm{d}z$$

$$\frac{M_x^2}{2EI_x^2}\int_A y^2\,\mathrm{d}A\,\mathrm{d}z = M_x^2/(2EI_x)\,\mathrm{d}z \qquad U_i = \int_0^l M^2/(2EI_x).\mathrm{d}z$$

If all the effects are existent at the same time, then the energy density can be expressed as follows:

$$U_i = 1/2\int_0^l \left[M_x^2/(EI_x) + M_b^2/(GI_o) + N^2/(EA) + k'\,T^2/(G.A)\right]\mathrm{d}z$$ The signs are always positive.

$$U_i = 1/2 \int_0^l (M_x^2/(EI_x) + M_b^2/(GI_o) + N^2/(EA) + k\ .T^2/(G.A)).\mathrm{d}z$$

The share of the bending moments and torsion are greater than the others in the terms given above. Later comes normal force and shear force. The effect of the shear force is in the order of %1-%2 of the effect of the bending moment.

If the energies of the same kind of effects are calculated, then the separate energies of each effect can not be summed. Namely the sum of the separate energies of the moments M_1 and M_2 is not equal to the energy of the moments when they act at the same time. The sum of separate energies is $M_1^2 + M_2^2$, and when they act together it is $(M_1 + M_2)^2$

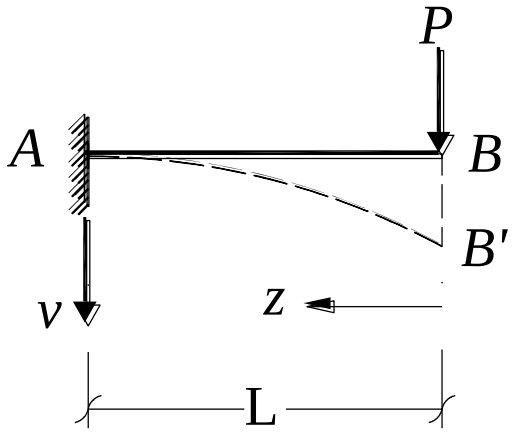
If the mechanical energy is conserved, then the work of external forces is stored as internal elastic potential energy in the elastic system. $W_d = U_i$

W_d : *Work of external forces*

U_i : Elastic potential energy because of internal forces

Problem

The cantilever beam shown on the right hand side is loaded with the force P at the free end. Find the displacement v_B at the end B by equating the work of external forces to the elastic potential energy of the internal forces. Omit the elastic potential of the shear forces.



Solution

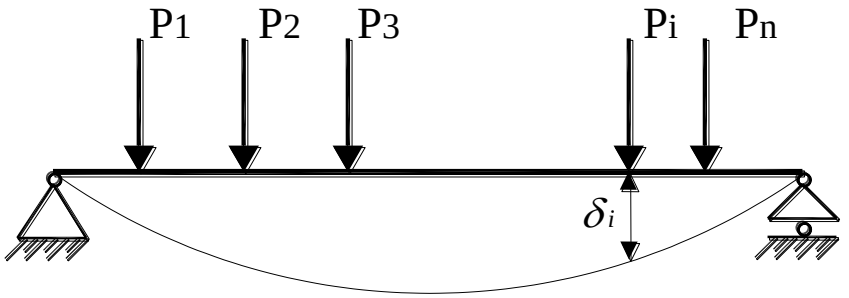
$$U_i = \int_0^l \frac{M^2}{2EI}\,dz = \int_0^l \frac{(-Pz)^2}{2EI}\,dz$$

$$U_i = \frac{P^2 l^3}{6EI}; \qquad W_d = \frac{Pv_B}{2} \quad (\text{static loading})$$

$$W_d = U_i \quad \Rightarrow \quad \frac{Pv_B}{2} = \frac{P^2 l^3}{6EI} \Rightarrow v_B = \frac{Pl^3}{3EI}$$

General expression of the displacements

Due to Hooke' law, forces are proportional to displacements and therefore a typical displacement δ_i can be expressed as follows:



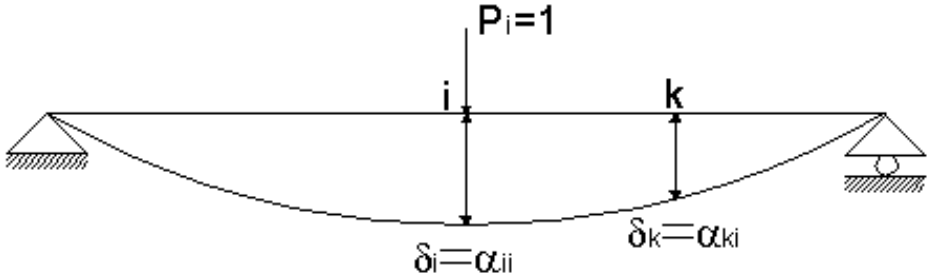
$$\delta_i = c_1 P_1 + c_2 P_2 + c_3 P_3 + ... + c_n P_n$$

$c_1, c_2, ..., c_n$ constants are called proportionality constants which depend on the geometry and elastic properties of the body.

By using the designation given above, if we want to determine the displacements δ_i at n points, we must use n symbols. To avoid this situation, we define the proportionality constants with two indices.

$$\delta_i = \alpha_{i1} P_1 + \alpha_{i2} P_2 + ... + \alpha_{ii} P_i + ... + \alpha_{in} P_n = \sum_{m=1}^{n} \alpha_{im} P_m$$

$$\delta_1 = \alpha_{11} P_1 + \alpha_{12} P_2 + ... + \alpha_{1n} P_n$$
$$\delta_2 = \alpha_{21} P_1 + \alpha_{22} P_2 + ... + \alpha_{2n} P_n$$
$$\delta_i = \alpha_{i1} P_1 + \alpha_{i2} P_2 + ... + \alpha_{in} P_n$$
$$\delta_n = \alpha_{n1} P_1 + \alpha_{n2} P_2 + ... + \alpha_{nn} P_n$$



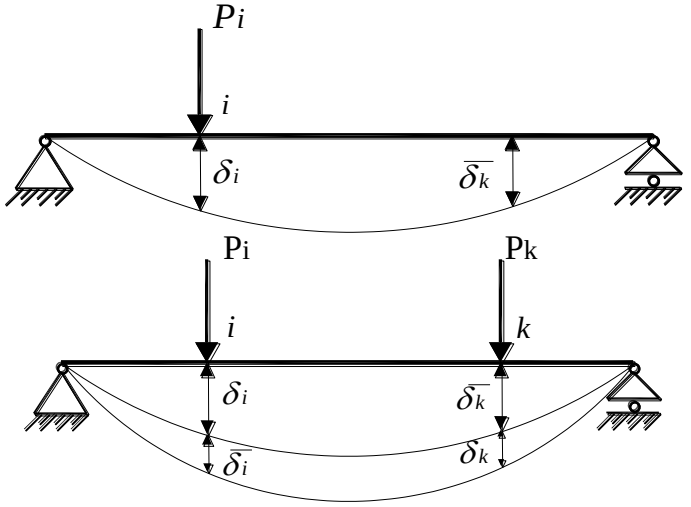
$$\delta_i = \sum_{m=1}^{n} \alpha_{im} P_m = \alpha_{ii} 1 = \alpha_{ii} \qquad \delta_k = \sum_{m=1}^{n} \alpha_{km} P_m = \alpha_{ki} 1 = \alpha_{ki}$$

$$\delta_i = \alpha_{ii} \qquad \delta_k = \alpha_{ki}$$

Maxwell's Reciprocity Theorem

$$\delta_i = \sum_{k=1}^{n} \alpha_{ik} P_k \qquad \delta_k = \sum_{k=1}^{n} \alpha_{ki} P_i$$

In a beam which is in equilibrium, the displacement at the point k and in the direction k caused by a unit force at the point i and in the direction i (α_{ki}) is equal to the displacement at the point i and in the direction i caused by a unit force at the point k and in the direction k (α_{ik}).



Proof: First the force Pi is applied at the point i;

Work at the point i: $\delta_i = \alpha_{ii} P_i$ Work= $\frac{1}{2} P_i \delta_i = \frac{1}{2} \alpha_{ii} P_i^2$

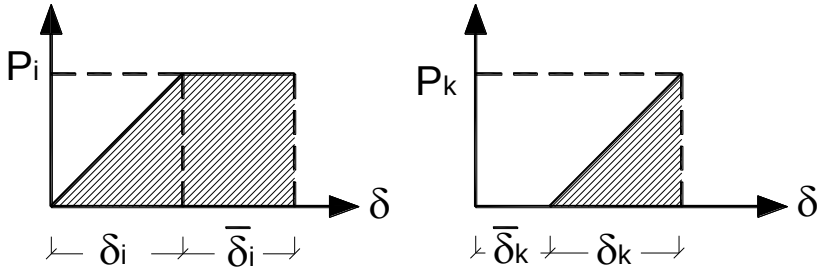
Work at he point k: $\bar{\delta}_k = \alpha_{ki} P_i$ $\text{Work} = 0$

When the force Pi is existent on the beam, the force Pk is applied at the point k:

$\bar{\delta}_i = \alpha_{ik} P_k$ Work at the point i: $Work = P_i \times \bar{\delta}_i = \alpha_{ik} \times P_i \times P_k$

Work done by Pi during the displacement caused by Pk is passive work hence there is no coefficient of ½ .

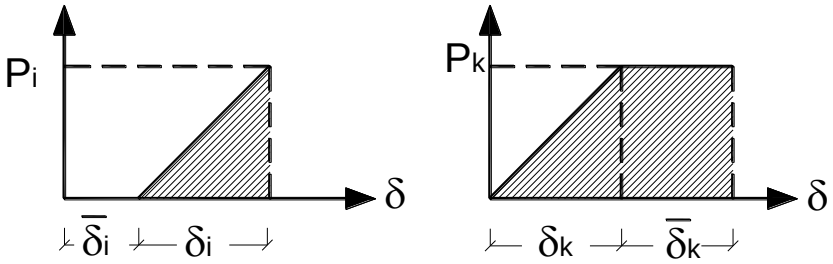
Work at he point k: $\delta_k = \alpha_{kk} P_k$, Work= $\frac{1}{2} P_k \delta_k = \frac{1}{2} \alpha_{kk} P_k^2$



Total work: $\frac{1}{2} \alpha_{ii} P_i^2 + \alpha_{ik} P_i \; P_k + \frac{1}{2} \alpha_{kk} P_k^2$

If the same procedure is followed by reversing the application sequence of the forces yields

Total work: $\frac{1}{2} \alpha_{ii} P_i^2 + \alpha_{ki} P_i \; P_k + \frac{1}{2} \alpha_{kk} P_k^2$



Work expressions must be equivalent:

$$\frac{1}{2} \alpha_{ii} P_i^2 + \alpha_{ik} P_i \; P_k + \frac{1}{2} \alpha_{kk} P_k^2 = \frac{1}{2} \alpha_{ii} P_i^2 + \alpha_{ki} P_i \; P_k + \frac{1}{2} \alpha_{kk} P_k^2 \rightarrow \alpha_{ik} = \alpha_{ki}$$

This property is known as Maxwell's reciprocity theorem.

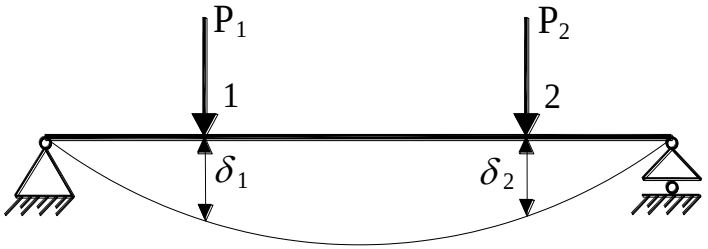
Generalized expression of work of external forces:

Lets calculate the work of external forces also by considering Maxwell's reciprocity theorem. $(\alpha_{ik} = \alpha_{ki})$

Let the forces seen in the figure be affected on the beam statically and at the same time. İn this case the work of external forces can be expressed as follows:

$$W_d = \frac{1}{2} P_1 \delta_1 + \frac{1}{2} P_2 \delta_2 = \frac{1}{2} P_1 \left(\alpha_{11} P_1 + \alpha_{12} P_2 \right) + \frac{1}{2} P_2 \left(\alpha_{21} P_1 + \alpha_{22} P_2 \right)$$

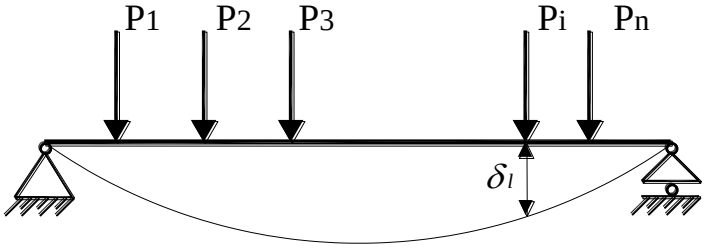
$$W_d = \frac{1}{2} \alpha_{11} P_1^2 + \frac{1}{2} \alpha_{12} P_1 P_2 + \frac{1}{2} \alpha_{21} P_1 P_2 + \frac{1}{2} \alpha_{22} P_2^2$$



By comparing the work expressions for the forces applied at the same time and for the forces applied in sequence, it is seen that the sum of the works is equal to each other.

If there are n forces affecting on the beam, the work expression becomes as follows:

$$W_d = \frac{1}{2} \sum_{i=1}^{n} P_i \delta_i = \frac{1}{2} \sum_{i=1}^{n} P_i \sum_{k=1}^{n} \alpha_{ik} P_k = \frac{1}{2} \sum_{k=1}^{n} \sum_{i=1}^{n} \alpha_{ik} P_i P_k$$



First Castigliano's theorem: In an elastic system which is in equilibrium under external loads, partial derivative of the external work with respect to one of the external loads gives the displacement in the direction of the load at the application point of the load.

$$\partial W / \partial P_i = \delta_i$$

Proof: By considering also $\alpha_{ik} = \alpha_{ki}$

$$\frac{\partial W_d}{\partial P_1} = \frac{\partial\left(\frac{1}{2}\alpha_{11}P_1^2 + \frac{1}{2}\alpha_{12}P_1P_2 + \frac{1}{2}\alpha_{21}P_1P_2 + \frac{1}{2}\alpha_{22}P_2^2\right)}{\partial P_1} = \frac{2}{2}\alpha_{11}P_1 + \frac{1}{2}\alpha_{12}P_2 + \frac{1}{2}\alpha_{21}P_2 + 0 = \alpha_{11}P_1 + \alpha_{12}P_2 = \delta_1 \qquad K = \frac{1}{2}\alpha_{22}P_2^2$$

The first theorem of Castigliano is valid only for the linear elastic bodies which obey Hooke's law.

$$W = \frac{1}{2}[(2\alpha_{i1}P_iP_1 + 2\alpha_{i2}P_iP_2 + \ldots\ldots\ldots + 2\alpha_{ii\text{-}1}P_iP_{i\text{-}1} + \alpha_{ii}P_i^2 + \ldots\ldots + 2\alpha_{in}P_iP_n) + K]$$

$$\frac{\partial W}{\partial P_i} = \frac{1}{2}[(2\alpha_{i1}P_1 + 2\alpha_{i2}P_2 + \ldots\ldots\ldots + 2\alpha_{in}P_n)] = \sum_{k=1}^{n}\alpha_{ik}P_k = \delta_i \qquad \partial K / \partial P_i = 0$$

Application of the first theorem of Castigliano to a beam under the effect of bending moment and shear force My=0, Qx=0, Principal coordinate axes are chosen

When mechanical energy is conserved,

$$\frac{\partial W}{\partial P_i} = \frac{\partial U_i}{\partial P_i} = \delta_i$$

Second solution way,

$$\delta_k = \frac{1}{2}\frac{\partial}{\partial P_k}\int_0^l\left(\frac{M_x^2}{EI_x} + \frac{k'Q_y^2}{GA}\right)\mathrm{d}z = \int_0^l\left(\frac{M_x}{EI_x}\frac{\partial M_x}{\partial P_k} + \frac{k'Q_y}{GA}\frac{\partial Q_y}{\partial P_k}\right)\mathrm{d}z$$

First Castigliano's Theorem:

$$W = W^*$$

$$dW = dW^*$$

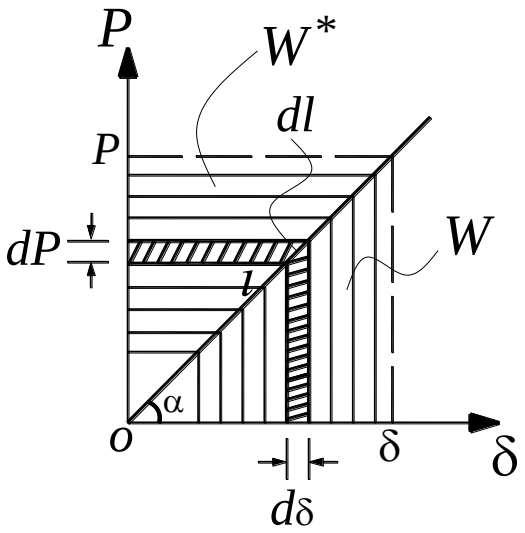
$$P.d\delta = \delta.dP$$

$$(\ell.\sin\alpha).(d\ell.\cos\alpha) = P.d\delta$$

$$(\ell.\cos\alpha).(d\ell.\sin\alpha) = \delta.dP$$

$$\delta.dP = dW$$

$$\frac{dW}{dP} = \delta$$

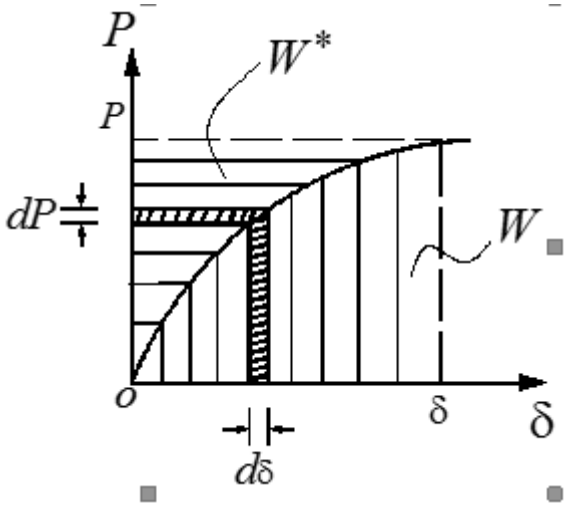


Second Castigliano's theorem: Partial derivative of the external work with respect to one of the displacements gives the external load which is in the same direction and point with the displacement.

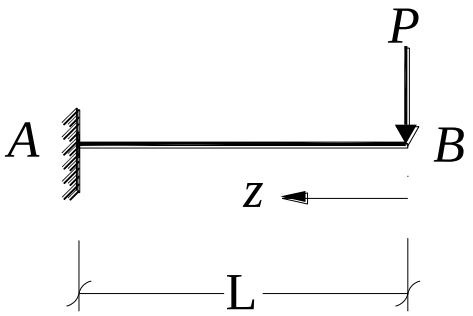
$W_d = \frac{1}{2}\sum_{i=1}^{n} P_i\ \delta_i$ Second Castigliano's theorem is general. Proof: $P_j = \frac{\partial W_i}{\partial \delta_j}$ $\mathrm{d}W_i = P\,\mathrm{d}\delta \rightarrow P = \frac{\mathrm{d}W_i}{\mathrm{d}\delta}$

According to this result, the second Castigliano's theorem is valid whether or not there is a proportionality relation between P and δ.

$$dW = P.d\delta \rightarrow \frac{dW}{d\delta} = P$$



Problem : Find the displacement v_B of the cantilever beam loaded with P.

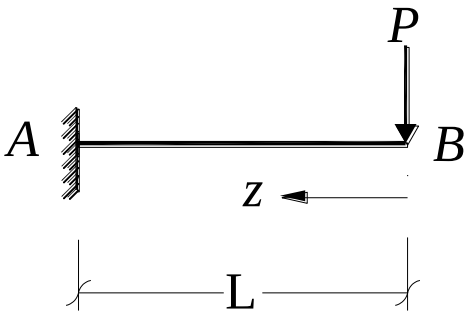


First solution way:

$$U_i = \int_0^l \frac{M^2}{2EI} dz \qquad M = -Pz \Rightarrow U_i = \frac{1}{2EI} \int_0^l P^2 z^2 dz$$

$$U_i = \frac{P^2 . l^3}{6EI}; \; v_B = \frac{\partial U_i}{\partial P} = \frac{Pl^3}{3EI}$$

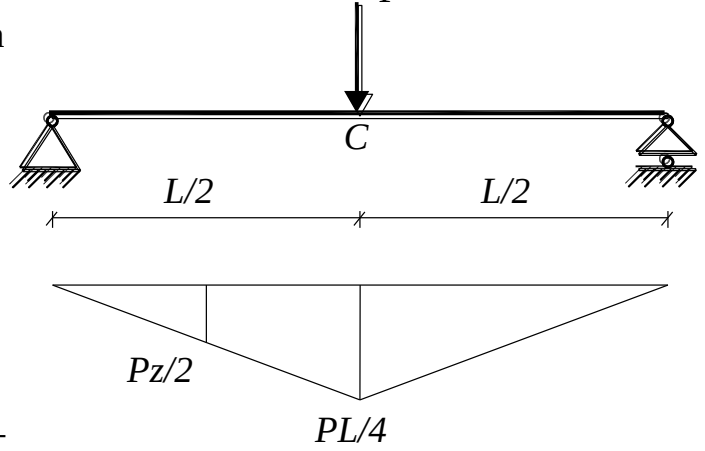
$$\delta_k = \frac{1}{2} \frac{\partial}{\partial P_k} \int_0^l \left(\frac{M_x^2}{EI_x} + \frac{k' Q_y^2}{GA} \right) \mathrm{d}\, z = \int_0^l \left(\frac{M_x}{EI_x} \frac{\partial M_x}{\partial P_k} + \frac{k' Q_y}{GA} \frac{\partial Q_y}{\partial P_k} \right) \mathrm{d}\, z$$



Second solution way:

$$M = -Pz \Rightarrow \delta_k = \int_0^L \left(\frac{M_x}{EI_x} \frac{\partial M_x}{\partial P_k} \right) \mathrm{d}\, z = \frac{1}{EI} \int_0^L -P\, z(-z) dz = \frac{P}{EI} \int_0^L z^2 dz = \frac{PL^3}{3EI}$$

Problem: Find the displacement of the midpoint of the simple beam given on the right hand side.



Solution: $M - P\frac{z}{2} = 0 \Rightarrow M = P\frac{z}{2}$

$$U_i = 2\frac{1}{2}\int\left(\frac{M^2}{EI}\right)dz = \frac{1}{EI}\int\left(\frac{Pz}{2}\right)^2 dz$$

$$U_i = \frac{P^2}{4EI}\int_0^{l/2} z^2 dz = \frac{P^2 z^3}{12EI}\Big|_0^{l/2} = \frac{P^2 l^3}{96EI} \rightarrow \delta = \frac{\partial U_i}{\partial P} = \frac{Pl^3}{48EI}$$

Second solution way: $between\ z = 0\ and\ L/2 \quad M = P\,z/2$

$$between\ z = L/2\ and\ L \quad M = mx + n,\ at\ z = L/2 \rightarrow M = m\frac{L}{2} + n = \frac{PL}{4}$$

$$at\ z = L \rightarrow M = mL + n = 0 \rightarrow m = -n/L$$

$$-\frac{n}{L}\frac{L}{2} + n = \frac{PL}{4} \rightarrow -\frac{n}{2} + n = \frac{PL}{4} \rightarrow n = PL/2 \quad m = -\frac{n}{L} = \frac{-PL/2}{L} = -\frac{P}{2}$$

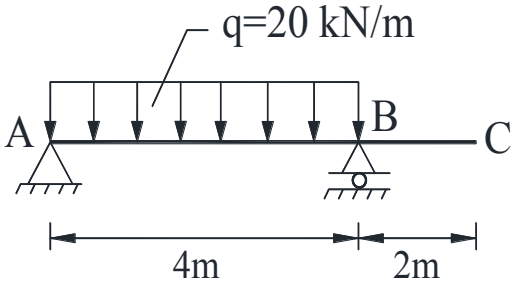
$$M = -\frac{P}{2}z + \frac{PL}{2}$$

$$\delta_C = \int_0^L \left(\frac{M_x}{EI_x}\frac{\partial M_x}{\partial P_C}\right)\mathrm{d}z = \frac{1}{EI}\int_0^{L/2}\frac{Pz}{2}\left(\frac{z}{2}\right)dz + \frac{1}{EI}\int_{L/2}^{L}\left(-\frac{P}{2}z + \frac{PL}{2}\right)\left(-\frac{z}{2} + \frac{L}{2}\right)dz = \frac{1}{EI}\left|\frac{Pz^3}{12}\right|_0^{L/2} + \frac{1}{EI}\left|\frac{Pz^3}{12} - \frac{PLz^2}{4} + \frac{PL^2 z}{4}\right|_{L/2}^{L} = \frac{PL^3}{48EI}$$

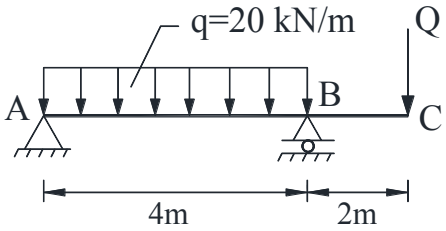
Due to symmetry, we can use the factor 2 to obtain the displacement by using only one half of the beam and then multiplying the obtained value by 2. Also, we can take the origin of the coordinate axes at the right end of the beam for the second region.

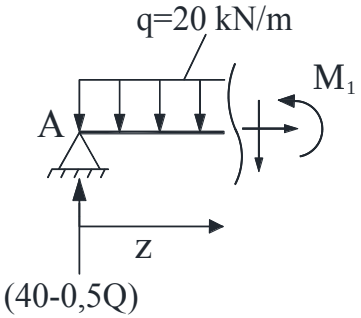
$$\delta_C = \int_0^L \left(\frac{M_x}{EI_x}\frac{\partial M_x}{\partial P_k}\right)\mathrm{d}z = \frac{2}{EI}\int_0^{L/2}\frac{Pz}{2}\left(\frac{z}{2}\right)dz = \frac{2}{EI}\left|\frac{Pz^3}{12}\right|_0^{L/2} = \frac{PL^3}{48EI}$$

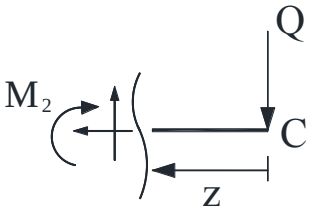
Problem: The overhanging beam shown in the figure has a constant bending rigidity of *EI*. Find the displacement at the C point by employing **CASTIGLIANO**'s theorem.



Solution:







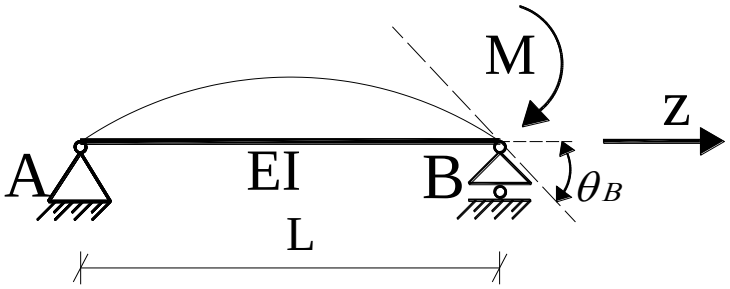
$$\sum M_B = 0 \rightarrow A_y\,4 - 20.4.2 + Q.2 = 0$$
$$\rightarrow A_y = 40 - 0,5Q$$
$$\sum F_y = 0 \rightarrow A_y + B_y - 20.4 - Q = 0$$
$$\rightarrow B_y = 40 + 1,5Q$$

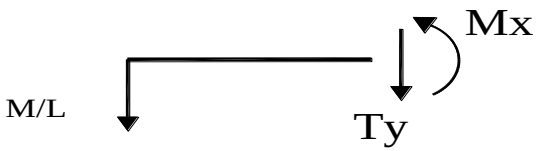
$$M_1 = -10\,z^2 + 40z - 0,5Q\,z$$
$$\frac{\partial M_1}{\partial Q} = -0,5\,z$$

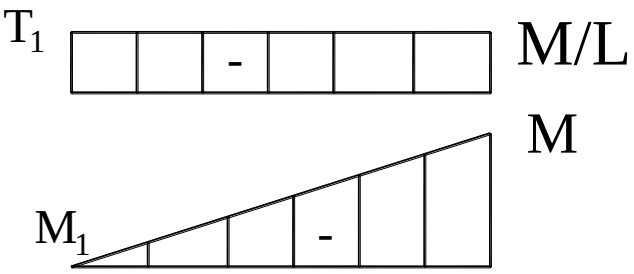
$$M_2 = -Q\,z$$
$$\frac{\partial M_2}{\partial Q} = -z$$

$$\delta_C = \frac{1}{EI}\int_0^4 (-10\,z^2 + 40z - 0,5Q\,z).(-0,5\,z)\,dz + \frac{1}{EI}\int_0^2 (-Q\,z).(-z)\,dz = -\frac{106,67}{EI}$$

Problem 1.7: The simple beam given below is loaded with a concentrated moment at the point B. Find the rotation angle of the end point B of the beam by using Castigliano's theorem..





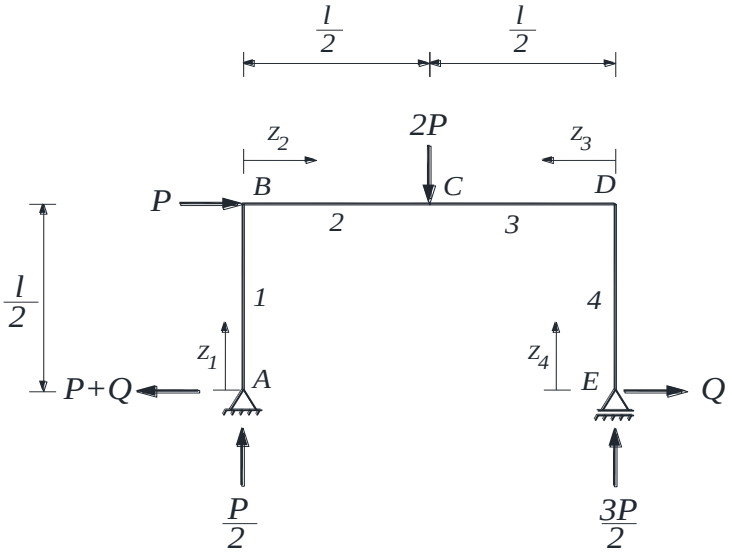


$$-\frac{M}{L}z - M_x = 0 \rightarrow M_x = -\frac{M}{L}z$$

$$\theta_B = \int_0^L \left(\frac{M_x}{EI_x}\frac{\partial M_x}{\partial M} \right) \mathrm{d}z = \frac{1}{EI}\int_0^L -\frac{M\,z}{L}(-\frac{z}{L})dz = \frac{M}{EIL^2}\int_0^L z^2 dz = \frac{ML}{3EI}$$

EXAMPLE

Find the horizontal displacement at the roller support E by using the Castigliano method for the frame given with the dimensions and loading.



$$U_i = \int \frac{M^2}{2EI}\, dz$$

SOLUTION

$$M_1 = (P+Q)\, z_1 \quad U_1 = \frac{1}{2EI}\int_0^{\ell/2} (P+Q)^2\, z_1^2 dz_1 =$$

$$\frac{1}{2EI}(P+Q)^2\, \frac{z_1^3}{3}\bigg|_0^L = \frac{1}{2EI}\cdot\frac{(P+Q)^2\cdot l^3}{24}$$

$$M_2 = (P+Q)\,\frac{\ell}{2} + \frac{P}{2} z_2 \qquad U_2 = \frac{1}{2EI}\int_0^{\ell/2}\left[(P+Q)\,\frac{l}{2} + \frac{P}{2} z_2\right]^2 dz_2 = \frac{1}{2EI}\left[(P+Q)^2\,\frac{l^3}{8} + (P+Q)\,\frac{Pl^3}{16} + \frac{P^2 l^3}{96}\right]$$

$$M_3 = Q\,\frac{\ell}{2} + \frac{3P}{2} z_3 \qquad U_3 = \frac{1}{2EI}\int_0^{\ell/2}\left(Q\,\frac{\ell}{2} + \frac{3P}{2} z_3\right)^2 dz_3 = \frac{1}{2EI}\left(\frac{Q^2 l^3}{8} + \frac{3QP\, l^3}{16} + \frac{3P^2\, l^3}{32}\right)$$

$$M_4 = Q\, z_4 \qquad U_4 = \frac{1}{2EI}\int_0^{l/2} (Q z_4)^2 dz_4 = \frac{1}{2EI}\left(\frac{Q^2 l^3}{24}\right)$$

Total strain energy:

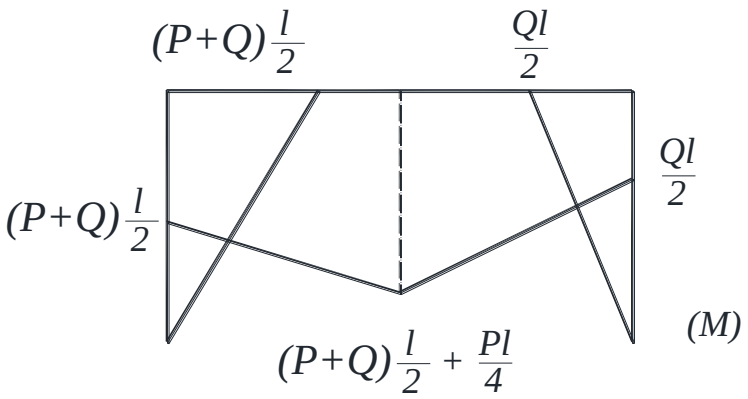
$$U = \frac{1}{2EI}\left[(P+Q)^2\,\frac{l^3}{24} + (P+Q)^2\,\frac{l^3}{8} + (P+Q)\,\frac{Pl^3}{16} + \frac{P^2 l^3}{96} + \frac{Q^2 l^3}{8} + \frac{3QPl^3}{16} + \frac{3P^2 l^3}{32} + \frac{Q^2 l^3}{24}\right]$$

$$\frac{\partial U}{\partial Q} = \frac{1}{2EI}\left[2(P+Q)\,\frac{l^3}{24} + 2(P+Q)\,\frac{l^3}{8} + \frac{Pl^3}{16} + 0 + \frac{2Q l^3}{8} + \frac{3P l^3}{16} + 0 + \frac{2Q l^3}{24}\right]$$

δ_{HE} displacement can be found by substituting Q = O in the equation.

$$\delta_{HE} = \frac{1}{2EI}\left(\frac{Pl^3}{12} + \frac{P l^3}{4} + \frac{P l^3}{16} + \frac{3P l^3}{16}\right) = \frac{1}{2EI}\left(\frac{7P l^3}{12}\right) \rightarrow \delta_{HE} = \frac{7P l^3}{24EI}$$

This problem can also be solved by using the tables as below.



	k	k	k	k	k
i	lik	$1/2\,lik$	$1/2\,li(k_1+k_2)$	$2/3\,lik$	
i	$1/2\,lik$	$1/3\,lik$	$1/6\,li(k_1+2k_2)$	$1/3\,lik$	
i	$1/2\,lik$	$1/6\,lik$	$1/6\,li(2k_1+k_2)$	$1/3\,lik$	
i_1	$1/2\,l(i_1+i_2)k$	$1/6\,l(i_1+2i_2)k$	$1/6\,l(2i_1k_1+i_1k_2+i_2k_1+2i_2k_2)k$	$1/3\,l(i_1+i_2)k$	
i	$2/3\,lik$	$1/3\,lik$	$1/3\,li(k_1+k_2)$	$8/15\,l\,i\,k$	

$$U_1 = \frac{1}{2EI}\cdot\frac{1}{3}\cdot\frac{l}{2}(P+Q)\,\frac{l}{2}\cdot(P+Q)\,\frac{l}{2} = \frac{1}{2EI}(P+Q)^2\,\frac{l^3}{24}$$

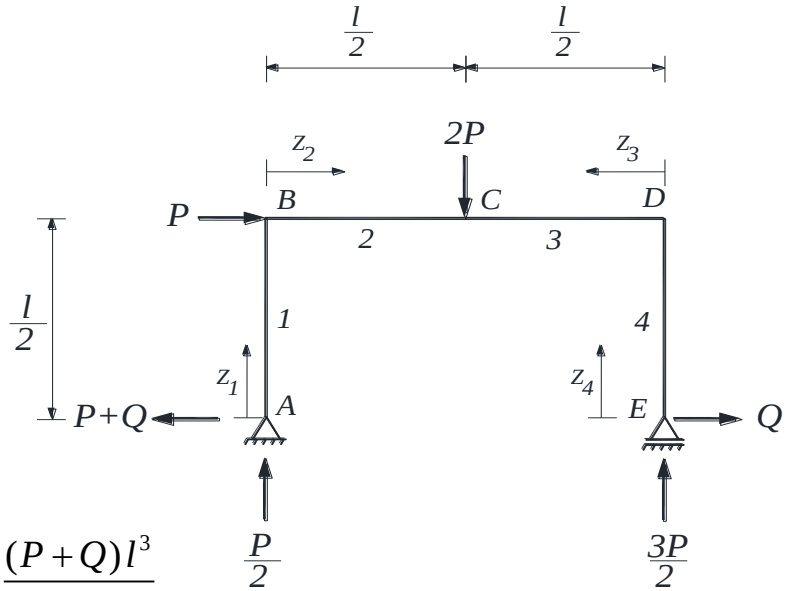
$$U_2 = \frac{1}{2EI}\cdot\frac{1}{6}\cdot\frac{\ell}{2}\left[2(P+Q)\,\frac{l}{2}\cdot(P+Q)\,\frac{l}{2}+2\cdot(P+Q)\,\frac{l}{2}\cdot\left((P+Q)\,\frac{l}{2}+\frac{Pl}{4}\right)+2\cdot\left((P+Q)\,\frac{l}{2}+\frac{Pl}{4}\right)^2\right]$$

$$U_3 = \frac{1}{2EI}\cdot\frac{1}{6}\cdot\frac{l}{2}\left[2(Q\,\frac{l}{2})^2+2\cdot Q\,\frac{l}{2}\cdot\left((P+Q)\,\frac{l}{2}+\frac{Pl}{4}\right)+2\cdot\left((P+Q)\,\frac{l}{2}+\frac{Pl}{4}\right)^2\right]$$

$$U_4 = \frac{1}{2EI}\cdot\frac{1}{3}\cdot\frac{l}{2}(Q\,\frac{l}{2})^2 = \frac{1}{2EI}\cdot\frac{Q^2l^3}{24}$$

EXAMPLE

Find the horizontal displacement at the roller support E by using the Castigliano method for the frame given with the dimensions and loading.



SOLUTION

$$\delta_Q = \int_0^L \left(\frac{M_x}{EI_x} \frac{\partial M_x}{\partial Q} \right) \mathrm{d}z$$

$$M_1 = (P+Q)\, z_1 \qquad \frac{\partial M_1}{\partial Q} = z_1 \qquad \delta_{Q_1} = \int_0^{l/2} (P+Q)\, z_1^2 dz = \frac{(P+Q)\, z_1^3}{3}\Big|_0^{l/2} = \frac{(P+Q) l^3}{24}$$

$$M_2 = (P+Q)\frac{l}{2} + \frac{P}{2} z_2 \qquad \frac{M_2}{Q} = \frac{l}{2} \qquad \delta_{Q_2} = \int_0^{L/2} \left[(P+Q)\frac{l}{2} + \frac{P}{2} z_2 \right] \times \frac{l}{2} dz_2 = \frac{(P+Q)\, l^2\, z_2}{4} + \frac{Pl}{4}\frac{z_2^2}{2}\Big|_0^{l/2} = \frac{(P+Q) l^3}{8} + \frac{Pl^3}{32}$$

$$M_3 = Q\frac{l}{2} + \frac{3P}{2} z_3 \qquad \frac{M_3}{Q} = \frac{l}{2} \qquad \delta_{Q_3} = \int_0^{l/2} \left[(Q\frac{l}{2} + \frac{3P}{2} z_3 \right] \times \frac{l}{2} dz_3 = \frac{Q\, l^2\, z_3}{4} + \frac{3Pl}{4}\frac{z_3^2}{2}\Big|_0^{l/2} = \frac{Q l^3}{8} + \frac{3P l^3}{32}$$

$$M_4 = Q\, z_4 \qquad \frac{\partial M_4}{\partial Q} = z_4 \qquad \delta_{Q_4} = \int_0^{L/2} Q\, z_4 \times z_4 dz_4 = \frac{Q\, z_4^3}{3}\Big|_0^{l/2} = \frac{Q l^3}{24}$$

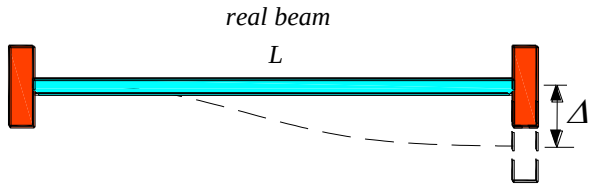
$$\delta_{HE} = \delta_{Q_1} + \delta_{Q_2} + \delta_{Q_3} + \delta_{Q_4} = \frac{1}{EI}\left[\frac{(P+Q) l^3}{24} + \frac{(P+Q) l^3}{8} + \frac{P l^3}{32} + \frac{Q l^3}{8} + \frac{3P l^3}{32} + \frac{Q l^3}{24} \right]$$

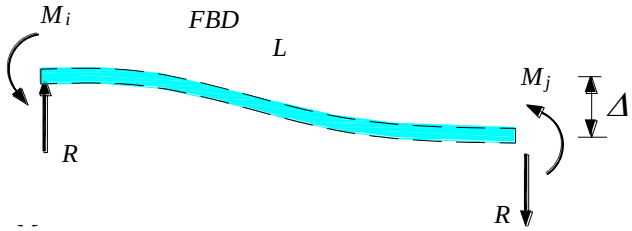
δ_{HE} displacement can be found by substituting Q = O in the equation.

$$\delta_{HE} = \frac{1}{EI}\left[\frac{Pl^3}{24} + \frac{Pl^3}{8} + \frac{Pl^3}{32} + \frac{3Pl^3}{32} \right] = \left[\frac{4Pl^3}{96} + \frac{12Pl^3}{96} + \frac{3Pl^3}{96} + \frac{9Pl^3}{96} \right] = \frac{1}{EI}\left(\frac{28Pl^3}{96}\right) \rightarrow \delta_{HE} = \frac{7Pl^3}{24EI}$$

Problem 1.1

Determine support reactions of the beam due to the unit vertical support displacement of the support B by using Castigliano method. EI=Bending stiffness (or flexural rigidity)





Solution: We choose the origin of the coordinate axes at the right support and the z axis is from right to left.

$$\theta_B = \int_0^L \left(\frac{M_x}{EI_x} \frac{\partial M_x}{\partial M_1} \right) \mathrm{d}z$$

$$M = M_1 - Rz \qquad \frac{\partial M}{\partial M_1} = 1 \qquad \theta_B = \frac{1}{EI}\int_0^{l/2} (M_1 - Rz)(1)\,dz = \frac{1}{EI}\left(M_1 z - \frac{Rz^2}{2} \right)\Bigg|_0^L = \frac{1}{EI}\left(M_1 z - \frac{Rz^2}{2} \right) = 0 \rightarrow M_1 = \frac{RL}{2}$$

$$\delta_B = \int_0^L \left(\frac{M_x}{EI_x} \frac{\partial M_x}{\partial R} \right) \mathrm{d}z$$

$$M = M_1 - Rz \qquad \frac{\partial M}{\partial R} = -z \qquad \delta_R = \frac{1}{EI}\int_0^{l/2} (M_1 - Rz)(-z)\,dz = \frac{1}{EI}\left(\frac{-M_1 z^2}{2} + \frac{Rz^3}{3} \right)\Bigg|_0^L = \frac{1}{EI}\left(\frac{-M_1 L^2}{2} + \frac{RL^3}{3} \right) = \Delta$$

$$\frac{1}{EI}\left(\frac{-(RL/2)L^2}{2} + \frac{RL^3}{3} \right) = \frac{1}{EI}\left(\frac{-RL^3}{4} + \frac{RL^3}{3} \right) = \frac{1}{EI}\left(\frac{-3RL^3}{4} + \frac{4RL^3}{3} \right) = \frac{RL^3}{12EI} = \Delta \rightarrow R = \frac{12EI}{L^3}\Delta$$

$$For\ unit\ displacement\ R_{unit} = \frac{12EI}{L^3}$$

$$M_1 = \frac{RL}{2} = \frac{\left[\left(12EI/L^3 \right)\Delta \right] L}{2} = \frac{6EI}{L^2}\Delta \rightarrow For\ unit\ displacement\ M_{1unit} = \frac{6EI}{L^2}$$

Virtual work equation

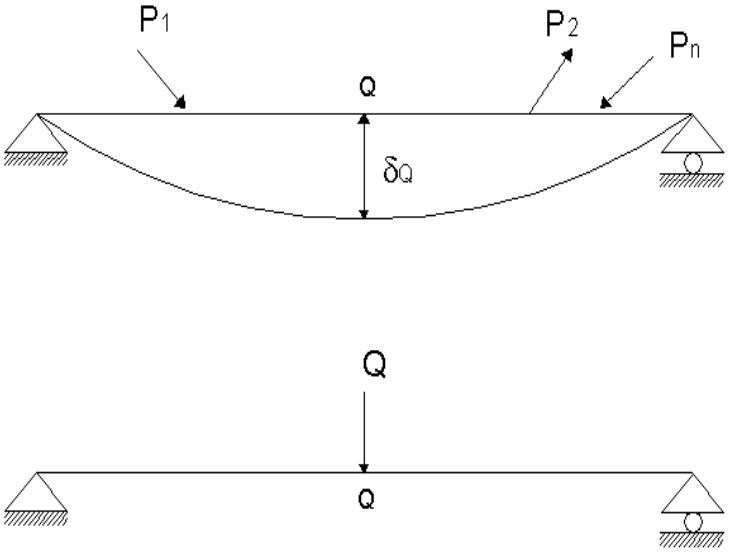
$$U_i = 1/2 \int_0^l (M_x^2/(EI_x) + M_b^2/(GI_o) + N^2/(EA) + k\ .T^2/(G.A)).\mathrm{d}z$$

$$N = N_0 + Q\bar{N}$$

$$T = T_0 + Q\bar{T}$$

$$M_b = M_{bo} + Q\bar{M}_b$$

$$M_e = M_{eo} + Q\bar{M}_e$$



Potential energy of the internal forces in the case when both loads are existent at the same time:

$$U_i = \int \left[\frac{(N_0 + Q\bar{N})^2}{2EA} + \frac{k'(T_0 + Q\bar{T})^2}{2GA} + \frac{(M_{bo} + Q\bar{M}_b)^2}{2GI_0} + \frac{(M_{eo} + Q\bar{M}_e)^2}{2EI_x} \right] dz$$

$$\delta_Q = \frac{\partial U_i}{\partial Q} = \int_0^l \left[\frac{(N_0 + Q\bar{N})\bar{N}}{EA} + \frac{k'(T_0 + Q\bar{T})\bar{T}}{GA} + \frac{(M_{b0} + Q\bar{M}_b)\bar{M}_b}{GI_o} + \frac{(M_{e0} + Q\bar{M}_e)\bar{M}_e}{EI_x} \right] \mathrm{d}z$$

$$\delta_Q = \left(\frac{\partial U_i}{\partial Q} \right)_{Q=0} = \int_0^l \left[\frac{N_0\bar{N}}{EA} + \frac{k'T_0\bar{T}}{GA} + \frac{M_{b0}\bar{M}_b}{GI_o} + \frac{M_{e0}\bar{M}_e}{EI_x} \right] \mathrm{d}z$$

Virtual work equation is obtained.

Principle of virtual work

For a body which is in equilibrium under the effect of external loads, the work of the external loads is equal to the virtual potential energy of the internal forces during virtual displacements and deformations.

$$W_{vd} = U_{vi} \qquad W_{vd} = \textit{virtual work of external loads}$$

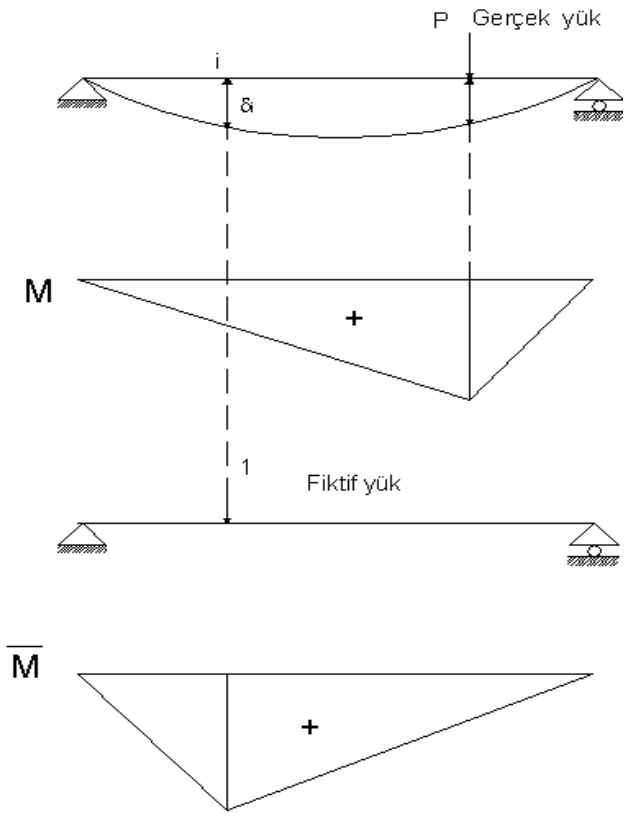
$$U_{vi} = \textit{Virtuel potential energy due to internal forces}$$

$$\delta_i = \int \left(\frac{\mathrm{M}}{\mathrm{EI}}\right)\bar{M}\mathrm{dz} \qquad N = N_0 + Q\bar{N} \qquad T = T_0 + Q\bar{T}$$

$$M_b = M_{bo} + Q\bar{M}_b \qquad M_e = M_{eo} + Q\bar{M}_e$$

$$U_i = 1/2\int_0^l (M_x^2/(EI_x) + M_b^2/(GI_o) + N^2/(EA) + k\,.T^2/(G.A)).\mathrm{d}z$$

$$\delta_{\mathrm{Q}} = \left(\frac{\partial U_i}{\partial Q}\right)_{Q=0} = \int_0^l \left[\frac{N_0\bar{N}}{EA} + \frac{k'T_0\bar{T}}{GA} + \frac{M_{b0}\bar{M}_b}{GI_o} + \frac{M_{e0}\bar{M}_e}{EI_x}\right]\mathrm{d}z$$



It is very easy to calculate these integrations by using standard tables:

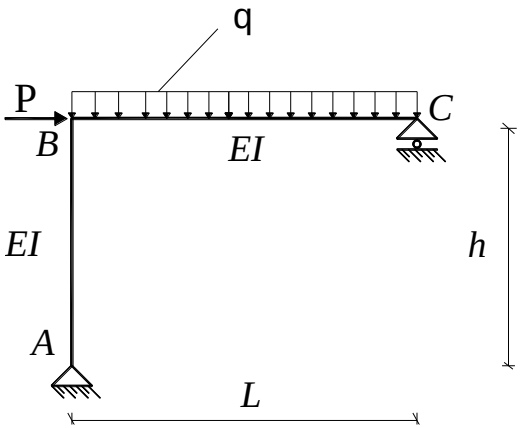
	k k	k	k_1 k_2	k
i i	lik	$1/2\,lik$	$1/2\,li(k_1+k_2)$	$2/3\,lik$
i	$1/2\,lik$	$1/3\,lik$	$1/6\,li(k_1+2k_2)$	$1/3\,lik$
i	$1/2\,lik$	$1/6\,lik$	$1/6\,li(2k_1+k_2)$	$1/3\,lik$
i_1 i_2	$1/2\,l(i_1+i_2)k$	$1/6\,l(i_1+2i_2)k$	$1/6\,l(2i_1k_1+i_1k_2+i_2k_1+2i_2k_2)k$	$1/3\,l(i_1+i_2)k$
i	$2/3\,lik$	$1/3\,lik$	$1/3\,li(k_1+k_2)$	$8/15\,l\,i\,k$

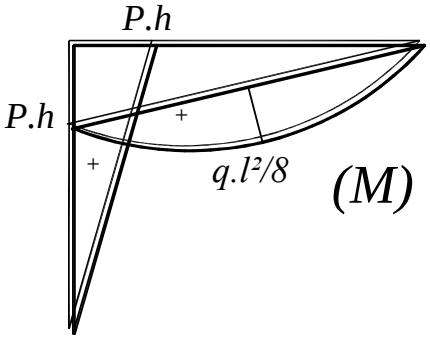
Problem: Find the horizontal displacement of the point C of the frame of given dimensions under the loads shown in the figure by using the virtual work theorem. Omit the works of normal forces and shear forces.

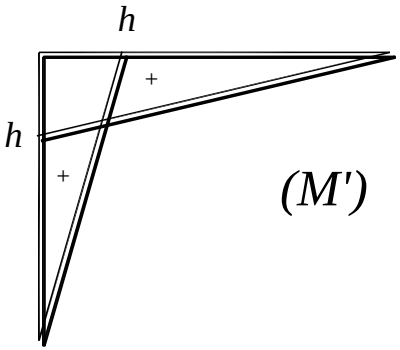
Solution: The moment diagrams for the real and fictitious loads must be drawn. For fictitious system, a unit force is affected to the point C.

Standard multiplication table will be used for the solution of this problem.

	k k	k	k_1 k_2	k
i i	lik	$1/2\,lik$	$1/2\,li(k_1+k_2)$	$2/3\,lik$
i	$1/2\,lik$	$1/3\,lik$	$1/6\,li(k_1+2k_2)$	$1/3\,lik$
i	$1/2\,lik$	$1/6\,lik$	$1/6\,li(2k_1+k_2)$	$1/3\,lik$
i_1 i_2	$1/2\,l(i_1+i_2)k$	$1/6\,l(i_1+2i_2)k$	$1/6\,l(2i_1k_1+i_1k_2+i_2k_1+2i_2k_2)k$	$1/3\,l(i_1+i_2)k$
i	$2/3\,lik$	$1/3\,lik$	$1/3\,li(k_1+k_2)$	$8/15\,l\,i\,k$



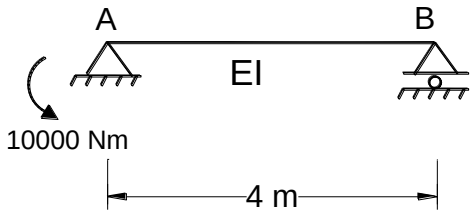




$$\delta_{HC}=\int\frac{M_x\bar{M}_x}{EI}dz=\frac{1}{EI}\left(\frac{h.Ph.h}{3}+\frac{L.Ph.h}{3}+\frac{L.qL^2}{3.8}.h\right)$$

$$\delta_{HC}=\frac{1}{EI}\left(\frac{Ph^3}{3}+\frac{Ph^2L}{3}+\frac{qL^3h}{24}\right)$$

Problem: Find the rotation of support B of the beam shown in Figure 3 using Virtual Work method.



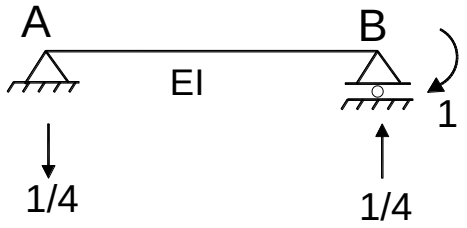
Solution:

.

Real loading

$$M_x(z) = 2500z - 10000$$

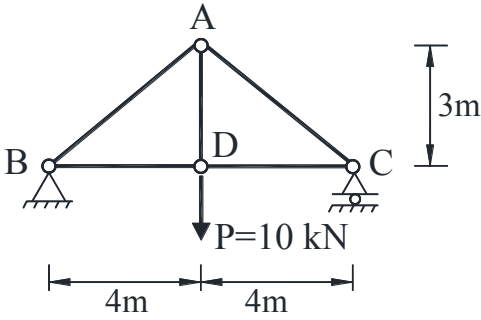
Virtual unit loading,



$$\bar{M}_x(z) = -\frac{z}{4}$$

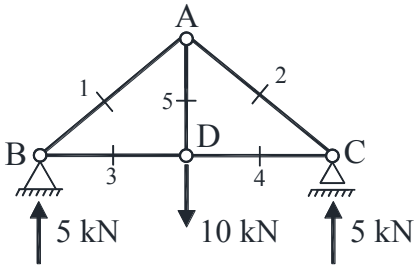
$$\theta_B = \int_0^4 \frac{2500z - 10000}{EI}\left(\frac{-z}{4}\right)dz = \frac{6666,67}{EI}$$

Problem: All the bars of the simple truss structure shown in the figure have the same, constant elongation rigidity EA. Find the horizontal displacement at point **C** and vertical displacement at point D by using **VIRTUAL WORK** equation.



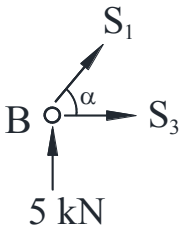
Solution:

Actual loading



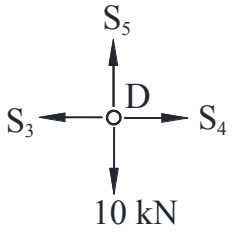
$\sum M_B = 0 \rightarrow C_y.8 - 10.4 = 0 \rightarrow C_y = 5kN$

$\sum F_y = 0 \rightarrow B_y + C_y - 10 = 0 \rightarrow B_y = 5kN$



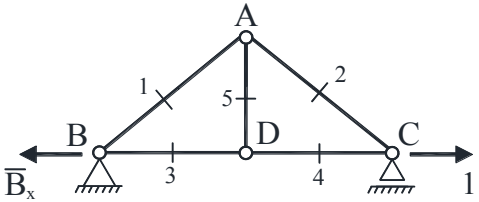
$\sum F_y = 0 \rightarrow S_1 \sin\alpha + 5 = 0 \rightarrow S_1 = -8,33kN = S_2$

$\sum F_x = 0 \rightarrow S_1 \cos\alpha + S_3 = 0 \rightarrow S_3 = +6,66kN = S_4$

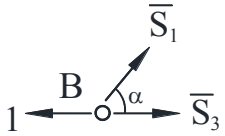


$\sum F_y = 0 \rightarrow S_5 - 10 = 0 \rightarrow S_5 = +10$

Virtual loading

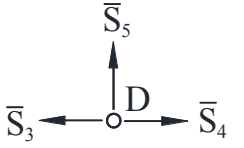


$\sum F_x = 0 \rightarrow B_x - 1 = 0 \rightarrow B_x = 1kN$



$\sum F_y = 0 \rightarrow \bar{S}_1 \sin\alpha = 0 \rightarrow \bar{S}_1 = 0 = \bar{S}_2$

$\sum F_x = 0 \rightarrow \bar{S}_1 \cos\alpha + \bar{S}_3 - 1 = 0 \rightarrow \bar{S}_3 = +1 = \bar{S}_4$



$\sum F_y = 0 \rightarrow \bar{S}_5 = 0$

Horizontal displacement of C point:

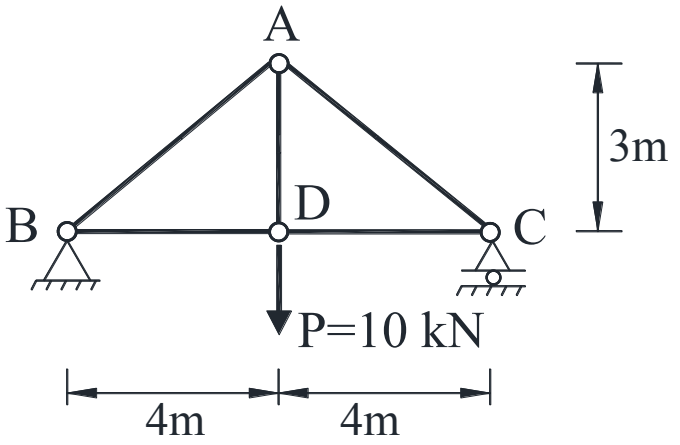
$$(\delta_C)_{horizontal} = \frac{(-8,33).(0).5}{EA}.2 + \frac{(6.66).(1).4}{EA}.2 + \frac{(10).(0).3}{EA} \rightarrow (\delta_C)_{horizontal} = \frac{53,28}{EA}$$

Unit Loading;

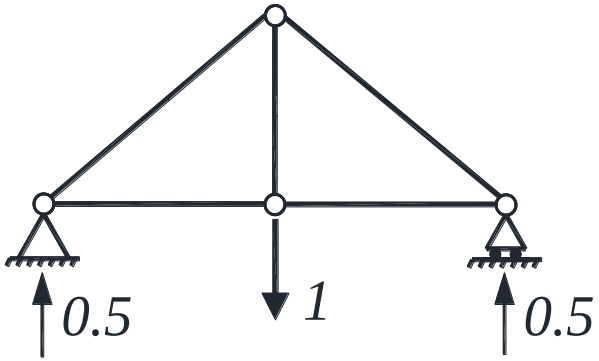
$S_1 = -8,33kN = S_2$

$S_3 = +6,66kN = S_4$

$S_5 = +10$



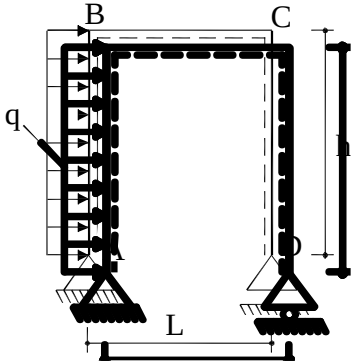
Bar forces will be 1/10th of the real values.



$\bar{S}_1 = \bar{S}_2 = -0,833 \quad \bar{S}_3 = \bar{S}_4 = -0,667 \quad \bar{S}_5 = 1$

$$S_D = \frac{(-8,33).(-0,833)}{EA}.5 + \frac{(-8,33).(-0,833)}{EA}.5 + \frac{(6,67).(0,667)}{EA}.4 + \frac{(6,67).(0,667)}{EA}.4 + \frac{10.1}{EA}.3 \Rightarrow S_D = \frac{135}{EA}$$

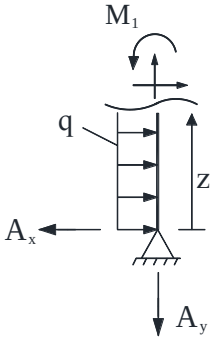
Problem: Find the horizontal displacement of D point of the system shown in the figure $(\delta_D)_{horizontal}$ by using the **VIRTUAL WORK THEOREM**.

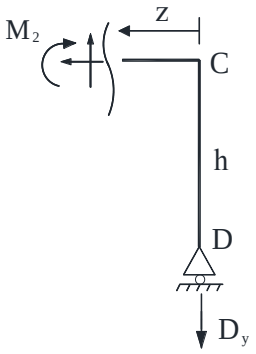


Solution:

$$\sum M_A = 0 \rightarrow D_y L - qh\frac{h}{2} = 0 \rightarrow D_y = \frac{qh^2}{2L} \qquad \sum F_y = 0 \rightarrow A_y + D_y = 0 \rightarrow A_y = -\frac{qh^2}{2L} \qquad \sum F_x = 0 \rightarrow A_x - qh = 0 \rightarrow A_y = qh$$

Actual loading:



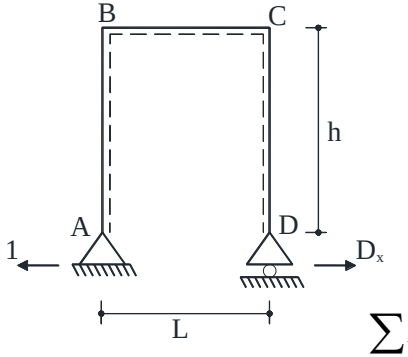


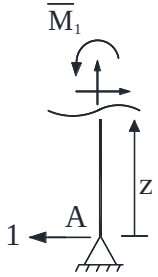


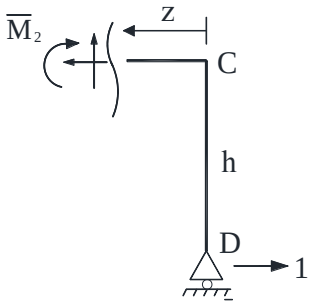
$$\sum M = 0 \rightarrow M_1 + qz\frac{z}{2} - A_x z = 0 \qquad \sum M = 0 \rightarrow M_2 - D_y.z = 0 \rightarrow M_2 = \frac{qh^2}{2L}z \qquad \sum M = 0 \rightarrow M_3 = 0$$

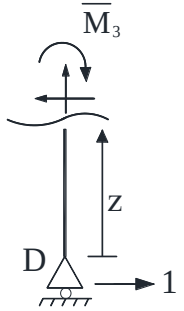
$$M_1 = qhz - \frac{qz^2}{2}$$

Virtual loading:





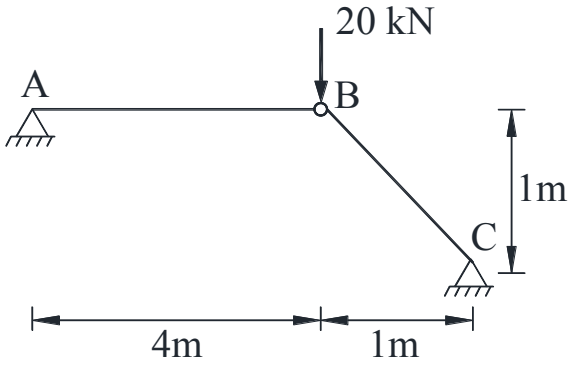




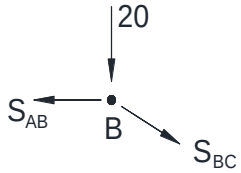
$$\sum M = 0 \rightarrow \overline{M}_1 - 1.z = 0 \rightarrow \overline{M}_1 = z \qquad \sum M = 0 \rightarrow \overline{M}_2 - 1.h = 0 \rightarrow \overline{M}_2 = h \qquad \sum M = 0 \rightarrow \overline{M}_3 - 1.z = 0 \rightarrow \overline{M}_3 = z$$

$$(\delta_D)_{horizontal} = \frac{1}{EI}\int_0^h (qhz - \frac{qz^2}{2}).(z)\,dz + \frac{1}{EI}\int_0^L (\frac{qh^2}{2L}z).(h)\,dz + \frac{1}{EI}\int_0^h (0).(z)\,dz = \frac{5qh^4}{24EI} + \frac{qh^3L}{4EI}$$

Problem: Find the vertical and horizontal displacement at the B point of the system shown in the figure by employing virtual work theorem. If necassary express the results in terms of EA, GA', EI . **HINT**: You mignt not need some of these terms *in your expressions*.

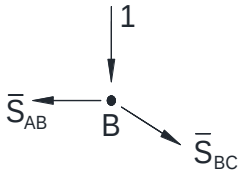


Solution



$S_{BC} = -20\sqrt{2}$

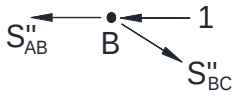
$S_{AB} = -20$



$\bar{S}_{BC} = -\sqrt{2}$

$\bar{S}_{AB} = -1$

Unit loading for vertical displacement



$S''_{BC} = 0$ $\quad *Q = 0$

$S''_{AB} = -1$ $\quad M = 0$

Unit loading for horizontal displacement

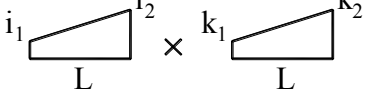
Vertical displacement of B point:

$$(\delta_B)_{düşey} = \frac{(-20).(-1)}{EA}.4 + \frac{(-20\sqrt{2}).(-\sqrt{2})}{EA}.\sqrt{2} \rightarrow (\delta_B)_{düşey} = \frac{136,57}{EA}$$

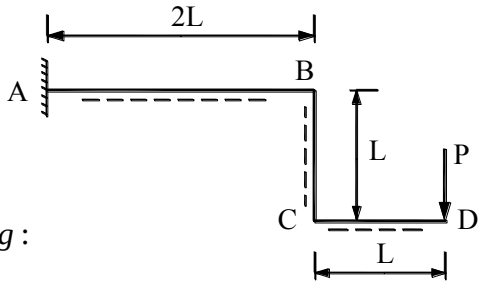
Horizontal displacement of B point:

$$(\delta_B)_{yatay} = \frac{(-20).(-1)}{EA}.4 + \frac{(-20\sqrt{2}).(0)}{EA}.\sqrt{2} \rightarrow (\delta_B)_{yatay} = \frac{80}{EA}$$

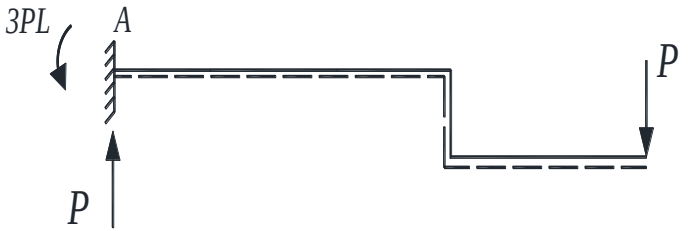
Problem: Find the vertical deflection of the point D in the system below by using the Castigliano theorem and find the horizontal deflection using the Virtual Work method. (*EI* is constant, and consider only the bending moment effect in the system).

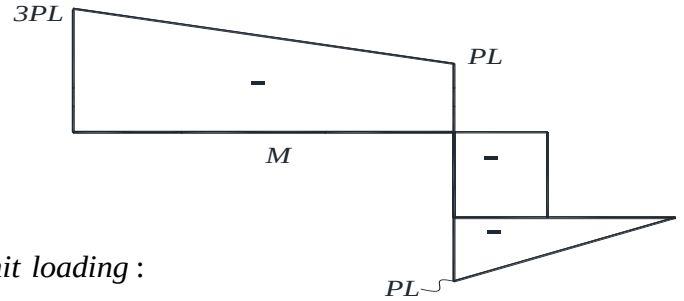


$$\frac{1}{6}L\left[2i_1k_1+i_1k_2+i_2k_1+2i_2k_2\right]$$



Actual loading :





Solution:

Real loading

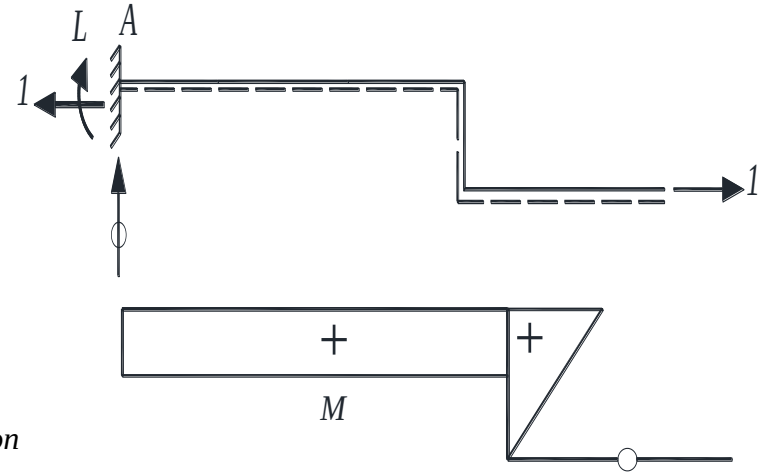
$$\sum F_y = 0 \Rightarrow A_y = P \qquad \sum M_A = 0 \Rightarrow M - 3PL = 0 \Rightarrow M = 3PL$$

Castigliano Theorem :

$$U_i = \int_L \frac{M^2}{2EI}dz$$

$$U_i = \frac{1}{2EI}\left\{\frac{1}{6}.2L\left[2.(-3PL)(-3PL)+(-3PL)(-PL)+(-PL)(-3PL)+2.(-PL)(-PL)\right]+ L(-\mathrm{PL})(-PL)+\frac{1}{3}L(-PL)(-PL)\right\} = \frac{5P^2L^3}{EI} \qquad \delta_{VD} = \frac{\partial U_i}{\partial P} = \frac{10PL^3}{EI}$$

Unit loading :



Virtual unit loading

$$\sum F_x = 0 \Rightarrow A_x = 1 \qquad \sum M_A = 0 \Rightarrow M - 1.L = 0 \Rightarrow M = L \qquad \sum F_y = 0 \Rightarrow A_y = 0$$

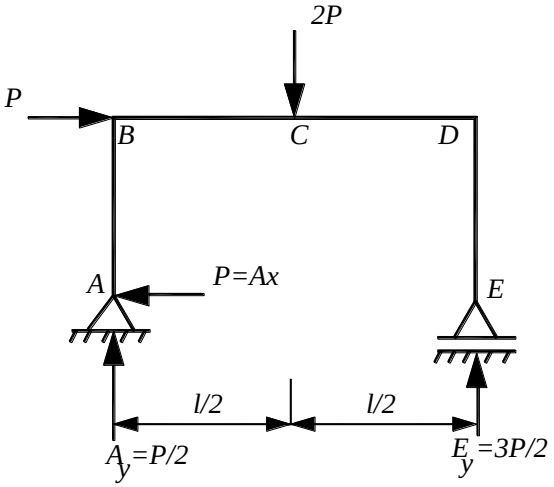
$$\textit{Virtual Work} \Rightarrow \delta = \int_L \frac{M\bar{M}}{EI}dz$$

$$\delta_{VH} = \frac{1}{EI}\left\{\frac{1}{6}2L\left[2(-3PL)(\mathrm{L})+(-3PL).\mathrm{L}+(-\mathrm{PL}).\mathrm{L}+2(-\mathrm{PL})\,\mathrm{L}\right]+ \frac{1}{6}L\left[2.(-\mathrm{PL})(\mathrm{L})+(-\mathrm{PL}).0+(-\mathrm{PL}).\mathrm{L}+0\right]\right\} \Rightarrow \delta_{VH} = \frac{9PL^3}{2EI} \rightarrow \textit{horizontal deflection}$$

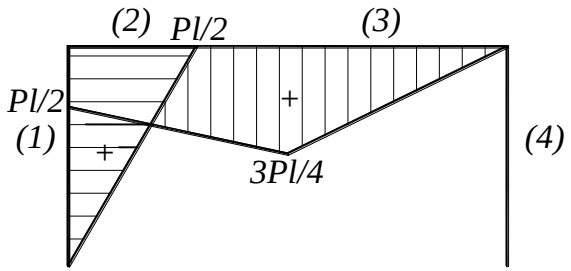
Problem 2: Find the horizontal displacement of the roller support E of the system under the given loads by using virtual work method.

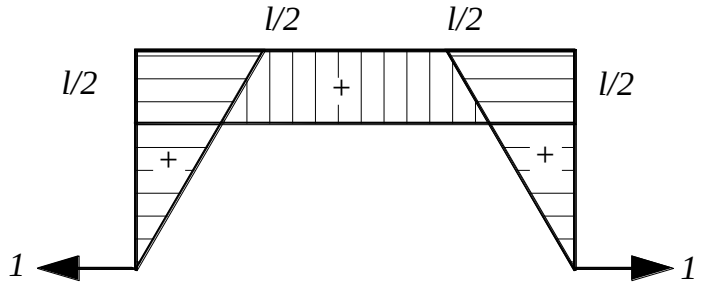
$$M_1 = Pz \quad M_2 = \frac{Pl}{2} + \frac{P}{2}z \quad M_3 = \frac{3}{2}Pz \quad M_4 = 0$$

$$\bar{M}_1 = z \quad \bar{M}_2 = \frac{l}{2} \quad \bar{M}_3 = \frac{l}{2} \quad M_4 = z$$



Moment diagram for real loading & Moment diagram for virtual loading

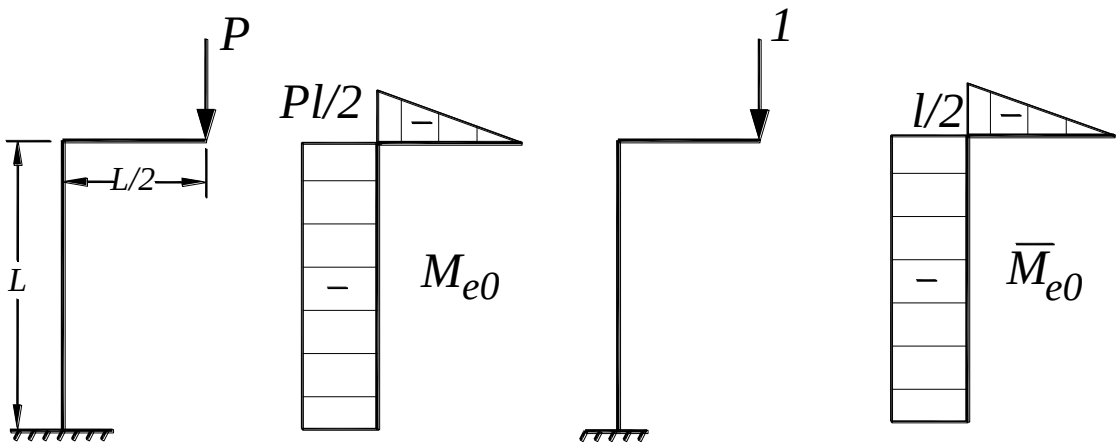




$$\delta_{HE} = \int \frac{M}{EI}\bar{M}dz = \frac{1}{EI}\left(\int_0^{l/2} Pz^2 dz + \int_0^{l/2}\left(\frac{Pl^2}{4} + \frac{Pl}{4}z\right)dz + \int_0^{l/2}\frac{3}{4}Plz + 0\right)$$

$$\delta_{HE} = \frac{1}{EI}\left(\frac{Pl^3}{24} + \frac{Pl^3}{8} + \frac{Pl^3}{32} + \frac{3Pl^3}{32}\right) = \frac{7}{24}\frac{Pl^3}{EI}$$

Örnek: Find the vertical and horizontal displacements of the point C of the system under the given loading condition by using virtual work method.

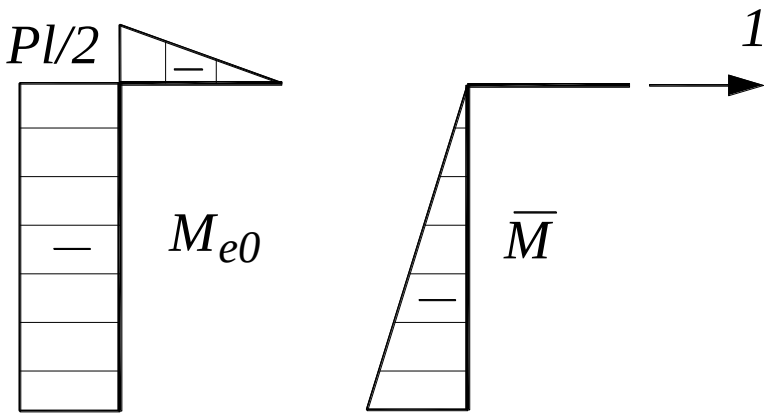


We will omit the contributions due to normal and shear forces.

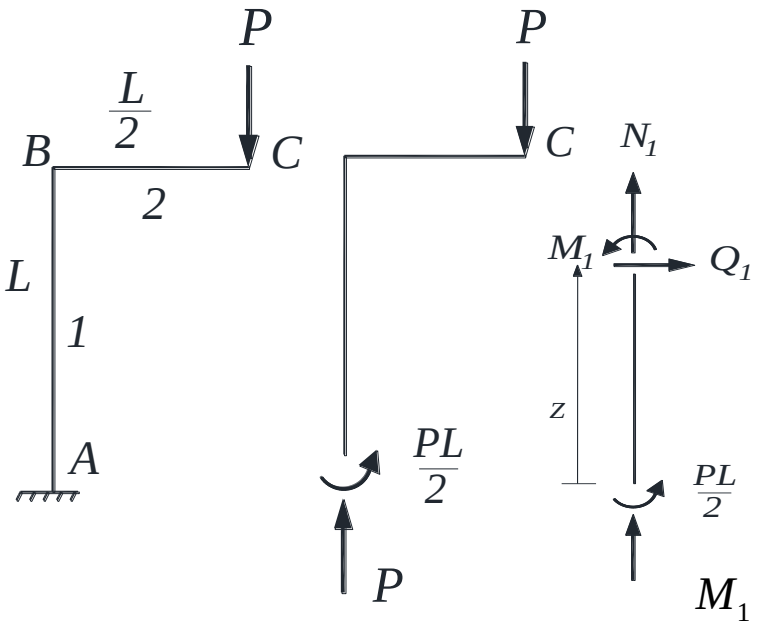
$$\delta_{VC} = \int \frac{M}{EI}\bar{M}dz = \frac{1}{EI}\left[l\frac{Pl}{2}\frac{l}{2} + \frac{1}{3}\frac{l}{2}\frac{Pl}{2}\frac{l}{2}\right] = \frac{7Pl^3}{24EI}$$

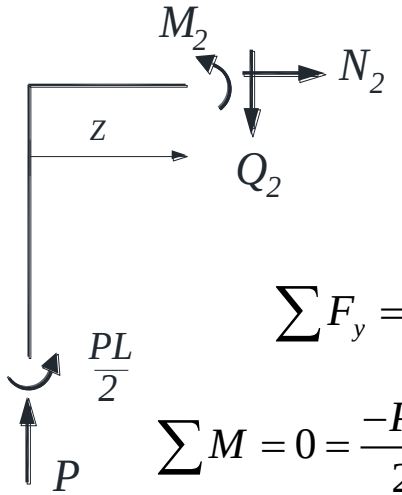
$$\delta_{HC} = \frac{1}{EI}\left(\frac{1}{2}l\frac{Pl}{2}l\right) = \frac{Pl^3}{4EI}$$

	k — k	k	k_1 — k_2	k
i — i	lik	$1/2\,lik$	$1/2\,li(k_1+k_2)$	$2/3\,lik$
i	$1/2\,lik$	$1/3\,lik$	$1/6\,li(k_1+2k_2)$	$1/3\,lik$
i	$1/2\,lik$	$1/6\,lik$	$1/6\,li(2k_1+k_2)$	$1/3\,lik$
i_1 — i_2	$1/2\,l(i_1+i_2)k$	$1/6\,l(i_1+2i_2)k$	$1/6\,l(2i_1k_1+i_1k_2+i_2k_1+2i_2k_2)k$	$1/3\,l(i_1+i_2)k$
i	$2/3\,lik$	$1/3\,lik$	$1/3\,li(k_1+k_2)$	$8/15\,l\,i\,k$



We will solve the same problem by obtaining the moment functions for the related regions:





$$\sum F_y = 0 = P - Q_2 = 0 \;\rightarrow\; Q_2 = P$$

$$\sum M = 0 = \frac{-PL}{2} + Pz - M_2 \;\rightarrow M_2 = Pz - \frac{PL}{2}$$

$$M_1 = -\frac{PL}{2}$$



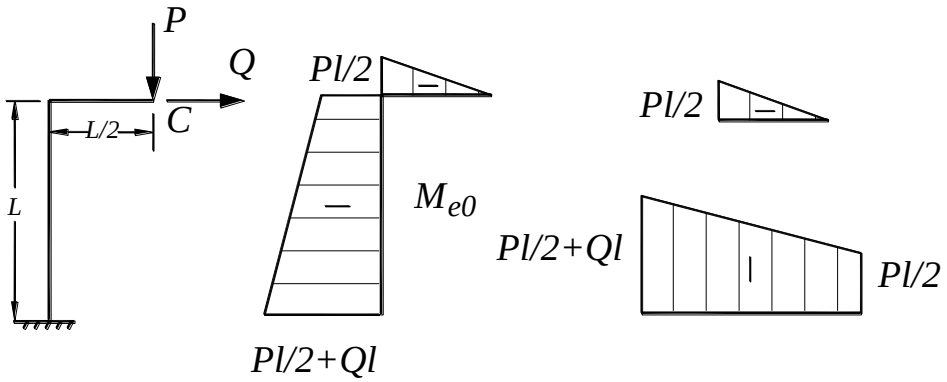
$$M_1^* = -\frac{L}{2} \quad ; \quad M_2^* = z - \frac{L}{2}$$

$$\delta_{VC} = \int \frac{M}{EI}\overline{M}\,dz \qquad EI\,\delta_{VC} = \int_0^L \left(-\frac{PL}{2}\right)\left(-\frac{L}{2}\right)dz + \int_0^{L/2} \left(Pz - \frac{PL}{2}\right)\left(z - \frac{L}{2}\right)dz$$

$$EI\,\delta_{VC} = -\frac{PL^2}{4}\cdot z\Big|_0^L + \left(\frac{Pz^3}{3} - \frac{PLz^2}{4} - \frac{PLz^2}{4} + \frac{PL^2 z}{4}\right)\Bigg|_0^{L/2}$$

$$EI\,\delta_{VC} = -\frac{PL^3}{4} + \left(\frac{PL^3}{24} - \frac{PL^3}{16} - \frac{PL^3}{16} + \frac{PL^3}{8}\right) \longrightarrow EI\,\delta_{VC} = \frac{PL^3}{4} + \frac{PL^3}{24} = \frac{7PL^3}{24}$$

Find the horizontal displacement of the system given in the previous example by using Castigliano's theorem. Use standard multiplication table in the calculations.



	k	k	k_1 k_2	k
i i	lik	$1/2\,lik$	$1/2\,li(k_1+k_2)$	$2/3\,lik$
i	$1/2\,lik$	$1/3\,lik$	$1/6\,li(k_1+2k_2)$	$1/3\,lik$
i	$1/2\,lik$	$1/6\,lik$	$1/6\,li(2k_1+k_2)$	$1/3\,lik$
i_1 i_2	$1/2\,l(i_1+i_2)k$	$1/6\,l(i_1+2i_2)k$	$1/6\,l(2i_1k_1+i_1k_2+i_2k_1+2i_2k_2)k$	$1/3\,l(i_1+i_2)k$
i	$2/3\,lik$	$1/3\,lik$	$1/3\,li(k_1+k_2)$	$8/15\,l\,i\,k$

$$U^I=\frac{1}{2EI}\frac{1}{3}\frac{L}{2}\left(\frac{PL}{2}\right)^2=\frac{P^2L^3}{48EI}$$

$$U^{II}=\frac{1}{2EI}\frac{L}{6}\left[2\left(\frac{PL}{2}+QL\right)^2+\left(\frac{PL}{2}+QL\right)\left(\frac{PL}{2}\right)+\left(\frac{PL}{2}\right)\left(\frac{PL}{2}+QL\right)+2\left(\frac{PL}{2}\right)^2\right]$$

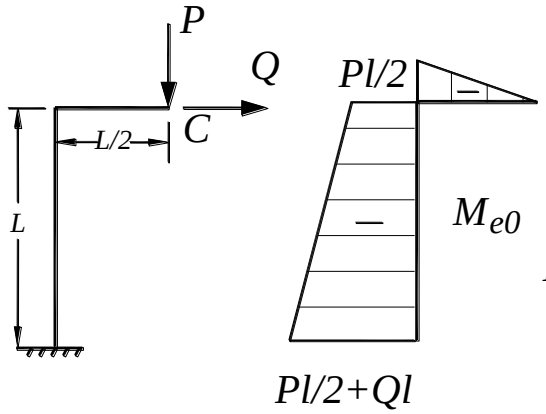
$$U^{II}=\frac{L}{6EI}\left(\frac{P^2l^2}{4}+PQl^2+Q^2l^2+\frac{P^2l^2}{4}+\frac{PQl^2}{2}+\frac{P^2l^2}{4}\right)$$

$$U=U^I+U^{II}=\frac{P^2L^3}{48EI}+\frac{3P^2L^3}{24EI}+\frac{3PQL^3}{12EI}+\frac{Q^2L^3}{6EI}$$

$\delta_{HC}=\dfrac{\partial U}{\partial Q}=\dfrac{Pl^3}{4EI}+\dfrac{Ql^3}{3EI}$ is obtained. We put $Q=0$ in the equation.

$\delta_{HC}=\dfrac{Pl^3}{4EI}$ This result is the same obtained by using virtual work method.

$$\delta_k = \int_0^l \left(\frac{M_x}{EI_x} \frac{\partial M_x}{\partial P_k} \right) \mathrm{d}\, z$$



$$-\left(\frac{PL}{2} + QL \right) + Qz - M_1 = 0 \rightarrow M_1 = -\left(\frac{PL}{2} + QL \right) + Qz \qquad \frac{\partial M_1}{\partial Q} = z - L$$

$$-\left(\frac{PL}{2} + QL \right) + QL + Pz - M_2 = 0 \rightarrow M_2 = Pz - \frac{PL}{2} \qquad \frac{\partial M_2}{\partial Q} = 0$$

$$\delta_{HC} = \int_0^L \left(\frac{M_1}{EI_x} \frac{\partial M_1}{\partial Q} \right) \mathrm{d}\, z + \int_0^{L/2} \left(\frac{M_2}{EI_x} \frac{\partial M_2}{\partial Q} \right) \mathrm{d}\, z = \frac{1}{EI_x} \int_0^L \left[-\left(\frac{PL}{2} + QL \right) + Qz \right] (z - L)\, dz$$

$$\delta_{HC} = \frac{1}{EI_x} \int_0^L \left\{ \left[-\left(\frac{PL}{2} + QL \right) + Qz \right] z + \left[-\left(\frac{PL}{2} + QL \right) + Qz \right] \right\} (-L)\, dz$$

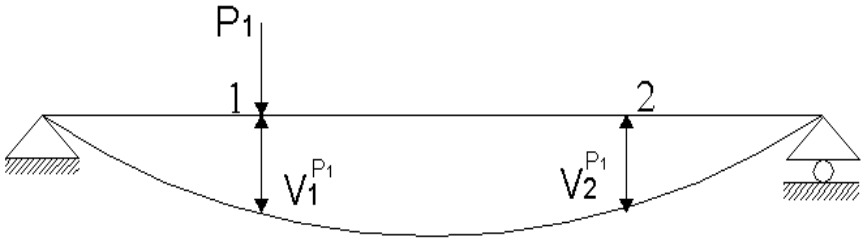
$$\delta_{HC} = \frac{1}{EI_x} \int_0^L \left\{ \left[-\left(\frac{PLz}{2} + QLz \right) + Qz^2 \right] z + \left[\left(\frac{PL^2}{2} - QL^2 \right) - QzL \right] \right\} dz$$

$$\delta_{HC} = \frac{1}{EI_x} \left\{ \left[-\left(\frac{PLz^2}{4} + \frac{QLz^2}{2} \right) + \frac{Qz^3}{3} \right] + \left[\left(\frac{PL^2 z}{2} - QL^2 z \right) - \frac{Qz^2 L}{2} \right] \right\} \Bigg|_0^L = \frac{1}{EI_x} \left(-\frac{PL^3}{4} + \frac{PL^3}{2} \right) = \frac{PL^3}{4EI_x}$$

Betti reciprocity theorem

The virtual work of any set of forces during the displacements due to another set of forces is equal to the virtual work of the second set of forces during the displacements due to the first set of forces.

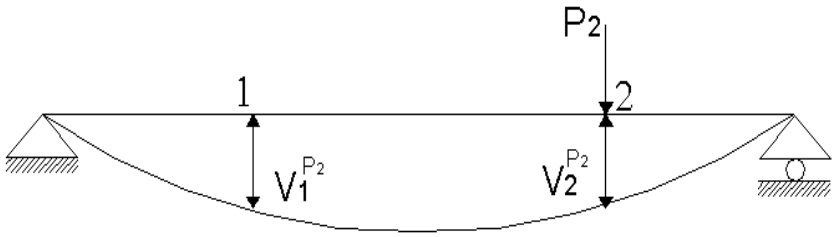
$$V_2^{P1} = \alpha_{21} P_1 \quad ; \qquad V_1^{P2} = \alpha_{12} P_2$$



$$V_2^{P1} P_2 = \alpha_{21} P_1 P_2 \quad ; \qquad V_1^{P2} P_1 = \alpha_{12} P_1 P_2$$

Şekil 1.11a

$$\alpha_{12} = \alpha_{21} \qquad P_1 V_1^{P2} = P_2 V_2^{P1}$$



Şekil 1.11b

Total potential energy principle

The equilibrium problem of elastic systems can be dealt with in two different ways.

a) Investigation by dividing the system into parts

i) Differential equilibrium equations**:**

$$\frac{\mathrm{d}T_y}{\mathrm{d}z} = -p_y \qquad \frac{\mathrm{d}M_x}{\mathrm{d}z} = T_y$$

ii) Compatibility equations :

$$\frac{1}{\rho_x} \cong -\frac{\mathrm{d}^2 v}{\mathrm{d}\,z^2}$$

iii) Constitutive relations: Stress-strain relations
Hooke's law between bending moment and curvature.

$$\frac{1}{\rho_x} \cong \omega_x = \frac{M_x}{E\,I_x}$$

If we eliminate the shear force and bending moment between these expressions, we get the following expression:

$$\frac{\mathrm{d}^4 v}{\mathrm{d}\,z^4} = \frac{p_y}{E\,I_x}$$

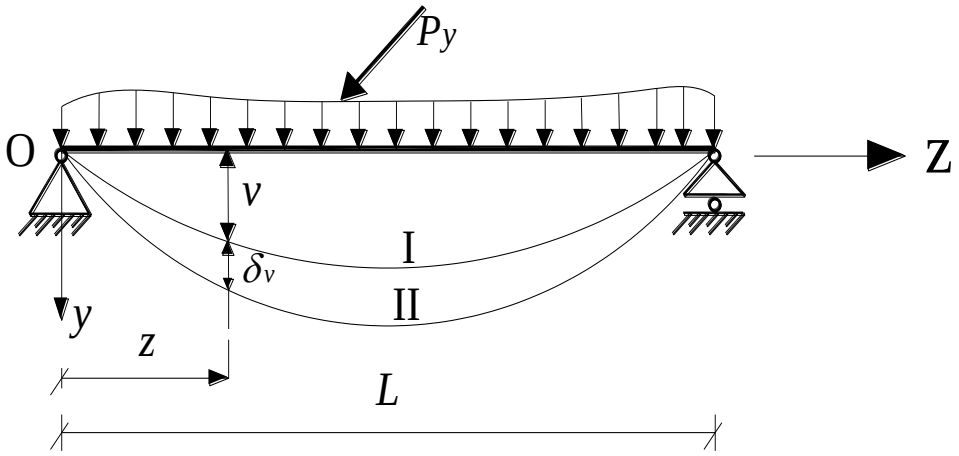
***iv)* Boundary conditions**: $v(0) = v(l) = 0$ geometrical boundary conditions (essential boundary conditions)

$v''(0) = v''(l) = 0$ Dynamical boundary conditions (natural boundary conditions)

We can obtain the integration constants by using these boundary conditions**.**

Principle of the extremity of the total potential energy: In this method, an integral expression called the total potential energy which depend on the coordinates defining the position of system is defined. Later a position in which the potential energy is extremum between the positions which satify some boundary conditions is sought.

***i)* Potential energy of the internal forces:** This energy is the deformation energy. If we omit the energy caused by shear forces, we get the following expression:



$$U_i = \Pi_i = \int_0^l \frac{M_x^2}{2EI_x}\mathrm{d}z = \int_0^l \frac{EI_x}{2}\left(\frac{\mathrm{d}^2 v}{\mathrm{d}z^2}\right)^2 \mathrm{d}z$$

***ii)* Potential energy of the external forces:** As it is known from the mechanics, potential energy of the external forces is equal to the external forces' work with opposite sign:

$$\Pi_d = -\int_0^l p_y\, v\, \mathrm{d}z$$

In this case the total potential energy is $\Pi = \Pi_i + \Pi_d = \int_0^l \left[\left(\frac{EI_x}{2}\frac{\mathrm{d}^2 v}{\mathrm{d}z^2}\right)^2 - p_y\, v\right]\mathrm{d}z$

As a result, in an elastic system, the position which make the potential energy minimum is the real equilibrium position between all the possible position which satisfy the geometrical boundary conditions.

To illustrate the preceding principles somewhat further, let us consider the one-degree-of-freedom spring-mass system shown in Fig. 2-6a. If the weight of the mass is 200 lb and if the stiffness of the spring is 50 lb/in., the system will come to rest with the spring stretched 4 in. Letting the elongation of the spring be given by x, the strain energy stored in the spring is

$$U = 25x^2$$

and the potential energy of the external loads, as defined by Eq. (2.20), is

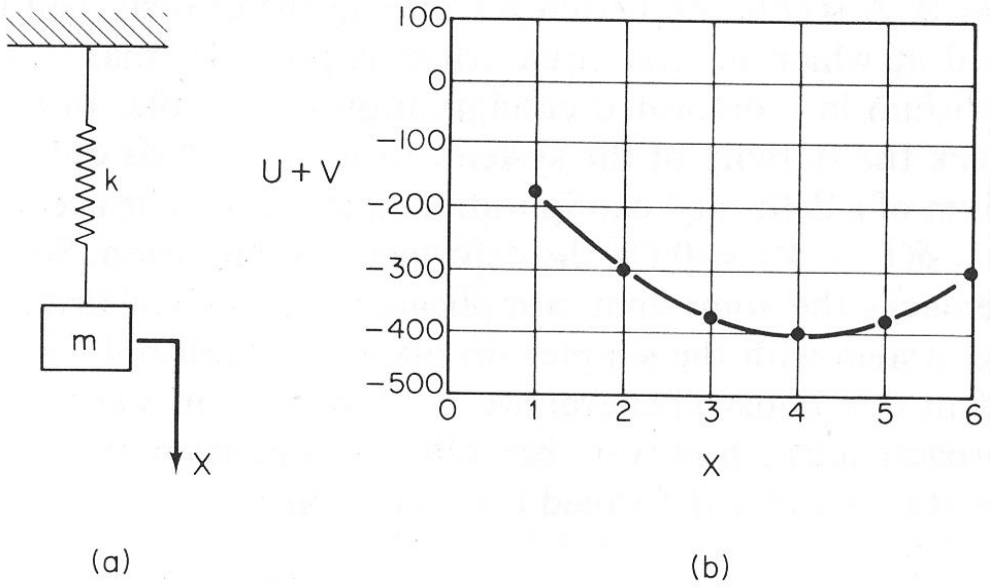


Fig. 2-6 Potential energy of one-degree-of-freedom system.

$$V = -200x$$

Hence the total potential energy of the system is

$$U + V = 25x^2 - 200x$$

A plot of U + K versus x, shown in Fig. 2-6b, indicates that the total potential energy has a horizontal tangent when x = 4. At the equilibrium configuration x = 4, the total potential energy of the system is thus shown to have a stationary value. In this instance the stationary value corresponds to a minimum point, signifying that the equilibrium configuration x = 4 is a stable one.

By a different way, lets obtain the potential energy expression. In equilibrium state it shou be

$$\sum F = k\,x - P = 0$$

By multiplying this equilibrium expression with $d\,x$ and then by integrating it from 0 to x we get

$k\frac{x^2}{2} - 200x$ here $k\frac{x^2}{2}$ is the elastic potential energy or spring potential energy. $-200x$ is called as external potential energy. As seen here the external potential $-200x$ is different from the work done in the case of static loading which is $200\,x/2$. In the expression -200x, final value of the force is considered. It will be suitable to make a difference between the potential energy concept and the work used in the principle of conservation of energy. The work of the force used in the concept of conservation of total energy is related with the whole work done during the deformation. In this case, the magnitude of the force increases from zer oto its final value and therefore there the work is the half of the multiplication of final displacement by force. But if the force is constant during the displacement, then the work is the multiplication of final displacement by force.

Potantiel energy of the external force is not related to how the load changes during the deformation, and therefore it is the multiplication of the final displacement by the final load. The factor ½ which is used in the concept of work when there is a linear relation between loading and the displacement is never used in the concept of the potential of external forces. In theoretical mechanics, it is customary to call the variation of the potential energy as minus the work of external forces due to a virtual displacement δW_e and to designate it by δV . The total of the external potential defined above and the internal potential is as follows:

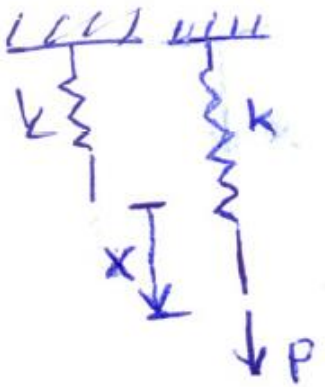
$$U + V = k\frac{x^2}{2} - 200x$$

First variation of this expression is

$$\delta\left(U+V\right) = \frac{\partial\left(U+V\right)}{\partial x} = \frac{\partial}{\partial x}\left(k\frac{x^2}{2} - 200x\right)\delta x$$

δx is arbitrary and is different from zero. In this case the sum of terms enclosed in parenthesis, which is explicitly the equilibrium equation, must be zero. In this case, it can be concluded that in the case of equilibrium, the potential energy is minimum.

Toplam Potansiyel Enerjideki Değişim



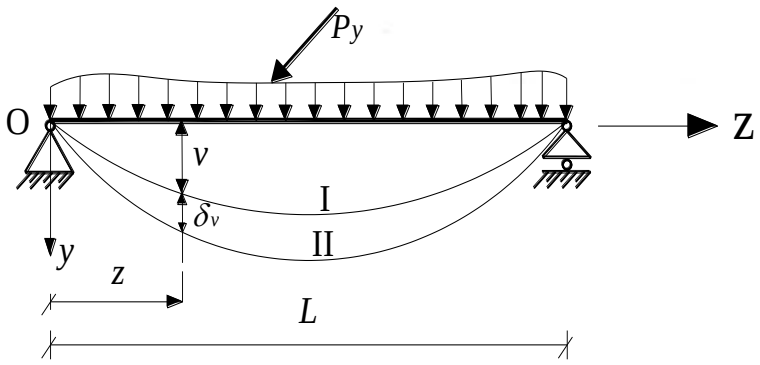
$$\delta\pi = \left[\frac{1}{2}k(x+\delta x)^2 - P(x+\delta x)\right] - \left[\frac{1}{2}kx^2 - Px\right]$$

$$\delta\pi = \left[\frac{1}{2}k(x^2 + 2x\delta x + \cancel{\delta x^2}) - P(x+\delta x)\right] - \left[\frac{1}{2}kx^2 - Px\right]$$

(ihmal)

$$\delta\pi = kx\delta x - P\delta x = (kx - P)\delta x$$

B. ifade de denge denklemini verir.



Ritz metodu: If the solution of a variational problem is reduced to the solution of a differential equation, this differential equation is the same with the one obtained by cutting the system into small parts.

Such a way is inconvenient and difficult (hard) as it is known from calculation of the elastic curve by integration method.

There are other methods which solve the variation problem without reducing it to a differential equation. These are called as direct variation methods.

The main idea in such methods which give approximate solutions is to express the total potential energy, which must be extremum, with respect to some parameters, and in this way to express the problem as an ordinary minimum problem with lots of variables.

In a beam problem the quantity which will be varied is the displacement v . For doing so first we make the following assumption:

$$v = a_1 A_1(z) + a_2 A_2(z) + ... + a_n A_n(z) = \sum_{i=1}^{n} a_i A_i(z)$$

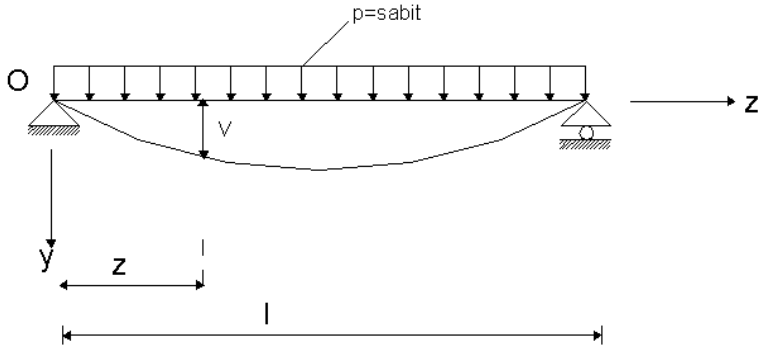
$A_i(z)$ Satisfy only the geometrical boundary conditions, and other then that is totally arbitrary, called as coordinate functions and they are chosen at the start of the problem.

a_i Factors or multipliers are initially unknown n coefficients.

Now, total potential energy can be expressed in terms of n variables as follows: $\Pi = \Pi(a_1, a_2, ..., a_n)$

$$\frac{\partial \Pi}{\partial a_i} = 0 \qquad i = 1,\ 2,\ 3, ..., n$$ Minimum condition.

Example



Assumption for the elastic curve;

$$v = a_1 \sin\left(\frac{\pi z}{l}\right) + a_3 \sin\left(\frac{3\pi z}{l}\right) + ... + a_n \sin\left(\frac{n\pi z}{l}\right) = \sum_{k=1,3,5} a_k \sin\left(\frac{k\pi z}{l}\right)$$

v is the approximate solution for the elastic curve.

$\sin\left(\frac{k\pi z}{l}\right)$ Coordinate functions.

All these functions must satisfy *v(0)=v(l)=0* geometric conditions; also because the problem is symmetric with respect to the midpoint of the beam, the odd terms k=1, 3, 5,… is taken into consideration in the calculations.

By using the total energy equation $\Pi = \int \left[\frac{1}{2} EI_x \left(\frac{\mathrm{d}^2 v}{\mathrm{d}z^2} \right)^2 - pv \right] \mathrm{d}z$

And using the condition that $\partial \Pi / \partial a_k = 0$

We obtained the unknown coefficients and then substitute them in the displacement equation and get the following expression for the displacement function:

$$v(z) = \left(\frac{4}{\pi^5} \right) \frac{pl^4}{EI_x} \sum \left(\frac{1}{k^5} \right) \sin\left(\frac{k\pi z}{l} \right) \quad k = 1,\ 3,\ 5...$$

By using v(z), the displacements, rotations can be calculated. The error amount for each quantity decreases with increase in the terms. Also, with increase in the order of derivative of v(z), the order of the error increases too.

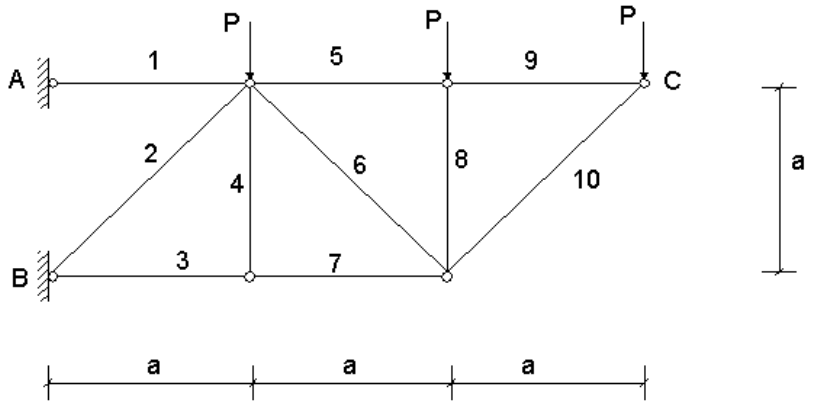
Problems related with energy methods

Problem : By using the virtual work theorem, find the vertical displacement of the point C of the plane truss seen in the figure. The cross-section areas and Young modulus of the bars are equal to each other and are A and E respectively.

There are only normal forces in the bars:

$$\delta = \sum_{i=1}^{n} \left(S_i \bar{S}_i \right) / \left(E_i A_i \right) L_i$$

$$\delta = \int \left[\left(N_0 \bar{N} \right) / EA + \left(k' T_0 \bar{T} \right) / GA + \left(M_{b0} \bar{M}_b \right) / GI_0 + \left(M_{e0} \bar{M}_e \right) / EI \right] dz$$

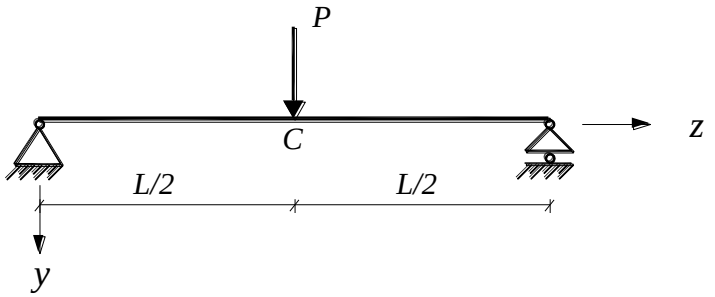


$$\delta_{VC} = \frac{\left(32 + 12\sqrt{2} \right) Pa}{EA}$$

Çubuk No	S_i	$\bar{S}_i$	L_i	$E_i A_i$	$\left(S_i \bar{S}_i \right) / \left(E_i A_i \right) L_i$
1	$6P$	3	a	EA	$18Pa / EA$
2	$-3\sqrt{2}P$	$-\sqrt{2}$	$a\sqrt{2}$	EA	$6\sqrt{2}Pa / EA$
3	$-3P$	-2	a	EA	$6Pa / EA$
4	0	0	a	EA	0
5	P	1	a	EA	Pa / EA
6	$2\sqrt{2}P$	$\sqrt{2}$	$a\sqrt{2}$	EA	$4\sqrt{2}Pa / EA$
7	-3P	-2	a	EA	$6Pa / EA$
8	-P	0	a	EA	0
9	P	1	a	EA	Pa / EA
10	$-\sqrt{2}P$	$-\sqrt{2}$	$a\sqrt{2}$	EA	$2\sqrt{2}Pa / EA$

Problem 1.2: By using k=3 and assuming the coordinate function as given below, find the displacement function and later by using the displacement function obtain the rotation function and investigate the results.

$$v = a_1 \sin\left(\frac{\pi z}{l}\right) + a_3 \sin\left(\frac{3\pi z}{l}\right) + ... + a_n \sin\left(\frac{n\pi z}{l}\right) = \sum_{k=1,3,5} a_k \sin\left(\frac{k\pi z}{l}\right)$$



The coordinate function must satisfy the conditions *v(0)=v(l)=0* .

$$k = 3 \text{ için } v(z) = a_1 \sin(\pi z / l) + a_3 \sin(3\pi z / l)$$

$$\Pi = \Pi_i + \Pi_d = \int_0^L \frac{1}{2} EI_x \left(\frac{d^2 v}{dz^2}\right)^2 dz - P\, v(l/2)$$

$$v'(z) = a_1 \pi / l \cdot \cos(\pi z / l) + a_3 3\pi / l \cdot \cos(3\pi z / l)$$

$$v''(z) = a_1 \pi^2 / l^2 \cdot \sin(\pi z / l) + a_3 9\pi^2 / l^2 \cdot \sin(3\pi z / l)$$

$$\Pi = \int_0^L \left\{ EI_x / 2 \left[a_1^2 \pi^4 / l^4 \cdot \sin^2(\pi z / l) + 2 a_1 a_3 9\pi^4 / l^4 \cdot \sin(\pi z / l) + a_3^3 81\pi^4 / l^4 \cdot \sin^2(3\pi z / l) \right] dz \right.$$

$$\left. - P\left\{ a_1 \sin(\pi / 2) + a_2 \sin(3\pi / 2) \right\} \right\}$$

By calculating this integral, we get

$$\Pi = \frac{EI_x}{2}\left(\frac{a_1^2 \pi^4}{2l^3} + \frac{81 a_3^2 \pi^4}{2l^3}\right) - \frac{1}{2} P \left[a_1 \sin\left(\frac{\pi}{2}\right) + a_3 \sin\left(\frac{3\pi}{2}\right) \right]$$

For the extremity of the total potential energy, the following condition must be satisfied: $\partial\Pi / \partial a_i = 0, \quad i = 1,3$

$$\partial\Pi / a_1 = EI / 2a_1\pi^4 / l^3 - P\left(\sin(\pi/2)\right) = 0$$

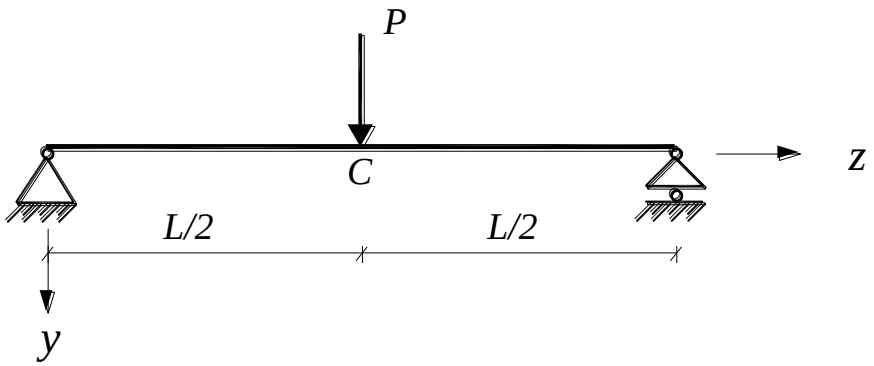
$$\partial\Pi / a_3 = EI / 2a_3 81\pi^4 / l^3 - P\left(\sin(3\pi/2)\right) = 0$$

$$EI / 2\pi^4 / l^3 a_1 - P = 0$$

$$\Rightarrow a_1 = 2Pl^3 / \pi^4 EI$$

$$EI / 2\cdot 81\pi^4 / l^3 a_3 - (-P) = 0$$

$$\Rightarrow a_3 = -2Pl^3 / 81\pi^4 EI$$



If the obtained constants are put in the displacement equation, we get

$$v(z) = \left(2Pl^3 / \pi^4 EI\right)\sin\left(\pi z / l\right) - \left(2Pl^3 / 81\pi^4 EI\right)\sin\left(3\pi z / l\right)$$

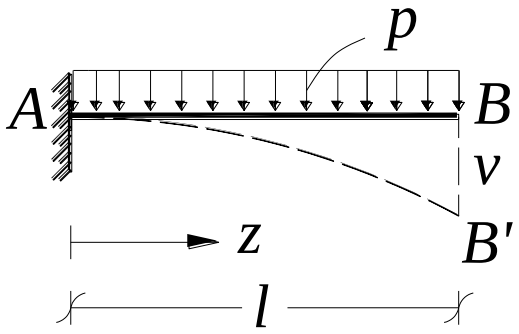
The displacement at C is $v(l/2) = \left(2Pl^3 / \pi^4 EI\right)\sin(\pi/2) - \left(2Pl^3 / 81\pi^4 EI\right)\sin(3\pi/2) = 164Pl^3 / 81\pi^4 EI$

This value is very close to the exact value: $Pl^3 / 48EI$

The difference between the two solutions is only equal to*=((48,11-48)/48).100=%0,27.* The rotation function is the derivative of the displacement function with respect to the z axis:

$$v'(z) = \left(2Pl^2 / \pi^3 EI\right)\cos\left(\pi z / l\right) - 2/27\left(2Pl^2 / \pi^3 EI\right)\cos\left(3\pi z / l\right)$$

Problem Find the displacement BB' of the beam by using the total potential energy principle.



Solution $\Pi = \Pi_i + \Pi_d = \int \left(\frac{1}{2EI_x} \left(\frac{d^2 v}{dz^2} \right) - p\,v \right) dz \quad v(0) = v'(0) = 0$

$$v(z) = a_2 z^2 + a_3 z^3 + \ldots + a_n z^n = \sum a_k z^k \quad v''(z) = \sum a_k k(k-1) z^{k-2}$$

$$\Pi = \int \frac{1}{2EI_x} \left(\left(\sum_{k=2}^{n} k(k-1) a_k z^{k-2} \right)^2 - p \sum_{k=2}^{n} a_k z^k \right) dz$$

For minimizing the total potential energy we choose the trial function as:

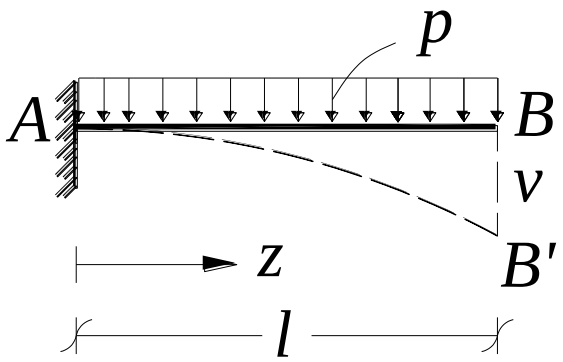
$$n = 4 \text{ için } v(z) = a_2 z^2 + a_3 z^3 + a_4 z^4 \qquad \frac{\partial \pi}{\partial a_i} = 0, \quad i = 2, \ldots, n$$

$$v'(z) = 2a_2 z + 3a_3 z^2 + 4a_4 z^3 \qquad v''(z) = 2a_2 + 6a_3 z + 12a_4 z^2$$

$$(v'')^2 = 4a_2^{\,2} + 24a_2 a_3 z + 48a_2 a_4 z^2 + 36a_3^{\,2} z^2 + 144a_3 a_4 z^3 + 144a_4^{\,2} z_4$$

$$\Pi = \int \frac{EI_x}{2} \left(\left(4a_2^{\,2} + 24a_2 a_3 z + 48a_2 a_4 z^2 + 36a_3 z^2 + 144a_3 a_4 z^3 + 144a_4^{\,2} z^4 \right) - p \left(a_2 z^2 + a_3 z^3 + a_4 z^4 \right) \right) dz$$

$$\Pi = \frac{EI_x}{2} \left(4a_2^{\,2} l + 12a_2 a_3 l^2 + 16a_2 a_4 l^3 + 12a_3^{\,2} l^3 + 36a_3 a_4 l^4 + \frac{144}{5} a_4^{\,2} l^5 \right) - p \left(\frac{l^3}{3} a_2 + \frac{l^4}{4} a_3 + \frac{l^5}{5} a_4 \right)$$



$$\frac{\partial \Pi}{\partial a_i} = 0, \quad i = 2,3,4$$

$$\frac{\partial \Pi}{\partial a_2} = \frac{EI_x}{2}\left(8a_2 l + 12a_3 l^2 + 16a_4 l^3\right) - \frac{pl^3}{3} = 0$$

$$\frac{\partial \Pi}{\partial a_3} = \frac{EI_x}{2}\left(12a_2 l^2 + 24a_3 l^3 + 36a_4 l^4\right) - \frac{pl^4}{4} = 0$$

$$\frac{\partial \Pi}{\partial a_4} = \frac{EI_x}{2}\left(16a_2 l^3 + 36a_3 l^4 + 288/5\, a_4 l^5\right) - \frac{pl^5}{5} = 0$$

$$\frac{EI_x}{2}\begin{bmatrix} 8l & 12l^2 & 16l^3 \\ 12l^2 & 24l^3 & 36l^4 \\ 16l^3 & 36l^4 & 288/5\, l^5 \end{bmatrix}\begin{Bmatrix} a_2 \\ a_3 \\ a_4 \end{Bmatrix} = \begin{Bmatrix} p\,L^3/3 \\ p\,L^4/4 \\ p\,L^5/5 \end{Bmatrix}$$

$$\left.\begin{aligned} a_2 &= p\ l^2/(4E\,I_x) \\ a_3 &= -p\,l/(6E\,I_x) \\ a_4 &= p/(24E\,I_x) \end{aligned}\right\} \quad v(z) = \frac{p\,l^4}{24EI_x}\left(6\left(\frac{z}{l}\right)^2 - 3\left(\frac{z}{l}\right)^3 + \left(\frac{z}{l}\right)^4\right)$$

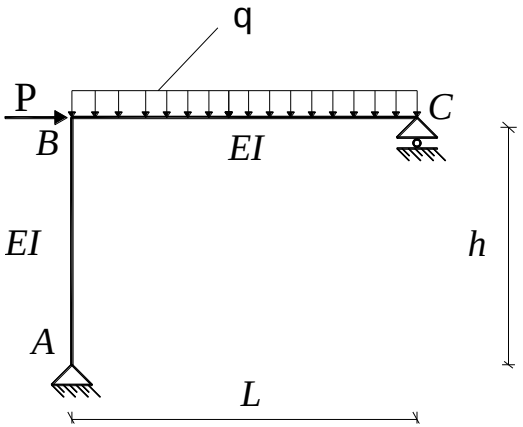
$$v(l) = f_B = \frac{p\,l^4}{8EI_x}$$

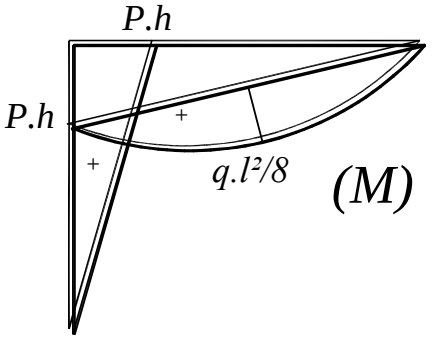
Problem: Find the horizontal displacement of the point C of the frame of given dimensions under the loads shown in the figure by using the virtual work theorem. Omit the works of normal forces and shear forces.

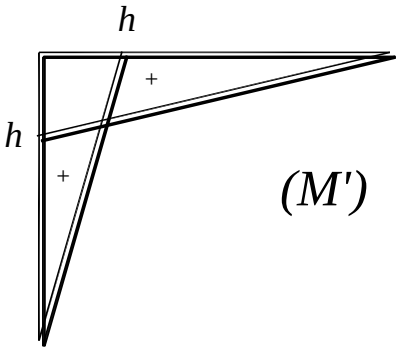
Solution: The moment diagrams for the real and fictitious loads must be drawn. For fictitious system, a unit force is affected to the point C.

Standard multiplication table will be used for the solution of this problem.

	k	k	k	k_1 k_2	k
i i	lik	$1/2\,lik$	$1/2\,li(k_1+k_2)$	$2/3\,lik$
i	$1/2\,lik$	$1/3\,lik$	$1/6\,li(k_1+2k_2)$	$1/3\,lik$
i	$1/2\,lik$	$1/6\,lik$	$1/6\,li(2k_1+k_2)$	$1/3\,lik$
i_1 i_2	$1/2\,l(i_1+i_2)k$	$1/6\,l(i_1+2i_2)k$	$1/6\,l(2i_1k_1+i_1k_2+i_2k_1+2i_2k_2)k$	$1/3\,l(i_1+i_2)k$
i	$2/3\,lik$	$1/3\,lik$	$1/3\,li(k_1+k_2)$	$8/15\,l\,i\,k$







$$\delta_{HC}=\int\frac{M_x\bar{M}_x}{EI}dz=\frac{1}{EI}\left(\frac{h.Ph.h}{3}+\frac{L.Ph.h}{3}+\frac{L.qL^2}{3.8}.h\right)$$

$$\delta_{HC}=\frac{1}{EI}\left(\frac{Ph^3}{3}+\frac{Ph^2L}{3}+\frac{qL^3h}{24}\right)$$

Finite Element Method

Element nodes 1 and 2 are located at the ends of the element, and the nodal variables are the transverse displacements v_1 and v_2 at the nodes and the slopes (rotations) 1 and 2. The nodal variables as shown are in the positive direction, and it is to be noted that the slopes are to be specified in radians. The displacement function $v(x)$ is to be discretized such that

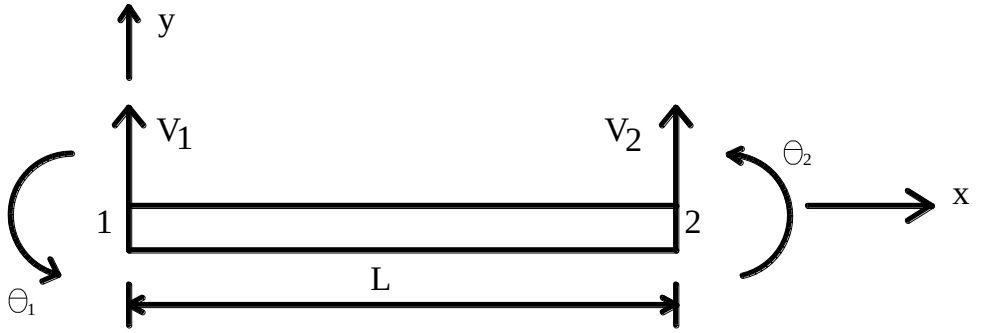


Fig. Beam element nodal displacements shown in a positive sense.

$$v(x) = f(v_1, v_2, \theta_1, \theta_2, x) \qquad \textbf{(4.12)}$$

subject to the boundary conditions

$$v(x = x_1) = v_1 \qquad \textbf{(4.13)} \qquad v(x = x_2) = v_2 \qquad \textbf{(4.14)}$$

$$\left.\frac{dv}{dx}\right|_{x=x_2} = \theta_2 \qquad \textbf{(4.15)} \qquad \left.\frac{dv}{dx}\right|_{x=x_1} = \theta_1 \qquad \textbf{(4.16)}$$

Considering the four boundary conditions and the one-dimensional nature of the problem in terms of the independent variable, we assume the displacement function in the form

$$v(x) = a_0 + a_1 x + a_2 x^2 + a_3 x^3 \qquad \textbf{(4.17)}$$

$$a_0 = v_1 \qquad a_1 = \theta_1$$

$$a_2 = \frac{3}{L^2}(v_2 - v_1) - \frac{1}{L}(2\theta_1 + \theta_2) \qquad a_3 = \frac{2}{L^3}(v_1 - v_2) + \frac{1}{L^2}(\theta_1 + \theta_2)$$



Figure 4.5 Loads for the bending element

$$v(x) = \left(1 - \frac{3x^2}{L^2} + \frac{2x^3}{L^3}\right)v_1 + \left(x - \frac{2x^2}{L} + \frac{x^3}{L^2}\right)\theta_1 + \left(\frac{3x^2}{L^2} - \frac{2x^3}{L^3}\right)v_2 + \left(\frac{x^3}{L^2} - \frac{x^2}{L}\right)\theta_2$$

$$v(x) = N_1(x)\, v_1 + N_2(x)\, \theta_1 + N_3(x)\, v_2 + N_4(x)\, \theta_2$$

$$v(x) = \begin{bmatrix} N_1 & N_2 & N_3 & N_4 \end{bmatrix} \begin{Bmatrix} v_1 \\ \theta_1 \\ v_2 \\ \theta_2 \end{Bmatrix} = [N]\{\delta\}$$

$$v(x) = N_1(x)\ v_1\ +\ N_2(x)\ \theta_1 +\ N_3(x)\ v_2 +\ N_4(x)\ \theta_2$$

$$\sigma_z = \frac{M(x)y}{I_z} = yE\frac{d^2v}{dx^2} \qquad \textbf{(4.11)}$$

$$v(x) = \begin{bmatrix} N_1 & N_2 & N_3 & N_4 \end{bmatrix} \begin{Bmatrix} v_1 \\ \theta_1 \\ v_2 \\ \theta_2 \end{Bmatrix} = [N]\{\delta\} \qquad \textbf{(4.27b)}$$

Using Equation (4.11) in conjunction with Equation (4.27b), the normal stress distribution on a cross section located at axial position x is given by

$$\sigma_x(x,y) = yE\frac{d^2[N]}{dx^2}\{\delta\} \qquad \textbf{(4.30)}$$

$$\sigma_x(x,y) = y\ E\left[\left(\frac{12x}{L^3} - \frac{6}{L^2}\right)v_1 + \left(\frac{6x}{L^2} - \frac{4}{L}\right)\theta_1 + \left(\frac{6}{L^2} - \frac{12x}{L^3}\right)v_2 + \left(\frac{6x}{L^2} - \frac{2}{L}\right)\theta_2\right] \qquad \textbf{(4.32)}$$

4.4 Flexural element stiffness matrix

We may now utilize the discretized approximation of the flexure element displacement to examine stress, strain, and strain energy exhibited by the element under load. The total strain energy is expressed as

$$U_e = \frac{1}{2}\int_V \sigma_x \varepsilon_x dV \qquad \textbf{(4.35)}$$

$$\varepsilon_x = y\frac{d^2v}{dx^2} \qquad \textbf{(4.5)}$$

$$\sigma_x = E\varepsilon_x = Ey\frac{d^2v}{dx^2} \qquad \textbf{(4.6)}$$

where V is total volume of the element. Substituting for the stress and strain per Equations 4.5 and 4.6, we get

$$U_e = \frac{E}{2}\int_V y^2\left(\frac{\mathrm{d}^2v}{\mathrm{d}x^2}\right)^2 dV \qquad \textbf{(4.36)}$$

$$U_e = \frac{E}{2}\int_0^L (\frac{\mathrm{d}^2v}{\mathrm{d}x^2})^2\left(\int_A y^2 dA\right)dx \qquad \textbf{(4.37)}$$

Again recognizing the area integral as the moment of inertia I_z about the centroidal axis perpendicular to the plane of bending, we have

$$U_e = \frac{EI_z}{2}\int_0^L\left(\frac{d^2v}{dx^2}\right)^2 dx \qquad \textbf{(4.38)}$$

$$v(x) = N_1(x)\ v_1\ +\ N_2(x)\ \theta_1 +\ N_3(x)\ v_2 +\ N_4(x)\ \theta_2$$

Equation 4.38 represents the strain energy of bending for any constant cross-section beam that obeys the assumptions of elementary beam theory. For the strain energy of the finite element being developed, we substitute the discretized displacement relation of Equation 4.27 to obtain

$$U_e = \frac{EI_z}{2}\int_0^L\left(\frac{d^2N_1}{dx^2}v_1 + \frac{d^2N_2}{dx^2}\theta_1 + \frac{d^2N_3}{dx^2}v_2 + \frac{d^2N_4}{dx^2}\theta_2\right)^2 \mathrm{d}x \qquad \textbf{(4.39)}$$

as the approximation to the strain energy. We emphasize that Equation 4.39 is an approximation because the discretized displacement function is not in general an exact solution for the beam flexure problem.

Applying the first theorem of Castigliano to the strain energy function with respect to nodal displacement $v1$ gives the transverse force at node 1 as

$$U_e = \frac{EI_z}{2}\int_0^L \left(\frac{d^2N_1}{dx^2}v_1 + \frac{d^2N_2}{dx^2}\theta_1 + \frac{d^2N_3}{dx^2}v_2 + \frac{d^2N_4}{dx^2}\theta_2 \right)^2 \mathrm{d}x$$

$$\frac{\partial U_e}{\partial v_1} = F_1 = EI_z \int_0^L \left(\frac{d^2N_1}{dx^2}v_1 + \frac{d^2N_2}{dx^2}\theta_1 + \frac{d^2N_3}{dx^2}v_2 + \frac{d^2N_4}{dx^2}\theta_2 \right) \frac{d^2N_1}{dx^2}\,dx \qquad \textbf{(4.40)}$$

while application of the theorem with respect to the rotational displacement gives the moment as

$$\frac{\partial U_e}{\partial \theta_1} = M_1 = EI_z \int_0^L \left(\frac{d^2N_1}{dx^2}v_1 + \frac{d^2N_2}{dx^2}\theta_1 + \frac{d^2N_3}{dx^2}v_2 + \frac{d^2N_4}{dx^2}\theta_2 \right) \frac{d^2N_2}{dx^2}\,dx \qquad \textbf{(4.41)}$$

For node 2, the results are

$$\frac{\partial U_e}{\partial V_2} = F_2 = EI_z \int_0^L \left(\frac{d^2N_1}{dx^2}v_1 + \frac{d^2N_2}{dx^2}\theta_1 + \frac{d^2N_3}{dx^2}v_2 + \frac{d^2N_4}{dx^2}\theta_2 \right) \frac{d^2N_3}{dx^2}dx \qquad \textbf{(4.42)}$$

$$\frac{\partial U_e}{\partial \theta_2} = M_2 = EI_z \int_0^L \left(\frac{d^2N_1}{dx^2}v_1 + \frac{d^2N_2}{dx_2}\theta_1 + \frac{d^2N_3}{dx^2}v_2 + \frac{d^2N_4}{dx^2}\theta_2 \right) \frac{d^2N_4}{dx^2}\,dx \qquad \textbf{(4.43)}$$

Equations 4.40–4.43 algebraically relate the four nodal displacement values to the four applied nodal forces (here we use *force* in the general sense to include applied moments) and are of the form

$$\begin{Bmatrix} F_1 \\ M_1 \\ F_2 \\ M_2 \end{Bmatrix} = \begin{bmatrix} k_{11} & k_{12} & k_{13} & k_{14} \\ k_{21} & k_{22} & k_{23} & k_{24} \\ k_{31} & k_{32} & k_{33} & k_{34} \\ k_{41} & k_{42} & k_{43} & k_{44} \end{bmatrix} \begin{Bmatrix} v_1 \\ \theta_1 \\ v_2 \\ \theta_2 \end{Bmatrix} \qquad \textbf{(4.44)}$$

where k_{mn}, m, n = 1, 4 are the coefficients of the element stiffness matrix.

By comparison of Equations 4.40–4.43 with the algebraic equations represented by matrix Equation 4.44, it is seen that

$$k_{mn} = k_{nm} = EI_z \int_0^L \frac{\mathrm{d}^2 N_m}{\mathrm{d}x^2} \frac{\mathrm{d}^2 N_n}{\mathrm{d}x^2} \mathrm{d}x \qquad \textbf{(4.45)} \qquad \textbf{m, n =1, 4}$$

and the element stiffness matrix is symmetric, as expected for a linearly elastic element. The stiffness coefficients are then evaluated as follows:

$$k_{11} = \frac{12EI_z}{L^3} \qquad k_{13} = k_{31} = -\frac{12EI_z}{L^3} \qquad k_{14} = k_{41} = \frac{6EI_z}{L^3} \qquad k_{12} = k_{21} = \frac{6EI_z}{L^2}$$

$$k_{22} = \frac{4EI_z}{L} \;;\; k_{23} = k_{32} = -\frac{6EI_z}{L^2} \;;\; k_{24} = k_{42} = \frac{2EI_z}{L} \qquad k_{33} = \frac{12EI_z}{L^3} \;;\; k_{34} = k_{43} = -\frac{6EI_z}{L^2} \;;\; k_{44} = \frac{4EI_z}{L}$$

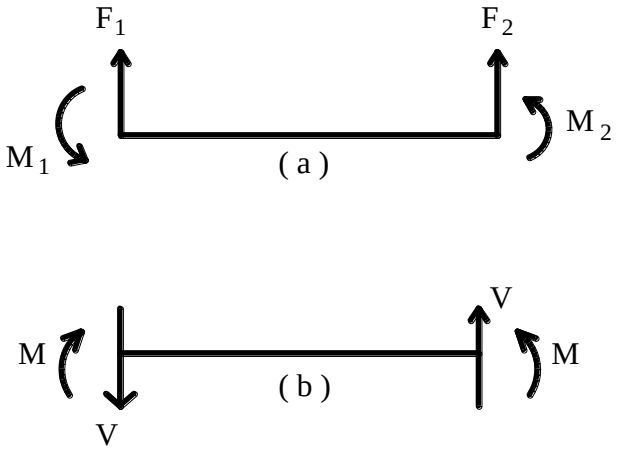
The complete stiffness matrix for the flexure element is then written as

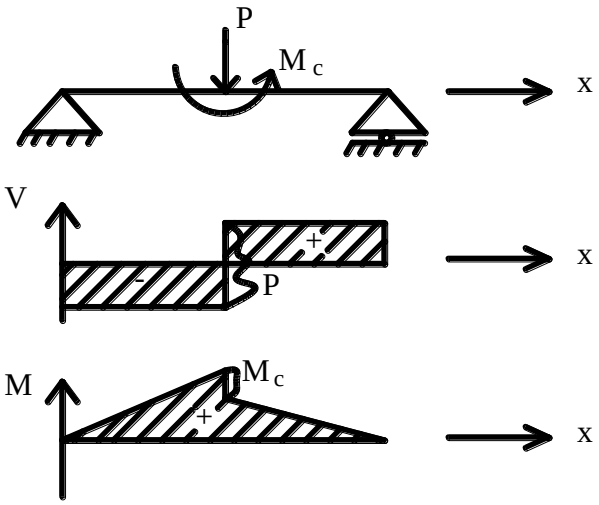
$$[k_e] = \frac{EI_z}{L^3} \begin{bmatrix} 12 & 6L & -12 & 6L \\ 6L & 4L^2 & -6L & 2L^2 \\ -12 & -6L & 12 & -6L \\ 6L & 2L^2 & -6L & 4L^2 \end{bmatrix} \qquad \textbf{(4.49)}$$

Symmetry of the element stiffness matrix is apparent, as previously observed.

4.5 Element Load Vector

In Equations 4.40–4.43, the element forces and moments were treated as required by the first theorem of Castigliano as being in the direction of the associated displacements. These directions are in keeping with the assumed positive directions of the nodal displacements. However, as depicted in Figures 4.6a and 4.6b, the usual convention for shear force and bending moment in a beam are such that

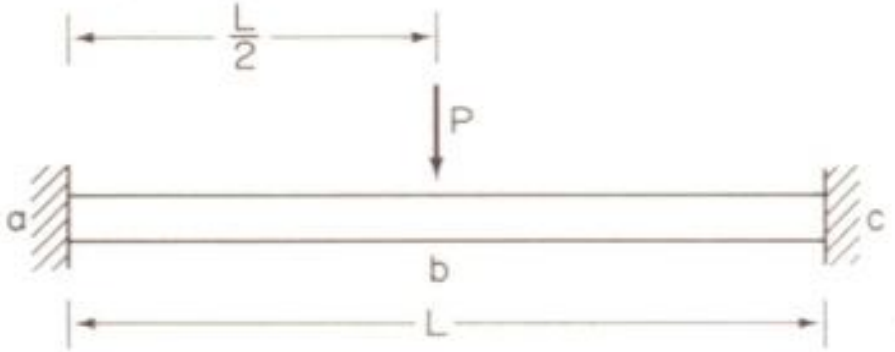


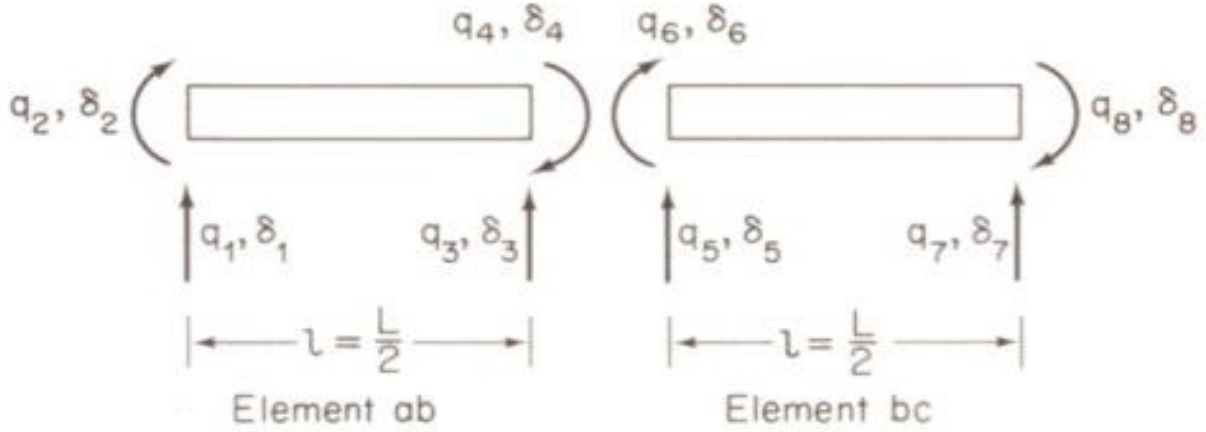


$$\begin{Bmatrix} F_1 \\ M_1 \\ F_2 \\ M_2 \end{Bmatrix} \Rightarrow \begin{Bmatrix} -V_1 \\ -M_1 \\ V_2 \\ M_2 \end{Bmatrix} \qquad \textbf{(4.50)}$$

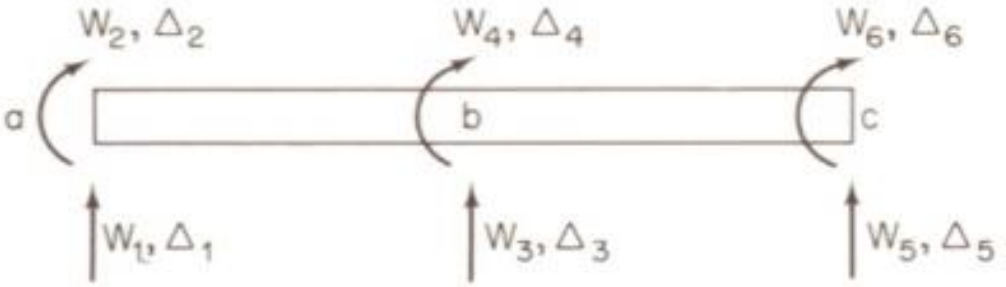
(a) Nodal load positive convention.
(b) Positive convention from the strength of materials theory.
(c) Shear and bending moment diagrams depicting nodal load effects.

In Equation 4.50, the column matrix (vector) on the left represents positive nodal forces and moments per the finite element formulations. The right-hand side contains the corresponding signed shear forces and bending moments per the beam theory sign convention.





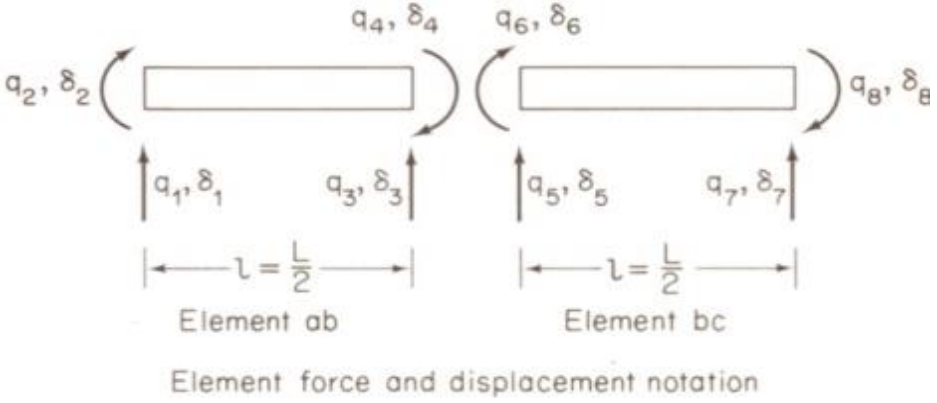
Element force and displacement notation



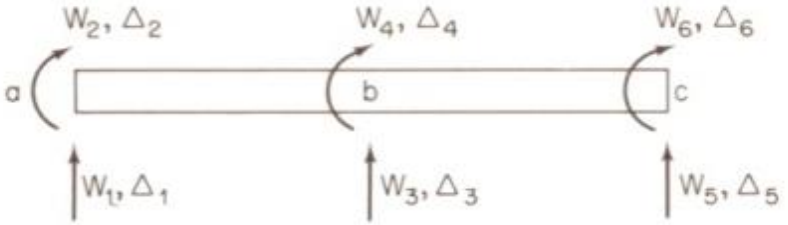
Structure force and displacement notation

$$[K]_{ab} = \frac{EI}{l}\begin{matrix} \\ \frac{12}{l^2} \\ -\frac{6}{l} \\ -\frac{12}{l^2} \\ -\frac{6}{l} \end{matrix}\begin{matrix} \Delta_1 & \Delta_2 & \Delta_3 & \Delta_4 \\ -\frac{6}{l} & -\frac{12}{l^2} & -\frac{6}{l} & W_1 \\ 4 & \frac{6}{l} & 2 & W_2 \\ \frac{6}{l} & \frac{12}{l^2} & \frac{6}{l} & W_3 \\ 2 & \frac{6}{l} & 4 & W_4 \end{matrix}$$

$$[K]_{bc} = \frac{EI}{l}\begin{matrix} \\ \frac{12}{l^2} \\ -\frac{6}{l} \\ -\frac{12}{l^2} \\ -\frac{6}{l} \end{matrix}\begin{matrix} \Delta_3 & \Delta_4 & \Delta_5 & \Delta_6 \\ -\frac{6}{l} & -\frac{12}{l^2} & -\frac{6}{l} & W_3 \\ 4 & \frac{6}{l} & 2 & W_4 \\ \frac{6}{l} & \frac{12}{l^2} & \frac{6}{l} & W_5 \\ 2 & \frac{6}{l} & 4 & W_6 \end{matrix}$$



Element force and displacement notation



$$[K]_{ab} = \frac{EI}{l}\begin{matrix} \\ \frac{12}{l^2} \\ -\frac{6}{l} \\ -\frac{12}{l^2} \\ -\frac{6}{l} \end{matrix}\begin{matrix} \Delta_1 & \Delta_2 & \Delta_3 & \Delta_4 \\ -\frac{6}{l} & -\frac{12}{l^2} & -\frac{6}{l} & W_1 \\ 4 & \frac{6}{l} & 2 & W_2 \\ \frac{6}{l} & \frac{12}{l^2} & \frac{6}{l} & W_3 \\ 2 & \frac{6}{l} & 4 & W_4 \end{matrix}$$

$$[K]_{bc} = \frac{EI}{l}\begin{matrix} \\ \frac{12}{l^2} \\ -\frac{6}{l} \\ -\frac{12}{l^2} \\ -\frac{6}{l} \end{matrix}\begin{matrix} \Delta_3 & \Delta_4 & \Delta_5 & \Delta_6 \\ -\frac{6}{l} & -\frac{12}{l^2} & -\frac{6}{l} & W_3 \\ 4 & \frac{6}{l} & 2 & W_4 \\ \frac{6}{l} & \frac{12}{l^2} & \frac{6}{l} & W_5 \\ 2 & \frac{6}{l} & 4 & W_6 \end{matrix}$$

$$[K] = \frac{EI}{l}\begin{bmatrix} \frac{12}{l^2} & -\frac{6}{l} & -\frac{12}{l^2} & -\frac{6}{l} & 0 & 0 \\ -\frac{6}{l} & 4 & \frac{6}{l} & 2 & 0 & 0 \\ -\frac{12}{l^2} & \frac{6}{l} & \frac{24}{l^2} & 0 & -\frac{12}{l^2} & -\frac{6}{l} \\ -\frac{6}{l} & 2 & 0 & 8 & \frac{6}{l} & 2 \\ 0 & 0 & -\frac{12}{l^2} & \frac{6}{l} & \frac{12}{l^2} & \frac{6}{l} \\ 0 & 0 & -\frac{6}{l} & 2 & \frac{6}{l} & 4 \end{bmatrix}$$

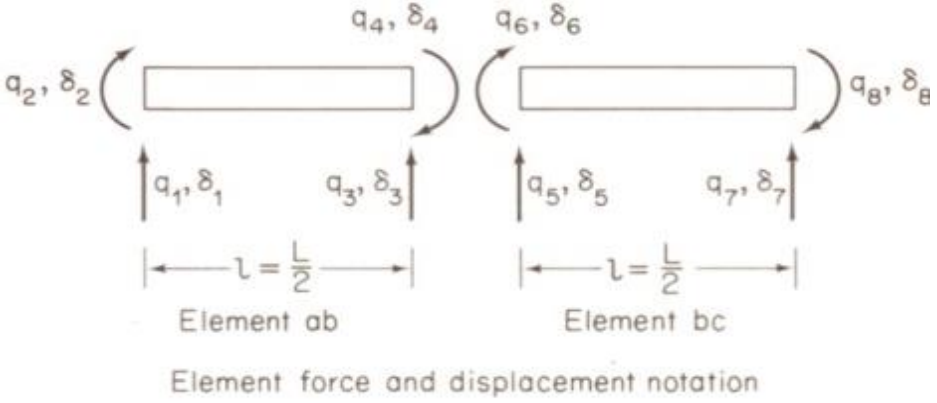
$$W_3^1 = \frac{EI}{l}\left\{-\frac{12}{l^2}\Delta_1 + \frac{6}{l}\Delta_2 + \frac{12}{l^2}\Delta_3 + \frac{6}{l}\Delta_4\right\}$$

$$W_3^2 = \frac{EI}{l}\left\{\frac{12}{l^2}\Delta_3 - \frac{6}{l}\Delta_4 - \frac{12}{l^2}\Delta_5 - \frac{6}{l}\Delta_6\right\}$$

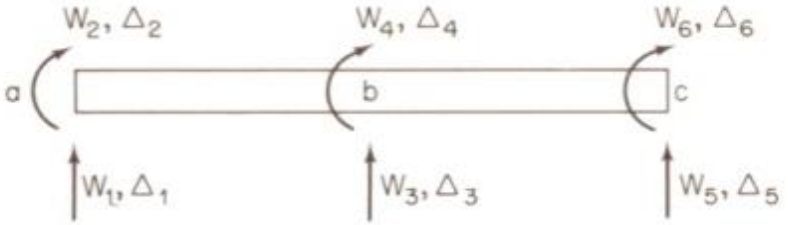
$$W_3 = W_3^1 + W_3^2 = \frac{EI}{l}\left\{-\frac{12}{l^2}\Delta_1 + \frac{6}{l}\Delta_2 + \left(\frac{12}{l^2} + \frac{12}{l^2}\right)\Delta_3 + \left(\frac{6}{l} - \frac{6}{l}\right)\Delta_4 - \frac{12}{l^2}\Delta_5 - \frac{6}{l}\Delta_6\right\}$$

$$[K]_{ab} = \frac{EI}{l}\begin{matrix} \\ \frac{12}{l^2} \\ -\frac{6}{l} \\ -\frac{12}{l^2} \\ -\frac{6}{l} \end{matrix}\begin{matrix} \begin{matrix} \Delta_1 & \Delta_2 & \Delta_3 & \Delta_4 \end{matrix} \\ \begin{bmatrix} \frac{12}{l^2} & -\frac{6}{l} & -\frac{12}{l^2} & -\frac{6}{l} \\ -\frac{6}{l} & 4 & \frac{6}{l} & 2 \\ -\frac{12}{l^2} & \frac{6}{l} & \frac{12}{l^2} & \frac{6}{l} \\ -\frac{6}{l} & 2 & \frac{6}{l} & 4 \end{bmatrix} \end{matrix}\begin{matrix} \\ W_1 \\ W_2 \\ W_3 \\ W_4 \end{matrix}$$

$$[K]_{bc} = \frac{EI}{l}\begin{matrix} \\ \frac{12}{l^2} \\ -\frac{6}{l} \\ -\frac{12}{l^2} \\ -\frac{6}{l} \end{matrix}\begin{matrix} \begin{matrix} \Delta_3 & \Delta_4 & \Delta_5 & \Delta_6 \end{matrix} \\ \begin{bmatrix} \frac{12}{l^2} & -\frac{6}{l} & -\frac{12}{l^2} & -\frac{6}{l} \\ -\frac{6}{l} & 4 & \frac{6}{l} & 2 \\ -\frac{12}{l^2} & \frac{6}{l} & \frac{12}{l^2} & \frac{6}{l} \\ -\frac{6}{l} & 2 & \frac{6}{l} & 4 \end{bmatrix} \end{matrix}\begin{matrix} \\ W_3 \\ W_4 \\ W_5 \\ W_6 \end{matrix}$$



Element force and displacement notation



$$[K]_{ab} = \frac{EI}{l}\begin{matrix} \\ \frac{12}{l^2} \\ -\frac{6}{l} \\ -\frac{12}{l^2} \\ -\frac{6}{l} \end{matrix}\begin{matrix} \begin{matrix} \Delta_1 & \Delta_2 & \Delta_3 & \Delta_4 \end{matrix} \\ \begin{bmatrix} \frac{12}{l^2} & -\frac{6}{l} & -\frac{12}{l^2} & -\frac{6}{l} \\ -\frac{6}{l} & 4 & \frac{6}{l} & 2 \\ -\frac{12}{l^2} & \frac{6}{l} & \frac{12}{l^2} & \frac{6}{l} \\ -\frac{6}{l} & 2 & \frac{6}{l} & 4 \end{bmatrix} \end{matrix}\begin{matrix} \\ W_1 \\ W_2 \\ W_3 \\ W_4 \end{matrix}$$

$$[K]_{bc} = \frac{EI}{l}\begin{matrix} \\ \frac{12}{l^2} \\ -\frac{6}{l} \\ -\frac{12}{l^2} \\ -\frac{6}{l} \end{matrix}\begin{matrix} \begin{matrix} \Delta_3 & \Delta_4 & \Delta_5 & \Delta_6 \end{matrix} \\ \begin{bmatrix} \frac{12}{l^2} & -\frac{6}{l} & -\frac{12}{l^2} & -\frac{6}{l} \\ -\frac{6}{l} & 4 & \frac{6}{l} & 2 \\ -\frac{12}{l^2} & \frac{6}{l} & \frac{12}{l^2} & \frac{6}{l} \\ -\frac{6}{l} & 2 & \frac{6}{l} & 4 \end{bmatrix} \end{matrix}\begin{matrix} \\ W_3 \\ W_4 \\ W_5 \\ W_6 \end{matrix}$$

$$[K] = \frac{EI}{l}\begin{bmatrix} \frac{12}{l^2} & -\frac{6}{l} & -\frac{12}{l^2} & -\frac{6}{l} & 0 & 0 \\ -\frac{6}{l} & 4 & \frac{6}{l} & 2 & 0 & 0 \\ -\frac{12}{l^2} & \frac{6}{l} & \frac{24}{l^2} & 0 & -\frac{12}{l^2} & -\frac{6}{l} \\ -\frac{6}{l} & 2 & 0 & 8 & \frac{6}{l} & 2 \\ 0 & 0 & -\frac{12}{l^2} & \frac{6}{l} & \frac{12}{l^2} & \frac{6}{l} \\ 0 & 0 & -\frac{6}{l} & 2 & \frac{6}{l} & 4 \end{bmatrix}$$

$$W_4^1 = \frac{EI}{l}\left\{-\frac{6}{l}\Delta_1 + 2\Delta_2 + \frac{6}{l}\Delta_3 + 4\Delta_4\right\}$$

$$W_4^2 = \frac{EI}{l}\left\{-\frac{6}{l}\Delta_3 + 4\Delta_4 + \frac{6}{l}\Delta_5 + 2\Delta_6\right\}$$

$$W_4 = W_4^1 + W_4^2 = \frac{EI}{l}\left\{-\frac{6}{l}\Delta_1 + 2\Delta_2 + \left(\frac{6}{l} - \frac{6}{l}\right)\Delta_3 + (4+4)\Delta_4 + \frac{6}{l}\Delta_5 + 2\Delta_6\right\}$$

For the fixed-end beam in Fig. 2-23, $\Delta_1 = \Delta_2 = \Delta_5 = \Delta_6 = 0$

Consequently, $[\Delta_a]$ consists of Δ_3 and Δ_4 and Eq. (2.159) tal the form

$$[W_a] = [K_{aa}][\Delta_a]$$

$$\begin{bmatrix} W_3 \\ W_4 \end{bmatrix} = \frac{EI}{l} \begin{bmatrix} \frac{24}{l^2} & 0 \\ 0 & 8 \end{bmatrix} \begin{bmatrix} \Delta_3 \\ \Delta_4 \end{bmatrix} \qquad (2.162)$$

Inverting the stiffness matrix and noting that $W_4 = 0$, one obtains

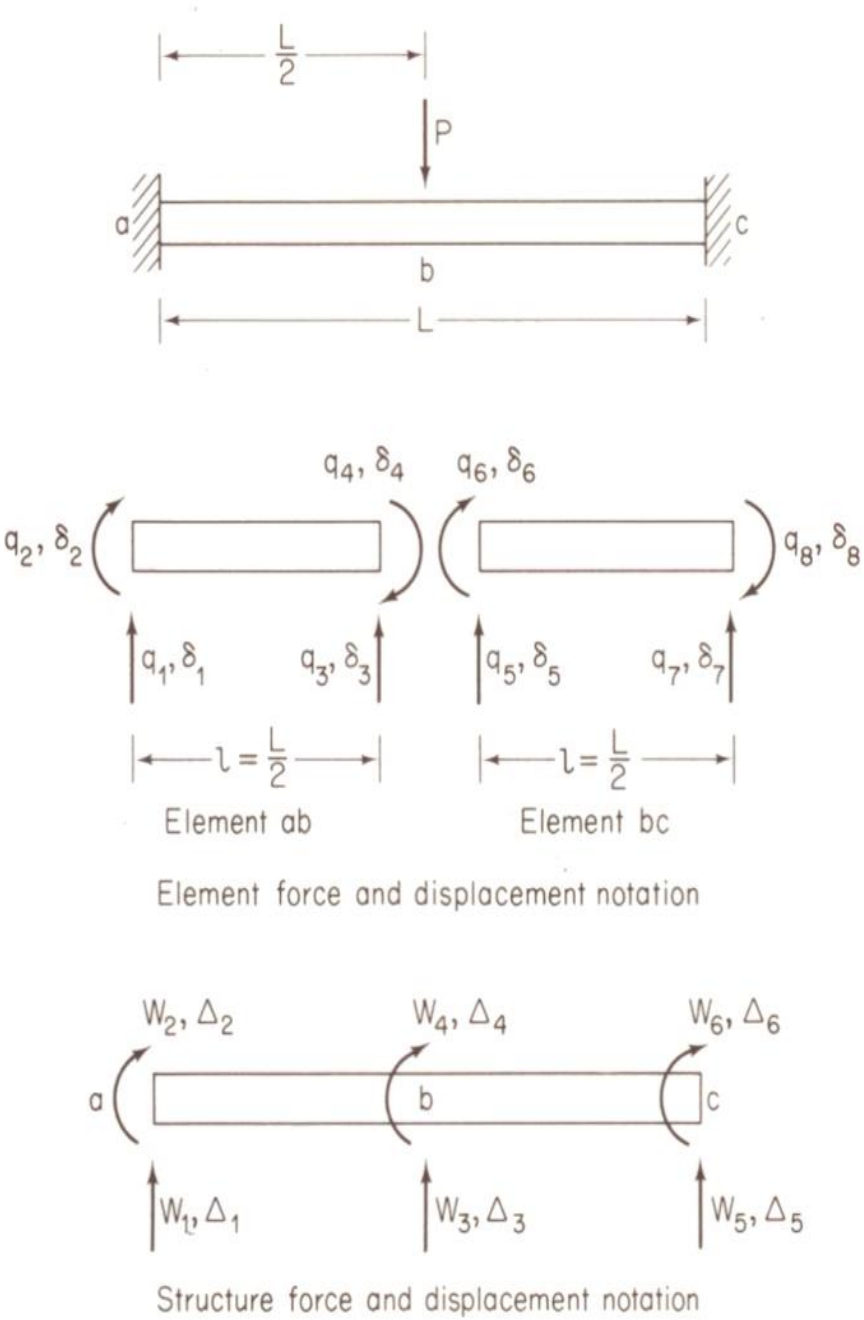
$$\begin{bmatrix} \Delta_3 \\ \Delta_4 \end{bmatrix} = \frac{l}{24EI} \begin{bmatrix} l^2 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} W_3 \\ 0 \end{bmatrix} \qquad (2.163)$$

from which the midspan deflection Δ_3 is found to be

$$\Delta_3 = \frac{W_3 l^3}{24EI} = \frac{W_3 L^3}{192EI} \qquad (2.164)$$

Having obtained $[\Delta_a]$ the reactions W_b can be determined using Eq. (2.160). Thus

$$[W_b] = [K_{ba}][\Delta_a]$$



Şekil 2.23

Example :

Figure 4.7a depicts a statically indeterminate beam subjected to a transverse load applied at the midspan. Using two flexure elements, obtain a solution for the midspan deflection.

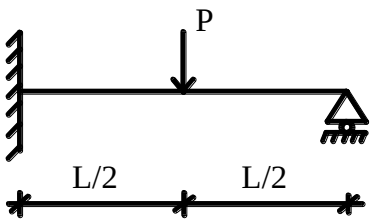


Fig. 4.7a Loaded beam

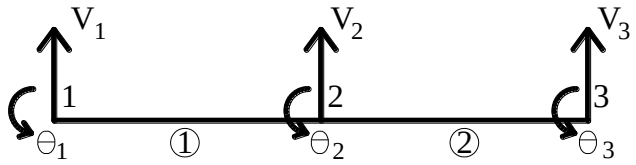


Fig. 4.7b Element and displacement designations

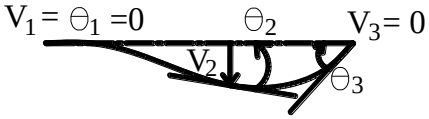


Fig. 4.7c Displacement solution

Global displacement .	Eleman 1	Eleman 2
1	1	0
2	2	0
3	3	1
4	4	2
5	0	3
6	0	4

By using element-system displacement relations, the set of system equations are obtained as follows:

$$\frac{EI_z}{L^3}\begin{bmatrix} 96 & 24L & -96 & 24L & 0 & 0 \\ 24L & 8L^2 & -24L & 4L^2 & 0 & 0 \\ -96 & -24L & 192 & 0 & -96 & 24L \\ 24L & 4L^2 & 0 & 16L^2 & -24L & 4L^2 \\ 0 & 0 & -96 & -24L & 96 & 24L \\ 0 & 0 & 24L & 4L^2 & 24L & 8L^2 \end{bmatrix} \begin{Bmatrix} v_1 \\ \theta_1 \\ v_2 \\ \theta_2 \\ v_3 \\ \theta_3 \end{Bmatrix} = \begin{Bmatrix} F_1 \\ M_1 \\ F_2 \\ M_2 \\ F_3 \\ M_3 \end{Bmatrix}$$

Boundary conditions: $v_1 = \theta_1 = v_3 = 0$

Invoking the boundary conditions $v_1 = \theta_1 = v_3 = 0$, the reduced equations become

$$\frac{EI_z}{L^3}\begin{bmatrix} 192 & 0 & 24L \\ 0 & 16L^2 & 4L^2 \\ 24L & 4L^2 & 8L^2 \end{bmatrix}\begin{Bmatrix} v_2 \\ \theta_2 \\ \theta_3 \end{Bmatrix} = \begin{Bmatrix} -P \\ 0 \\ 0 \end{Bmatrix} \rightarrow v_2 = -\frac{7PL^3}{768EI_z} \quad , \quad \theta_2 = -\frac{-PL^2}{128EI_z} \quad , \quad \theta_3 = \frac{PL^3}{32EI_z}$$

Substitution of the nodal displacement values into the constraint equations gives the reactions as

$$F_1 = \frac{11P}{16} \quad , \quad F_3 = \frac{5P}{16} \quad , \quad M_1 = \frac{3PL}{16}$$

Checking the overall equilibrium conditions for the beam, we find

$$\sum F_y = \frac{11P}{16} - P + \frac{5P}{16} = 0 \qquad \sum M = \frac{3PL}{16} - \frac{PL}{2} + \frac{5PL}{16} = 0$$

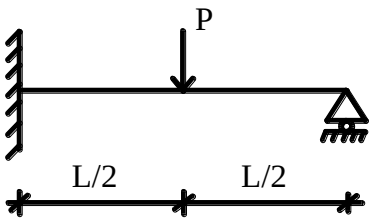


Fig. 4.7a Loaded beam

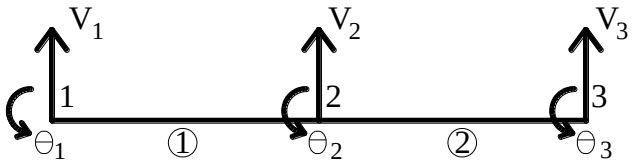


Fig. 4.7b Element and displacement designations

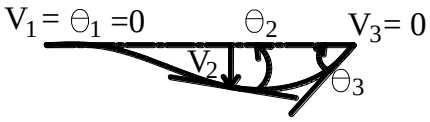


Fig. 4.7c Displacement solution

YILDIZ TEKNİK ÜNİVERSİTESİ
İNŞAAT MÜHENDİSLİĞİ BÖLÜMÜ
MEKANİK ANABİLİM DALI

ELASTIC STABILITY-BUCKING

Eighth Chapter

Two Weeks, Eighth in line

PROF. DR. TURGUT KOCATÜRK

1.1 Introduction

In the mechanics of deformable bodies, there are three critical cases for a body under the effect of external loads which is in equilibrium:

1). The stresses in the investigated element mustn't exceed the allowable stresses.

2). The displacements in the investigated element mustn't exceed the allowable displacements.

3). The equilibrium must be stable.

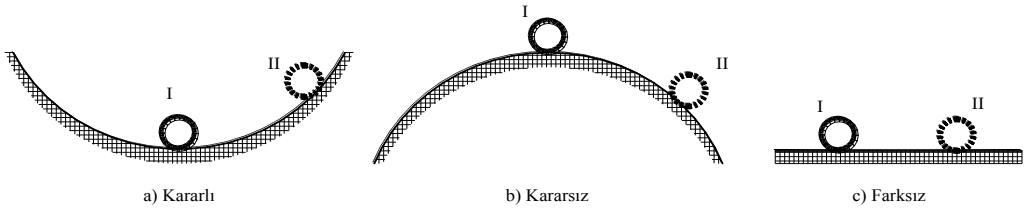
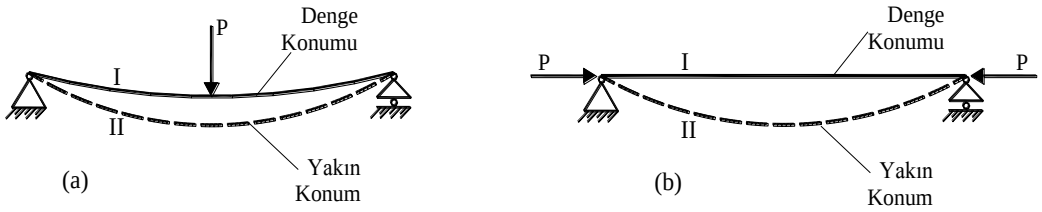


Fig. 1.1. Equilibrium cases

Whether or not an equilibrium state is stable can be achieved by giving a transient displacement to the system which is in equilibrium.

If the system goes back to its original equilibrium state when the transient displacement is removed then it is stable, otherwise it is unstable.

In the stress problems, we want to find the equilibrium position and the stresses in this position. But the equilibrium position in the stability problems is explicit generally. What is important is whether or not it is stable.



(a) Stress problem **(b) Stability problem**

Fig. Equilibrium and near equilibrium positions

1.2 Various Stability Problems

It is possible to classify the stability problems according to the relation between the force and deviation. These groups are as follows:

a) Bifurcation buckling problems,

b) Snap-through buckling problems.

1.2a Bifurcation buckling problems

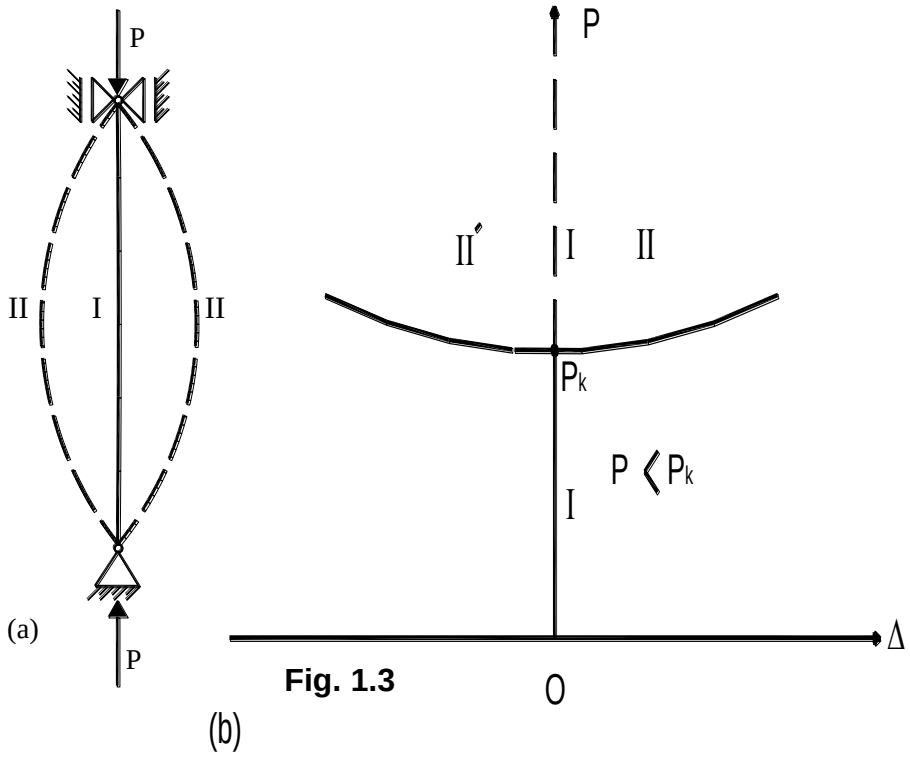


Fig. 1.3

In the bar given by Fig. 1.3a,

For $P < P_k$, if the system is forced by a transient effect to displace, after the force is removed, the bar makes a vibration movement about the equilibrium position I. Namely the systems is stable. Stable condition

For $P = P_k$, the equilibrium is called as neutral equilibrium.

For $P > P_k$, the equilibrium is unstable. Unstable condition

Also, the tangents of the branches II and II' which correspond to the displaced equilibrium situations, are horizontal at the point $P = P_k$; then, it is possible to pass near equilibrium positions without any load increment. But in moving these near equilibrium positions, displacements increase excessively. Although there are equilibrium positions like II ve II′ in the case **$P > P_k$, displacements become very large in these equilibrium positions. These large displacements cause large deformations and large stresses and therefore the system fails.**

1.2b Snap-Through Stability Problems:

The cause of the multi-valueness is not the bifurcation in the curve of load-displacement. In this type, the curve in question includes extremum points and therefore different equilibrium configurations can be obtained for the same loading and some of these can be stable while others can be unstable. A simple example for this case is shown in Fig. 1.4. In this example, there are two elastic compression bars.

By using vertical equilibrium equation on the displaced configuration we can obtain the relation P = P(δ) as follows:

$$P = 2EA\left(\frac{h-\delta}{l}\right)\left(\frac{l}{\sqrt{l^2-2h\delta+\delta^2}}-1\right) \quad \textbf{(1.1)}$$

EA : Extension rigidity

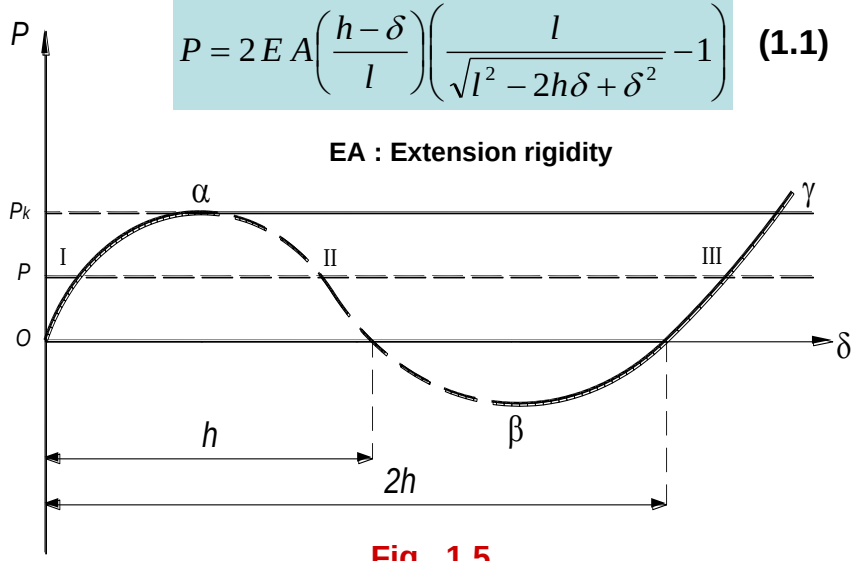


Fig. 1.5

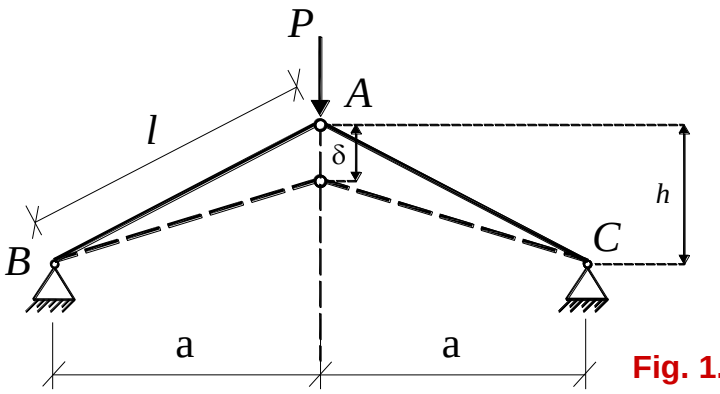


Fig. 1.4

The relation between P and δ is no longer linear and the superposition principle is no more valid: This is because the equilibrium equations are obtained on the displaced configuration of the system.

If the equation (1.1) is represented on the plane (P, δ) graphically, then the figure 1.5 is obtained.

$P_\alpha = P_k$ value is callled as critical snap-through load and this type of stability problem is called as snap-through stability. For $P < P_k$, As it can be seen from Fig. **1.5, there are three equilibrium configurations like I, II, III. I and III in these configurations are stable although II is unstable. In Fig. 1.6, these three configurations are shown.**

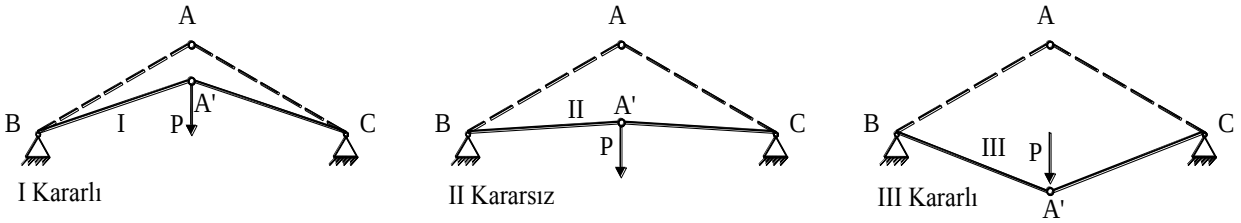


Fig. 1.6

The problems encountered in practice are generally bifurcation stability problems. Therefore the bifurcation stability type problems will be dealt here.

The results obtained until now will be given below:

1). Elastic stability is not a linear theory because of (due to, owing to) the equilibrium equations are written on the displaced configuration. Hence there is no proportionality between load-displacement.

2). Some theorems or principles which are based on linear theory are no longer valid. For example Superposition law, Betti reciprocity law etcetera.

1.3 Solution Methods in Stability

There are three methods to decide on whether an equilibrium is stable or not.

a). Dynamic method
b). Static Method
c). Energy method

These methods will be investigated on a very simple example problem:

The rigid bar with length *l* shown in Fig. 1.7 is attached to the ground by an elastic hinge.

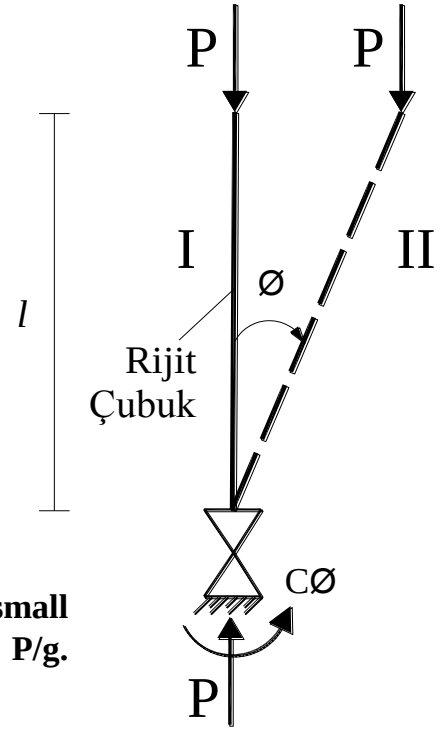


Fig. 1.7

c is the elastic hinge constant; Which condition must the load P satisfy that for this case the configuration I is stable? We will give answer to this question by using the dynamical method first.

1.3a Dynamic Method

Dynamic method can be applied to all kinds of systems.

The change of movement of the displaced configuration is investigated in the dynamic method.

The only mass in Fig. 1.7 is assumed to be at the end of the bar : The mass of the bar is very small compared to the mass at the end and therefore neglected. The value of the mass is assumed as P/g. The equation of motion in this case is

$$(P/g)l^2\frac{d^2\phi}{dt^2} = -c\phi + Pl\sin(\phi) \quad \textbf{veya} \quad \frac{d^2\phi}{dt^2} + \frac{g.(c - pl)\phi}{pl^2} = 0 \qquad \textbf{(1.3)}$$

The type of the motion depends on the sign of the second term. If this term is positive, the equation is as follows:

$$\frac{d^2\phi}{dt^2} + w^2\phi = 0 \quad (2.4)\ ;\ w^2 = \frac{g(c - pl)}{pl^2} > 0 \qquad \textbf{(1.4)}$$

This shows a simple harmonic motion around $\phi = 0$ and the solution of it is

$$\phi = A\sin(\omega t) + B\cos(\omega t)$$

A and B are dependent on the initial conditions. These initial effects can be made as small as we want. In other words we can take these disturbing effects as small as we want. For example if we choose for t = 0, $\phi = 0$ and $\phi' = a$ then $A = a/\omega$ and $B = 0$.

$$\phi(t) = a\frac{\sin(\omega t)}{\omega} \qquad \textbf{(1.5)}$$

Here α is the angular velocity which can be chosen as small as we want. The graphical presentation of Eq. (1.5) is shown by Fig. 1.8.

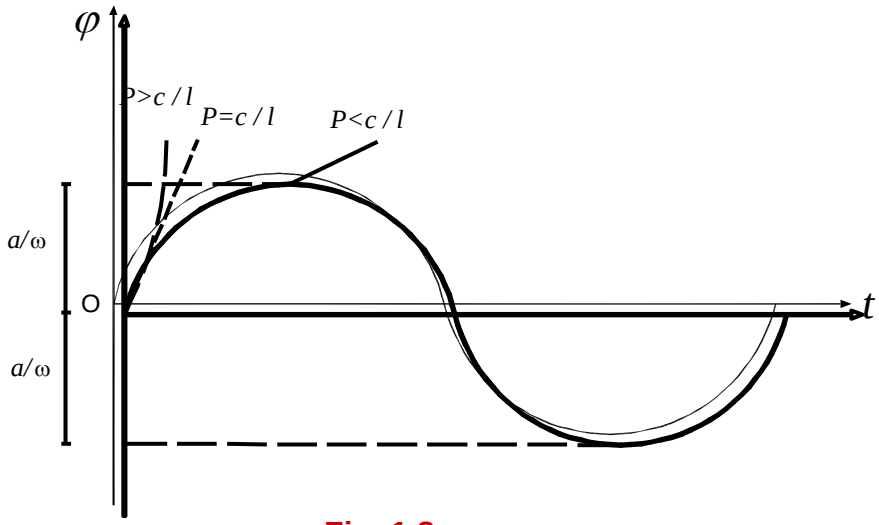


Fig. 1.8

According to the definition of the stability, the condition that the motion always remains around $\phi = 0$, it is sufficient that the condition (1.4) is satisfied. In that case stability condition is as follows;

$$P < \frac{c}{l} \qquad \textbf{(1.6)}$$

1.3b Static Method

Lets investigate the same problem by static method and look for the answer of that question:

Is there a near equilibrium configuration II which is very near near to the equilibrium configuration I?

To answer this question lets assume that the configuration II is an equilibrium configuration. The equilibrium condition for II is

$$Pl\sin(\phi) - c\phi = 0 \quad \textbf{(1.7)}$$

$$\Rightarrow \quad P = \frac{c}{l}\frac{\phi}{\sin(\phi)} \quad \textbf{(1.8)}$$

We are seeking a second equilibrium configuration which is very close to the configuration I; this is possible only for $\phi \Rightarrow 0$ in Eq. (1.8);

$$P = \frac{c}{l}\frac{\phi}{\sin(\phi)}\bigg|_{\phi\Rightarrow 0} = \frac{c}{l}$$

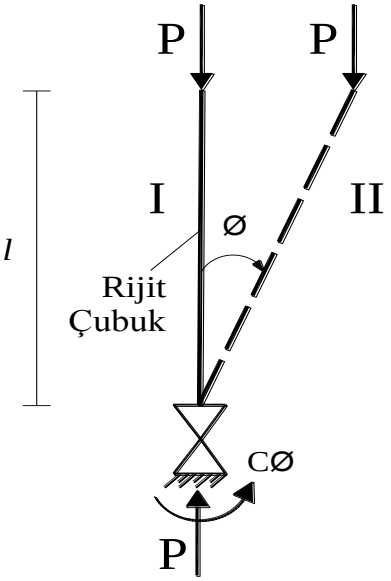


Fig. 1.7

In that case if the configuration I is neutral equilibrium configuration then the load is;

$$P_k = \frac{c}{l} \quad \textbf{(1.9)}$$

This load is called critical load. If the load P satisfy the following condition;

$$P < P_k = \frac{c}{l} \quad \textbf{(1.10)}$$

then there is no equilibrium configuration which is near to or far from the configuration I. The only equilibrium configuration is I. (1.10) is the stability criteria which is the same with the one we obtained by Eq. (1.6)

It must be noted importantly that the value of the ratio $\phi / \sin\phi$ is a little different from unity in the vicinity of $\phi = 0$; This difference is only in the second order and therefore there will be no difference in terms of loads between the neutral equilibrium configuration and other near equilibrium conditions. In other words while $P = P_k$ =constant, the system can pass to near equilibrium configurations.

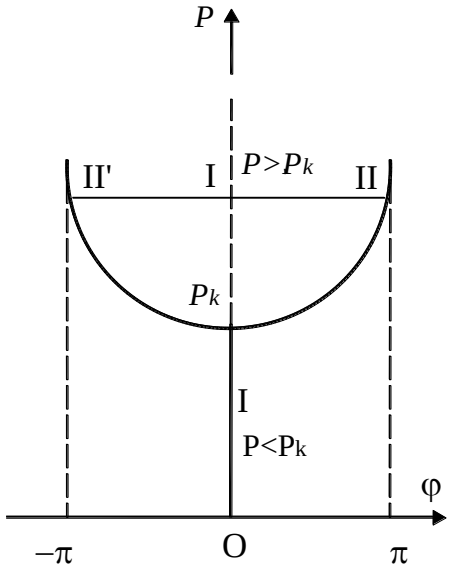


Fig. 1.9

For apprehending the problem better, lets show the function (1.8), which describes equilibrium configurations, graphically in the (P,ϕ) plane (Fig. 1.9).

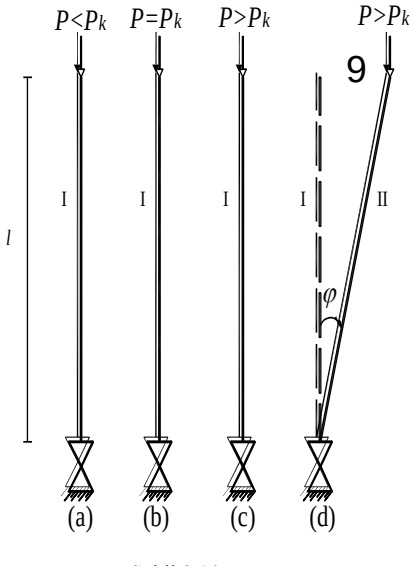


Fig. 1.10

1.3c Energy Method

Elastic potential energy of spring is; $\prod_i = \frac{1}{2} c\varphi^2$

External potential of P is $\prod_d = p.l.\cos(\varphi)$

Total potential energy is

$$\pi(\phi) = \prod_i + \prod_d = \frac{1}{2} c\varphi^2 + p.l.\cos(\varphi) \qquad \textbf{(1.11)}$$

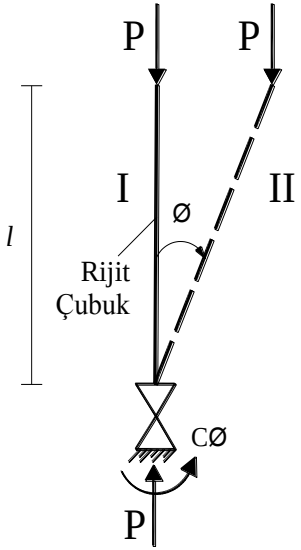


Fig. 1.7

It is a general principle of mechanics that energyy is extremum in the equilibrium configuration of the systems. If the exstrem value is a minimum, then the equilibrium configuration is a stable one. In that case, the problem is related to the following question:

Does the equation (1.11) have a minimum at $\varnothing \neq 0$.

This is only possible if the following conditions are satisfied for $\varnothing \neq 0$ **:**

$$\frac{d\pi}{d\varphi} = 0 \qquad \frac{d^2\pi}{d\varphi^2} > 0 \qquad \textbf{(1.12)}$$

The following condition is always satisfied for $\varnothing = 0$ **, namely I is always an equilibrium configuration:**

$$\frac{d\pi}{d\varphi} = c\phi - Pl\sin(\phi) = 0$$

But the answer to the question when is I stable can be found by using the second condition of Eq. (1.12).

$$\frac{d^2\pi}{d\varphi^2} > 0 \Rightarrow \quad c - pl > 0 \qquad \textbf{(1.13)}$$

This expression too is the stability condition which was obtained by using other methods.

The equilibrium configurations which correspond to the maximum of the potential energy are unstable equilibrium configurations. If the system is conservative, then the three methods give the same result. If the external forces don't have a potential, we can't use energy method, and therefore only the static method and the dynamic method are left to use. The values obtained by using these two methods become different from each other when the system is not conservative. In this case the true result can be obtained only by using the dynamic method.

LINEAR COLUMN THEORY-EIGENVALUE PROBLEM

The small-deflection column theory is based on a linear differential equation and is, therefore, known as the linear column theory. By comparison, the large-deflection theory of columns to be studied later is based on a nonlinear differential equation and is referred to as a nonlinear theory. Although the small-deflection column theory can be considered linear when contrasted with large-deflection theories, it is not linear in the same sense of the word as simple beam bending theory. In the latter, equilibrium is based on undeformed geometry and deflections are proportional to the applied loads, which is certainly not true for the Euler column.

Thus linear theory is evidently different from both simple-flexure and large-deformation column theory. As a matter of fact, it does belong to an entirely different class of problems known as eigenvalue problems. These problems are characterized by the fact that nonzero solutions for the dependent variable exists only for certain discrete values of some parameter. The values of the parameter, for which nonzero solutions exist, are known as eigenvalues, and the solutions as eigenvectors. Only the shape and not the amplitude of the eigenvector can be determined in an eigenvalue problem. In a stability problem such as the axially loaded column, the loads

$$\frac{n^2\pi^2 EI}{L^2}$$

At which nonzero deflections are possible are the eigenvalues, and the deflected shapes that can exist at these loads are eigenvectors. The smallest eigenvalue is the critical load and the corresponding eigenvector is the buckling mode shape.

1.4 Buckling load of an elastic rod whose both ends are pinned:

As seen in Fig. 1.11, the elastic column with length l and flexural rigidity EI is under the effect of end loads P. We assume that there is a displaced equilibrium configuration II of the column near to the equilibrium configuration I, namely lets assume that the equilibrium configuration I is neutral configuration. We can prove that the configuration II can not be satisfied for all the values of the load.

Lets seek the function v(z) which defines the equilibrium configuration II. The stress resultants at any section of this configuration are,

$$N = -P\cos(\phi), T = P\sin(\phi) \text{ ve } M = Pv$$

Due to the configuration II is very close to the configuration I, we can assume that

$$\varphi \cong tg\varphi \quad \cos\varphi \cong 1 \quad \sin\varphi \cong \varphi$$

and then the stress resultants can be obtained as follows:

$$N = -P, \quad T = pv', \quad M = p.v \qquad \textbf{(1.14)}$$

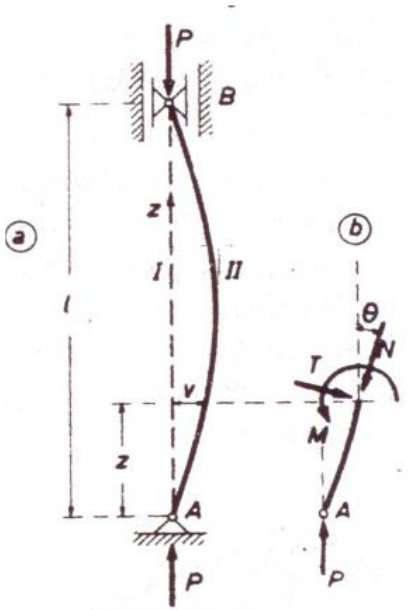


Fig. 1.11

By neglecting the effect of shear force on the elastic curve, differential equation of the curve II can be obtained under the given conditions as follows:

$$\frac{d^2v}{dz^2} = -\frac{M}{EI} = -\frac{Pv}{EI} \quad \text{or} \quad \frac{d^2v}{dz^2} + \frac{Pv}{EI} = 0 \qquad \textbf{(1.15)}$$

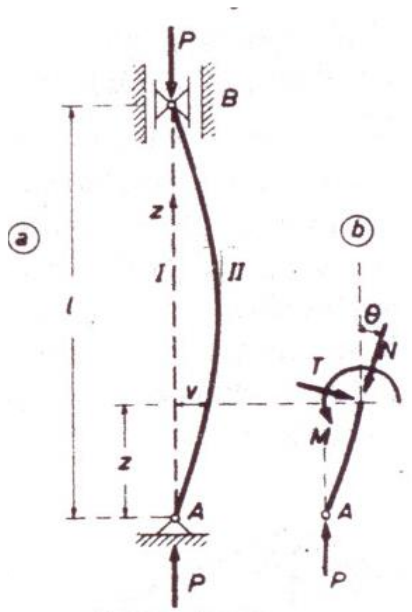


Fig. 1.11

Boundary conditions are as follows:

$$v(0) = 0 \quad ve \quad v(l) = 0 \qquad \textbf{(1.16)}$$

For the neutral equilibrium configuration, the solution of linear homogeneous differential equation (1.15) will be investigated under homogeneous boundary conditions.

There are two solutions for these types of problems:

a) Obvious (explicit) or trivial solution; we don't deal with this solution which corresponds to the equilibrium configuration I.

b) The solution which is not obvious; v(z) ≠ 0 corresponds to the equilibrium configuration II. This solution comes up as follows:

$$v(z) = A\sin\lambda z + B\cos\lambda z \quad ; \qquad \lambda^2 = \frac{P}{EI} > 0 \qquad \textbf{(1.17)}$$

A and B are the constants which will be obtained by using the boundary conditions:

From the boundary condition v(0) = 0, B = 0 and therefore the elastic curve is a sine curve as follows:

$$v(z) = A\sin(\lambda z) \qquad \textbf{(1.18)}$$

$v(z) = A\sin(\lambda z)$ **(1.18)** $v(z) = A\sin\lambda z + B\cos\lambda z \quad ; \quad \lambda^2 = \frac{P}{E\,I} > 0$ **(1.17)** …

From the other boundary condition v(l) = 0 we get;

$$A\sin\lambda l = 0 \Rightarrow \sin\lambda l = 0 \Rightarrow \lambda l = n\pi; n = 1, 2, 3 \qquad \textbf{(1.19)}$$

(1.17) and (1.19) dan $\lambda^2 = \frac{n^2\pi^2}{l^2} = \frac{P}{EI}$ **veya** $P = \frac{n^2\pi^2 EI}{l^2}$ **(1.20)**

$$P_k = \frac{n^2\pi^2 EI}{l^2} \qquad \textbf{n = 1, 2, 3, …} \quad \textbf{(1.21)}$$

The meaning of this is that the load P must be equal to the critical load P_k for possible equilibrium configurations which are very near to the equilibrium configuration I. These numerous particular values are called Euler critical loads. Technically the least critical load for n = 1 is important in practical applications:

$$P_k = \frac{\pi^2 EI}{l^2} \qquad \textbf{(1.22)}$$

The straight eauilibrium configuration I is;

Stable for $P < P_k$
Neutral for $P = P_k$
Unstable for $P > P_k$

The equilibrium configurations II for n = 1, n = 2 and n = 3 are drawn in Fig. 1.12.

All the equilibrium configurations except the one n = 1 involve intermediary inflection points and they are unstable configurations.

The critical loads for n > 1 is called as high critical loads and they are 4, 9, 16,… times the load given by (1.22).

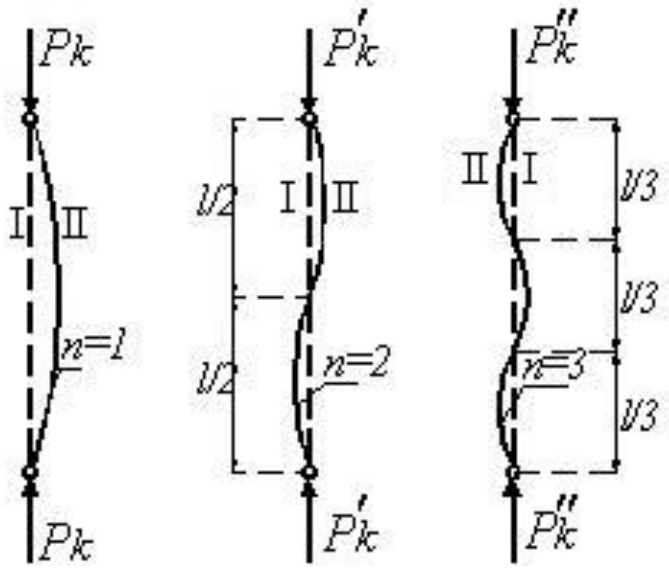


Fig. 1.12

In the bifurcation stability problems of elastic columns, bifurcations always start from critical points. The absis in Fig. 1.33 shows the biggest displacement which is the magnitude A. The first bifurcation starts at the least critical load

$$P_k = \frac{\pi^2 EI}{l^2}$$

and A becomes different from zero in the configurations II. The configuration I is now unstable for P > Pk loading. However there are again bifurcations at the higher values given below.

$$P = P_k' = \frac{4\pi^2 EI}{l^2} \quad \textbf{ve} \quad P = P_k'' = \frac{9\pi^2 EI}{l^2}$$

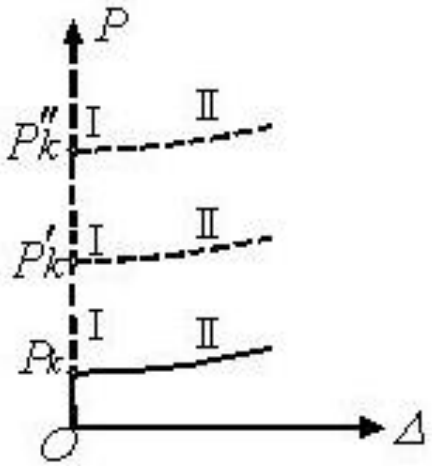


Fig. 1.13

But all of these bifurcations, like their starting points, corresponds to unstable points and are not important technically. Although A is unclear, we assumed that it is very small; because only the equilibrium configurations II which are very close to the equilibrium configurations I are considered. Small unclear valueness of A shows that bifurcation curves starting from $P=P_k$ have horizontal tangents (Fig. 1.13). Forms of bifurcation curves can not be found by using the equations given until now.

When $P < P_k$, the equilibrium configuration I is stable while it is neutral if the load reaches to P_k, and it may take another configuration shown by II; the phenomena is called buckling if the column passes to a curved or bended configuration while the load is constant.

The important point for an engineer is to be careful that the column doesn't buckle and to take necessary precautions not to let the column buckle. If the column buckles, the stresses due to bending form may be so greater that the column fails immediately under the effect of these.

Example:

Find the critical loads for the column shown in Fig. 1.14 for the lengths l = 25 cm, 50 cm and 100 cm.

$$P_k = \frac{\pi^2 EI}{l^2} = \frac{\pi^2 2{,}1.10^7.1.1^3}{12l^2} = \frac{17254300}{l^2}$$

$l = 25 \Rightarrow P_k = 27607$ N

$l = 50 \Rightarrow P_k = 6902$ N

$l = 100 \Rightarrow P_k = 1725$ N

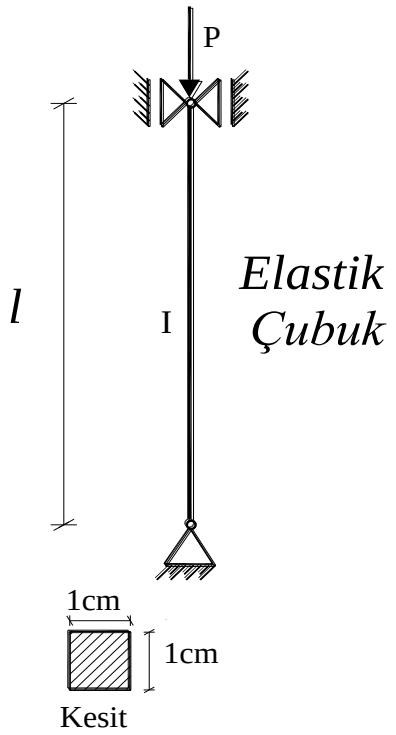


Fig. 1.14

It can be seen from the loads P_k that with decrease in the length of the column the stress problem gains importance instead of buckling.

If we summarize, an engineer firstly should think that the failure cause is buckling in long and slender columns while it is crushing due to the compression in short columns.

1.5 General Theory of Elastic Columns

Let's generalize the previous problem with respect to its boundary conditions. By increase in the load lets assume that there is a neutral equilibrium configuration II which is very close to the equilibrium configuration I. For obtaining a differential equation which is independent of the boundary conditions, let's investigate the equilibrium of a small element cut out from the form II of column:

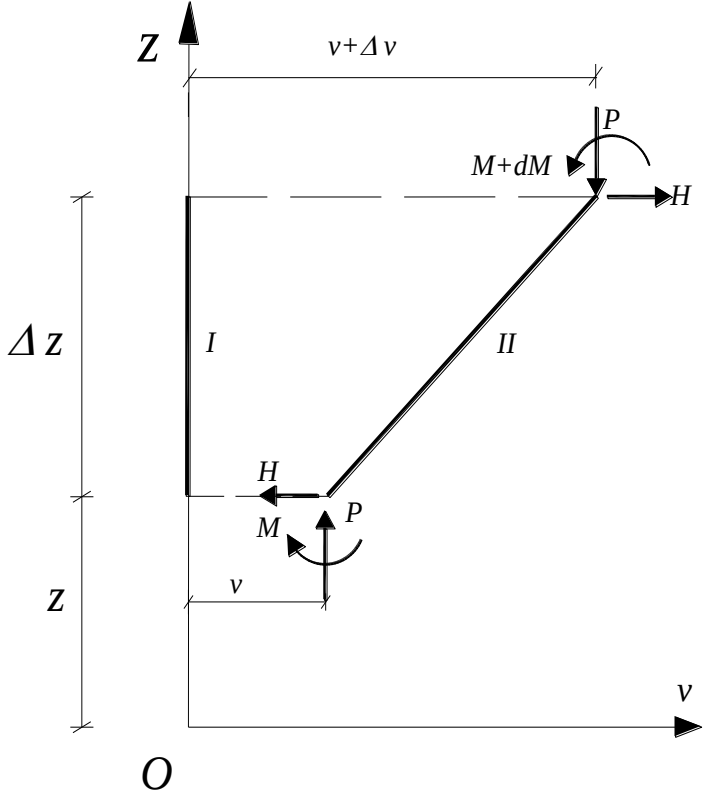


Fig. 1.15

An equation which is not zero identically is the equation of equilibrium of moments.

$$\sum M = 0 = M + dM - M - P.\Delta v - H.\Delta z = 0 \quad \textbf{(1.23)} \quad \textbf{or}$$

$$\frac{dM}{dz} = H + P\frac{dv}{dz} \Rightarrow H = \frac{dM}{dz} - P\frac{dv}{dz} \quad \textbf{(1.24)}$$

By considering the relation between curvature and moment;

$$\frac{M_x}{EI_x} = \frac{1}{\rho_x} = -\frac{d^2v}{dz^2} \Rightarrow M_x = -EI_x.v'' \quad \textbf{(1.25)}$$

By substituting Eq. (1.25) in Eq. (1.24); $H = -[(EI_x v'')' + pv']$ **(1.26)**

By differentiating this equation with respect to z , we can eliminate the horizontal force H which is constant

$$EI_x\frac{d^4v}{dz^4} + P\frac{d^2v}{dz^2} = 0 \quad \textbf{(1.27)} \quad \textbf{If we define that} \quad P/EI_x = \lambda^2$$

then Eq. (1.27) becomes $\frac{d^4v}{dz^4} + \lambda^2\frac{d^2v}{dz^2} = 0$ **(1.28)**

The solution of this equation is as follows: $v(z) = c_1.\sin\lambda z + c_2.\cos\lambda z + c_3.z/l + c_4$ **(1.29)**

We will write now the possible boundary conditions for various boundary conditions:

a). For hinged (pinned) supports $v = 0;\ v'' = 0$

b). For fixed supports $v = 0;\ v' = 0$

c). For a free end loadad by a concentrated force $v'' = 0;\ (EI\ v'')' + P v' = 0$

We can write two boundary conditions for each end. A fourth order linear homogeneous differential equation can be solved under the again homogeneous boundary conditions. This situation brings us to the eigenvalue problem which is known from the theory of differential equation.

We will find the critical load of a column fixed at one end and free at the other

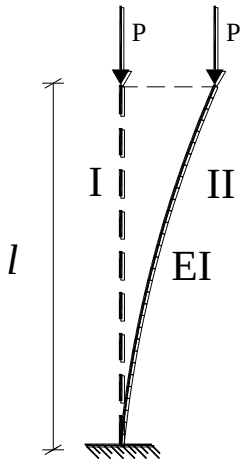


Fig. 1.16

Boundary conditions:

$$z = 0 \Rightarrow v = 0$$
$$z = 0 \Rightarrow v' = 0$$
$$z = l \Rightarrow v'' = 0$$
$$z = l \Rightarrow v''' + \lambda^2 v' = 0$$

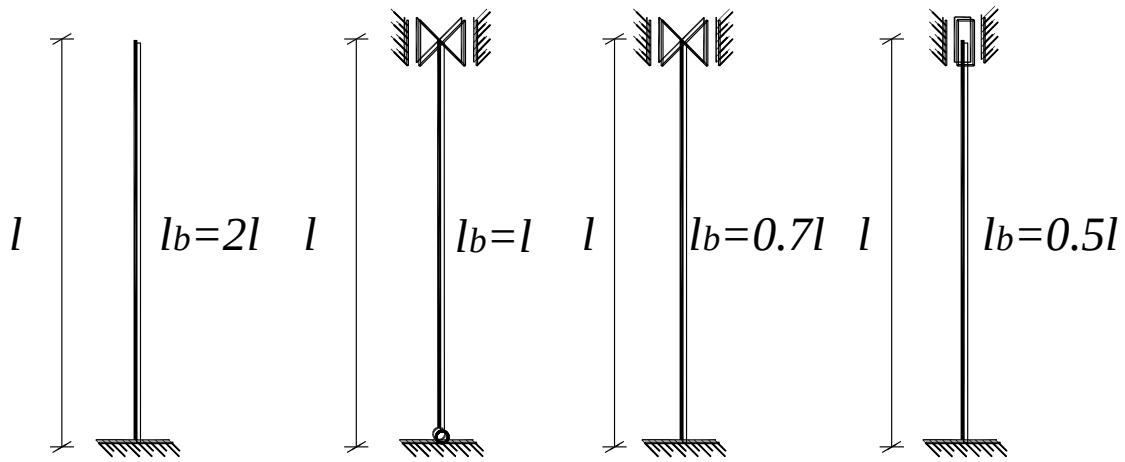
$$D = \begin{vmatrix} 0 & 1 & 0 & 1 \\ \lambda & 0 & 1/l & 0 \\ -\lambda^2 \sin \lambda l & -\lambda^2 \cos \lambda l & 0 & 0 \\ 0 & 0 & \lambda^2 / l & 0 \end{vmatrix} = \frac{\lambda^5}{l} \cos \lambda l = 0$$

The roots of the equation give the eigenvalues: $\lambda\, l = \pi/2;\ 3\pi/2;\ 5\pi/2;.....;\ (2n+1)\,\pi/2$ **(1.30)**

The load with respect to the least root is $\lambda^2 = \frac{P_k}{EI} \Rightarrow P_k = EI(\pi/2l)^2 \Rightarrow P_k = \frac{\pi^2 EI}{4l^2}$ **(1.31)**

Critical loads for other boundary conditions are determined by using the solution method mentioned before and given collectively in the table below:

Table 1



$P_k = \pi^2 EI/(2l)^2$ $P_k = \pi^2 EI/l^2$ $P_k = \pi^2 EI/(0.7l)^2$ $P_k = \pi^2 EI/(0.5l)^2$

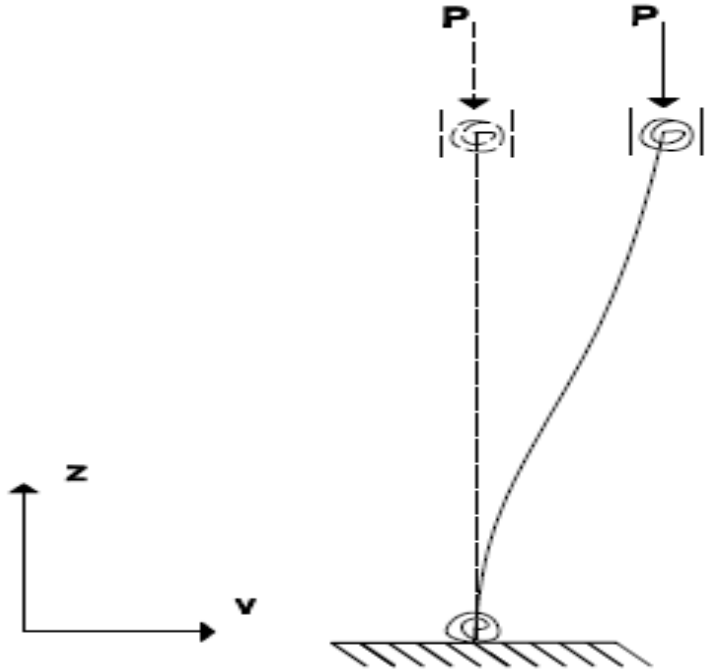
We can collect all the critical load formulas for various boundary conditions in only one form. This form is the form of critical load obtained for pinned-pinned column. The general expression is as follows:

$$P_k = \frac{\pi^2 EI}{l_b^2} \quad \textbf{(1.32)}$$

Here l_b is called as effective length (buckling length) of the column. This is the lenghth between the two inflection points of the elastic curve of equilibrium configuration II. In the inflection points M = 0 or v" = 0. The same reasoning can be used for all the other possible support conditions.

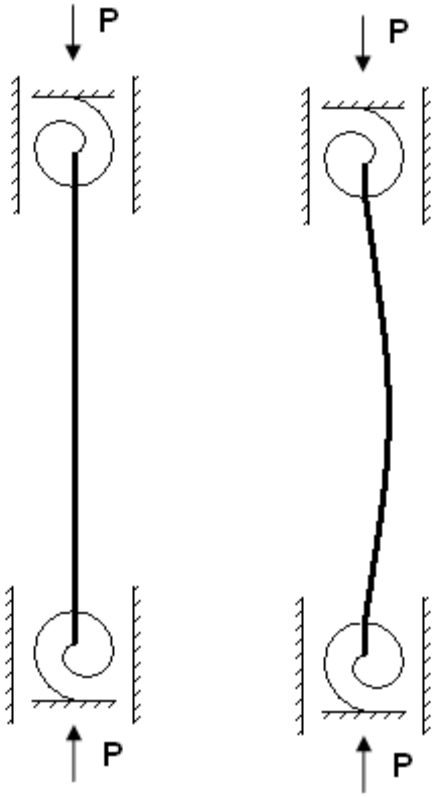
Elastically supported columns

Sideway permitted

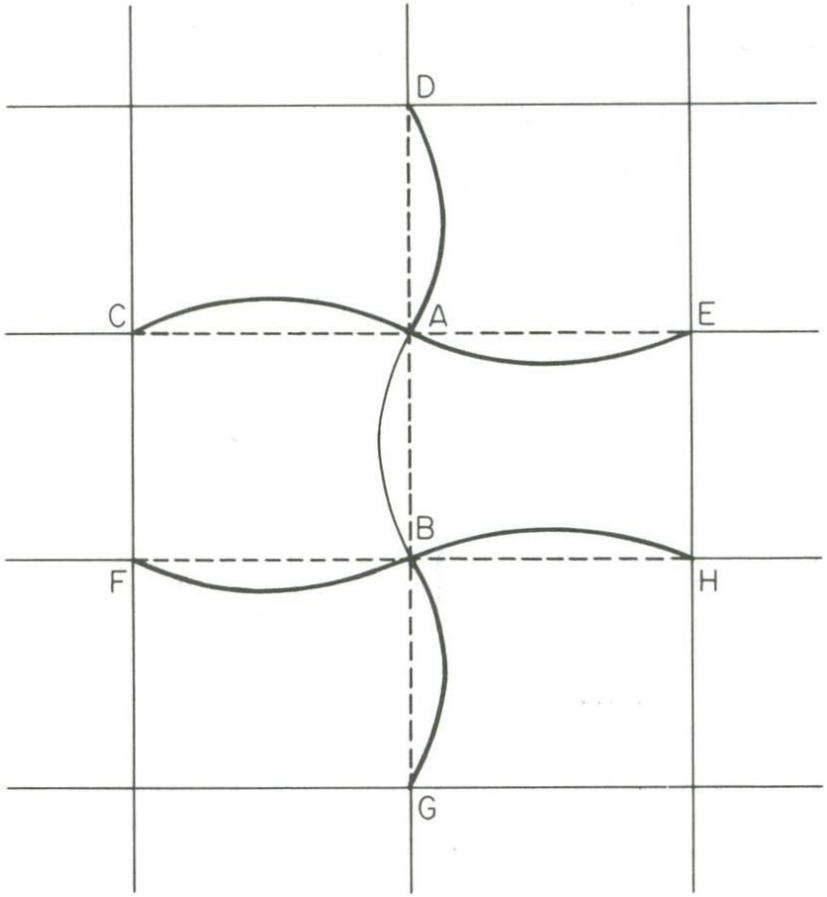


Rotational spring

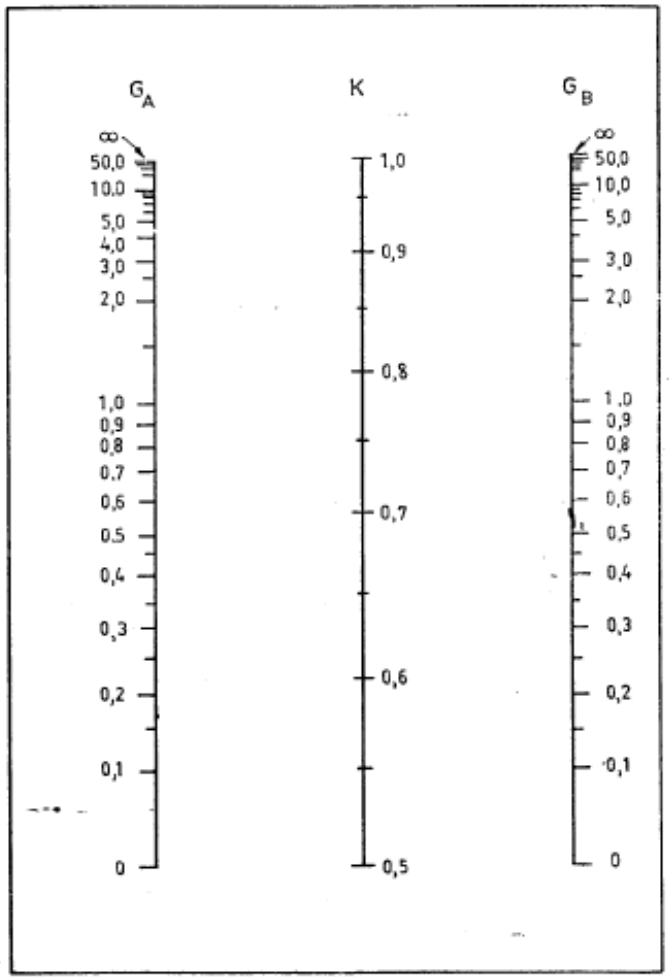
Sideway prevented



Effective (or buckling) length of framed columns



(a) Framed column and subassemblage of adjacent members assumed to resist its deformation



$$G = \frac{\sum I_c / S_c}{\sum I_g / S_g}$$

After the coefficients GA and GB at column ends are determined, K values is obtained from Charts 4-5. The effective length is obtained as $S_k = K.s$

The effective (buckling) length of framed columns can be determined from Charts 4-5 on the condition that sideway is permitted or prevented. The G coefficient for the ends A and B of framed columns can be obtained from Charts 4-5 as

$$G = \frac{\sum I_c / S_c}{\sum I_g / S_g}$$

where ;

G = A coefficient used in determining the effective length,

Ic = The inertia moment of columns, rigidly connected to the considered point, lying in the plane of buckling of the column considered (cm4),

Ig = The inertia moment of beams, rigidly connected to the considered point, lying in the plane of buckling of the column considered (cm4),

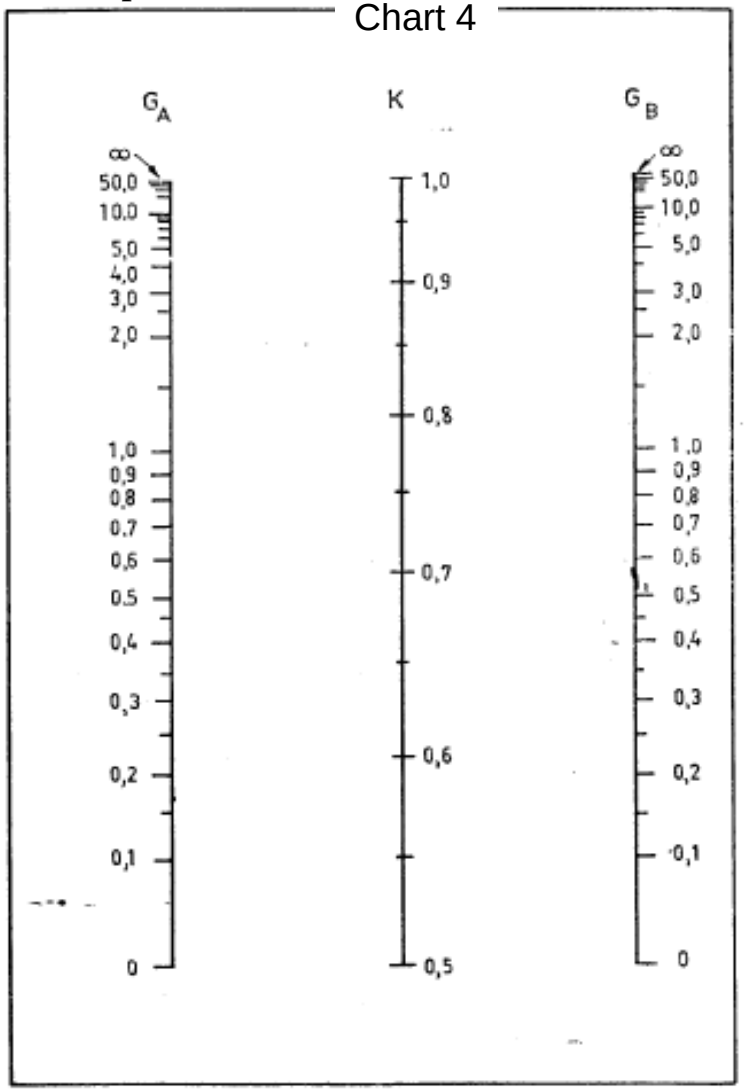
Sc = The length of the columns rigidly connected to the considered point (cm),

Sg = The length of the beams rigidly connected to the considered point (cm). If the column is rigidly connected to foundation, G = 1,0 is taken at that point. If the column is connected by a hinge, G = 10,0 is taken.

After the coefficients GA and GB at column ends are determined, K values is obtained from Charts 4-5. The effective length is obtained as $S_k = K.s$

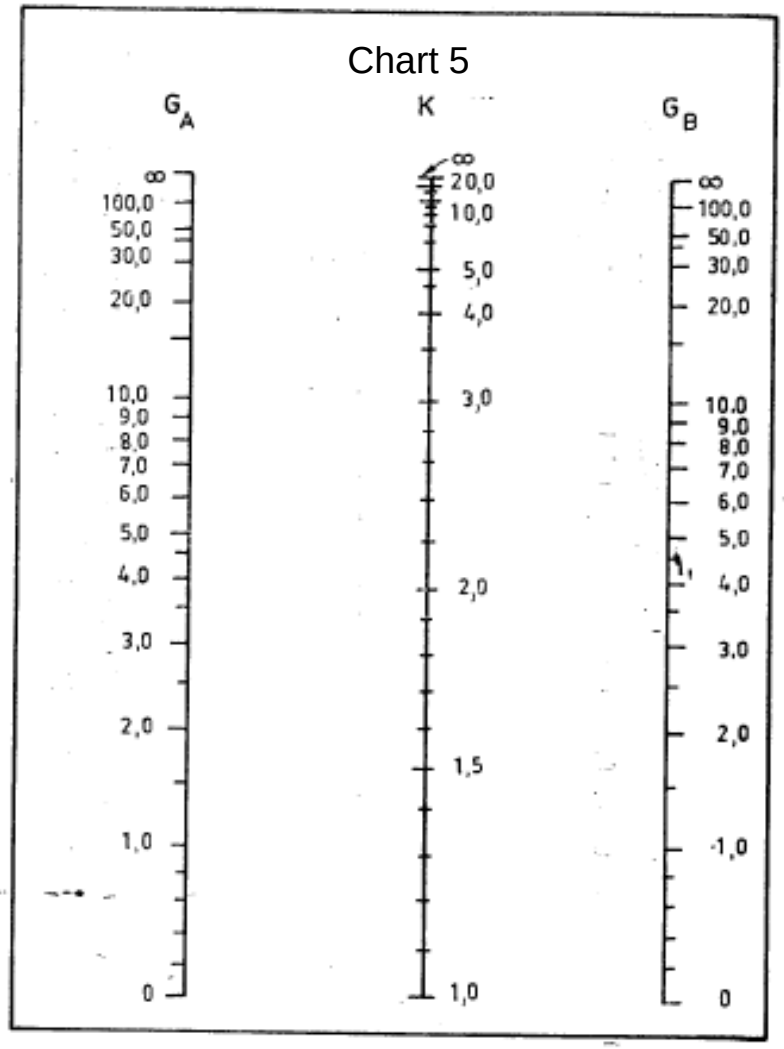
Nomograms for K value that using in the determination of the effective length of columns whose sideway is prevented and permitted.

Chart 4

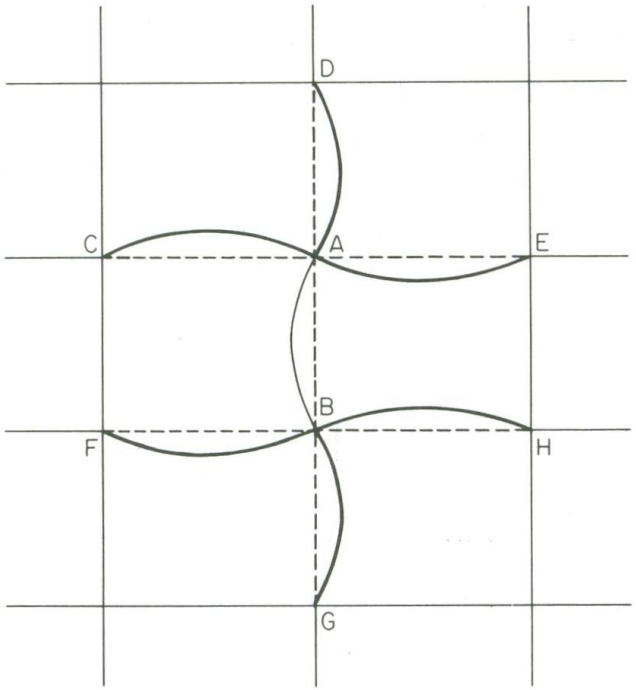


Sideway prevented

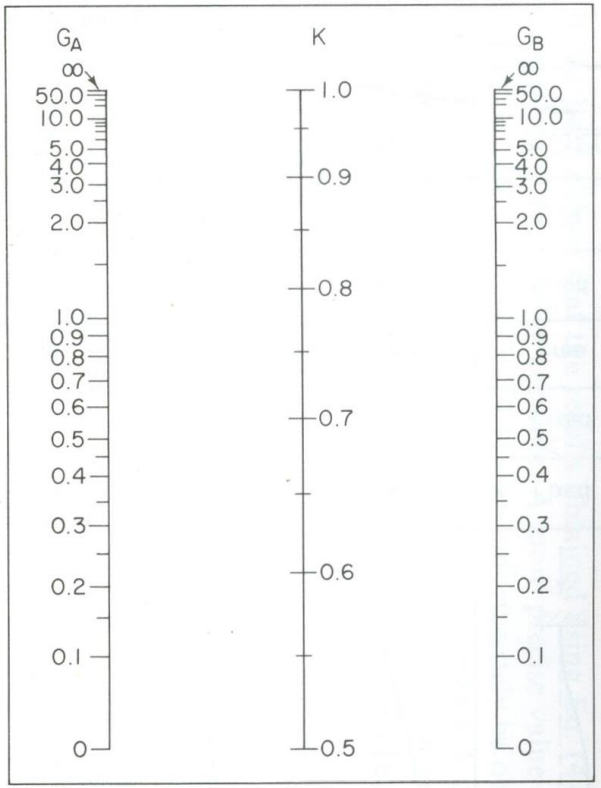
Chart 5



Sideway permitted



(a) Framed column and subassemblage of adjacent members assumed to resist its deformation



(b) Jackson and Moreland nomograph for effective length of framed columns

Fig. 4-8 Julian and Lawrence method for estimating effective length of framed column. (Adapted from Ref. 4.9.)

stiffnesses of the adjacent members. The latter are accounted for by means of the parameters

$$G_A = \frac{I_{AB}/L_{AB} + I_{AD}/L_{AD}}{I_{CA}/L_{CA} + I_{AE}/L_{AE}} \tag{4.45}$$

$$G_B = \frac{I_{AB}/L_{AB} + I_{BG}/L_{BG}}{I_{FB}/L_{FB} + I_{BH}/L_{BH}} \tag{4.46}$$

in which I and L are the moment of inertia and unbraced length of the members lying in the plane of buckling of the column being considered. The nomograph shown in the figure is for a laterally braced frame. A similar

1.6 Limitations of Euler's Formula

All the stability formulas which assumes that the material is elastic and obeys Hooke's law are limited. Because the structural materials can have these properties only within certain limitations. Let's consider Eq. (1.32) for defining the limitation of the Euler's formula:

$$P_k = \frac{\pi^2 EI}{l_b^2} \qquad \textbf{(1.32)}$$

E : Young modulus of the material which we assume that obeys Hooke' law.

I : Least moment of inertia

L_b: Effective (buckling) length of the column

A : Cross section area

λ : Slenderness ratio

Let's do the calculations in terms of critical stresses instead of critical loads. Critical stress is as follows;

$$\sigma_k = \frac{P_k}{A} = \frac{\pi^2 EI}{Al_b^2}$$

$I / A = i^2$ **is the square of the radius of gyration. In this case;**

$$\sigma_k = \frac{\pi^2 E}{(l_b / i)^2}, \lambda = l_b / i \Rightarrow \sigma_k = \frac{\pi^2 E}{\lambda^2} \qquad \textbf{(1.33)}$$

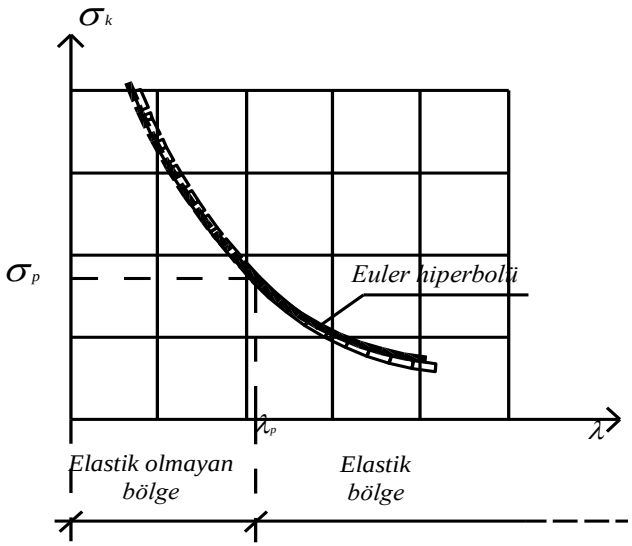


Fig. 1.17

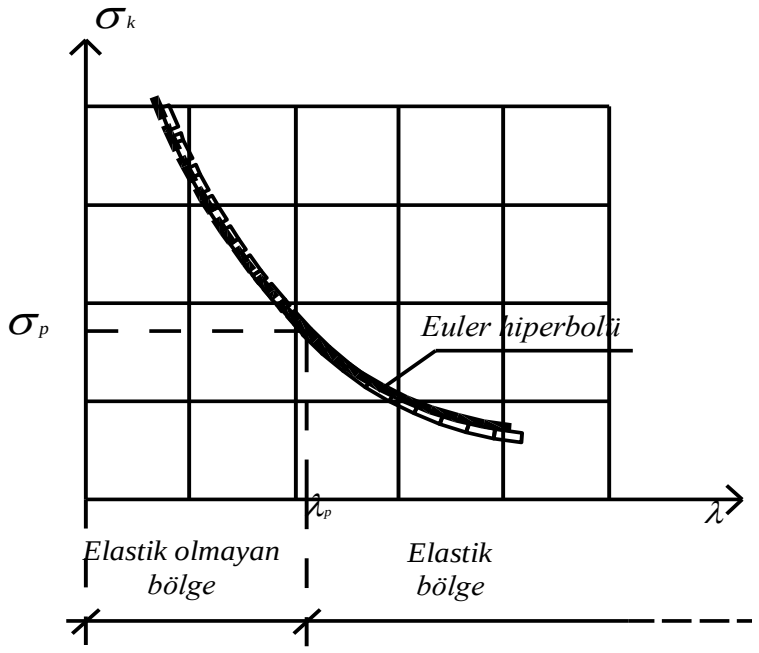
Critical stress is inversely proportional with the square of slenderness ratio. This (σ_k, λ) relation is given by Fig. 1.17 for structural steeel. This formula is valid only when the material is elastic. $\sigma_k = \frac{\pi^2 E}{\lambda^2}$

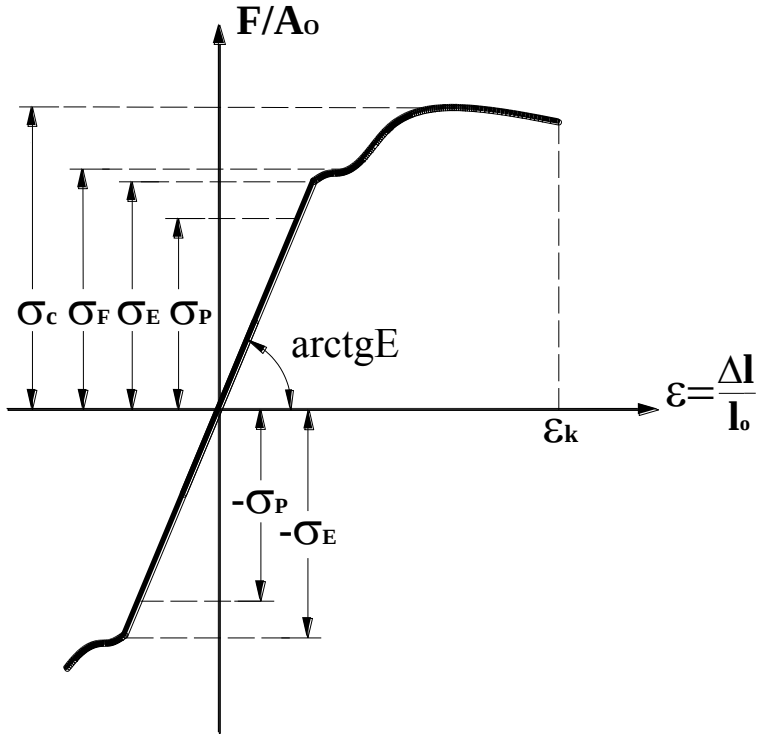
Because outside of the elastic region, the elasticity module changes. If the proportionality limit of the material is shown by σ_p, then Eq. (1.33) is only valid when $\sigma_k < \sigma_p$. Therefore Euler hyperbola seen in Fig. 1.17 is not valid in the region when $\sigma_k > \sigma_p$ and because of this it is shown by a dashed line.

If we know the elastic limit σ_p , then we can define the slenderness ratio λp which defines the region in which the Euler's formula is no more valid;

$$\lambda_p = \pi\sqrt{(E / \sigma_p)} \qquad \textbf{(1.34)}$$

For example, for structural steeel whose proportionality limit is $\sigma_p = 19000$ N/cm^2 , this value is about $\lambda p = 105$ for **$E = 2{,}1.10^7$ N/cm^2 . For $\lambda < \lambda_p$, Euler's formula is no longer valid. Instead of it theoretical or emprical formulas which are obtained on the experiments are used.**


σk
σp
Euler hiperbolü
λp
λ
Elastik olmayan bölge
Elastik bölge


F/Ao
σc σF σE σP
arctgE
ε=Δl/lo
εk
-σP
-σE

The foremost theoretical formula is the tangent modulus formula proposed by Engesser.

Tangent Modulus Formula: The stress-strain relation obtained by tensile test is not linear for the stresses which are greater than the proportionality limit σ_p and in this region the elasticity modulus E which is defined as the tangent of the linear part is no longer valid.

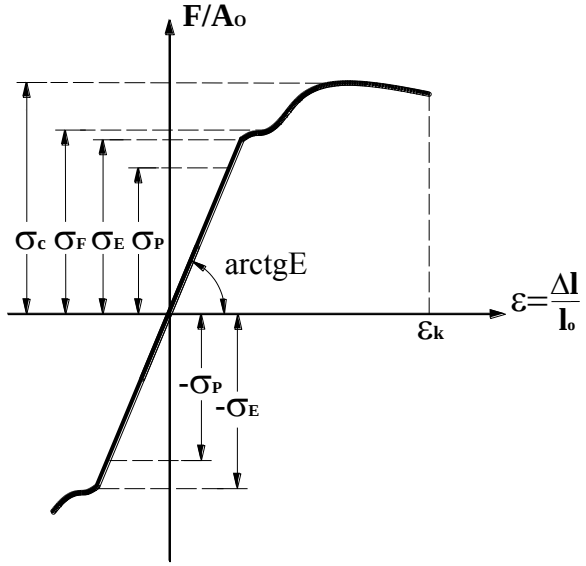
But in this section, for any critical stresses $\sigma_k > \sigma_p$, the tangent at the point which corresponds to the critical stress value, the tangent of this point is called as tangent modulus and defined as follows;

$$\sigma_k > \sigma_p \text{ için } E_t = \frac{d\sigma}{d\varepsilon}$$

This value is assumed as the elasticity modulus of the material for the chosen critical stress and an analysis which is similar to the obtaining the Euler formula gives us the following formula;

$$\sigma_k = \frac{\pi^2 E_t}{\lambda^2}$$

This formula is called as tangent modulus formula.



Experimental Studies:

The formulas which are obtained by using experiments and used in European countries for a long while are given below:

For structural steel: $\lambda < \lambda_p = 105 \rightarrow \sigma_k = 31000 - 114\,\lambda$ N/cm^2

For timber : $\lambda < \lambda_p = 100 \rightarrow \sigma_k = 2930 - 19.4\,\lambda$ N/cm^2

For cast iron : $\lambda < \lambda_p = 80 \rightarrow \sigma_k = 77600 - 1200\,\lambda + 5{,}3\,\lambda^2$ N/cm^2

While doing calculations for axial tensile load, we want to stay a certain distant from the yield stress σ_a for safety considerations. This yield stress for structural steel is**;**

$$\sigma_a = 24000\ N/cm^2$$

And if we take the safety factor as n = 1.71, then the working stress (safety stress) or allowable stress is obtained as

$$\sigma_{em} = 24000/1,71 = 14035\ N/cm^2$$

;

When the column is very short, $(\lambda \Rightarrow 0)$ **there are some methods that consider the critical stress as yield stress of the material. For the steeel ST37, for λ_p = 20, the critical stress is taken as constant namely as a horizontal line. Lets choose an elips curve which is tangent to the horizontal line at λ_p = 20 and tangent to the Euler hyperbola at λ_p = 105 as in Fig. 1.19 and with the conditions given above and taking the proportionality stress as σ_p = 19000 N/cm^2 we get the critical load when the critical stress is greater then the proportionality stress:**

$$20 \le \lambda \le 105 \rightarrow \sigma_k = 1778 + 7,178\sqrt{7515 - (\lambda - 20)^2}\ \ N/cm^2 \qquad \textbf{(1.37)}$$

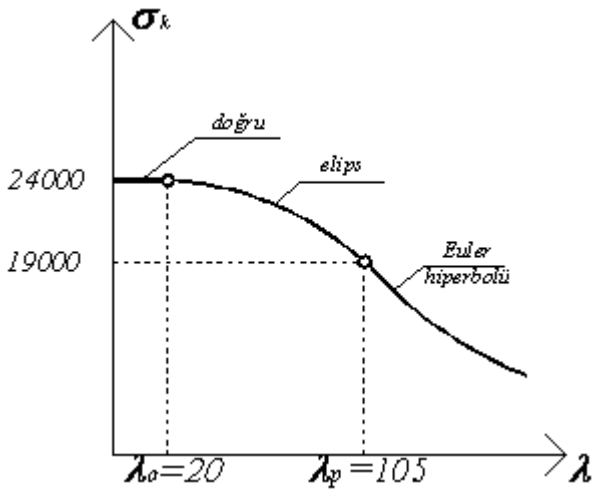
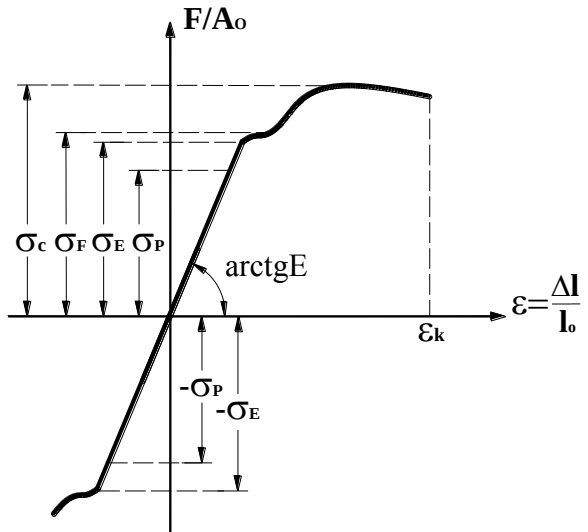


Fig. 1.19. Stress-slenderness ration graph for steel ST37



We use safety factor in buckling calculations: Reasons for the usage of safety factors are given below:

a) Increase in the load which can not be predicted initially

b) Difficulties in loading the column axially. There can be some eccentricity

c) Eccentricity in the column axis

In slender columns, difficulties in exact axial loading and eccentricity of the axis of the columns gains importance. Therefore while the column becomes more slender, it is necessary to increase the safety factor. This factor is generally taken n=2.5 in practice for elastic columns. But in the inelastic region, it is between 1.7 < n < 2.5. In this case;

$$P_{mak} = \frac{P_k}{n} \quad \text{veya} \quad P_{\max} = \frac{\sigma_k A}{n} \qquad \textbf{(1.38)}$$

Practical Procedures in Calculation of Buckling of Columns:

a) Slenderness ratio is found

b) For $\lambda < \lambda_p$, buckling is out of the elastic region (it is inelastic stability), otherwise buckling is in the elastic region.

c) We use Euler formula for elastic region, and a suitable one of Tetmajer or Engesser formula for inelastic regin.

d) Dividing P_k by, *n* we calculate the working load (safety load) as $P_{max} = P_k / n$.

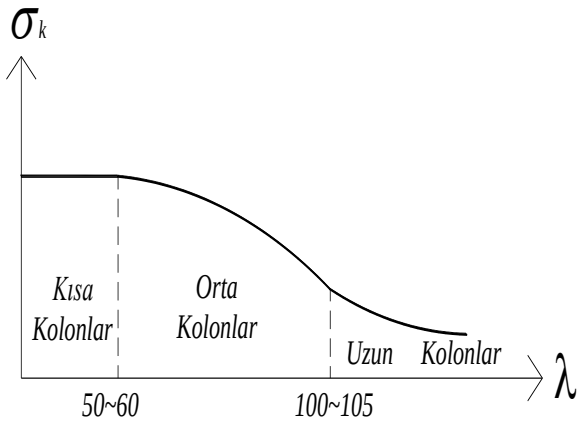


Fig. 1.20

In Fig. 1.20, it is shown in which slenderness ratios we can call the columns as long, medium (intermediate) or short.

In short columns, compression stresses due to axial compression load exceeds the allowable stresses and the column fails because of this. Therefore short columns must be designed for allowable stresses.

2.7 Calculation by Using Buckling Factor ω

For simplifying the bukling calculations and considering the stability problems as if they are stress problems, these method has been improved.

Because a numerical value ω called buckling factor plays the main role in this calculation, this method is called ω method. In the case of tensile stress, the stresses must be less than or equal to the allowable stress:

$$\frac{P}{A} \le \sigma_{allow}$$

If the problem is a compressive one, and if we still want to use the equation used above for tensile stress comparison, we introduce a new factor ω called Buckling Factor, and we can write the following equations:

$$\frac{P}{A} \le \frac{\sigma_c}{n} = \sigma_{allow\,buckling} \quad \frac{\sigma_{allow}}{\sigma_{allow\,buckling}} = \omega \rightarrow \sigma_{allow\,buckling} = \frac{\sigma_{allow}}{\omega} \rightarrow \frac{P}{A} \le \frac{\sigma_{allow}}{\omega} \rightarrow \frac{P\omega}{A} \le \sigma_{allow} \qquad \textbf{(1.39)}$$

Because there is buckling risk in the neutral equilibrium state, the load P must certainly satisfy the following condition:

$$P = \frac{P_c}{n} = \frac{\sigma_c A}{n} \qquad \textbf{(1.40)}$$

By eliminating P between (1.39) and (1.40), we obtain the following expression for ω;

$$\frac{P\omega}{A} = \frac{\sigma_c A \omega}{n\,A} \le \sigma_{allow} \rightarrow \frac{\sigma_c\,\omega}{n} \le \sigma_{allow} \rightarrow \omega = n\frac{\sigma_{allow}}{\sigma_c} \qquad \textbf{(1.41)}$$

Although σ_{allow} are constant, the critical stress σ_c and safety factor n depend on the slenderness ratio and therefore also ω depends on it.

(ω, λ) relation can be shown in a table by using Eq. (1.41). Mostly the safety factor is taken constant in the elastic region. However, in the inelastic region, this factor is reduced with respect to decreasing slenderness ratio. Therefore, while preparing the ω table, lots of points must be determined beforehand:

a) The choice of σ_c,
b) Determination of n,
c) Definition of σ_c formula in the inelastic region

After preparing the table, dividing the safety stress of the case of simple compression or tension by the factor ω which is greater than unity(ω > 1), the greatest average normal stress can be obtained as follows:

$$\sigma_{max} \le \frac{\sigma_{allow}}{\omega} \qquad \textbf{(1.42)}$$

For every value of λ , the factor ω is obtained by using Euler formula when $\lambda > \lambda_p$ and an assumed approximate formula in the case when $\lambda < \lambda_p$. Later by arranging the table using these ω values, definition of the dimensions of cross section in the buckling problem can be converted to a simple compression or tension problem in which the safety stress changes with respect to slenderness ratio λ .

Because we take into account the elastic and inelastic buckling in the first place (initially) in the calculation of ω, there is no need to think about this subject any more, namely the subject that whether the buckling is elastic or not.

As an example, ω table is prepared for St37 steel by using Eq. (1.37) in the inelastic region and the Euler formula in the elastic region for the parameters given below:

$$\sigma_F = 24000\ N/cm^2,\ \sigma_P = 19000\ N/cm^2,\ \sigma_{em} = 14000\ N/E = 2{,}1.10^7\ N/cm^2\ ve\ \lambda_p = 105$$

λ	0	10	20	30	40	50	60	70	80	90	100	110	120	130	140	150
ω	1.00	1.00	1.00	1.05	1.10	1.16	1.23	1.30	1.38	1.49	1.64	1.96	2.34	2.74	3.18	3.60

For $20 \le \lambda_p \le 105$ the safety factor is changed between $1.71 \le n \le 2.4$

$$20 \le \lambda \le 105 \rightarrow \sigma_c = 17780 + 71{,}78\sqrt{7515 - (\lambda - 20)^2}\ \ N/cm^2 \qquad \textbf{(1.37)}$$

As an example, let's calculate the bucklin factor ω for the slenderness ratio λ =80. When the slenderness ratio is less than 105, we are in the inelastic region:

$$\left.\begin{array}{ll} 20 & 1.71 \\ 80 & n \\ 105 & 2.4 \end{array}\right\} \frac{105-20}{2.4-1.71} = \frac{80-20}{n-1.71} \rightarrow 123.188 = \frac{60}{n-1.71} \rightarrow 123.188\,n = 270.652 \rightarrow n = 2.197$$

$$\omega = 2.197 \times \frac{14000}{17780 + 71{,}78\sqrt{7515 - (80-20)^2}} = 1.381$$

1.8 Problems related with buckling

Problem 1.1:

$$\lambda = \lambda_p = 100$$

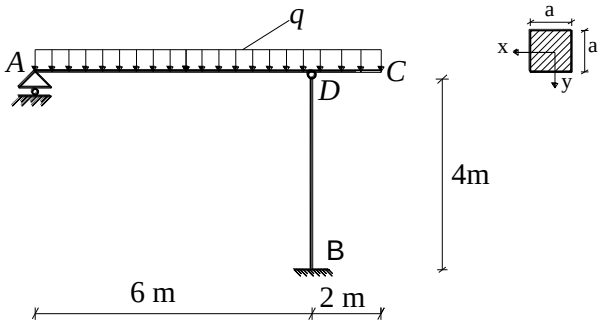


Fig. 1.21

a) Design the cross section of the column DB shown in Fig. 1.21 as square section.

b) Find the maximum working load that the column can bear by taking the safety factor as n=2.

c) What is the uniformly distributed load on the beam which causes to this load?

$$\lambda_p = 100,\ \sigma = \frac{\pi^2 E}{\lambda^2},\ E = 2{,}1.10^7\ \text{N/cm}^2\ \ \lambda = \frac{l_b}{i}\ \ i_x^{\,2} = \frac{I_x}{A}$$

Solution 1.1:

a) $$i_x^{\,2} = \frac{a\,a^3}{12a^2} = \frac{a^2}{12} \Rightarrow i = \frac{a}{2\sqrt{3}},\ l_b = 2l = 2.400 = 800\,cm$$

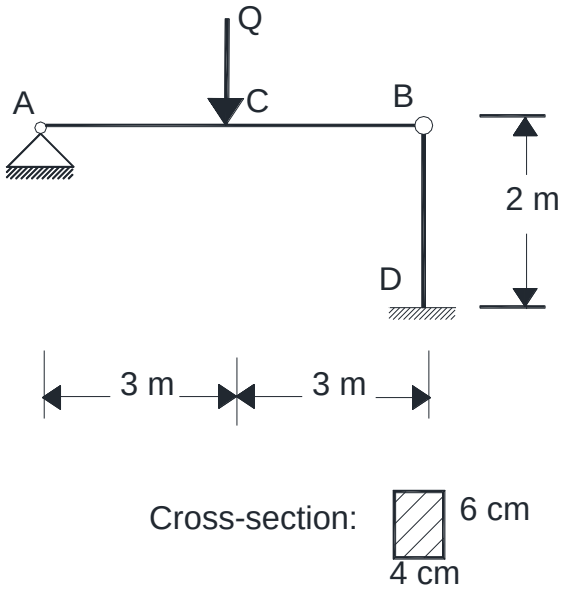
$$\lambda = \frac{l_b}{i} \Rightarrow 100 = \frac{2\sqrt{3}.800}{a} \Rightarrow \text{a} = 27{,}71\ \text{cm}$$

b) $$\sigma_k = \frac{\pi^2.2{,}1.10^7}{100^2} = 20726{,}17\ \text{N/cm}^2 \Rightarrow \text{P}_k = \sigma_k \text{A} = 20726{,}17.27{,}71^2 = 15914{,}47\ k\text{N}$$

$$P_{em} = \frac{P_k}{n} = \frac{15914{,}47}{2} = 7957{,}24\ \text{kN}$$

c) $$\sum \text{M}_\text{A} = 0 \quad 7957{,}24.6 - 8.\text{q}.4 = 0 \rightarrow \text{q} = 1491{,}98\ \text{kN/m}$$

Problem: For the system shown in the figure find the maximum allowable Q load which can be carried with a safety factor of n=2. , $\lambda_p = 105$, $E = 2.1x10^7\ N/cm^2$, $\lambda = \frac{l_b}{i_{\min}}$, $\lambda > \lambda_p \rightarrow \sigma_{cr} = \pi^2 E / \lambda^2$, $P_{allowable} = \frac{\pi^2 EI}{n\, l_b^2}$



Solution:

$$l_B = 0,7.200 = 140\ \text{cm}$$

$$I_{\min} = \frac{6.4^3}{12} = 32\ \text{cm}^4$$

$$i_{\min} = \sqrt{\frac{32}{24}} = 1,155\ \text{cm}$$

$$\lambda = \frac{140}{1,155} = 121 > \lambda_p = 105$$

$$P_{allowable} = \frac{\pi^2 2,1.10^7.32}{2.140^2} = 169021,7\ \text{N}$$

Problem 1.2:

Find the critical load P_c for the timber column given in the right side.

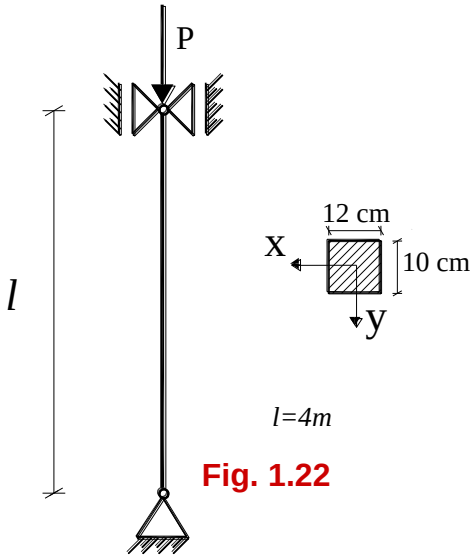


Fig. 1.22

$\lambda_p = 100,\ E = 10^6\ N/cm^2,\ n = 1$

$I_x = \frac{12.10^3}{12} = 1000\,cm^4,\ I_y = \frac{10.12^3}{12} = 1440\ cm^4 \qquad I_{\min} = I_x = 1000\ cm^4\ \ A = 12.10 = 120\ cm^2$

$i_{\min} = \sqrt{I_{\min}/A} = \sqrt{1000/120} \Rightarrow i_{\min} = 2,89\ cm,\ l_B = 1.400$ (Effective length or buckling length)

Slenderness ratio; $\lambda = l_B / i_{min} \Rightarrow \lambda = 400/2,89 = 138,41\ > \lambda_p = 100$

$\lambda > \lambda_p$ Therefore the buckling is in the elastic region. Critical load can be obtained by using Euler formula.

$P_k = \frac{\pi^2 EI}{nl_b^2} = \frac{\pi^2.10^6.1000}{1.400^2} \Rightarrow P_k = 616850N$ or $P_k = \frac{\pi^2 E}{n\lambda^2}A = \frac{\pi^2 \times 10^6}{1\times(138,41)^2}\times 12\times 10 \Rightarrow P_k = 616886,47N$

If it were [ ] the buckling would be in the inelastic region. Then we would use Tetmajer,

$\sigma_k = 2930 - 19,4\lambda$ or another suitable formula for the inelastic buckling.

Problem 1.3:

Find the load P for the safety factor n=5 for the column given in the right side.

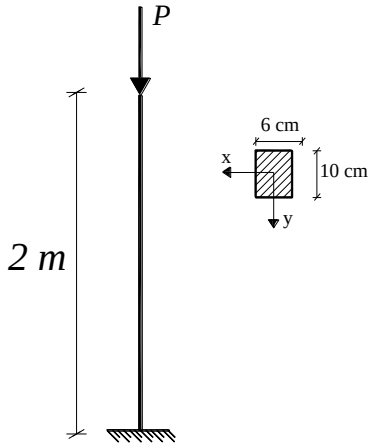


Fig. 1.23

$$\lambda_p = 100,\ E = 1.10^6\ N/cm^2$$

$$I_x = \frac{6.10^3}{12} = 500 cm^4 \quad I_y = \frac{10.6^3}{12} = 180\ cm^4 \qquad I_{\min} = I_y = 180 cm^4 \qquad A = 6.10 = 60\ cm^2$$

$$i_{\min} = \sqrt{\frac{I_{\min}}{A}} = \sqrt{\frac{18}{60}} = 1,73\ cm \qquad l_b = 2.200 = 400 cm$$

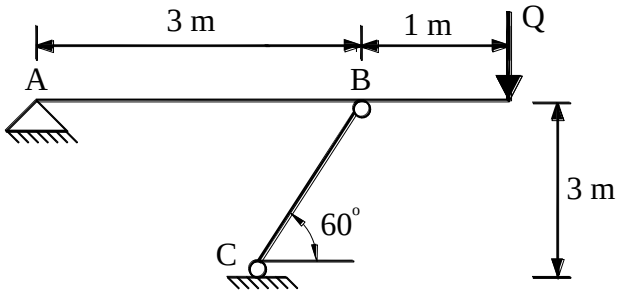
Buckling is in the elastic region. Euler formula is used.

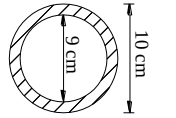
$$\lambda = l_B / i_{\min} = 400/1.73 = 231,2 > \lambda_p = 100 \Rightarrow \quad P_k = \frac{\pi^2 E I_{\min}}{n l_b^2} = 2220,66\ N$$

Problem: In the system shown below, BC bar is constructed as a hollow cylinder with an outer and inner diameter of 10cm and 9cm respectively. Find the Q force which can be carried by the system with a safety factor of n=3 with respect to BC bar. $E = 21000\,kN/cm^2$, $\lambda_p = 105$

$$\lambda = \frac{l_b}{i_{\min}},\ \lambda > \lambda_p \rightarrow \sigma_{cr} = \pi^2 E / \lambda^2,\qquad \lambda \le \lambda_p \rightarrow \sigma_c = 31000 - 114\lambda$$

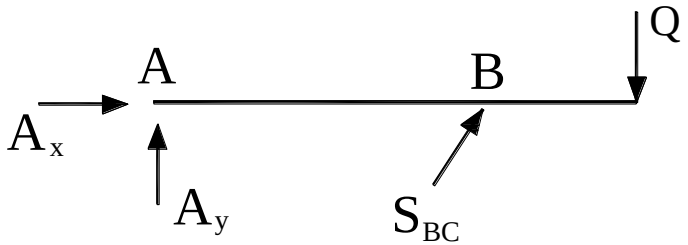
$$P_{allowable} = \frac{\sigma_c A}{n}$$





BC çubuğunun kesiti

Solution:



$$\sum M_A = 0 \rightarrow S_{BC}\sin 60.3 - Q.4 = 0 \rightarrow S_{BC} = 1.54Q$$

$$I_{\min} = \frac{\pi(10^4 - 9^4)}{64} = 168,73\,cm^4,\quad A = \frac{\pi(10^2 - 9^2)}{4} = 14,915\,cm^2,\quad i_{\min} = \sqrt{\frac{168,73}{14,915}} = 3,36\,cm$$

$$\lambda = \frac{346,4}{3,36} = 103 < \lambda_p = 105 \rightarrow \sigma_{kr} = 31000 - 114(103) = 19230\,N/cm^2$$

$$P_{max} = \frac{19,23.14,915}{3} = 95,61\,kN = 1,54Q \rightarrow Q = 62,1\,kN$$

Problem 1.4:

Design cross sections of the elements BC and BD of the system as circular sections given in Fig. 1.24 in the right side.

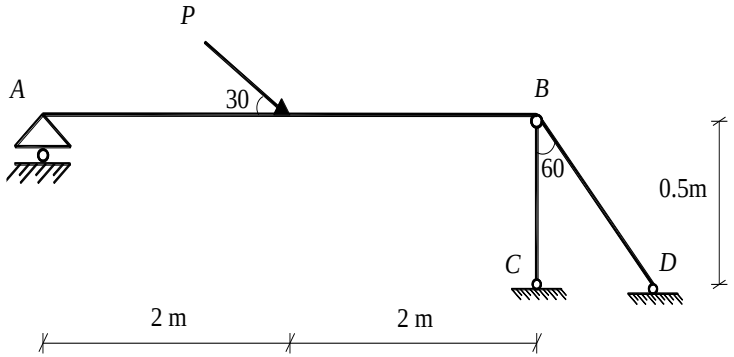


Fig. 1.24

$n=2,\ \lambda_p=105,\ E=2{,}1.10^7\ N/cm^2,\ \lambda<\lambda_p \Rightarrow \sigma_k=31000-114\lambda$

$\sigma_{allow}=14000\ N/cm^2 \quad P=100000N$

$\sum F_z=0 \Rightarrow -H_B+100000.\cos(30)=0 \Rightarrow H_B=86602.5\,N$

$\sum M_A=0 \Rightarrow -V_B.4+100000.\sin(30).2=0 \Rightarrow V_B=25000\,N$

$S_{BD}.\sin 60+86602{,}5=0 \Rightarrow S_{BD}=-100000\ N$ Compression

$S_{BC}+100000.\cos 60+25000=0 \Leftarrow S_{BC}=25000\ N$ Tension

Stress control for BD $\quad \sigma_{allow}=\dfrac{100000}{(\pi d^2/4)}\le 14000,\ d=3{,}02\,cm$

Stress control for BC $\quad \sigma_{allow}=\dfrac{25000}{(\pi d^2/4)}\le 14000,\ d=1{,}51\,cm$

Buckling control for BD $\quad i=\sqrt{\dfrac{I}{A}}=\sqrt{\dfrac{\pi d^4/64}{\pi d^2/4}}=\dfrac{d}{4},\ L_B=1\,m=100\ cm$

$\lambda=\dfrac{L_B}{i_{\min}}=\dfrac{100}{d/4}=\dfrac{400}{d}$ Lets assume $\lambda=l_b/i_{\min}=100/(d/4)=400/d$

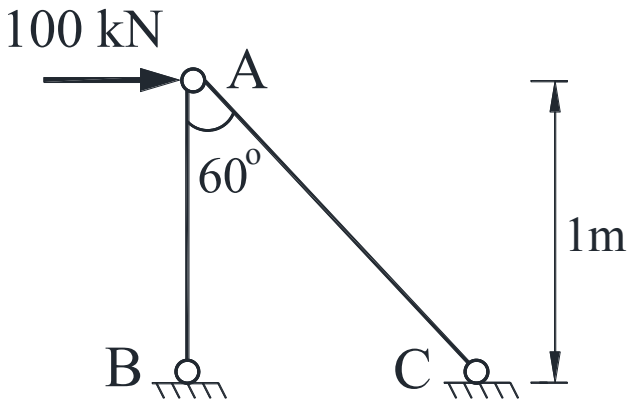
$\lambda>\lambda_p=105;\ P_{allowable}=\dfrac{\pi^2 EI}{n\,l_B^{\,2}}=100000 \Rightarrow \dfrac{\pi^2.2{,}1.10^7.I}{2.100^2} \Rightarrow I=9{,}65\ cm^4$

$\dfrac{\pi d^4}{64}=9{,}65 \Rightarrow d=3{,}75\ cm,\ \lambda=400/3{,}75=106{,}7>\lambda_p$ $\quad d = 3.75\ cm$

There is no need to buckling control analysis for BC. Because BC is under the tension effect.

Problem

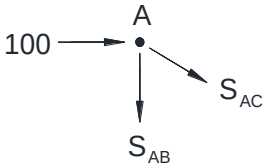
4) Determine the dimensions of the AB and AC structural members shown in the figure assuming circular cross-sections for both of them. $\lambda_p = 105$,



$n = 2$, $E = 2 \cdot 1 \times 10^7\ N/cm^2$, $\sigma_{safety} = 14000\ N/cm^2$,

$\lambda < \lambda_p \rightarrow \sigma_{cr} = 31000 - 114\lambda$, $\quad I_x = I_y = \dfrac{\pi r^4}{4}$

Solution



$S_{AC} = -100 / \sin 60 = -115,4\,kN\ (compression)$

$S_{AB} = -S_{AC}.\sin 60 = +57,73\,kN\ (tension)$

at AB member, $S_{AB} = \dfrac{57,74.10^3}{\pi r_{AB}^2} \le 14000 \rightarrow r_{AB} \ge 1,145\,cm$

at AC member, $S_{AC} = \dfrac{57,74.10^3}{\pi r_{AC}^2} \le 14000 \rightarrow r_{AC} \ge 1,62\,cm,$

$P = \dfrac{\pi^2 E I}{n L_b^2} \rightarrow I = \dfrac{n P L_b^2}{\pi^2 E}$, $L_b = 200cm$, $I = \dfrac{2.115,47.10^3.200^2}{\pi^2\, 2,1.10\pi^7} = 44,54\,cm^4$

$\dfrac{\pi r_{AC}^4}{4} = 44,54 \rightarrow r_{AC} \ge 2,74\,cm$

$A = \pi(2,74)^2 = 23,54cm^2$, $i_{\min} = \sqrt{44,54/23,54} = 1,37cm$, $\lambda = \dfrac{200}{1,37} = 145 > \lambda_p = 105$ (assumption valid)

$r_{AB} = 1,145cm, \quad r_{AC} = 2,74\,cm$

Problem 1.5:

Design the cross section of the bars which are under the compression force among the ones S1, S2, S3 as square cross section. Get the safety $E = 2{,}1.10^7\ N/cm^2\ \ \lambda_p = 105$

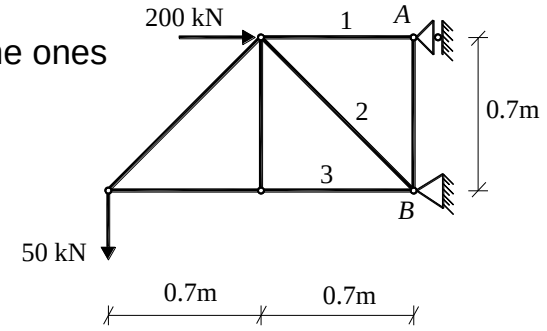


Fig. 1.25

$$\lambda < \lambda_p \ \Rightarrow\ \sigma_k = 31000 - 114\lambda\ \left(N/cm^2\right) \qquad \lambda > \lambda_p \ \Rightarrow\ \sigma_k = \frac{\pi^2 EI}{\lambda^2}$$

$$\sum M_B = 0 \ \Rightarrow\ 50.1{,}4 - 200.0{,}7 + H_A.0{,}7 = 0 \ \rightarrow\ \mathrm{H}_A = 100\ \mathrm{kN},\ \mathrm{H}_B = 100\ \mathrm{kN},\ \mathrm{V}_B = 50kN$$

$S_1 = -100\,kN$ Compression

$$S_2 = \frac{V_B}{\sin 45} = \frac{50}{\sin 45} = -70{,}71kN$$ Compression

$$S_3 = H_B - 70{,}71.\cos 45 = 100 - 70{,}71.\cos 45 = -50\,kN$$ Compression

$$I = \frac{a^4}{12},\ A = a^2,\ i = \sqrt{\frac{I}{A}} = \sqrt{\frac{a^4/12}{a^2}} = \frac{a}{2\sqrt{3}},\ l_B = l \qquad l_1 = 70\,cm,\ l_2 = 99\ cm,\ l_3 = 70\ cm \qquad \lambda = \frac{l_B}{\mathrm{i}_{\min}} \Rightarrow\ \lambda_1 = \frac{242{,}49}{a},\ \lambda_2 = \frac{342{,}95}{a},\ \lambda_3 = \frac{242{,}49}{a}$$

Let's investigate the bars one by one:

For S_1 we assume that: $\lambda > \lambda_p$ $\quad \mathrm{P} = \frac{\pi^2 \mathrm{E\,A}}{\mathrm{n}\lambda^2} = 100000 \Rightarrow \frac{\pi^2.2{,}1.10^7.\mathrm{a}^2}{2.\left(242{,}49/\mathrm{a}\right)^2} \Rightarrow\ \mathrm{a} = 2{,}74\ \mathrm{cm}$ $\quad a = 2{,}74\ cm$ için $\Rightarrow\ \lambda = \frac{242{,}49}{2{,}74} = 88{,}5 < \lambda_p = 105$

We must repeat the calculations for S_1 for the inelastic regin.

$$\frac{nP}{A} = 31000 - 114\lambda \Rightarrow \frac{2.100000}{a^2} = 31000 - 114\frac{242{,}49}{a} \qquad 31000\,a^2 - 27643{,}86\,a - 200000 = 0 \Rightarrow\ a = 3{,}02\ cm$$

For S_2 let's assume $\lambda > \lambda_p$ $\quad 70710 = \frac{\pi^2.2{,}1.10^7.a^2}{2.\left(342{,}95/a\right)^2} \Rightarrow\ a = 2{,}99\ cm\ \Rightarrow\ \lambda = 114{,}58\ > \lambda_p$

For S_3 let's assume $\lambda > \lambda_p$ $\quad 50000 = \frac{\pi^2.2{,}1.10^7.a^2}{2.\left(242{,}49/a\right)^2} \Rightarrow\ a = 2{,}31\ cm\ \Rightarrow\ \lambda = 104{,}97 < \lambda_p$

But we may assume that $\lambda \cong \lambda_p$

Problem 1.6:

Find the working load P for the safety factor n=3 for the timber column of length *l*=100cm given in the right side.

$$E = 10^6\ N/cm^2,\ \lambda_p = 100,\ \lambda < \lambda_p \Rightarrow \sigma_k = 2930 - 19{,}4\,\lambda\ \left(N/cm^2\right),\ n = 3$$

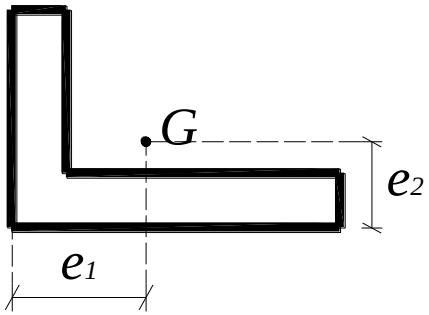
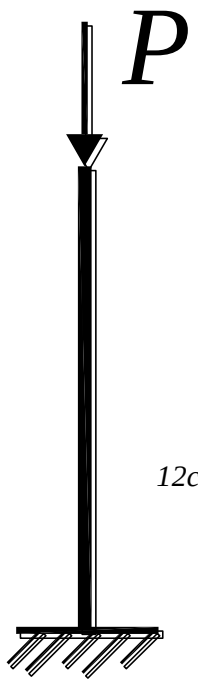


Fig. 1.27

$$e_1 = 6{,}5\ cm, \quad e_2 = 3{,}5\ cm$$

$$I_x = 870{,}75\,cm^4,\ I_y = 2490{,}75\ cm^4,\ I_{xy} = -810\ cm^4$$



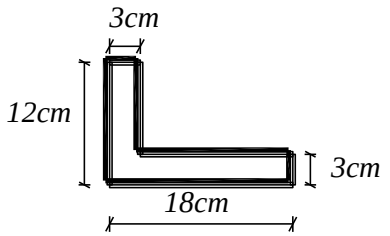


Fig. 1.26

$$I_{\min} = \frac{I_x + I_y}{2} - \sqrt{\left(\frac{I_x - I_y}{2}\right)^2 + I_{xy}^{\ 2}} = \frac{870{,}75 + 2490{,}75}{2} - \sqrt{\left(\frac{870{,}75 - 2490{,}75}{2}\right)^2 + (-810)^2} = 535{,}25\ cm^4$$

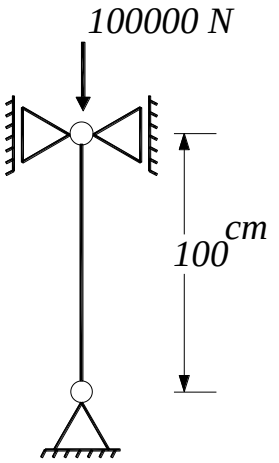
$$A = 12.3 + 15.3 = 81 cm^2 \qquad l_B = 2.100 = 200\,cm$$

$$i_{\min} = \sqrt{\frac{I_{\min}}{A}} = \sqrt{\frac{535{,}25}{81}} = 2{,}57\ cm \qquad \lambda = \frac{l_B}{i_{\min}} = \frac{200}{2{,}57} = 77{,}82 < \lambda_p = 100$$

$$\sigma_k = 2930 - 19{,}4.77{,}82 = 1420{,}29\ N/cm^2 \qquad P = (\sigma_k\ A)/n = \frac{1420{,}29.81}{3} = 38347{,}83\,N$$

Problem 7:

Design the cross section of the column as circular under the given loading by using the Omega method.



Solution:

First the minumum area which is necessary in terms of axial normal force is obtained:

$$100000 = 14000A \rightarrow A = 7{,}143\ cm^2 \rightarrow D = \sqrt{\frac{4A}{\pi}} = \sqrt{\frac{4.(7{,}143)}{3{,}141}} = 3{,}0165\ cm; \qquad l_b = 100\ cm$$

$$i = \sqrt{\frac{I}{A}} = \sqrt{\frac{\pi D^4/64}{\pi D^2/4}} = \frac{D}{4} \rightarrow i = \frac{3{,}0165}{4} = 0{,}754\ cm \quad \lambda = \frac{l_b}{i} = \frac{100}{0{,}754} = \frac{100}{D/4} = 132{,}605$$

$$\left.\begin{array}{ll} 130 & 2{,}74 \\ 132{,}605 & \omega \\ 140 & 3{,}18 \end{array}\right\} \frac{3{,}18 - 2{,}74}{140 - 130} = \frac{\omega - 2{,}74}{132{,}605 - 130} \rightarrow \omega = 2{,}85462 \rightarrow \frac{\omega P}{A} \le \sigma_{allow} \rightarrow \frac{2{,}855.100000}{A} \le 14000 \rightarrow A = 20{,}393\ cm^2 \rightarrow D = 5{,}097\ cm$$

$$D = 5{,}097 \rightarrow \lambda = \frac{400}{5{,}097} = 78{,}478 \rightarrow \textit{from}\ \text{interpolation} \quad \omega = 1{,}368 \rightarrow \frac{\omega P}{A} \le \sigma_{allow} \rightarrow \frac{(1{,}368).100000}{A} \le 14000 \rightarrow A = 9{,}77\ cm^2 \rightarrow D = 3{,}523$$

It is clear that D is between 3,523 ile 5,097 cm. Let's approximate it by

$$D = (3{,}523 + 5{,}097)/2 = 4{,}31\ cm \quad \lambda = \frac{400}{4{,}31} = 92{,}807 \rightarrow \text{int.} \quad \omega = 1{,}532 \quad \frac{\omega P}{A} \le \sigma_{allow} \rightarrow \frac{(1{,}532).100000}{A} \le 14000 \rightarrow A = 10{,}944\ cm^2 \rightarrow D = 3{,}733\ cm$$

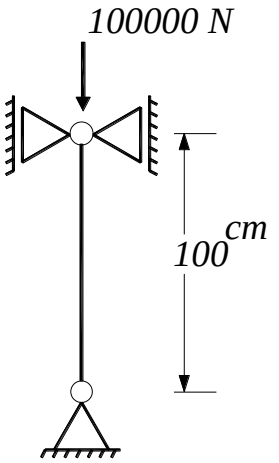
$$D = (3{,}733 + 4{,}31)/2 = 4{,}021\ cm \quad D = 4{,}021 \rightarrow \lambda = \frac{400}{4{,}021} = 99{,}47 \rightarrow \text{int.} \quad \omega = 1{,}632 \quad \frac{\omega P}{A} \le \sigma_{allow} \rightarrow \frac{(1{,}632).100000}{A} \le 14000 \rightarrow A = 11{,}6575\ cm^2 \rightarrow D = 3{,}853\ cm$$

$$D = (3{,}853 + 4{,}021)/2 = 3{,}937\ cm \quad D = 3{,}937 \rightarrow \lambda = \frac{400}{3{,}937} = 101{,}6 \rightarrow \text{int.} \quad \omega = 1{,}6912 \quad \frac{\omega P}{A} \le \sigma_{allow} \rightarrow \frac{(1{,}6912).100000}{A} \le 14000 \rightarrow A = 12{,}08\ cm^2 \rightarrow D = 3{,}922$$

$$D = (3{,}922 + 3{,}937)/2 = 3{,}9295\ cm \quad \lambda = \frac{400}{3{,}9295} = 101{,}794 \rightarrow \text{int.} \quad \omega = 1{,}69741 \quad \frac{\omega P}{A} \le \sigma_{allow} \rightarrow \frac{(1{,}69741).100000}{A} \le 14000 \rightarrow A = 12{,}124\ cm^2 \rightarrow D = 3{,}9295\ cm$$

Problem :

$D = 3,9295\ cm$



Inverse problem: Checking the design.

Boundary conditions, the geometry of the cross section and the loading are given.

It is asked whether or not the column is safe.

Solution:

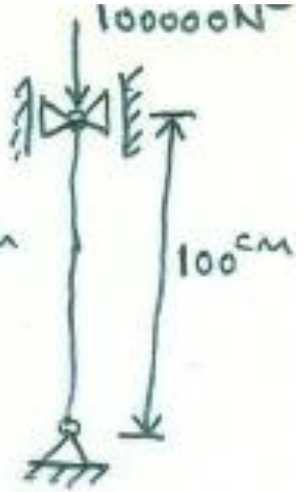
$l_b = 100\,cm$

$$i = \sqrt{\frac{I}{A}} = \sqrt{\frac{\pi D^4/64}{\pi D^2/4}} = \frac{D}{4} \rightarrow i = \frac{3,9295}{4} = 0,9824\ cm \quad \lambda = \frac{l_b}{i} = \frac{100}{0,9824} = \frac{100}{D/4} = 101,792$$

$$\left.\begin{array}{ll} 100 & 1,64 \\ 101,792 & \omega \\ 110 & 1,96 \end{array}\right\} \frac{1,96-1,64}{110-100} = \frac{\omega-1,64}{101,792-100} \rightarrow \omega = 1,697344 \rightarrow \frac{1,697344 \times 100000}{12,124} \le \sigma_{allow} \rightarrow \frac{2,855.100000}{A} = 13999.87 \le 14000$$

λ	0	10	20	30	40	50	60	70	80	90	100	110	120	130	140	150
ω	1.00	1.00	1.00	1.05	1.10	1.16	1.23	1.30	1.38	1.49	1.64	1.96	2.34	2.74	3.18	3.60

Problem 7: Omega metodunu kullanarak şekilde görülen çubuğu daire kesitli olarak boyutlandırınız.



Çözüm: Eksenel normal kuvvet açısından gerekli minimum alan $100000 = 14000 \cdot A \rightarrow A = 7{,}143\ cm^2 \rightarrow D = \sqrt{\frac{4A}{\pi}}$

$D = \sqrt{\frac{4 \cdot 7{,}143}{3{,}141}} = 3{,}0165\ cm$; $l_b = 100\ cm$

$i = \sqrt{I/A} = \sqrt{\frac{\pi D^4/64}{\pi D^2/4}} = \frac{D}{4} \rightarrow i = \frac{3{,}0165}{4} = 0{,}754\ cm$

$\lambda = \frac{l_b}{i} = \frac{100}{0{,}754} = \frac{100}{D/4} = \frac{400}{D} = 132{,}605$

130	2,74
132,605	w
140	3,18

$\frac{3{,}18 - 2{,}74}{140 - 130} = \frac{w - 2{,}74}{132{,}605 - 130} \rightarrow w = 2{,}85462$

$\frac{w.P}{A} \leq \sigma_{em} \rightarrow \frac{2{,}855 \cdot 100000}{A} \leq 14000 \rightarrow A = 20{,}393\ cm^2 \rightarrow D = 5{,}097\ cm$

$D = 5{,}097 \rightarrow \lambda = \frac{400}{5{,}097} = 78{,}478 \rightarrow$ Interpolasyondan $w = 1{,}368$

$\frac{w.P}{A} \leq \sigma_{em} \rightarrow \frac{1{,}368 \cdot 100000}{A} \leq 14000 \rightarrow A = 9{,}77\ cm^2 \rightarrow D = 3{,}523$

D min 3,523 ile 5,097 cm arasında olacağı açıktır.

Bir yaklaşım olarak $D = (3,523 + 5,097)/2 = 4,31$ cm alınsın.

$D = 4,31 \rightarrow \lambda = \frac{400}{4,31} = 92,807 \rightarrow$ int. $w = 1,532$

$\frac{w.P}{A} \leq \sigma_{em} \rightarrow \frac{1,532 \cdot 100000}{A} \leq 14000 \rightarrow A = 10,944\ cm^2 \rightarrow D = 3,733$

$D = (3,733 + 4,31)/2 = 4,021$ cm alınsın

$D = 4,021 \rightarrow \lambda = \frac{400}{4,021} = 99,47 \rightarrow$ int. $w = 1,632$

$\frac{w.P}{A} \leq \sigma_{em} \rightarrow \frac{1,632 \cdot 100000}{A} \leq 14000 \rightarrow A = 11,6575 \rightarrow D = 3,853$ cm

$D = (3,853 + 4,021)/2 = 3,937$

$D = 3,937 \rightarrow \lambda = \frac{400}{3,937} = 101,6 \rightarrow w = 1,6912$

$\frac{w.P}{A} \leq \sigma_{em} \rightarrow \frac{1,6912 \cdot 100000}{A} \leq 14000 \rightarrow A = 12,08 \rightarrow D = 3,922$ cm

$D = (3,922 + 3,937)/2 = 3,9295$ cm

$D = 3,9295 \rightarrow \lambda = \frac{400}{3,9295} = 101,794 \rightarrow w = 1,69741$

$\frac{w.P}{A} \leq \sigma_{em} \rightarrow \frac{1,69741 \cdot 100000}{A} \leq 14000 \rightarrow A = 12,124 \rightarrow D = 3,929$ cm

Tamam